PROBABILITY AND STOCHASTICS FOR FINANCE

Prof. Joydeep Dutta
Mathematics
IIT Kanpur

INDEX

S. No	Topic	Page No.
	Week 1	
1	Basic Probability	1
2	Interesting Problems In Probability	16
3	Random variables,distribution function and independence	29
4	Chebyshev inequality, Borel-Cantelli Lemmas and related issues	47
	Week 2	
5	Law of Large Number and Central Limit Theorem	61
6	Conditional Expectation-I	72
7	Conditional Expectation-II	86
8	Martingales	97
9	Brownian Motion-I	108
10	Brownian Motion-II	119
	Week 3	
11	Brownian Motion-III	132
12	Ito Integral-I	143
13	Ito Integral-II	156
14	Ito Calculus-I	169
15	Ito Calculus-II	180
	Week 4	
16	Ito Integral In Higher Dimension	190
17	Application to Ito Integral I	199
18	Application to Ito Integral II	209
19	Black Scholes Formula I	222
20	Black Scholes Formula II	234

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 01
Basic Probability

Welcome to this course on probability and stochastics for finance. I will not start by just blabbering some technical things at you. I want to tell you that, everybody nowadays seems to be interested in finance and I do not find the reason, why everybody should be interested in finance. Obviously, people are always interested in making some easy bucks, but let me tell you, it is not so easy, to make easy buck in a financial market. Of course, there are lot of stories around how people like, Warren Buffet has done it or people like x and y and z has done it.

But those are all exceptions and exceptions do not prove the rule, or rather exceptions are proving the rule that it is not so easy to make money in the stock market. Because the financial market is filled with uncertainties and unless you really known the language by which uncertainty is understood by human beings. There is no way, that you can have a better understanding of financial market or whether you actually get expose yourself to the risk of investing in a financial market thus, rather than investing in something like a fixed deposit in a bank.

So, before we do so, before we get into, actual issues of finance, we need to know the language by which uncertainty is quantified, specifically those tools from the theory of probability and stochastics which are actually needed to study finance. Now, without understanding probability, it is almost no chance for anyone to truly understand the intricacies of the financial market, the financial products and how to price them.

So, before we, this is a two-part course, the first part is on probability and stochastics, second part is on the issues of pricing of financial commodities. Now let me, start by talking about probability. What is probability?

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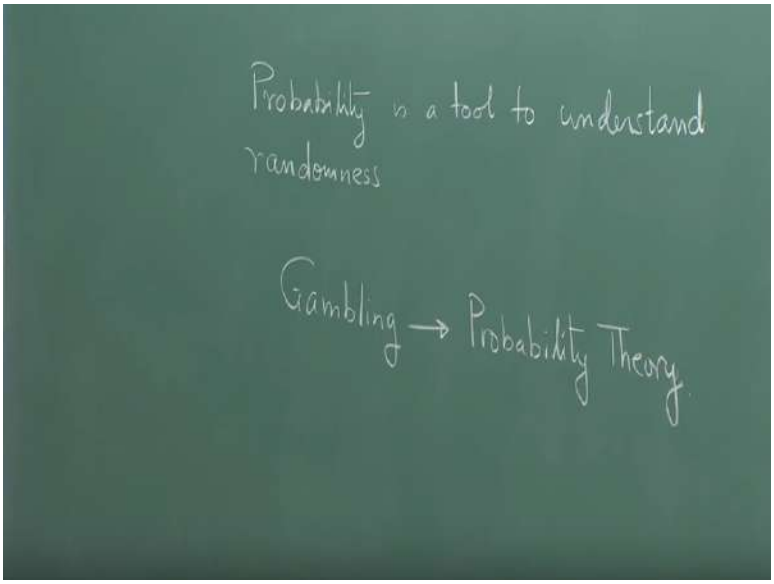

So, probability is a tool to understand randomness. So, I am writing in the board, rather than on the screen, so that as you listen to the lecture it will be advisable for you to makes the notes yourself. So, what is randomness? It is a very difficult thing. It is so difficult to really understand what randomness is. Randomness in a much more crude way, we can say that, something is random if we have no prior knowledge of what outcome would certain activity give.

For example, if you toss a coin, you do not know whether it will come out heads or tails, but you are sure that only that there are only two possibilities if you toss a coin, either it comes on head or comes out to be tail. Now the real mathematical ideas of probability came out by looking at the problems of gambling, so it is gambling to probability theory. So, probability theory did came out of a wise, rather than some virtue.

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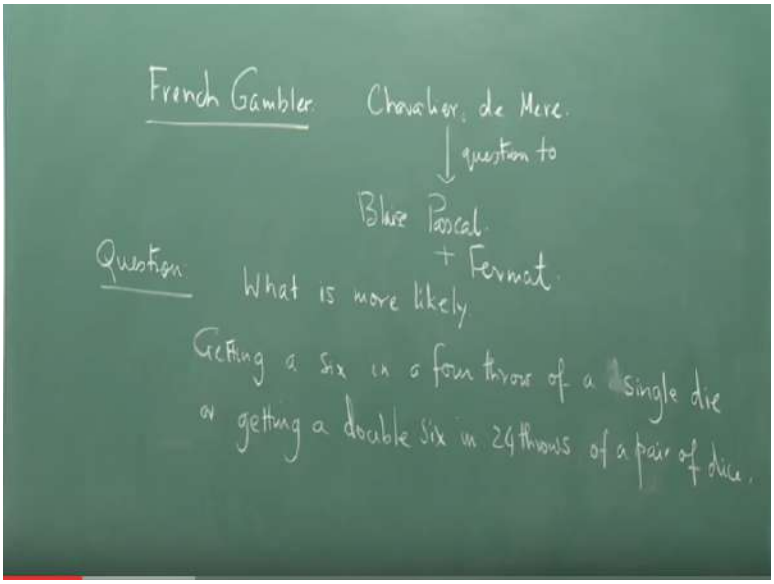

There was a very famous French gambler by the name of Chevalier de Mere. He had a mathematician friend called Blaise Pascal. So, Chevalier posed the question, following question to Blaise Pascal. Most of you have heard of about Pascal's triangle, when you studied binomial theorem. So, he posed the following question to Blaise Pascal. So, what was his question?

So, they were gambling. So, in those days, gambling were done through throwing out dice rather than nowadays the sophisticated rule. So, the question was the following. The question was that, what is more likely, getting 6 in a 4 throw of a single die. So, if you just throw 1 die 4 times, what is the probability of getting 6 or getting a double 6 in 24 throws; of a pair of dice. Okay, so you throw, take 2 dice and throw, we will do the experiments in the next class.

So, which is more likely. So, Blaise Pascal who had denounced mathematics has a bad form of perversion at that time. He had then got back to this question and in between him and Pierre de Fermat, who known for his famous Fermat's Last Theorem, they figured out what finally gave rise to the modern theory of probability. So, it is Blaise Pascal plus Pierre de Fermat. Now we will answer this question. In the next class, we will do the experiments or I will have my TA doing the experiment.

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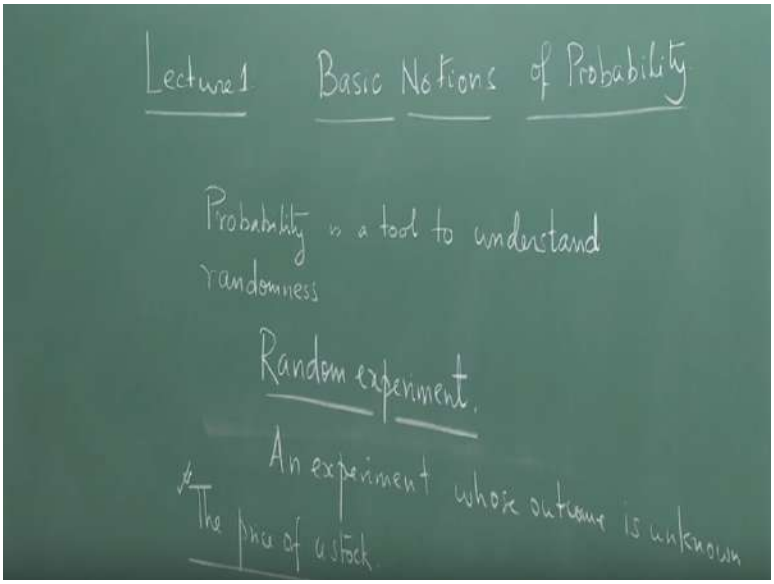

So, in order to more technically study probability, we should first study, what is called a random experiment. So, what is a random experiment? So, random experiment is a, experiment whose outcome is not known to us. Just like, throwing a die or pair of die or whether I will be involved in a car accident in the next 1 hour. In fact, I am going to drive in the next 1 hour, so it is really a fact, it is a question which, to which I also do not have an answer. I cannot say, it is half, I maybe or may not be, it depending on my driving experience, it is the way I am driving, my probability of my getting involved in accident will change.

This way of changing probability based on more information is what is called the Bayesian ideas, is now very important in mainline statistics, but we will not get into that right now. So, Random experiment, an experiment whose outcome is unknown. I am not writing down the tossing of coin or throwing of a die, I am telling it just repeatedly.

Another important thing is that, if you are in a stock market. You known the stock price today, you see it on the screen, you do not know the stock price after two hours or at the end of the day. So, the price of stocks, is random quantity. So, these are examples. So, price of a stock, so this is an example of a random experiment.

Suppose I am just looking at price of a stock every day, so I will not get the same price and so and I do not know what is the next day's price. Possibly there is a price range within which it

will vary, but I do not know what is the next day's price. So, I cannot predict before and exactly this will be the next day's price. So, the price of a stock in a stock market is actually, is a form of random experiment, if I want to look at it like that. So, this is something which is random quantity and hence has to be guided by the laws of probability to which we will soon start talking about.

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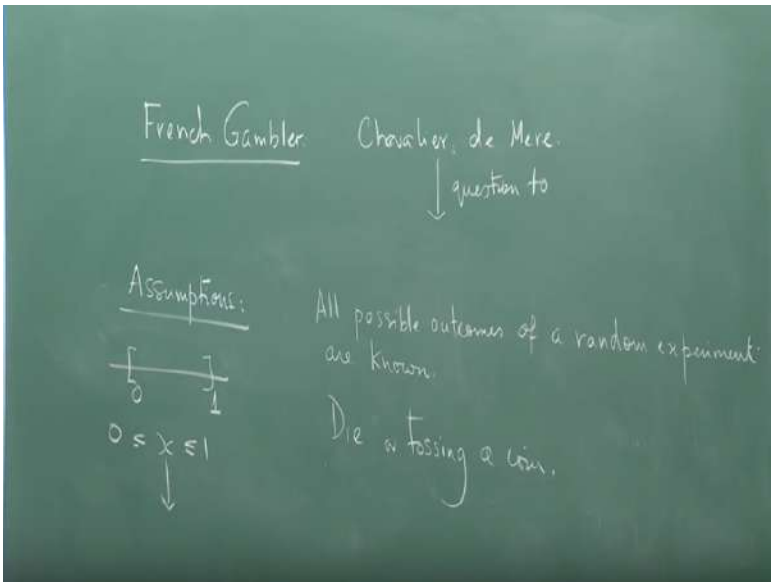

So, in a random experiment, what is known is that, I would at least have an idea of all the possible outcomes. All possible outcomes of a random experiment are known, that is something which is an assumption we always assume. Without making some basic assumptions which looks natural, science cannot proceed, so the assumptions is the following, that all possible outcomes, all possible outcomes of a random experiment are known.

For example, if I throw a die, I know there are only 6 possible outcomes on the face, there will be either dot 1, dot 2, 3 dots, 4 dots, 5 dots, or 6 dots; there cannot be anything else. So, the number of possible outcomes could be finite or could be infinite. If I am telling that, okay I am choosing 1 point randomly from the line 0 on 1, I am choosing 1 real number out of the interval 0 1, then my all possible outcomes is any number between 0 and 1.

So, any x between 0 and 1 is a possible outcome. Again, I have to draw any number out so you have infinite possibilities, but here for throwing of a die or tossing coin, I have finite possibilities once it is done.

Let us go to very basic definition of probability, that you even learn in your high school. So, let us see, what are the plus point of this definition and what are the minus point of this definition. And why we need to shed this definition and go to an axiomatic formalization of probability by understanding that we have some idea of what probability means in some.

It is something like a proportion, of occurrence of certain things, among a universe or certain things, but we do not really have a full and give a true definition of what probability is, so we basically make some rules of the game and we play the game by those rules. Axiomatization or trying to define certain rules, has been the standard in mathematics, since Euclid wrote his 13 Books of Elements.

You studied Euclidean's geometry at school and possibly, do not know about all these things, but it is very important that, as you learn more mathematics, you have to start appreciating that, many of the very basic things cannot be defined, just like a set cannot be defined, probability cannot be defined. It has to have a basic axiomatic framework and once you accept that axiomatic framework lot of other things would follow in a natural way.

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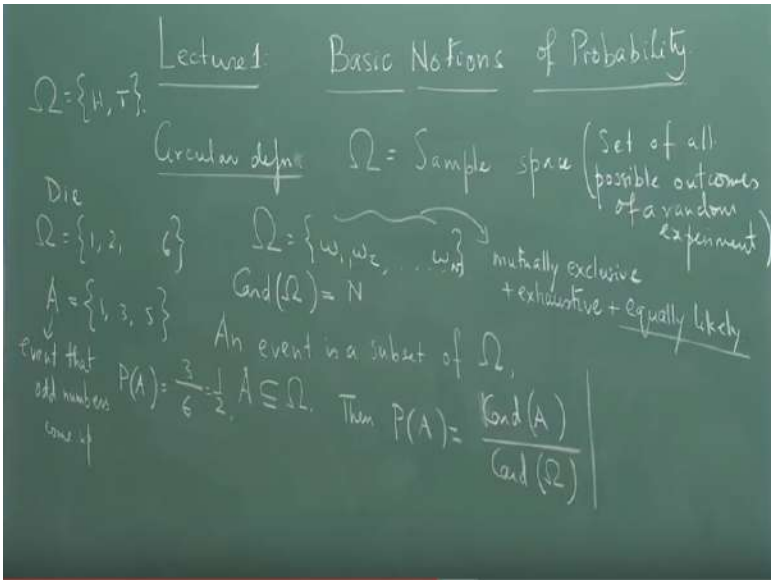

So, the circular definition, that is taught to all kids in the high school, what does it say? So, the set of all possible outcomes of a random experiment is collected in a set, which is called the sample space. So, in order to start any discussion about probability, we have to talk about the sample space, is a set of all outcomes, all possible outcomes.

Now, this set we can write something like, okay we are not, okay once I write like this, I am basically assuming, that I possibly think that, there are countable number of elements or infinite or finite number of elements, but okay let us just write it, for the heck of it, for the moment. So, it is assumed that, so these are outcomes of a random experiment which we have collected.

So, it consists of outcomes which are mutually exclusive, exhaustive, and equally likely that is the definition. So, these ω 1, ω 2, ω 3, ω 4, which are the outcomes of random experiment, they are mutually exclusive; means if ω 1 takes place ω 2 does not. If ω 3 takes place ω 4 does not and so and so forth.

Exhaustive means, these set of outcomes, are the all possible outcomes that you can have in the random experiment, like head and tail. If you take the tossing of a coin, then your sample space would be written like this head or tail. Now, suppose I just have finite number of them and then I talk about the cardinality of this which is N, N objects. And even it is defined, to be a collection

of some sort of a subset of these outcomes and even if it is a subset of these outcome, for example I say what is the occurrence of odd numbers when I throw a die, so it can be 1, it can be 3, or it can be 5; so, all these 1, 3, 5, they come collected in a set.

So, for this moment, an event, because now I just considered finite number of them, an event is a subset of, consider A to be a subset of ω, then the probability of A, asks you, how much proportion of A is in this big ω, so you write, this is nothing but the cardinality means the number of elements in A. Of course, this is finite.

One immediate thing you realize, that this quantity is greater than equal to 0 but lesser than equal to 1, because A being a subset and this is a finite set A would have lesser elements than ω has. This is a very useful definition. This gives you a very intuitive idea of, what probability is and it allows you to actually compute probability in most cases, but there are certain flaws in this definition.

The first flaw, is that we assume or we mention the term, which I will now write again, that all these events, these outcomes, are mutually exclusive plus exhaustive, but they are the only ones, plus equally likely; equally, likely simply means that, you are assuming that all of them can occur with, so you have actually assumed basically, a definition the idea of probability within the definition of probability. So, you cannot consider this as a fool proof definition of probability.

The second point is that, it this is alright, when you are talking about a finite sample space, but it is not alright, when you talk about infinite sample space. Now suppose I throw a pair of die, and say, what is the, throw just 1 die and I say what is the probability that I find an odd number on it.

So, I know that for a die, the sample space is 1, 2, 3, 4, 5, 6 and A is the event that odd numbers turn up and if this is my event then I have 1, 3, and 5. So, according to this definition, the probability of this event A is 3 by 6, which is half, which also looks natural because they are in equal proportions the even and odd numbers here.

Now, so this definition cannot be taken as a standard definition because it has a circularity, because it is assuming the notion of probability, within its description and also it cannot handle the case, when I am talking about infinite sample spaces.

So, we have to go to an axiomatic formalism, which was given by Kolmogorov in the 1940s. Kolmogorov the greatest Russian mathematicians, of all times, or rather one of the greatest mathematicians ever in this world. So, let us write down, so we are now writing the Kolmogorov's Axioms for probability.

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Tomorrow, or rather in the next class, we will first try to give an example, as to why this definition of probability does not work, which is, which will be called the Bertrand paradox. In that example, we will not give it now, we will tomorrow, in the next class, we will give you, we will do some examples.

What we will show is that, if we consider this definition of probability and in the Bertrand paradox, the same question will have two different probabilities, so which will simply say that this definition is not so useful. So, we will not get into doing Bertrand paradox right now.

But assuming, the difficulties that lie in here, first I will list down the Kolmogorov adjuncts of probability, that I have. To understand that, I understand probability is some sort of a proportion and then I need to go on about it. For example, if I take a coin, you can do the experiment at home and toss it 100 times and note down, how many times head and tail have come, then their occurrence is almost same; the number of time head comes and number of time tails comes is almost same.

The important issue, is that. When I am taking into account a sample space, which can have infinite outcomes, then the whole notion of events, the idea of events change, that every subset of such a, infinite set of outcomes, need not become an event.

This is a deep result from measured theory, because probability is viewed as some sort of a measure, so like measuring a length of something, but I am not going to stress on that. But certain things you accept from me just on faith for the time being, you can read up yourself but just for the heck of the course just for the faith you just accept that, if you have infinite number of sets, all possible subsets are not cannot be considered as events. But if you have finite number of sets like you have it here all possible subsets can be considered as events.

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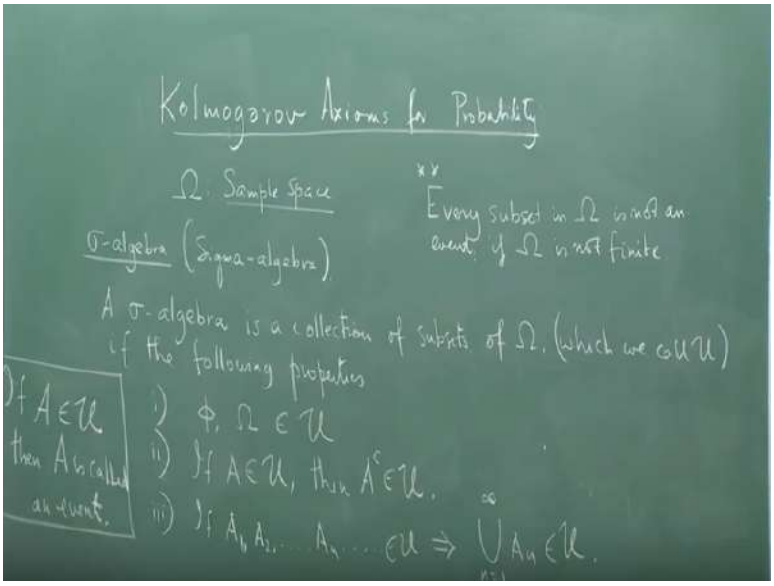

So, any discussion we will start with a sample space and to be more mathematically erudite, we will start with the definition of a σ-algebra, this is a sign called sigma sign in Greek. If you want to write in English, probably you can also call it σ field and I being truly an outsider to the field of probability. I would like to continue, what my math colleagues call it, as σ algebra.

So, σ algebra is a collection of subsets of ω, which we call U. If the following properties hold. Actually, those subsets, which follow these properties, which we will list down can be called as events. The empty set and the whole sample space, is an element of that collection of U. U is a set consisting of sets; it is a collection of subsets of the set ω.

If A Ɛ U, then A^c Ɛ U, means if an event A, if A is an event then its nonoccurrence is also an event. This is very natural way of looking at events. And the third which might look slightly counter intuitive, but that also is very important that, if you have a countably infinite sequence of events, then it implies and if all of them are element of U, so, if all of these are element of U, then it implies that if it is the union of this, set union, the infinite union, that would also belong to U.

But every such, every subset of this need not follow this property. If A, a subset, is a member of the σ -algebra U, then A is called an event, then A is called an event. So, this is something you have to keep in mind as you work on, that every element in U, so this is something very important, every subset in ω is not an event, if this is not finite. If it is finite, then it is an event then every subset is an event. If it is not finite then, finite means, it does not have finite number of elements, then you cannot say that this is an event.

Now, let us write down the Kolmogorov's laws, which are also very important. So, this is, so I am now defining a function which maps an event to the set the number between 0 and 1, because from this very crude definition.

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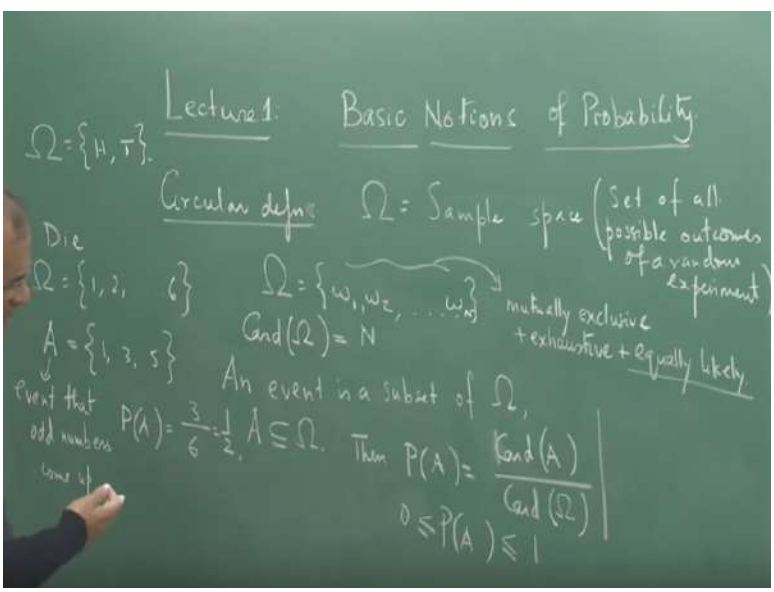

You can immediately see that, probability of any event A or at least finite event, when A comes out of a finite sample space has to lie between 1 and 0. So, P, the probability can be viewed itself as a function which is taking a set and putting that set, mapping that set, to a number between 0 and 1. So, that basic idea is already ingrained from the circular or not so good definition of probability.

So, once taking the clue from this, or rather a rough or a crude definition. We can now formulate, what are known in the literature or known to probability says the Kolmogorov axiom's. Okay, we should have put it that side but does not matter, I will just write down the axioms.

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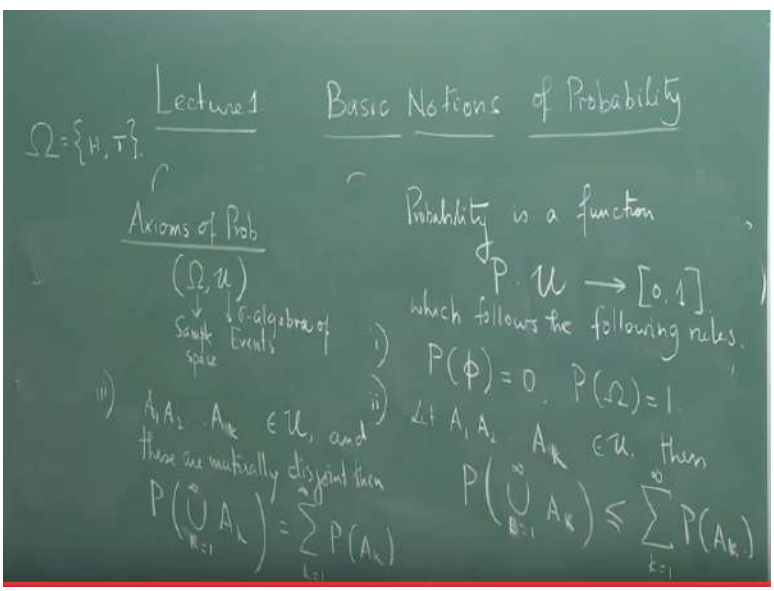

Axioms are probability. Sometimes I will write por or prob, prob for short because I am sure most of you are familiar with the word probs and stats. So, with this young generation I think in the SMS generation, we can always use short forms.

So, once you have a probability, whenever try to talk about probability. So, you have a sample space and a σ-algebra of events associated with it. So, to begin with, you should have a sample

space and you should have a σ-algebra of events, so this is given to you. This has to be known, without this nothing can be done. So, probability is a function P, which is, whose domain is U the σ-algebra. So, it takes an event, from the σ-algebra and maps it to some number between 0 and 1 which follows the following norms.

So, what are the following rules? Number 1, sorry, I think first we have to because, φ and this, φ or the null set is also called the null event impossible event, like I will live forever is an impossible event and I will die one day, is a sure event. So, this is this also comes out of from the crude definition this idea, but since we cannot use the crude definition for collection of infinite sets, infinite sample space, so we give them as rules.

Let, these are all elements of U, then probability of the union, this is we are just putting additional union, we did not put it, we can actually prove it, prove this value but okay forget it. Sorry, I should write k here, this is the second law.

And the third law, which is the more interesting one, that if you have Ak, I should write here, it is much more meaningful, is a element of U and so you are having a countable collection of sets, note these are not infinite collection of subsets or not arbitrary collection, but a countable collection and Ak element of U and these are mutually disjoint, then probability of the union, union of these countable collection is nothing but the sum of the probability.

So, this probability measure P, we call it a probability measure, this probability measure P, this sample space and U, all form what is called is should write it here the probability space.

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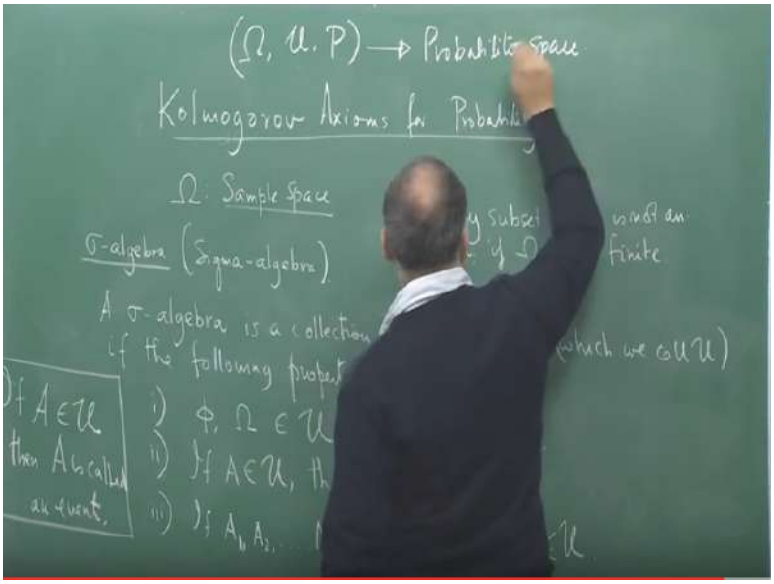

So, if you can have a sample space, a σ-algebra of events and a probability measure on the σ-algebra of events then this forms what is probability space. So, before we start talking about and we want to discuss or compute the probability of certain events, we have to be sure, what sort of probability space we are considering for that random experiment. Those are the things will come in the next lecture.

So, what does it tell you? It gives you certain, it takes ideas from that crude definition and puts them in form of laws, so that even if you are talking about infinite sample space, any measure or any function which takes events to this interval 0 and 1 and satisfies all these rules we will call it a probability measure on that.

We will soon come to its utility, very soon, that this sort of definitions are useful, and we will talk about Bertrand paradox in the next class and also next class we will do two problems. This Bertrand paradox will solve the Chevalier de Mere's question to Pascal. We will talk about the Buffon's needle problem, which actually I had been used to estimate pie by D' Alembert and we will also then talk about how to construct sample spaces.

When you are doing, no it is not sample space, how do you construct the probability spaces. If you know a sample space and if you can construct the algebra of events, σ-algebra of event, how do you, define this measure so that a probability space can be constructed.

See the term σ-algebra, why it is called σ-algebra because if I just had finite number of them and this had been occurring only for a finite number of events, then we would have called it just an algebra, it is called a σ-algebra because we are talking about a countable collection. That is why the term infinite sum, actually sounds sense that is why the term σ is used so with this very basic idea. I would stop today and then we will go to the next class where we will do this interesting sort of problem solving.

Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 02
Interesting problems in probability

So, we will start with whatever we had promised. We will start with something called the Bertrand's paradox, which shows the inadequacy of the classical definition, which is a circular definition. So, here we will just try to solve out the Bertrand's paradox and see that it will give two different answers for the same problem, two different probabilities.

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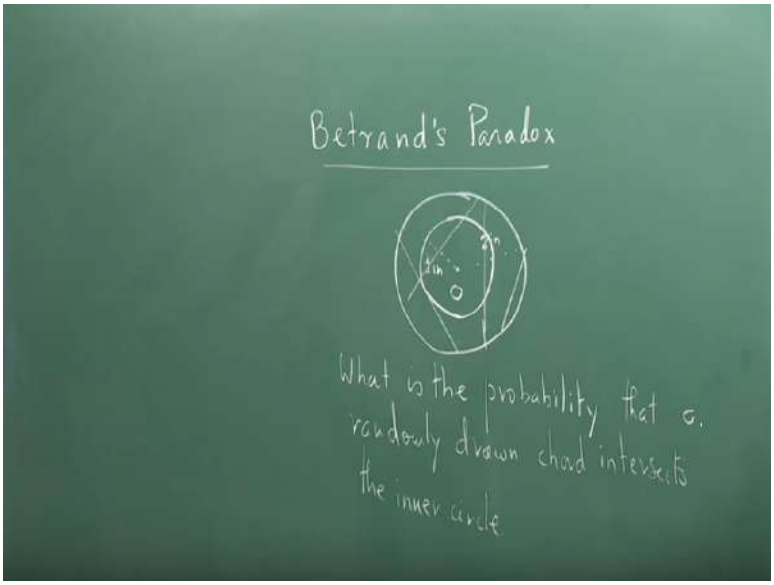

So, what does this Bertrand paradox says. The Bertrand paradox says the following; so, you have a circle, with a center say at o and a radius of 2 inches and then you have a concentric circle of radius 1 inch, 2 inch and 1 inch, the question is that, if I randomly draw a cord in the circle, if I randomly just draw a cord in the circle, what is the probability that it intersects the inner circle.

So, the question is, when does, what is the probability, what is the probability, that a randomly drawn cord intersects the inner circle.

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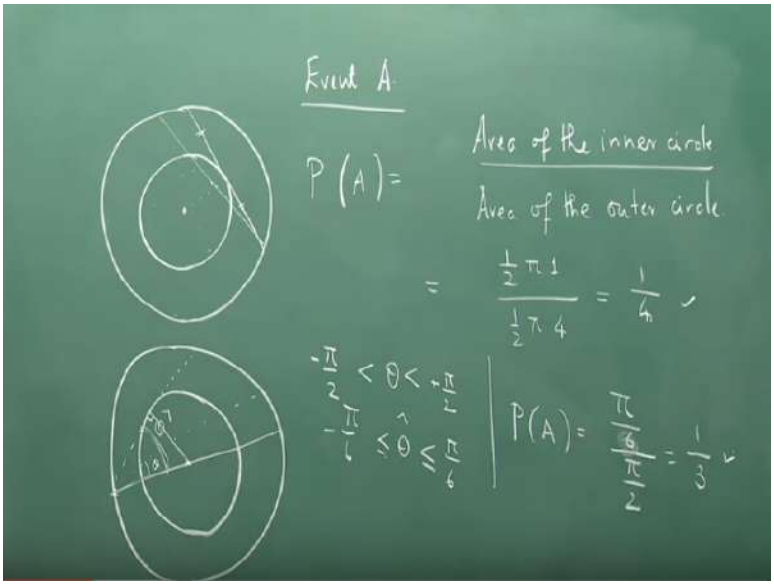

So, what does, if you look at it, let us look at it in the following way. So, by the fact that, it is intersecting the inner circle, whose radius is 1/2 of the original circle, when do I see a cord is intersecting the inner circle; if the midpoint of the cord lies in the inner circle. Where you see, even in the tangent one, if this is the tangent also, it does not matter because, if you join it by the radius of 1 inch that will be perpendicular and hence by very standard Euclidean geometry you will break it up into two equal parts, so the midpoint has to be inside the circle.

So, essentially the midpoint thus, can be here, here, here, here, anywhere in the circle and for any cord that is drawn the midpoint would be anywhere in the original circle. So basically, the probability, if I called that event as A, so let me call the event as A. Now you observe that essentially, I can say that my sample space is the space of all points in the bigger circle, all the points inside the circular disc and my possible position of midpoints are any point here on the inner circle.

So, the probability of an event A here we have an infinite sample space and here where the problem will come, the probability in the event A is thus, the area of the inner circle divided by the area of the outer circle.

So, it is the area of the inner circle divided by the area of the outer circle. So, area of the inner circle is 1/2 π into 1 square which is 1 and this is 1/2 π into 2 square which is 4 right, π r square, so this will just give me 1/4. So, that looks like to be a very feasible answer. Now I can look at the whole problem, in a very different way, which will also appear to you as logical.

So, here you again, take the outer circle and then draw the inner circle. So, what is the meaning of a cord; so take any point, I have taken any point. I can draw infinite cords from this point. Every cord need not hit the circle. So of course, you can draw the diameter, which is passing, a cord, which is passing through the center.

Now what would happen, this cord actually can vary, so from here, I can just sweep 90 degrees in this direction, 90 degrees in this direction and I can vary the cord. I can get all the possible cords; so basically, I can vary this thing by an angle θ right, so I can take any cord.

Suppose, this is the new position of the diameter, so I moved it by a θ angle but this θ angle can vary between - sorry + π by 2 to - π by 2 be it either here or it is here. Similarly, I can say that, if you look at the cord, what is the maximum amount θ can vary, if the cord has to touch the circle, as to cut the inner circle, the maximum it can vary is this amount, which let us call as θ hat.

Now, what is θ hat or what is the maximum amount θ hat can go, so this is the maximum amount hat can go. It starts from here and moves up and comes up to here, starts from here and moves and comes out to here; so, all the cords that are passing through this point.

Now, whatever probability I will have, for the cords cutting the circle from this point, this will have the same probability, all the points; so, that will be the probability of hitting, because you have to emanate from one of the points, start from one of the points. Now how much does this angle vary. So, this angle we have to observe that.

For example, if I draw again, by Euclidean geometry, if I draw the radius of 1 inch this would be perpendicular, so this is now 1 inch and this is 2 inch. So, sine of θ is 1/2, so θ hat is actually varying from, sine 1/2 is 30 degrees, it has 30, so it is π by 6 to okay. So, all the possible angles

to get the cords coming out of this point is varying from π by 2 to - π by 2 and all the cords which has to be inside the circle, in the sense cutting the circle, has to, have an angle variation from θ head from π by 6 to - π by 6.

So, then so, π by 2 - π by 2 what is the total possible angles we have, essentially; so, if I just look at this part only, which is symmetrically it will give the same thing on the other side. So basically, I am having 0 to π by 2 possibilities on this part, and 0 to π by 6 possibilities.

So, probability of the same event A, sorry π by 6 by π by 2, so π by 2 is the total possibilities here on this side and π by 6 is the possibilities that are favorable to you and then hence this is giving me the answer 2/ 3 which is 1/3. So, these 2 answers are varying and both of them are equally logical, looks equally logical.

So, this is the Bertrand paradox and hence that sort of approach, that we had the way, we have defined probability, the circular fashion that you learn at schools, that cannot really work in general and hence probability itself has to be thought of as, some sort of function which takes an event and maps it on to a real number between 0 and 1, which follows certain property, the Kolmogorov axioms, that we had studied in the last lecture.

So, let us go back to the true founder, the problem of the true founder of probability, Chevalier de Mere, the great gambler, who gave his friend Blaise Pascal, the question of how many times you get a 6 if you throw a die, 1 throw of a die 6 times or throwing 2 dies 24 times, throwing them simultaneously. So, let us just go and solve that problem

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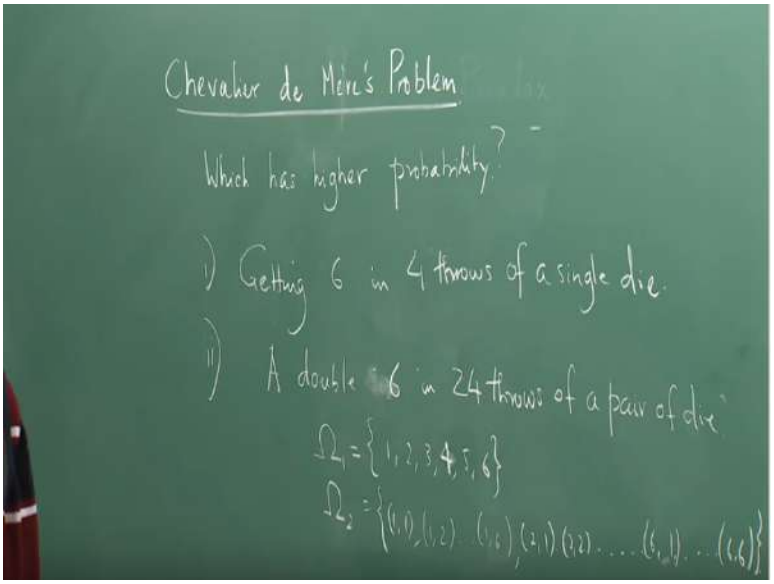

So, this is where stated the problem, we will state it again; of course, you will have practice problems with you, but this is something which you can enjoy, because that is what, how the history comes in so, which has higher probability, that is the question; getting 6 in 4 throws of a single die. So, first operation is, trying to getting 6 in 4 throws in 4 throws of a single die or No.2, getting 2 sixes in 24 throws of a pair of die, a double 6 in 24 throws of a pair of die. So, which has a higher probability that is the question.

So, in order to solve this, we have to use the language of compliment events. So, if A is an event, A compliment is the event, that A has not offered. So, if A is an event in the σ-algebra, and here you can understand in both the cases, here my ω has 6 elements, so it is a finite set, now, and the second one, this ω 1 and ω 2 has 36 elements they have the form (1, 1) (1, 2) (1, 6) (2, 1) goes like this (2, 2) (6, 1) (6, 6), so it has 36 elements. So, these are having finite sample spaces. So, they are σ-algebra associated with this u1 and u2, would have, would be nothing but subsets of this right.

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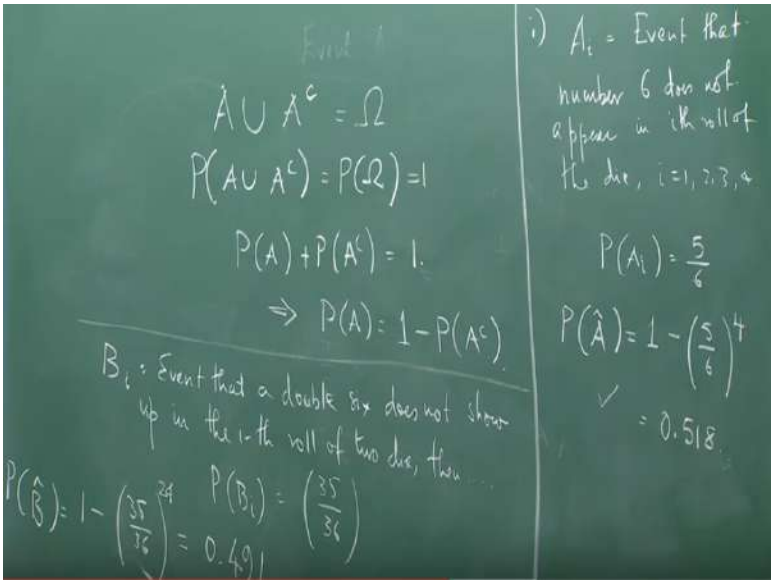

Now, here we will rely on this fact, that if A is a given event and if you consider A compliment and A union, A compliment where A is the subset of ω , this will give me back the whole sample space ω, then you know, why, now hence probability of A union A compliment is equal to the probability of ω and that is equal to 1, because that you know from the Kolmogorov's axiom and hence now here again we will apply the Kolmogorov's axiom here, because these are disjoint events, so this will immediately say that probability of A is 1 - probability of A compliment.

So, what I am now going to do is to, use this very idea that first I will talk about finding the probability of A compliment and then immediately you will know what is the probability of A by subtracting it from 1.

So, let us see where do we start; so, let us take the first case, case 1. Let us go by the first one. First one let us denote by Ai at that the number 6 does not appear, under rolling the die for the I th time, so Ai event that number 6 does not appear in the i th roll over i. Of course, i has only 4 possibilities and we will be allowed to roll the die 4 times.

Now what does it mean that 6 has not occurred, which means any of the remaining 5 has occurred? so probability of their occurrence is 5/6; so, probability of a i is 5/6. So, probability of

6 not occurring in the first place, first throw, and the probability that it does not also appear in the second throw are two independent events, they are not connected with each other.

Now we have not yet introduced what are independent events. So, what I am going to say, you look at it in this form that, I throw a die so what is the probability of this not occurring 5/ 6. Again, I throw a die, so if the probability of success is 1/ 6, the probability of failure is 5/6. So, I am just looking at the failures. So, if it fails at each and every time, so which means in all the 4 times, I have not got a 6. So, when shall I get a 6, is just the complimentary event.

So, this whole idea of independence of two events we have not yet discussed, but if two events are independent in the way as we can see it, because throwing a die now and not getting a 6 and throwing a die after two minutes and not getting a 6 are not connected anyway, then you multiply the probabilities. So basically, now what you do, the probability of my occurrence of the event say A hat, that a 6 occurs, in throw of a fourth die is, 1 - 5/6 to the power of 4 and that turns out to be 0.518. So, you see, that is how you do it.

Now you come to the case of, where you have, you are throwing a pair of die with a 36 possibility, 6 showing up in both the case is 6 is 6 is there; so which means if Bi is the event that a double 6 does not show up in the i th roll of 2 die then, what is probability of Bi, so if (6, 6) has not occurred, (6, 6) is just 1 of the occurrences, so there are 35 other occurrence which can come, so they are 35/36 a number of very near 1, but it is still less than 1.

Now this, suppose it happens for each of the 24 cases, 6 does not appear, so then probability of B head, that is a double 6 appears in a throw of 2 die, is nothing but 1 - 35/36 into 24 because, each of these 24 cases is independent. So, when you have independent events you multiply probabilities, which is something we have not told formally, which we will tell in the next class, but I did this, problem, because this is the historical problem so is good, histories are always a good guide to pedagogy and then what should be the answer.

The answer turns out to be, because of this power 24, which actually pulls down the value of this, not very far off, but this is obviously bigger and that is what the Chevalier de Mere had

actually observed in the actual gaming. When they actually played the die, actually threw the die in those casinos, he actually had seen that that was that 6 was appearing much more rapidly.

Now we will talk about a very interesting and important problem called the Buffon's needle problem and the answer, we are going to get a, fantastic answer to this problem, the probability is linked with the number π and that is what we are going to show. So, let us see what is Buffon's needle problem.

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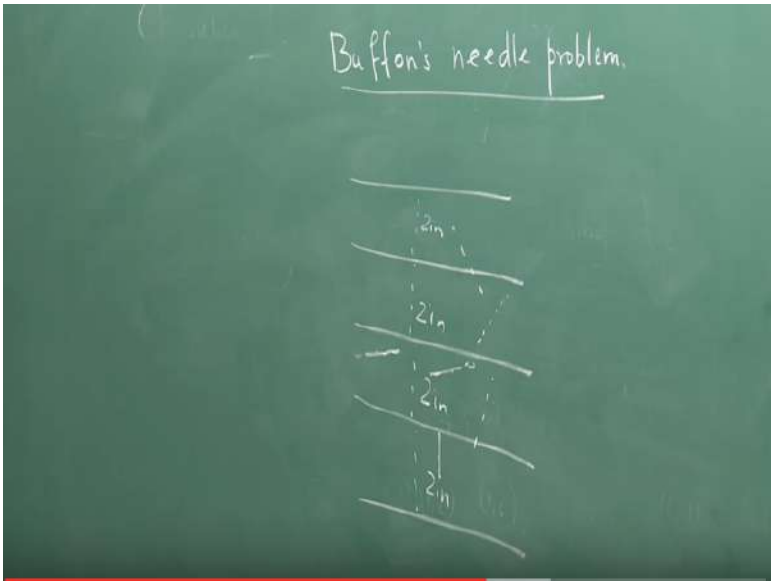

There is a lot of writing on Buffon's needle problem, but, not buffoon, but Buffon's. So, this would be the last problem. So here, you have seen in both the cases, we have been using the Kolmogorov axioms. In the first case, where you have finite number of outcomes, total outcomes, you see, you can use the so-called idea proportion, the other definition, but here, we need to we will use the similar sort of definition, but here we will see how to first, when the problem comes and when you have infinite possibilities you need to be very carefully construct the probability space. That is very fundamental

So, Buffon's needle problem is like this; it says that okay, let us take a plane, that is plane of the board and you are drawing parallel lines each say 2 inches apart, 2 cm apart does not matter, each

2 inches apart. Now, you are throwing on this so you have done it on the floor and you are throwing a 1-inch long needle on the floor randomly. So, the needle could land here, the needle could land like this, it could land like this, could land like this, could land like this, could land like this, could land like this straight touching at 1 point.

What is the probability that, the needle will cut or touch 1 of the parallel lines? What is that probability? How do we think about it? So, here our first step would be to know, how to, the answer is very beautiful, I do not want to tell you now, you will see when the answer comes out. So, first I have to realize how do I construct my probability space

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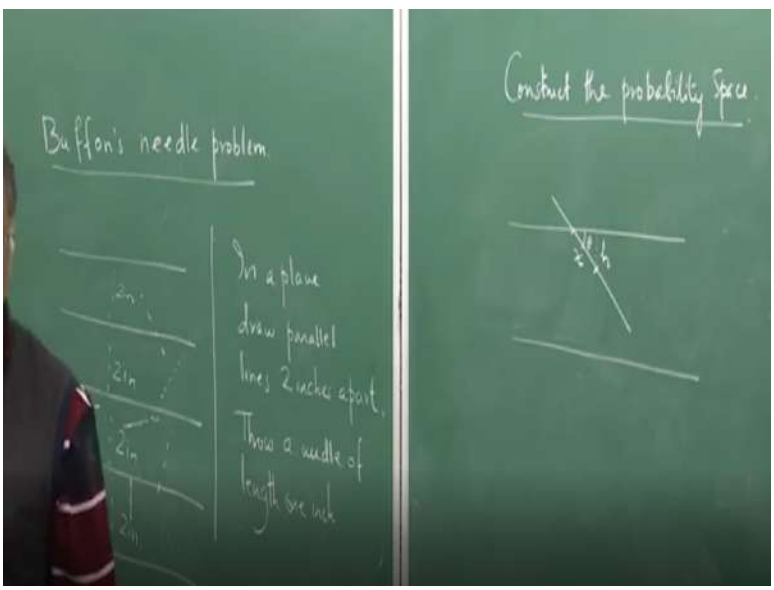

So, my first step is to construct the probability space and then how to use it, so let us construct the probability space. So, here is the line and suppose here is the needle, now take the midpoint of the needle, say here and find the distance of that needle from the nearest parallel line, so of course, there could be another parallel line here and then of course we will just look at the nearest one and the angle that it makes with the horizontal is the θ.

Now, this distance let me call x, this distance, so x, so this whole length is 1 inch, so what you have done, let me write down the problem clearly. In a plane draw parallel lines 2 inches apart, throw a needle of length 1 inch on the plane randomly. What is the probability that it touches one of the lines, parallel lines, one of the parallel lines?

My writing might not be very clear, but we will understand the problem. So, I have been telling, I have been reading it every time, that you have to throw the needle on the floor and see when does it touch one of these lines, so this is the situation. So how do I construct.

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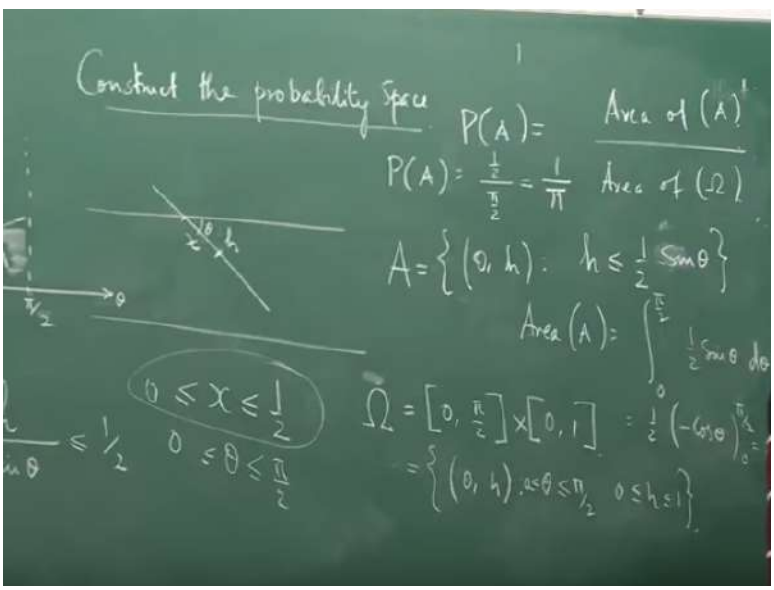

So, you see this x has to be less than or equal to 1/2 to touch, because it can be in this situation, this is the midpoint and this is this one. So, the angle and this angle is of course 90 degree. So, the angle can vary from 0 to π / 2, while this can vary from 0 to 1/2. So, this is possible if x is less than equal to 1/2 these are then equal to 0, because the midpoint can be exactly on this line where h becomes 0 and θ is θ and θ is of course varying from 0 to π / 2. So, I can define my space as, so I am not throwing it randomly. So, this is what happens, under this condition on x, it will touch the line, but I am just throwing it randomly.

So, my all possible outcomes need not, this is my favorable outcome. Angle will anyway remain from 0 to whatever you throw, it is either with the horizontal, it is either 0 or it has to vary between 0 or π / 2, either it is straight or something like that you know. You can always measure the angle acutely, but the length can be anywhere. The length x, the length of the needle is between, that x can be between 0 and 1 that, sorry this x can be between 0 and 1/2. So, this is the possibly these are the favorable situations for hitting the line.

But this length x so I can construct this as the distance of h, so the length x has to be between 0 and 1/2 this is by possible possibilities, there cannot be any other possibilities right, but the length h where can it vary. It can vary from 0 to 1 because it could be like this, the 2 inches' lines and it is just lying in the middle, this is where the needle comes and hits it, it stops and this is h and this is 1 inch.

So, though the condition on x is 0 to 1/2, my actual gameplay is happening with the 2 variables θ and h. So, θ can always vary between 0 and π / 2, but h will vary between 0 and 1. Now, so I can, but the interesting part is that, because I have θ and h and x is this line the hypotenuse, I can obviously convert x in terms of h and θ. So, the favorable situation is the following, that h by h and x is nothing but h, h / x is sinθ right, so h sinθ = x, so I can write so my favorable situation is when h and θ is linked with this inequality, this is the favorable situation.

So, I essentially my favorable event, my event that, I want to find probability of in this particular problem, is the collection of all θ h, such that h is less than equal to 1/2 of sine θ. So, you are just translating this x less than equal to 1/2 into this, h is obviously greater than equal to 0, which we are not writing down, which is obvious anyway, it is the length it is varying from 0 to 1. Now this is my event, so if I write down, so what I want can do? Now when you have infinite sample space, how do you really measure?

One way, of measuring or assigning probability, is assigning the standard measure of the sets, because you see, I can now find an area of the sample space sorry, I can write just ω; so I should be able to find the area of the sample space right and so basically, now what I am doing is I want

to find the area of this and the area of this. the standard measures, so I want to imply. Let us see if I take the area of this, if I want to compute the area of this. Can I use it as a probability?

Of course, the area of this is positive and area of this thing, so now how do I actually. know that total number of elements here and total number of elements here. that the only way of knowing is to calculate the areas, whether infinite elements, you want to capture the essence of total number is essentially, you have to capture it through the idea of an area. So, I find the area of this and I divide by the area of this, area of this is of course $\pi / 2$.

So, my probability of A is area of A/ area of ω; so how do I calculate the area of A. The area of A can be easily calculated in the following way, is that you draw a line put h and θ here, θ is along this line measured in radiance, so 0 to $\pi /2$. So, h is equal to $1/2 \sin\theta$, h equal to $\sin \theta$ you known would be something, that $\pi / 2 \sin \pi / 2$ is 1, so this is my 1, so 1/2 is 1/2, this is 1/2. So, this is your $1/2 \sin \theta$, so h has to be below that.

So, essentially my event is consisting of all possible points this one, this is my A; so how do I calculate the area of A? The area of A is simply, area of A is nothing but, 0 to $\pi / 2$ because θ was varying from 0 to $\pi / 2$, $1/2 \sin\theta \, d\theta$, okay, so that is exactly it.

So that would give me, what would be this one 1/2 of, if I integrate $\sin\theta$, the answer that would come out is, $- \cos\theta$, which I $\cos \pi / 2$ is 0, no it is 1 / 2; just a moment, so it is 0 to $\pi / 2 - \cos\theta$, so it is $\cos \pi /2$ is 0 and cos 0 is 1 - will cancel so it will become 1 so 1/ 2 so basically, $1/2 / \pi$, $1/2 \pi$, $\pi / 2$, probability of A is, $1/2 \pi$, $\pi / 2$. So, the answer is $1/ \pi$. So, we have a very beautiful answer, the answer is $1/ \pi$.

So, Buffon did a very interesting experiment. He took a plane like this, he drew those parallel lines and kept on dropпng, say for 1000 times he kept on dropпng a needle of length 1 inch and see, how many times it was hitting a line and then by doing it and increasing the number he could get better and better estimations of π.

So, with this very beautiful example, I stop our discussion here and in the next class we are going to talk about random variables and random vectors, their distributions, what is the meaning of independence and some relative issues.

Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 03
Random variables, distribution functions and independence

Welcome to the third lecture on series on probability and stochastics for finance. Let me tell you that from this lecture we are getting more and more immersed into the deeper theory associated with probability. The last 2 classes were much more of developing an insight. Of course, I expect that the course that you are now taking would be some sort of a second course for you rather than just a first course where the first course needs more detailing, which is not possible in the set of 10 hour lectures and it is essentially a pre-runner to the course on derivative pricing in financial markets.

So, it is better for me today to tell a book that I will be using from this class onwards till the end of this course, is a book called An Introduction to Stochastic Differential Equation by Lawrence C. Evans and this book is published by the American Mathematical Society.

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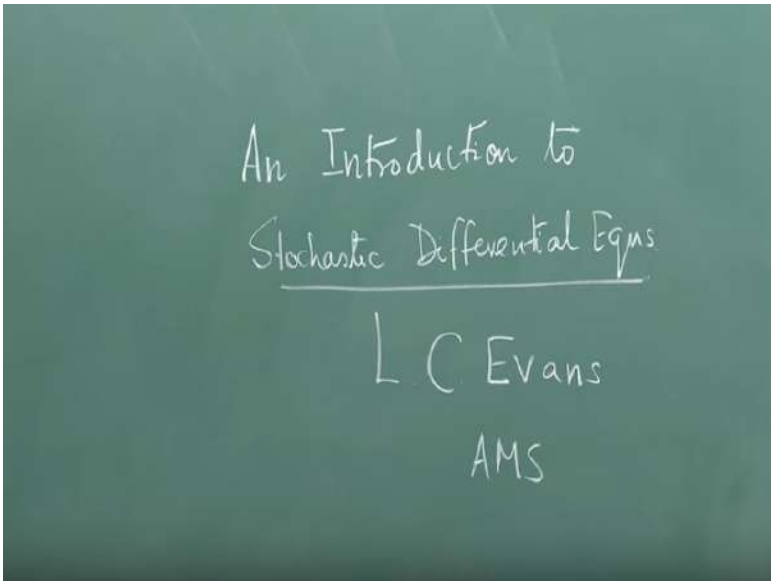

I do not expect you to have this book with you but I will in general follow this book except maybe a little bit of material taken from some other sources. Anyway, I should just show you the book in case you want to see the cover that it is better to show you the cover in detail.

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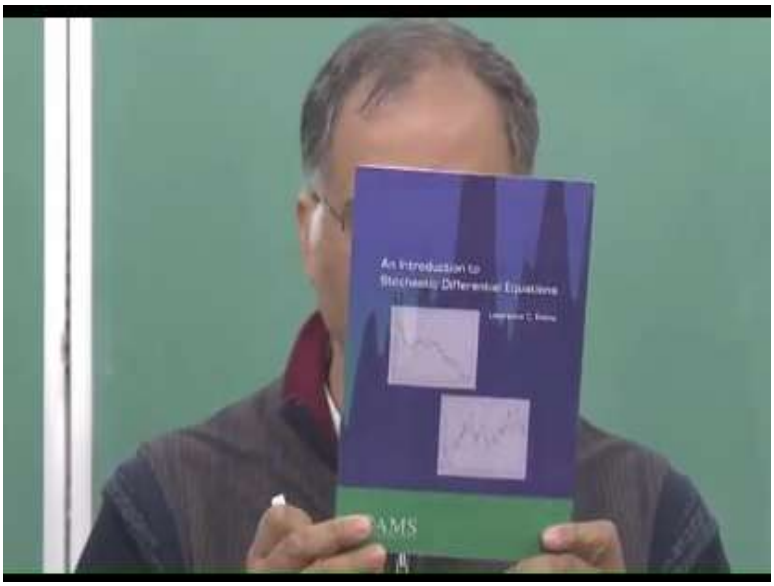

This is the book, which I will be largely using in this course and as I have told from today we are really going to get into the more theoretical issues. We will immerse more into the technicalities of probability theory.

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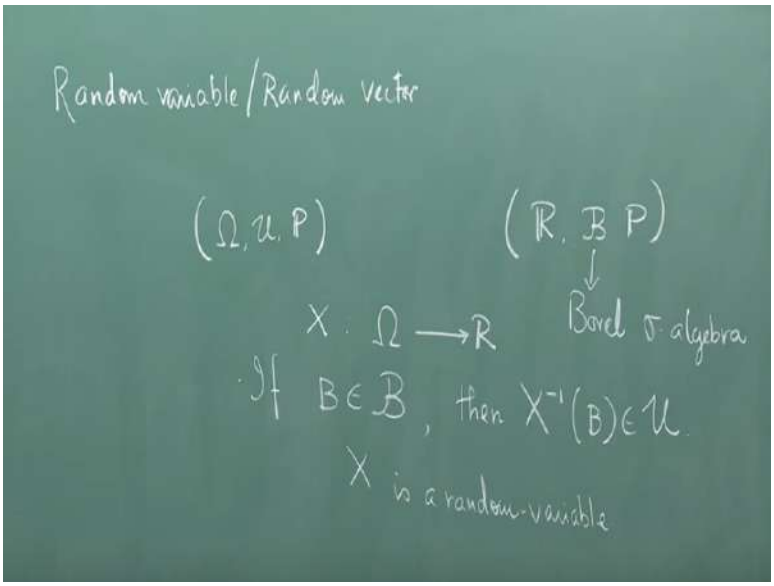

So one of the first set of technicalities that should come is the idea of a random variable or a random vector random variable/random vector. We will explain both these concepts. Of course, those of you have got a basic course in probability would know that random variable itself is slight misnomer because they are actually functions, but in a standard probability course you have been told that a random variable is nothing but a mapping from a sample space to the real line.

But there are certain other properties that needs to be attached to such a function in order to call it a random variable. So let me consider a probability space and a probability measure P and also consider a real line R and a Sigma-algebra B. On the real line R, the Sigma-algebra that we have associated with R is called the Borel Sigma-algebra, which is naturally related to its topology.

This Borel Sigma-algebra consists of all open sets; hence all closed sets, all half-open intervals, half-closed intervals, open intervals, closed intervals and so and so forth. So it is a collection of all, so it is the smallest Sigma-algebra which contains all open sets in R. Instead, is the smallest Sigma-algebra, which contains all open intervals in R, you can just think of it like that.

So, this is the information that we really need. Essentially, random variable is a way in which a outcome of a random experiment is translated into a number and a outcome of an event is translated into an interval, that is the essential idea, so the interval or union of intervals. So x is called a random variable if it is a mapping from $\omega \to R$.

But with the following property that if you take any B which belongs to the script B, the Borel Sigma-algebra, this is not the standard B, this is the script B. If B belongs to the Borel Sigma-algebra, then the inverse image of the set B must belong to the Sigma-algebra U. So, in this case x is called U measurable or just measurable. So this property has to hold. If can define a function for which this property hold, then we called x is a random variable.

So, this property is called the measurability property that inverse image of an event in the domains in the range space is also an event in the so if you take an event in the range space and

pull it back may be inverse image then it will map to some event in the map, to some event in the actual space, the decision space or the domain space.

So why, so that is why this, what does this idea tells you. This idea tells you that that every image, I am sorry, every event is actually related to some interval here or some union of interval, some set here. So that is why when you pull back, take a set from here and put it back, it will always map it back to an event.

So an event is taken to another event, but in the space R. So, this is something very important that, so every event is taken to some sort of an interval or an union of interval. That is the idea. That is the key idea behind introducing random variables. So that is the key idea behind the notion of random variable that an event here has to be always taken into an interval in the real line. So, we map an event to an interval.

So, we can now handle real numbers, which are much more easy to handle instead of handling events which may consist of arbitrary things. They may consist of set of all dice that you have thrown, outcomes of a die or outcomes of a coin throw or the arbitrary random, which was using points from a rectangle for example.

So, that arbitrariness of the events, because they seem, because these are very different different random experiments are being taken to a common thing or being mapped to something which is common. So all the studies of probability can be now done in the settings of real numbers. This idea can be extended and it is useful in some cases in finance to look at it this extended idea of a random vector.

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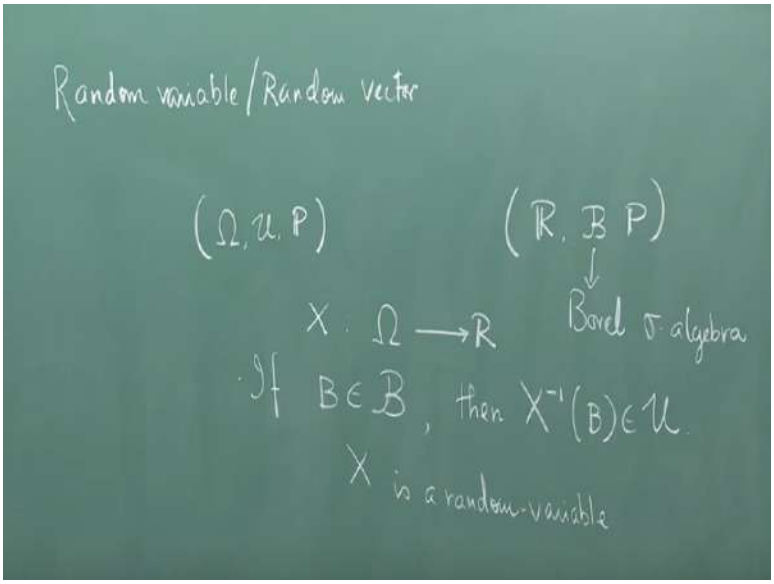

Where you have the same setting, okay, and in this case you even need not bother about any probability measure, you can just forget it, you can just look into this measured space and you can now talk about, may I just write it like this. So this sort of thing, so this B associated with R^n, so this this B is the smallest Sigma-algebra, which consists of all open sets in R^n.

What is the meaning of smallest Sigma-algebra? Take any Sigma-algebra which consists of all open sets. Take intersection of all possible Sigma-algebra which consists all open sets in R^n. Their intersection is a Sigma-algebra, which gives you this set. Now, can I visualize it, can I write them down clearly, no.

These are basically tools for you to understand what actually is happening. So, you take a mapping x. Again you take x. Now, so this is called a random vector, so random vector. Most of the studies would largely for the domain of random variables, but this is just to give you a generalization. So, in this case the same thing would occur that if you take any B, now it is set in R^n which goes to this sort of B, the Borel Sigma-algebra in R^n then it implies that X^{-1} (B) Ɛ U.

So the random variable concept is fundamental because it takes you arbitrariness from the notion of a random experiment. They are always in the setting of a real line, that is the key idea. So, what is the key? For example, I can give you 1 construct as 1 example, you can see it rather from standard books and many of you possibly know it.

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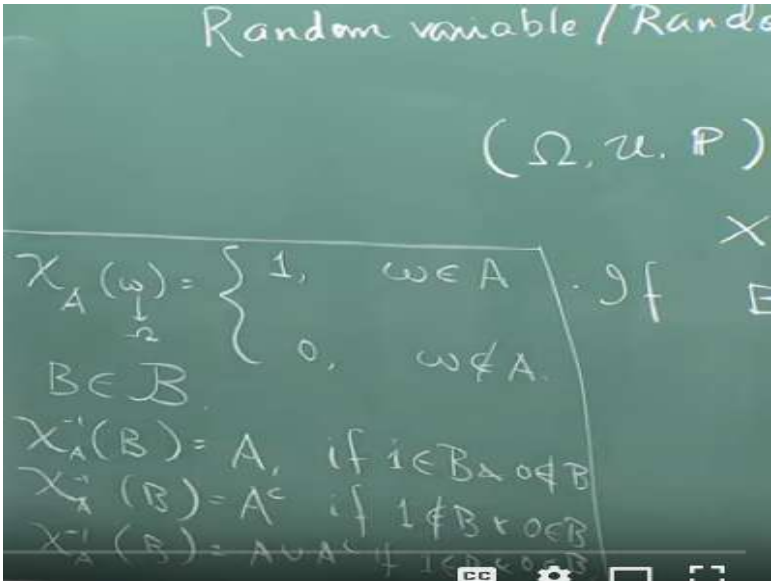

So you define this integrated function, so this is your set integrated function which says that if ω is in A that you mark it 1, you mark it 0 if ω is not in A, okay. Now, once that is done, how do I guarantee that this is a random variable? I say X_A is a random variable, so I am mapping taking each ω from this and making this definition.

So if I take, these in all so if take a B from this B, how do I calculate, that is the key question. See, if 1 is in B and 0 is not in B then this would be in A. If 1 is in B and 0 not in B. Of course, sorry, it will go into the compliment of A if 1 is not in B and sorry 1 is not in B and 0 is in B. So, what would happen if, so A is of course is an event.

This is in U. Since C is a Sigma-algebra, A compliment is in U. Similarly you can talk about what happens if 0, 1 and 1 is in B and 0 is also in that set B, then what would happen? Then it will become A union A compliment. So, and that is also again in A. So whatever way you say that okay this is not anywhere. 1 is not in B, 0 is not in B and A union A compliment's compliment so then that is the way so that will again be a Sigma-algebra. So you can try out. This is 1 example.

Other simple examples are head and tail; when you toss a coin you, let us denote by random variable the number of heads that come when I toss a coin. It can be either 0 or 1. That is the sort of thing. So that is the way you define random variable.

Another sort of important way of defining random variable is to construct some random variable out of these indicator function. This indicator functions play a major role in probability theory so this is something one needs to understand and try to get a hang of the ideas.

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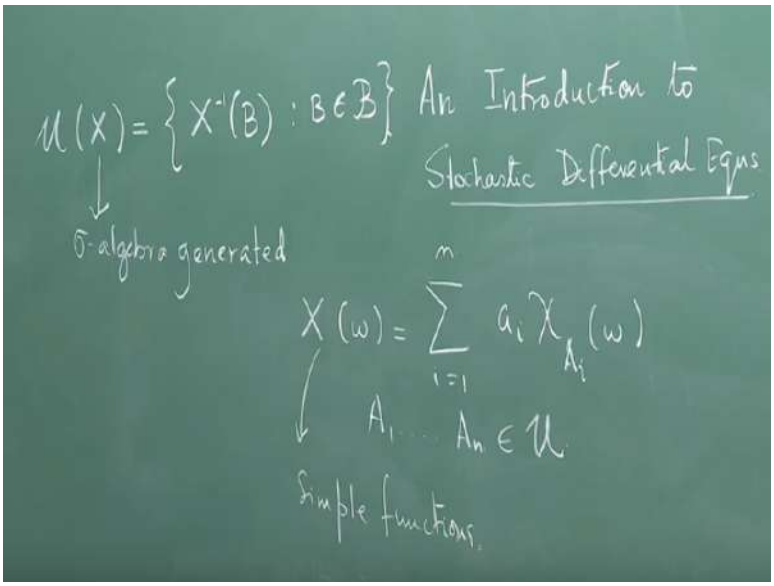

See what happens here is that for example I construct a random variable like this. Now this sort of thing, it is called a simple function. This also is random variable. So you have some set, say A 1 to A_n, so all of these are in U. Of course, you can completely, it will completely make sense that this is actually only non 0, it will if you take the union of A i's and take ω from there then it will always give you one of the A i's.

Otherwise, it will become 0. So this sort of function, this sort of random variables are sometimes called simple functions. A key thing that is sometimes required is the notion of this Sigma-algebra. This is called a Sigma-algebra generated by the random variable. You can think these are the random vector, but largely I am talking about the random variable. Does not matter if you think of it as random vector, you can think x is from ω to R^n where B is in the Borel Sigma-algebra.

This set is a Sigma-algebra itself and this is called a Sigma-algebra generated, so it is collecting all the inverse images of all elements in B. So you take the union of that set with collection of all

such things would actually form a Sigma-algebra, which is a sub Sigma-algebra of U basically. So, this is a Sigma-algebra. This is called a Sigma-algebra generated by x. See, x inverse B, need not be always giving me every element in U, you need some element in U.

So those are have a special significance. Because they are the ones by which the whole game is being done because they are linked to random variables, but they are the stuff that is really required when you calculate stuff. So this is called the Sigma-algebra generated. So I will leave it in the question of exercise to prove that it is truly a Sigma-algebra.

Our question would now turn to a notion of stochastic processes. So I am just rubbing this idea. So random variables can be also lined up in the form of a sequence and these stochastic processes play a very very fundamental role in finance because everything that happens in financial markets you see changes of certain things over time and though all these changes are stochastic in the sense that they are essentially random experiments.

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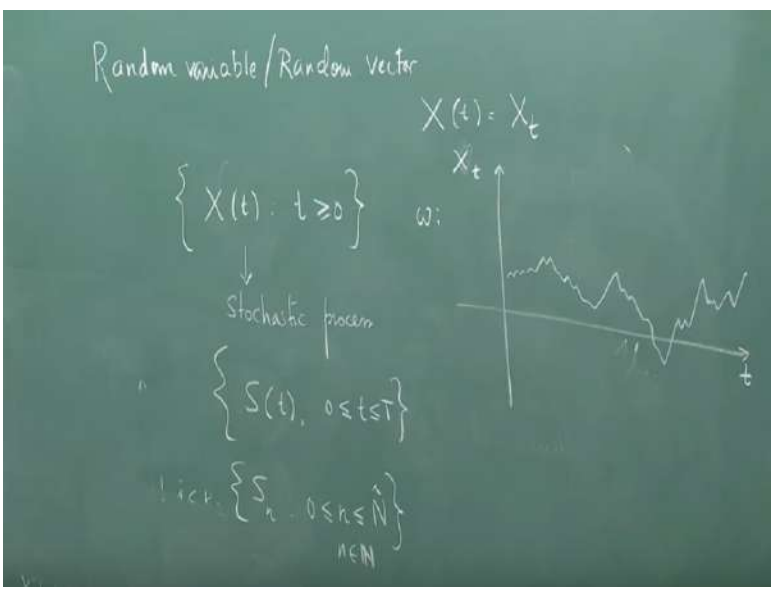

So you take a collection of random variables. For each t, t essentially denotes time, this X(t) is a random variable. Sometimes some authors write it as X(t) as X_t, but we will maintain the symbol X(t) here because it is much more helpful for handling certain things. So this collection, it could be infinite of course or it could be finite also. So this is called a stochastic process.

So for example, if you look at this stochastic process so that S(t) is the price of a given stock where 0 is the starting time where you know what is S(t) at this current moment and capital time T is the time when if your trading is over with that particular object and S(t) denotes the price of the stock of a company share, 1 share of a company, the unit price of the 1 single share of the company within this period.

Of course, the real issue that lies here is the following. The issue that lies here is that every S(t), S (0) you know, but you do not know what is S (t) which is beyond 0. Of course, you could think this is a very continuous setup. You could think of a discrete setup, you can think of S(n) 0 to say capital N where each n belongs to capital N.

So basically then you have s (1), s (2), s (3) so now 0 time I know the price but s (1) I do not know, I do not know what would be s (2), what would be s (3) etc., so that is the stochastic process. So if you take a fixed ω and even calculate for each of the different, see as you change the time your structure of the function X(t) changes; they are different random variable not the same random variable. So s (1), s (0) is of course a fixed, nothing is known. S (1) and s (2) are different random variables, s (2) and s (3) are different random variables.

So if you take a fixed ω from the sample space and then calculate that s (t) ω for every random variable for every t right for every t this the different random variable you press the ω. Then what you get is a sample path of a stochastic process.

So here what you have done is you have t and here you have say S(t) or X(t) it does not matter anyway X(t). So what you do you have you have basically for a fixed ω you can see how the value of each random variable changes at ω.

So in a market setup, if this was again the stock price x (t) then I am trying to see, obviously the stock price would not be negative. What I am trying to see is that if I am assuming that for a given some time period, I would have a particular type of scenario and how the stock prices would change, that is very unpredictable.

Assuming that I had the same scenario, a scenario does not change, market conditions are say at a particular are in a particular way then how does my stock price changes. That is something economists want to know, that something the market people want to know, the brokers want to know.

If I change my, at every t if I change my scenario then it is a true truly troubled market, it is truly a bizarre market which you cannot really handle mathematically. So the real mathematics is done when you are really thinking that over if this is my particular scenario now, how would my prices look like, how would my prices look like. So if you finally look into, so you know given the particular scenario what would be the price at every time T but you do not know what scenario would actually come.

So this is called this is called so what you have done is you have computed X(t) and ω sometimes written as x (t, ω). So this is called the sample path. A fixed ω a changing t, but changing, keeping t fixed if you change ω then it is bizarre basically. So this idea we have just introduced because they are linked to random variable which we will very soon come the use of stochastic process would increase in the next discussions.

So here now we are going to talk a little bit of what is integration and what is the meaning of expectation okay. So simple functions would play a very very fundamental role here. So suppose I have a simple function like this.

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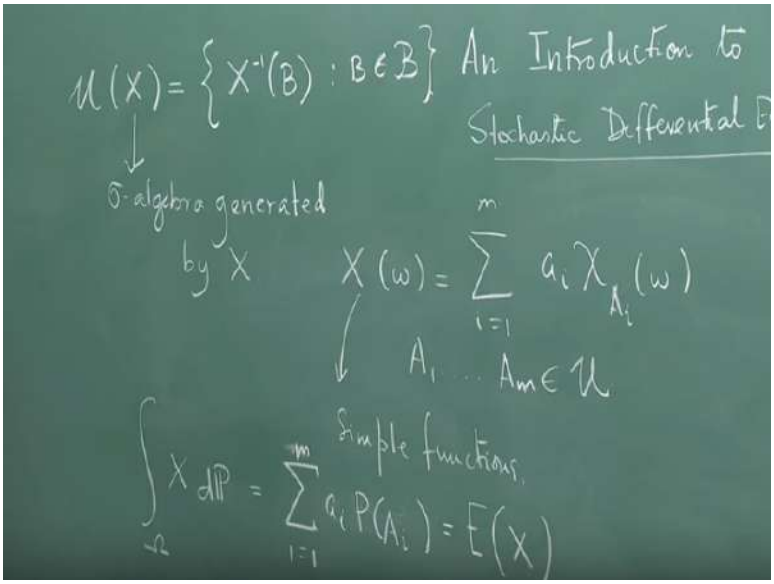

And I want to and I define that the integral of this simple function, every simple function of course is a random variable because you see actually I have defined a random variable. A random variable and also another random variable. So a simple function is, then we defined the integral of x with respect to the probability measure P as 1 to m, let me write this way. So this is the definition. Probability A_i is know because these are the events so you know the probabilities of that.

So then the integral is nothing but this. What essentially is this. If anybody has learnt some probability they will immediately recognize that this is nothing but the expected value but how do I do for, this is for a simple case, very simple setup essentially is a some sort of a discrete situation that if you breakup the whole thing A_1 into A_m then these are just taking the values A_1, A_2, rest of the values are 0.

So I do not need other parts except the parts which are within $A_1 \cup A_2 \cup \ldots A_m$. So this is nothing but the values being taken over those path and the probability. This is nothing but an expectation.

So this is essentially is the expectation of x or the average of x of the random variable x, but how do I define this when I when it comes to an arbitrary random variable and not just a simple function. We define this by using the simple function. This is the way lebesgue integral is defined. I would ask you to really look up because we had told in the beginning that we need

some mathematical maturity to do this course but does not matter that you can still follow by looking at what I am writing is that.

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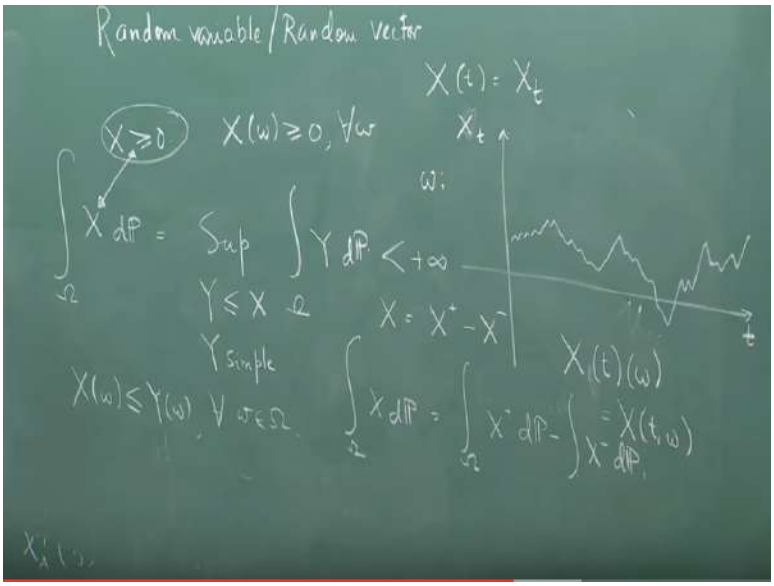

If you have any random variable and the integral of this random variable of course you can talk about a random vector I will not I am not getting, I will write to you at the end what one can do for a random vector but this is just for the random variable. So why I required simple functions? I am looking at all simple functions which are lying below x so this actually means I am writing it for all ω. It will be enough if it is for every if every ω other than a null event.

So let us not get into those technicalities; okay, assume this thing, so this is what is the definition that you can compute for all the simple cases and then take the supreme whatever be the value assuming it is finite so assuming that this is finite then we say that the integral exists. What this integral does is nothing but a generalization of this idea, you used the simple function idea and generalized it, so this integral is called expected value of a random variable x. This of course is defined for x greater than equal to 0 means a nonnegative random variable.

Here, I just want to mention that A_i's that I have taken they, so these are simple functions if so if I am talking about nonnegative random variables some of the Y's can be with the situation where all the A_i's are nonnegative. Some can be some nonnegative some negative. If all the A_i's are negative of course such a Y is less than this.

So this definition actually works for nonnegative random variable that is x(ω). This writing for all ω is not really technically fine but does not matter for this situation. That is exactly what it means. So now if you want to talk about what happens for a case which does not have such sign then you just have to write x as $x^+ - x^-$ so you can always express a real number as the difference of 2 nonnegative real numbers and then this is nothing but so that point I missed when I made the definition but you have to always remember that this has to satisfy this property is greater than equal to 0.

So for that we are defining this. Once you have nonnegative idea lot of good properties can be proved and once you have that then all those can be translated to the other things by this that is so much easier way of generating properties of integral. Of course here we assume that integrals are well known means this integrals you know you know what the properties of integrals.

You know how they add up. You know what happens if I multiply a function by a constant. The constant comes out of the integral sign and all those stuff. So that we will not get into such details here. What we are going to now talk about is that this is what we call expectation of a random variable.

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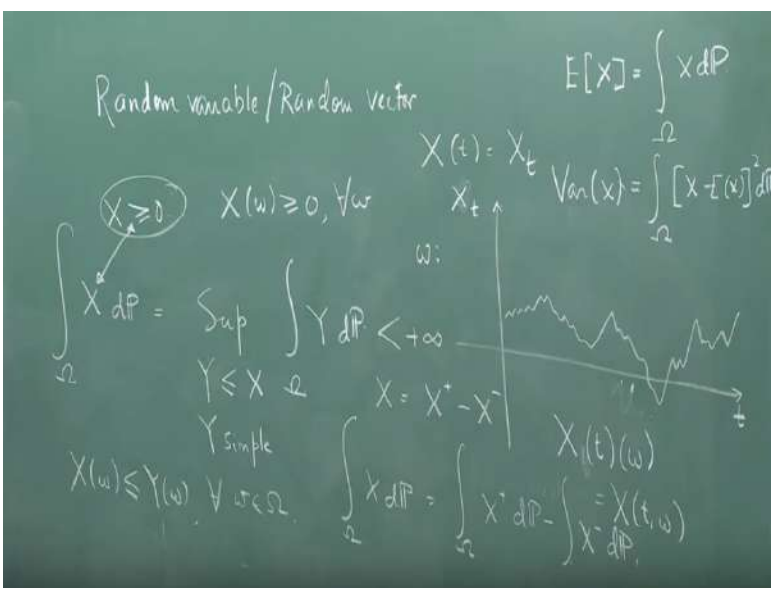

This is what is called the expectation of a random variable and this is what is called a variance. Of course these are all very abstract looking things, it does not make sense and those who have it probability might also start getting confused okay what is really going on but the key idea of really studying probability is to know the notion of a distribution function which tells you how a particular observed random variable would actually behave.

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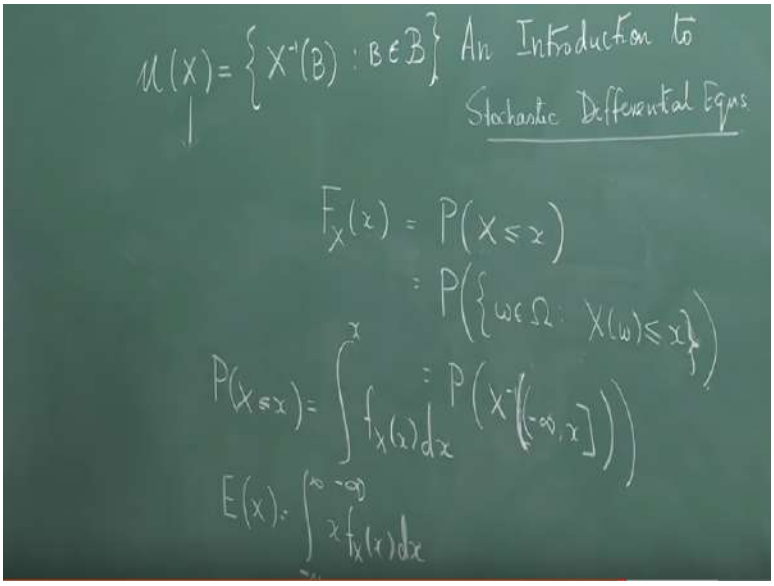

So, given a random variable, the distribution function is a real valued function which calculates the probability that once you want to evaluate this $F_X(x)$ this calculates the probability that X is less than equal to x which means calculating probability of the event a set of all ω in small ω in capital ω such that x(ω). I leave it to you to confirm that these are actually events, these sets are actually new because if you look at this what is this? This set is nothing but x inwards minus infinity to capital X.

So basically you are finding probability of x inwards of minus infinity X and that is why that particular Sigma-algebra is important. Why I need to know the distribution function because that allows us to understand, it is a cumulative thing just like you have studied mean median mode you have talked about cumulative effect of certain observation means we are looking at the growth of x how much with what probability it can come under x.

So once I am able to tell you, I am able to give you some sort of an idea about the stochastic behavior of that particular random variable. So random experiment is giving me some information but that is translated in a nice way in this set of our distribution function. I will not talk about the properties here, but I would just like to tell you that associated with every random variable, it is characteristic.

Usually it is either something called discrete random variable which it can take only countable number of values or finite number of values or a continuous random variable which takes you know non-discrete set of values and then how do I calculate this? The probability is usually calculated by the means of a density function. This I am just recalling the thing, this is something which has to be known in the first year of probability course and once you known this then expectation of x becomes an easy thing to calculate.

So why do you need the density function. It is very important to know when you have a continuous random variable you just cannot say that probability of x is equal to x is something. Probability of a continuous random variable picks up just 1 value is 0. How do you look at it? What is the intuition behind it? I am not going to write down the detail description, I will just tell you an intuition behind it. The intuition is the following. You take the real number 0, 1.

Now you are arbitrarily picking up a point from that set. So what is the probability that x equal to half. So what is my favourable outcome how many objects I have in that favourable outcome 1 but I have huge infinitely large all possible outcome, so you can immediately understand that probability is 0. This is not a mathematical explanation but this is just a rough intuitive idea but intuition is anyway right to mathematics so this density function plays a very very important role and now what I will do is I will introduce to you the notion of independence and by introducing to you the notion of independence I will actually stop my discussion.

So we will talk about conditional probability because we are going to talk about conditional expectation very soon so it is very important that we talk about conditional probability. I am writing, I have some notes written down which can be scanned and later on put into the web. So what is conditional probability?

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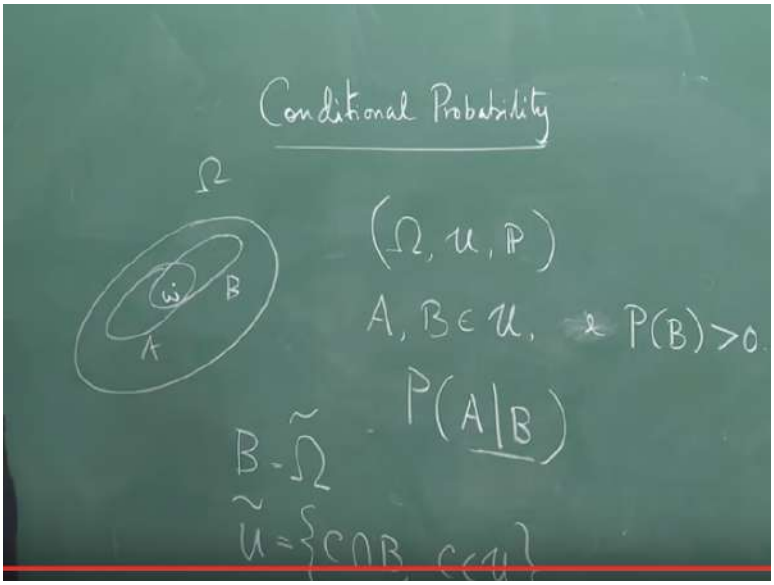

So let me now talk about this very important issue. Conditional probability asks how would be the probability of one given event A be, if I know that some other associated event has occurred. Mathematically it means the following. Mathematically it means that I have a probability space and I have 2 events and it implies that and also I know that probability of B is strictly greater than 0 that B can occur with a positive probability, right.

Now suppose B has actually occurred, so what is the probability of finding A knowing B. so which is symbolized like this P(A/B) and of course you have to assume this that B has a finite probability of occurrence sorry a positive probability of occurrence. So what does this actually mean?

It looks very fishy that you are just using language to talk about this. It means that if I chose an ω in B so B is a element in U. So it is a subset of the sample space so I can take some ω from B assuming they are non-empty of course, because U cannot be empty set because otherwise probability would be 0.

So I am choosing an ω from B. The question what we ask here by asking this A, P(A/B) is what is the probability that ω is also in A, that is the question. So here you have your ω and here you have your A and here you have your B. So if ω is in A what is the probability that ω is in B.

So ω must be in an area where A and B both can occur. So we then have to think of joint occurrence of events, right. Now, if I want to define this probability, this is not the same probability as the one I have defined there because that only talks about probability of event of probability of events in U, but what does this, where does this event stand, where is the Sigma-algebra where is the sample space for this.

So when I am looking at, so my question again is very important and I am talking about if ω is in B, that I know that this ω has occurred, what is the probability that it is also in A? So then A given B is an event which is not really a part of this Sigma-algebra. These are the sequence A and B; A, B, C, D; but here I am talking about something else so which means I really now have to consider a new Sigma-algebra and a new sample space here.

So I am only bothered about those ω 's which have occurred in B. If they have not occurred in B, I am not bothered about them. That is why the talk that I am looking for A given that B has occurred. So we are only looking at those ω 's which have occurred in B, which have come up in B.

We are not looking up at other ω's and thus B is taken as my new sample space which I mark as this and what is my U tilda. U tilda C ∩ B where C is element of U. So B has to be related to all such events. So you see joint occurrence must now come if you are going to talk about condition probability, okay.

Now we define by P tilda the following probability measure. P tilda they have to put some event in by P(B) which I know. But A ∩ B is actually an event in U.
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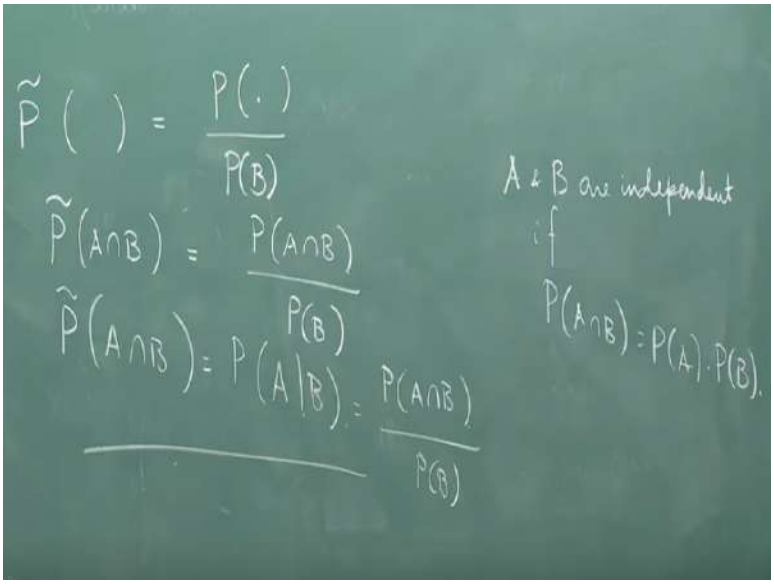

But here I am considering some sort of subset of the event U. Now if I want to consider P tilda of A ∩ B that is under this new probability measure what is the joint occurrence of A and B. That is exactly what we are seeking, a new probability measure.

Now, my sample space has changed. It is no longer ω, but it is ω tilda, which is nothing but equal to B, the event B, right. So then by this definition equal to P (A ∩ B) and we mathematicians define that P tilda of A ∩ B. P tilda of A ∩ B is defined as or said to be the probability of A event B and that is essentially the meaning of conditional probability.

So this is nothing but P (A ∩ B) / P(B). 2 events are said to be independent if it does not matter whether ω was in B or not. The fact that ω can be in A, that probability is independent of the fact that ω has come up in B. That is the meaning of independence of 2 events A and B. So if 2 events A and B are independent that is the probability of occurrence of one does not affect the probability of occurrence of the other.

Here the probability of occurrence of one affects the probability of the occurrence of other because the sample space gets completely changed. So A and B are independent if P (A ∩ B) = P (A) P (B) . So that is it and with this we stop our discussion today and next class we are going to talk about Chebyshev Borel–Cantelli Lemma and some surprisingly interesting facts, okay.
Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 04
Chebyshev Inequality, Borel-Cantelli lemmas and related issues

Yesterday we had spoken about random variable, their distribution function, their density functions. We have not gone into too much detail because we expect that you can look it up in any standard probability book. So, again just I want to recollect that distribution functions are essentially tools which allow you to view the range of the random variable, that is the whole idea and density functions specially for continuous random variable allows you to calculate the probability.

And let us now mention that certain random variables can behave in a certain typical way. That is they can follow a certain type of distribution function that is their distribution function would be actually known means you can actually know the form of the distribution function. Most of you who have done some basic probability in your basic engineering courses would know that there is a famous distribution for a random variable to follow is the normal distribution sometimes called the Gaussian distribution.

So one mathematician had told me that being a mathematician I should always call it to be a Gaussian variable, but I somehow want to, I always like to say that x follows a normal distribution with mean, μ and variant σ^2. So this distribution has a density function of this form.
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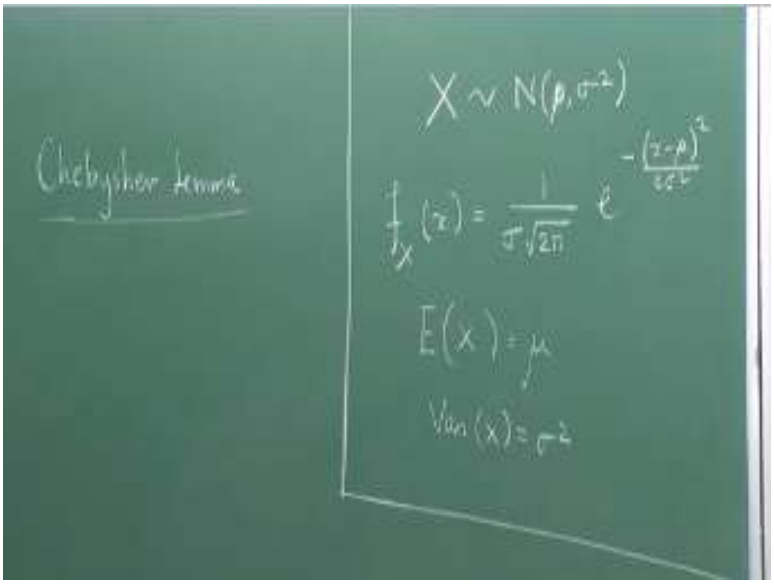

So this is a very famous one and you know that the expectation of x or the mean value is μ and the variants, so please remember our aim is not to teach you very basic probability and statistics. I am sure there are a lot of courses in MOOC which many couple of courses which will do it. Our aim is to very quickly with the very basic introduction to take you to the mathematics or probabilistic tools that are required for finance. Our aim is not just knowing probability theory then we should not have reached this point in this few lectures.

So what is our next step. Our next step is to learn this very important lemma called Chebyshev's lemma and then learn the Borel-Cantelli lemma. These 2 lemmas are useful throughout, they will be used throughout because they actually try to look at the behavior of sequences of events and sequences of events are very important when you talk about finance. So let us write down the Chebyshev's lemma.

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So X is a random variable. So we obviously distinguish random variable from a random vector. We had just given a definition of random vector in the last class, but we will just essentially consider random variable that is this X is from ω to R. So let this be a random variable and p is a number which is lying between + 1 and + infinity.

Then the probability, see a lot of things in mathematics is actually you are estimating certain quantities which you cannot calculate. So essentially this tells you how to estimate probability if you know if you can find an average, right. So it tells you that so if you take the modulus of the random variable that is basically you are essentially talking about non-negativity here and this is bigger than equal to lambda, a λ and for any λ > 0.

The fact the set of all ω's for which the modulus or absolute value of x(ω) is bigger than or equal to this positive quantity λ, this probability is always less than whatever p you choose between 1 to infinity does not matter, you will always have this to be true. This is called the Chebyshev's lemma. Now how do you prove it. Proof is very simple.

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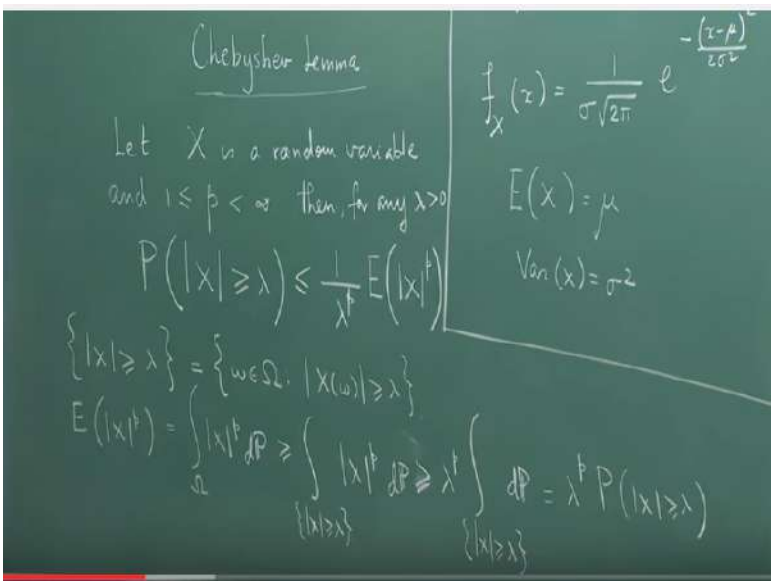

So let me denote in short this set which is the set of all such that, actually greater than equal to, so this, whenever I write this I am actually meaning this that has to be kept in mind. Now how do I formalize the steps. So again if I go by the standard definitions which we have already given in the last class and this is nothing but integral.

Of course, everything can be written in terms of the distribution function which you know, but here we are writing it in the more compact integral or measured theoretic form. This a whole is obviously bigger than the part so this is so in general either if I consider the ω is just for this set

obviously these are subset of this so the integral would be smaller because you are talking about non-negative functions.

So this would be observe that what does this mean on this particular set | x (ω)| is always bigger than λ so it will be greater than equal to λ so greater than equal to λ^p because here I have the raised the $| x |^P$ so what I would finally get from here is again I have to put the greater than equal to notation. So I have λ^p .

So what does this mean? This simply means the probability of this particular event, probability of this event so this is nothing but λ^p P (|x|>= λ) of and you see you have proved the Chebyshev's inequality. Of course, Chebyshev's inequality will have applications which we will come very soon.

Another interesting idea that we will come to is the Borel-Cantelli lemmas. So we are now going to talk about what is the meaning of the notion of a sample point occurring infinitely often in a given sequence of events. So if there is a sequence of events say 1 dot dot dot n the question is that if I have say ω element of the sample space what is the probability or what does it mean by telling that this ω occurs infinitely often in this sequence, right.

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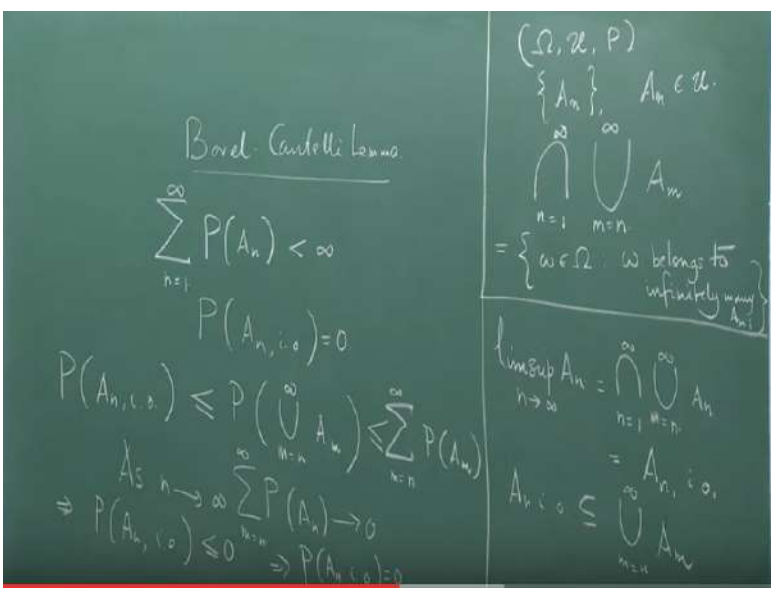

So what we will first do is to consider a probability space and then I will consider a sequence An where each An belongs to U and then I will write this I will construct make this construction. I will fix up a value of n and then take m = n infinity and then I will change and then I will shift the n so I will say I will take n =1 then I will take 1 to infinity. So I will take the union of all the events, all the events here, then sorry I made a mistake this is intersection.

So what I am going to do, first I will take say n =1 so take the union of all things union of all the sets here then I will take off take out A1 from there so I would start from A2 to Am, I will take the union of that and keep on doing it and then try to find a common intersection, right. This thing is often written as follows.

So what sort of final set it consists of. Now this is an element of the, question is, is this an element of the Sigma-algebra. So once you take the union the union is in the Sigma-algebra although unions are basically what you are taking you are taking the unions. So each of the union are in the Sigma-algebra. If you have countable union in the Sigma-algebra, then you know that the countable intersection are also in the Sigma-algebra, right because you can take the demorgan's law.

So this set is actually an element of the Sigma-algebra. So this is, what sort of sample points are here. So this is all those things where ω belongs to infinitely many An's. So what does this mean, belonging to infinitely many An's. So it might belong to A1 then A3 then A5, then A6, then A7, then A8 and A9 and then say from A100 to A101 and just goes on, something like this.
Suppose one ω belongs to A1 and A2 nothing else it does not come anywhere then we do not consider it here.

This is sometimes written as this set is also sometimes written this set is also sometimes written as because it satisfies the type of thing that we do with actual sequences of real numbers. It looks almost similar to that. I will allow you to ponder over what I am writing here whether it makes sense intuitively to call it lim Super An . So we also call it this.

But Borel-Cantelli lemma says that this gives you a more fascinating proposition. It says that now okay each of these P An's if this is there if this is then this is a null event, this is a null event, that is beautiful. That it says if your probabilities have a finite sum this infinite series then this event must be a null event. It is a huge conclusion you are drawing. That is there is no ω which will belong to infinitely many An's and all the ω's will belong to some finitely many A ns. That is a huge conclusion.

Sometimes you write this set also as some authors also write An infinitely often. So, this one is just a symbolism of this, it does not mean anything. So I will just use this shortcut symbol so ω belongs to An. So what would happen that there will be some An in which there will be infinite number of ω's coming. That is the key idea here. Okay, do not bother about this point. Just you are essentially constructing a set that ω belongs to infinitely many An's and we are trying to find the probability of this.

These types of sets become very important when we will study central limit theorems so we will not do much proves there, but we will give you a hint of what can happen and so let us draw the conclusion. so if this happens it tells you that P(An, i o) it is same as adding the lim sup that is 0, that is a null, set empty set. There is no ω which belongs to infinitely many An' s. So how do you prove this. Proof is very simple.

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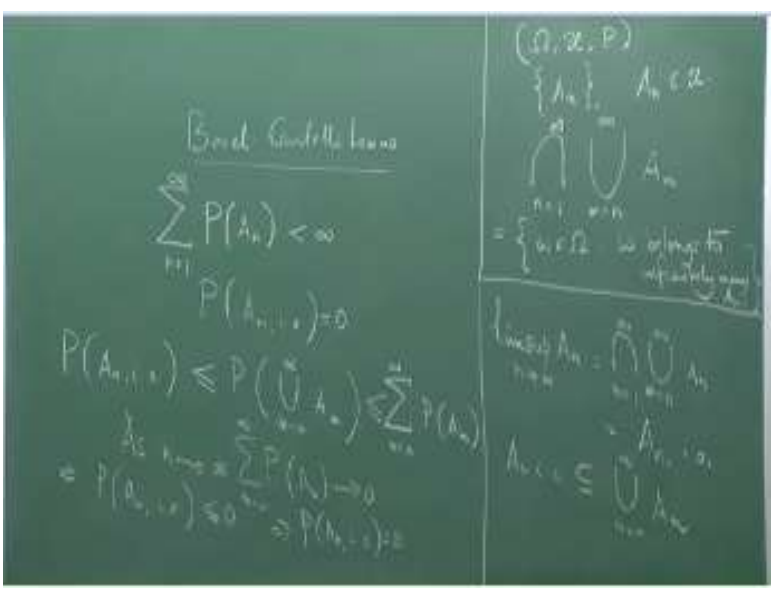

So if we look at this set, so let me take some n and look at this union m = n to infinity say n =3 say, m = 3 to infinity. So that set is one of the set. So if you have A1 intersection A2 the intersection must be smaller must be a subset of either A1 or A2 is a subset of both. See if I take any An it does not matter. This whole thing is a subset of union m = n to infinity because what I am trying to say is that.

This whole that is An i o is a subset of for some n whatever n you choose not for some n all n actually whatever n you choose does not matter. This is always true. So if that is true which will immediately tell you that the probability of course the larger set would have a larger event but event which has more possibilities of occurrence will have a larger probability. But what does this mean, the second means by the very Kolmogorov axioms, but if I look at this.

What is this, this is a tail of a sequence tail of a rather a tail of a series summation P A and n =1 to infinity is a tail of the series that is we are shutting off finite amount of elements cutting off finite number of terms of the series and looking at the remaining part. So the tail of the series will always go to 0 if this sequence right, if this sequence actually strictly less than infinity.

If this sequence is convergent then the tail always goes to so this is a notion from infinite series which I expect the viewers of this is supposed to know because that is the sort of requirement that we had that you have to have some understanding on mathematical analysis. Now if I look at this one., so what is happening as n tends to infinity this will go to 0. Now this is actually a fixed probability.

This is a fixed event. So which means as I take the limit from this side and this side this remains unaffected by the limit operation because this is already a fixed set, everything is done, so which from here, so we will have this, so we can conclude now that when any probabilities are always greater than equal to 0 and this will immediately tell me.

Now let me show you an application of all these things that we have just spoken, an application of the Borel-Cantelli lemma and application of the Chebyshev's theorem and that comes out when we are talking about convergence of random variables. If I am using a vector valued

random variable or random vector then I if I need to use it I will make an explicit mention of that.

Otherwise, please understand that we are always working with random variables that is functions which are mapping sample points to real numbers. Now there is something, there are 2 types of convergence which are very important; one is convergence in probability and convergence almost everywhere.

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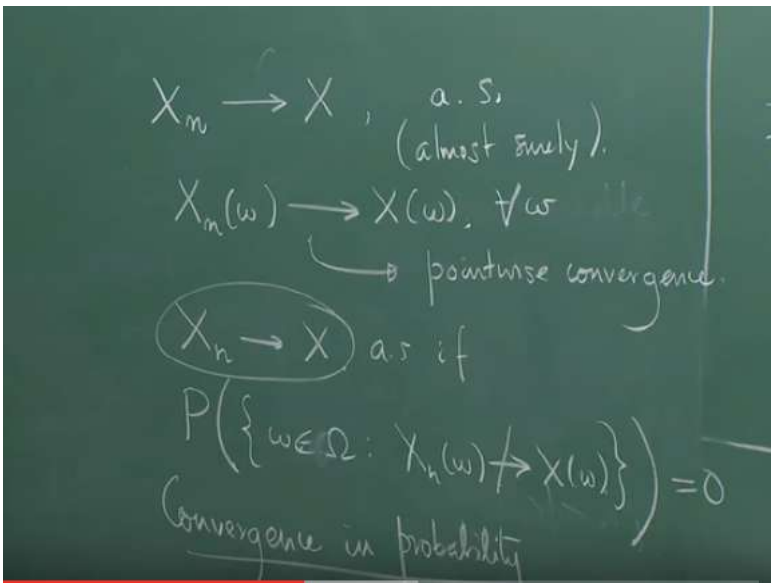

So, suppose I have a sequence of random variables Xn and I say Xn goes to another random variable X and we say it converges almost surely, almost everywhere convergence, this is called almost everywhere convergence, a.s. is a short form of the word almost surely. So when I am talking how, so these are random variables are functions, so a sequence of function converging to another function.

So when I would say when can I say that what is the meaning of a sequence of functions converging to another function. It means, technically it should mean that for whatever ω that I have in ω this is what it should mean if I am talking about pointwise convergence. So if this happens then this is what is called pointwise convergence, this is what is called pointwise convergence and then one needs to understand that in probability theory the tendency is always to throw out null events.

So we say Xn tends to X almost surely if probability of the event it is the collection of all ω's such that this is not true. Probability that Xn and ω does not convergence to ω. The collection of all such ω's is a null event. There can be some ω's in the sample point for which this may not happen, but their collection would form a set whose probability would be 0. If that happens then we say Xn converges to X almost surely and then there is a notion called convergence in probability. We will see how these are connected.

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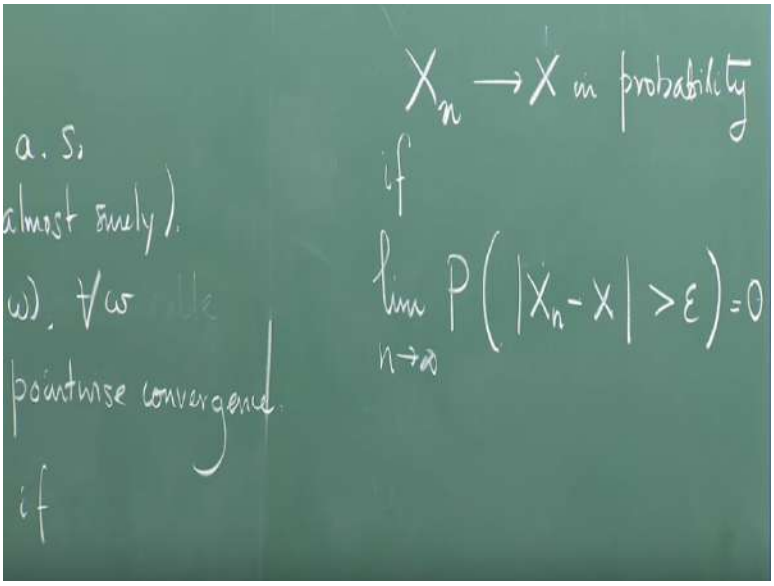

Convergence in probability, so what is the meaning of convergence in probability. Let us discuss this little fact. We will take 10 more minutes to finish everything so let me discuss this little fact, convergence in probability.

So convergence in probability means we say Xn converges to X in probability if, so we are telling that a set of all ω's or any given ϵ take any ϵ, if the set of all ω's so this is the probability that the set of all ω's such that the distance between x and x is strictly greater than ϵ keep on increasing the value of n right, see suppose I say x on minus x strictly greater than ω.

I collect all those ω that has a probability so you have x 2 minus x strictly greater than ϵ there could be some ω where for which that is true so that will have a positive probability but then as I increase the value of n this should go to 0; that this distance cannot be bounded away from 0.

There cannot be positive number such that this distance would be always greater than the positive number.

So the fact that that probability in the limit, the probability that this distance is bigger than some ϵ would be 0. So you are expected to, xn is expected to be coming towards x, that is the whole idea, that sort of explanation one might think.

Of course if you are satisfying this you are satisfying that but there is an interesting that okay if I am converging probability am I converging also almost surely. See if I am converging almost surely, I leave it to the listeners or viewers to decide whether they are converging probability, the answer is yes. Now the question is if I am converging in probability am I, is the same sequence also converging almost surely.

The answer is there is at least a subsequence which is converging surely to the same random variable. So this is a theorem if you want to write it like this.

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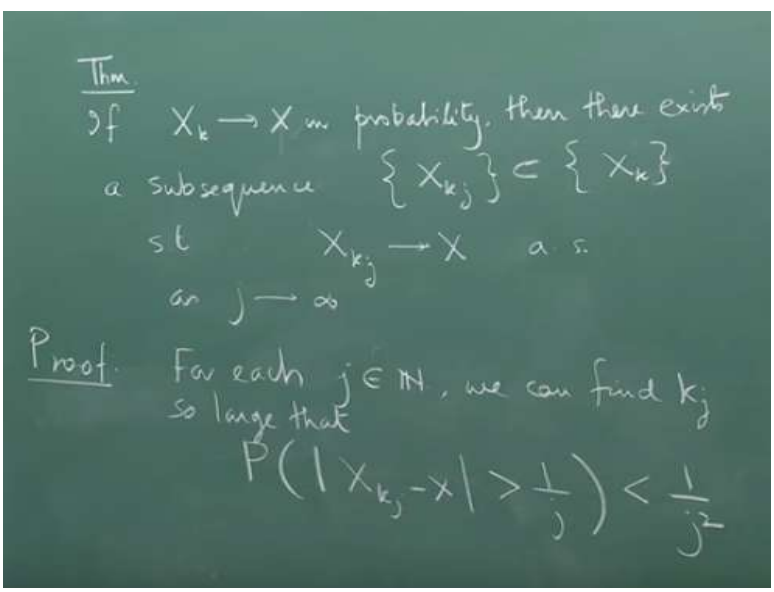

If X_k converges to X in probability, then there exists a subsequence X_{kj} of X, X_{kj} which is the subsequence of X_k such that this subsequence converges almost surely to X. Now, this definition of probability you must understand that this should be true for all ϵ >0. Whatever we have chosen ϵ this will always happen. It does not matter for some ϵ this will not happen.

This has to happen for all ϵ. So that is something you have to keep in mind when you are talking about convergence in probability. So what does it mean? So how do I prove this? So ideas that you take any integer j does not matter whatever integer you are taking, take an integer j.

We can always select k j or a index value so large that this difference, this difference the fact that the probability that the difference between X_{kj} and X is bigger than some $(1/j^2)$ or $1/j$ something like this can be made smaller than some quantity.

So the ideas that for each, why can we make it as smaller as I like, because I have this fact, because I know that it converges in probability that this what does this mean. This simply means that when n is large this probability can be made as small as I like, can be made smaller than some delta, right. So, given for each j in N we can find k j so large because this is where we are exactly applying the fact that we are talking about, the fact that x tends to x in probability that definition we are now applying.

So for each j we can find a k j so large that $P(|X_{kj} - X| > 1/j)$. This is my some sort of an ϵ. This is less than some delta, that delta may be chosen as $(1/j^2)$. So I can find a, k j if this is my delta. I can always find a, k j for which this is bigger is lesser than $(1/j^2)$. This is exactly I am applying this definition because this holds for whatever ϵ I want.

So once I have done that you might note that I can always chose $k_1 < k_2 < k_{j-1}$ because we are essentially taking a larger and larger k j. So this can always be done. These are very simple techniques, mathematical analysis which mathematicians use, so I am essentially giving a small warning, even to make money you need to know some good mathematics though many mathematicians do not have money but.

Now let me try to show you what does it mean. Now let me construct, so what you are constructing. You are always taking an increasing k j. So this k j is actually going to infinity. So you are increasing j, as you increase j you increase the k j, right. So this is the increasing function of k j increasing function of j.

This is exactly the definition of a subsequence because in subsequence when you chose the n k or k j you are actually constructing and increasing function increasing the function of the indexes because you have to maintain the order. That is the meaning of a subsequence.

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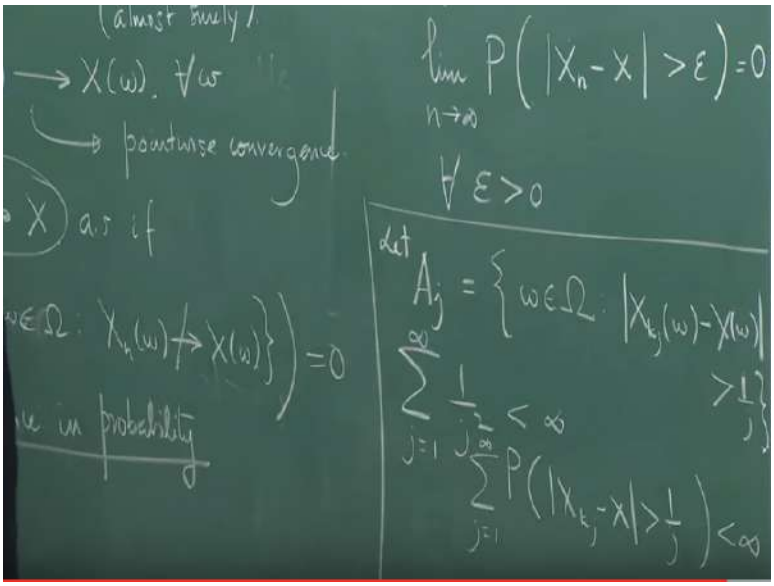

Now you construct, let us construct this sequences, this event. A set of all ω in ω such that $|X_{kj}(\omega) - X(\omega)| > 1/j$. so I am looking at all such for every j j equal to 1, 2, 3, 4 I am just trying to figure out. So for given Aj this should happen. So find, for every k, for take the k j and put that particular j as this Aj index, index will be the same j. So here I have an infinite sequence of events.

Now I know that probability of all them is strictly less than $(1/j^2)$ so you can sum it up from j =1 to infinity. The probability is strictly less than infinity because summation $1/j^2$ is a convergent series, right. So here I will get since summation $1/j^2$ is strictly less than infinity, it simply tells me that probability and this is a place where you apply the Borel-Cantelli lemma.

So what does the Borel-Cantelli lemma says that this thing this would imply that probability Aj infinitely often is 0. Sample points ω do not occur infinitely often. So what does this mean? That is for almost all sample points except a few of them, for almost all sample points what do I have. So here is something we now have to look into.

So for almost all sample points, so for almost all sample points we have, so this is this probability is 0. So for almost all sample points this has to be true.

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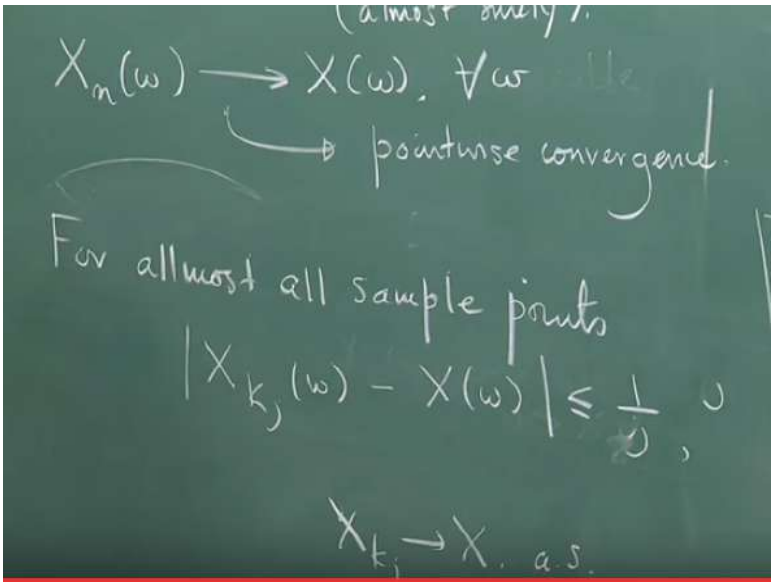

So ω cannot be occurring at infinite many Aj's. So for infinitely many ω's the opposite must occur. For infinitely many ω's this will not occur, only the opposite will occur.

So what does this mean where j is an integer greater than some particular j some index depending on the ω it does not matter. So what does this mean? This simply means that X_{kj} goes to X in probability. This simply means that $X_{kj}(\omega) - X(\omega)$ can be made as small as I like as j is increased. What does it mean but this is not true for every ω but almost all sample points ω. This simply means that X_{kj} - X a.s.

So with these ideas I would like to stop my discussion here and tomorrow we will see that how these ideas can be used for example to talk about central limit theorems and law of large numbers which are actually very useful while you study Brownian motion and Brownian motion is central to understanding the flow of stock markets and also the Ito calculus depends on Brownian motion.

The Ito calculus is the calculus of computing derivative pricing, pricing of derivative. So let us stop here and let us see tomorrow what we can discuss about central limit theorems. Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
IIT Kanpur

Lecture - 05
Law of Large Numbers and Central Limit Theorem

Today we are going to study something called law of large numbers and Central Limit Theorem. These are parts of any formal study of probability theory, but these are essentially about repeated experiments. We generate what is called random sample. Suppose we want to study something about the population of some you know population of population of the heights of the male members of a city.

You cannot really go and check the height of each and every male member. So what you really draw is a random sample so you really check the heights of certain number of male members of a city and then using that information, that data, you come to certain conclusion about the actual or the average height of the male members of that city or some more information about their heights.

So this information that you gather is essentially by this process called random sampling. So you do not look at the whole population, but you look at certain members of the population. So taking a random sample is same as taking a making a repeated experiment. So for example when you are going to measure the height of people, so you are making a repeated experiment. You are measuring the height of person 1, then person2, then person3, and so and so forth.

So your experiment is the same. You are measuring height and you do not know what is the outcome, but outcome is random so that is a random experiment. So this lecture is going to talk about repeated random experiments.

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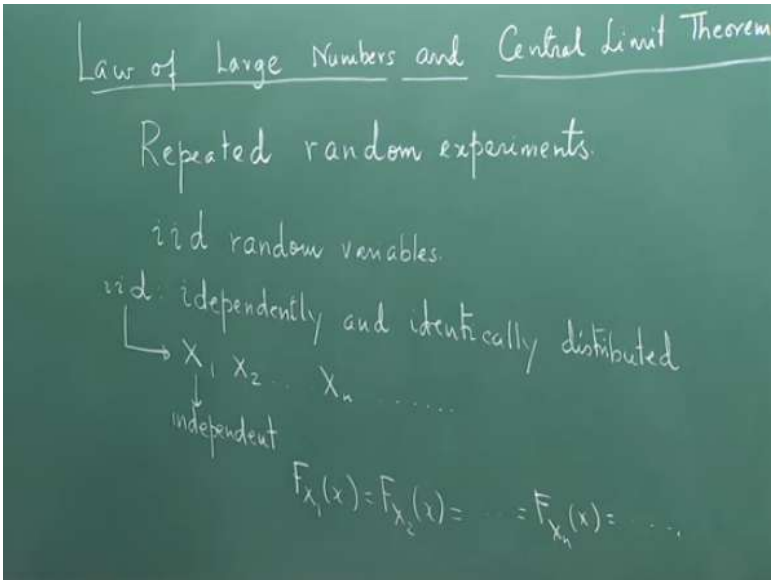

So to understand that we will first begin with the notion of iid random variables. So what does iid mean? IID essentially means independently and identically distributed. So the iid is a short form of independently and identically distributed. So the formal definition is that if you have a sequence of random variables, could be a finite sequence, could be an infinite sequence it does not matter.

If you have a sequence of random variable x1, x2, xn then iid means that these are independent, so iid means that iid this means that these are independent, so these are all independent and identically distributed means that distribution functions are same, they are the same functional format. IID random variables are the bread and butter of parametric estimation in statistics and so and so forth.

That is the meaning of iid. So the key idea of this repeated experiments leads to the idea of sampling or more precisely random sampling if you want to be more erudite.

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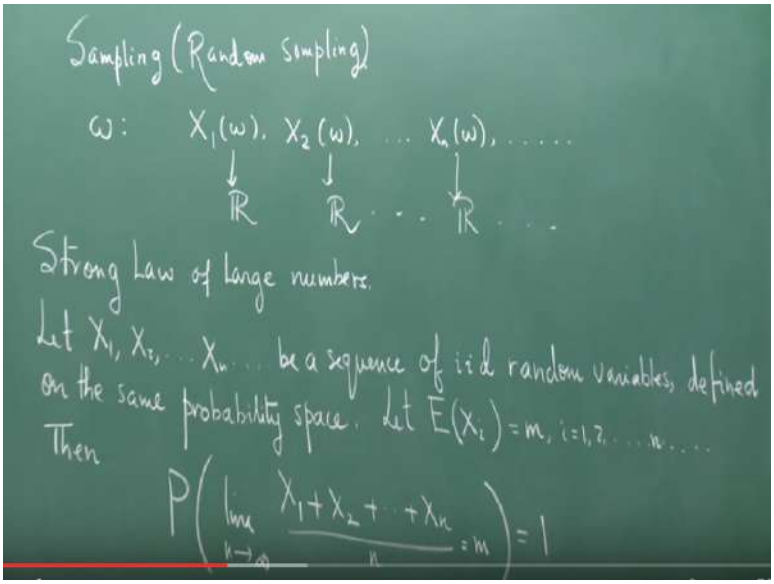

So what do we do? Given a scenario ω, I am taking the value of the random variables each one of them at ω. So I am creating the sample path. So this observations are numerical observations. These are real numbers. Each of them are real numbers. This is real, this is real, and this is real; so all are real numbers.

So these are observations and if you take a finite number of them then I say that I have taken a finite sample from the population. So there are say 5000 people whose heights I have I have to know that this city has say 5000 people and I want to know what is the average height of the people in the city. So I do not go and measure 5000 people, that is a waste of time. The whole key idea of statistics is to do stuff by taking less observations and giving almost a nice answer.

So, if you want measure the average height you take a random sample. Of course, there is a whole study of how to take random sample. So you take a random sample and then you look at the heights of those random samples and take an average of those guys and that if the random sample is taken properly then that would actually be a fairly good representation of the actual average height of the population.

Of course, if you take a random sample quite large then you would be giving much better approximation to the original actual average height which you would not know. So this if you take say only finite number of them then you are essentially doing random sampling. So

associated with this is first is the strong law of large numbers. You would ask me is there anything called weak law of large numbers, there is something called weak law of large numbers which will be a part of the exercise so here we are taking about strong law of large numbers.

So what does strong law of large numbers tell us. The strong law of large number tell us that if the size of your random sample increases then given a scenario ω, the average of the sample, so if you take a finite number of sample. So suppose you have taken 500 samples then you calculate the average height of those people, then this will be an approximation to the average height of the total population.

But if you keep on increasing the number of people, number of people in the sample then your sample average goes towards or tend towards the population average with probability 1. That is the idea. So sample average tends towards the population average with probability 1. So let me write down the strong law of large numbers.

Let X_1, X_2, X_n be a sequence of iid random variables okay, random variables defined on the same probability space. So because they are identically distributed they have the same average, they have the same mean. Assume that they have some distribution. So once you take this you are assuming that they have some density function and you can calculate the mean, so you have to make this more simplifying assumption.

So that is say n then the probability that the sample average leads to the population average which is m is 1 that is limit which is quite natural. It actually collaborates with our intuition so this is a very natural result. Of course, this simply means you are calculating this at every ω.
So given a scenario, so what is this event? So it is a set of all ω such that whenever I take the limit of this as n tends to infinity this becomes m.

That set is a sure event. So this probability this event is a set of all omega in a sample space, set of small ω such that limit n tends to infinity $x_1\ \omega + x_2\ \omega + \ldots\ x_n\ \omega / n = m$ as n tends to infinity. Such a event is a sure event and that is exactly what the theorem states.

I do not want to prove the theorem right now unless I have time at the end. You can of course ask for the proof in the forum if you want aware the proof can be given in the forum but the proof is slightly involved. It is better to have at the first stage many people would be possibly seeing this for the first time and so it is better to have a clear idea what it says than just going into the technical proof.

The technical proof here depend on 2 things. One is the Chebyshev's inequality and one is the Borel-Cantelli lemma, which we have done in the last section, in the last lecture. Now once I do this we have to move gradually to the notion of the strong law of large numbers. So in order to do that we introduce what is called the Bernoulli variables. The Bernoulli variables are random variables which takes either the value 1 or 0 means it denotes success or failure.

So if I toss a head so if X is a random variable which denotes the number of head then X(h) is 1 and X(t) is 0 that is by by success if I mean that I have head and if by failure I mean I have tail then X(h) the random variable when evaluated at h should give me 1 and random variable which is evaluated at t should give me 0. Such a random variable which actually encode success and failure that is called a Bernoulli random variable.

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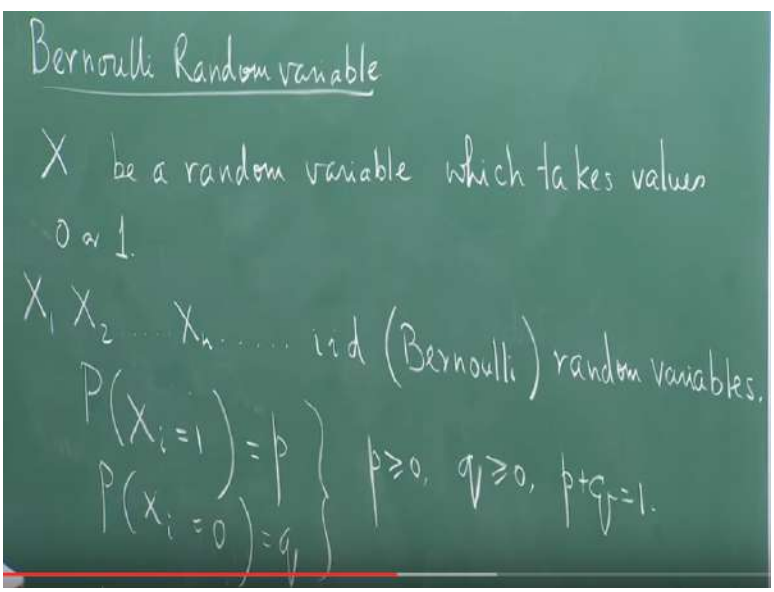

The Bernoulli random variables are random variables that take values either 0 or 1. So for every point in the sample space when you chose that point if it is considered as successful, will give

you the value 1 or will consider as a failure and give you the value 0. So let X be a random variable which takes value 0 or 1. So these are called Bernoulli random variables, right. Now take a sequence of so these are iid random variables which are Bernoulli, iid Bernoulli random variables and they either take value 0 or 1.

So for every i I have given that if there is a success the probability of success is p, if there is a failure the probability of failure which is 0 is q and p and q both are greater than 0 and their sum is of course 1 the total probability and any one of the two has happened it will surely happen either it is success or it is failure. $p > 0$, $q > 0$, $p+q=1$, okay.

So once I know this I can of course compute its expectation and compute its variance. So this will go as a homework to you. We will not compute this but we will write down the results. So here we are now continuing from this part of the board and going up to this part.

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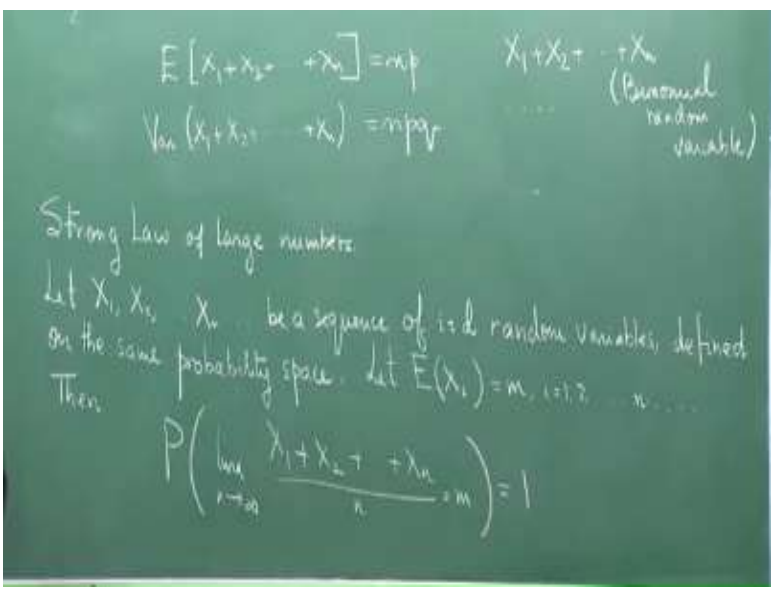

So if you have iid Bernoulli variables as stated since as we have stated the probabilities of success and failure and the expectation of the sum, so this sum actually means how many success I am having in a n trial, n repeated trials of the random experiment, how many are successes. So if toss a coin toss head and tail so say up to say 50 tosses how many coins are actually head because for those the random variable take the value 1.

So such variables which measures which gives you the number of successes in a given number of trials such variables are called binomial random variable so this x1, x2, xn are called binomial random variables. People who have studied binomial distribution would immediately catch this fact. So this is called a binomial random variable and the variance because these are all independent.

Of course I expect you to know this very basic way of calculating variance as an expectations as I have again repeatedly told you that this is not just the first course in probability theory then we would not have reached this part within this few lectures. Because of independence this is nothing but npq and those who know binomial distribution they perfectly know that these are the mean and variances.

So we will now provide some sort of a limit theorem which is called the Laplace de Moivre theorem for Bernoulli random variables only. Once that is settled we can, that is one of the precursors to what is called a central limit theorem. This is sometimes also written as Sn the sum of 2 n variables so I am removing the strong law of large numbers writing and write down the Laplace de Moivre theorem which we state for binary random variables which could be actually better.

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So this is also one form of central limit theorem but starting with iid random variables which are Bernoulli. So we consider X1, X2, Xn as iid random variables, iid Bernoulli random variables. So now we will define the way we have defined Sn so I am just defining for the heck of it, we could have just picked it up from there. Then for any a and b finite a and b what you have is the following.

So here I am normalizing the binomial random variable by subtracting the mean and dividing by the square root of the variance which is the standard deviation which is called the standard deviation. So whatever a and b you chose this limit as n tends to infinity will always provide you the probability that a standard normal variable is lying between a and b because here you see it is the density of the standard normal distribution, standard normal density function. So what does it mean? What is the conclusion from we can draw?

The rough conclusion that comes in is that if you increase your number of experiments repeated trials, as you increase the number of repeated trials, the Bernoulli and the normalized binomial variable this is called the normalized binomial variable; the normalized binomial variable starts looking or rather starts behaving like a standard normal variable which is very important because binomial variables are discrete random variables while that is a continuous random variable standard normal distribution.

So there is a deep link between the 2 and that is something which need to be seen. Now this is the beauty that your limit theorems are connecting the discrete and the continuous random variables and which is a very very important thing. So what are the interesting conclusions we can draw from the Laplace de Moivre theorem? We will actually later on modify the Laplace de Moivre theorem and write down the central limit theorem. The Laplace de Moivre theorem has the following.

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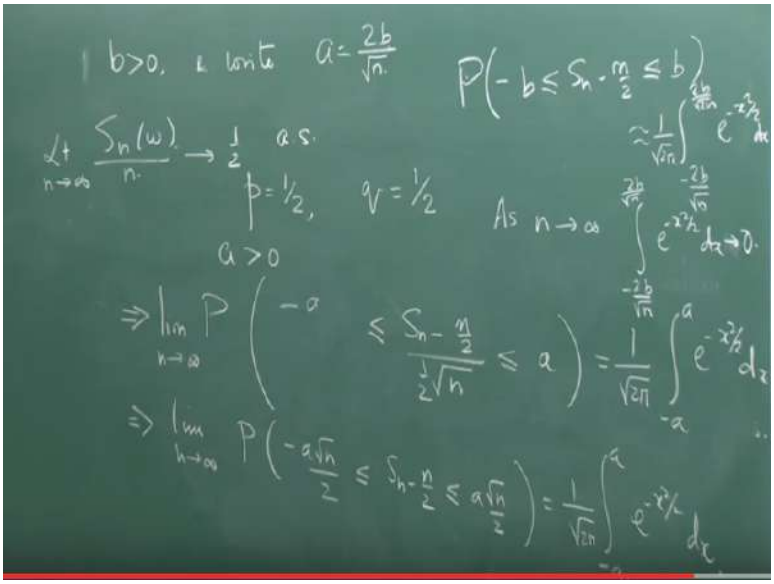

Suppose I have scenario where my probability of success is half just like a coin toss this is basically the coin tossing scenario. Now for any a greater than 0 it implies using the Laplace de Moivre theorem that limit n tends to infinity probability of so np , p is half here you have root npq which we have taken on this side, this has come one fourth, so I have taken this here.

So we are basically looking at the whole thing between a and - a so I would write it with 1 more step so basically it is this one and this okay. Now this can be put in like this so this can be written also as limit n tends to infinity P(-a √n / 2 ≤ Sn –n/2 ≤ a √n/2). This probability is in the limit is same as a standard normal variable lying between - a and + a.

Now write down take b to be strictly greater than 0 take b to be strictly greater than 0 and write a=2b/ √n then what I get from here is so what I get from here is that probability of course there is a limit thing I am not writing the limit I am just writing in a very rough sense this is very roughly so I am just plugging in a with this.

So from here I get a=2b/ √n so I get this is almost similar. So there is a limit so for very large n this is almost very very good crude approximation of the probability not very crude but alright, acceptable. So what does this show. As n tends to infinity this goes to 0 this goes to 0 so they tend to 0. So as n tends to infinity this whole thing it is a bell-shaped curve it finally just becomes a line and then it goes to 0.

This goes to 0. As n tends to infinity not equal to 0 just goes to 0. So essentially what I am telling that even though I know that the average that the average Sn (ω)/n so if take the average this goes to half in this case using the strong law of large numbers with probability 1 that is almost ω almost surely. So means for every ω except a few this goes to half. So probability of this event is equal to 1.

That is what we get from a strong law of large number, but what we are seeing here is something very interesting. It says that you cannot bound this difference. The fact that Sn, this actually means probability this one. Means probability of |Sn- n/ 2 | ≤ b this is somehow going to 0 is decreasing. So the fact that you can bind Sn - the Bernoulli the binomial random variable the deviation of the binomial random variable from its mean that you can bind it within certain value is that probability is less.

So it can fluctuate outside the value infinitely often or almost everywhere. That is for almost every ω, Sn- n/ 2 this should be strictly bigger than b which is a very very, so this can fluctuate with probability 1 of any value. This deviation can go over any value with probability 1. This is a very strange conclusion of the Laplace de Moivre theorem.

Now we are going to remove the Laplace de Moivre theorem the title and end it with the central limit theorem. So what is the central limit theorem? Central limit theorem which I write for short as CLT considers iid random variables but does not consider a Bernoulli random variable so you forget the Bernoulli.

So you define Sn which is no longer a binomial variable just but a random variable which is a sum of n random variables where E (xi) = μ for all i and the variance because they are identically distributed so they have same mean variance we assume that they have mean variance. Of course there could be random variables which are distributed by some distribution it has some density but does not have a mean like Cauchy distribution but we are here assuming all those things. So, it is sorry σ^2. So then the CLT says that then for any n s we have to replace this by μ and replace

this by sorry replace this not by μ but you have sum of it will be nμ and replace this by n σ^2. So it will be σ √n.

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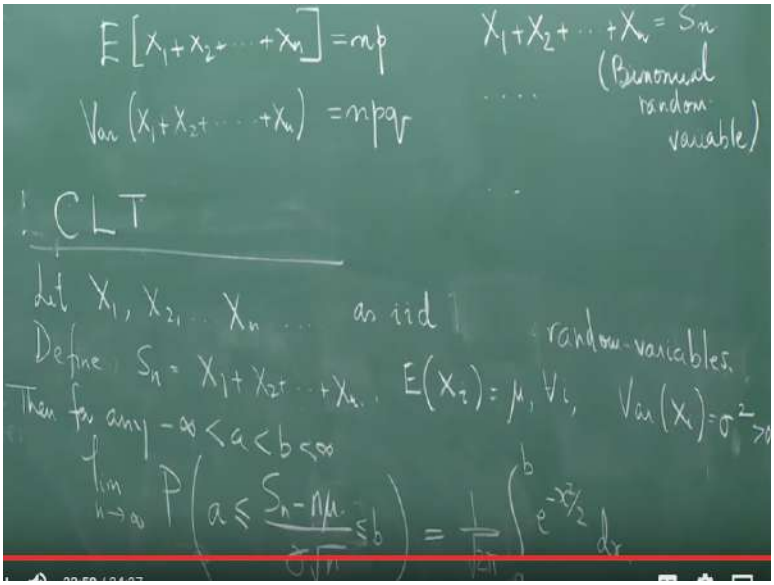

So this will be n means σ √n so if you take any iid random variable it does not matter and if you take the sum and normalize it in this fashion in terms of the mean and variable so nμ is the expectation of Sn and the standard deviation of Sn is σ √n right. So if you take a random variable subtract from its mean and divide by standard deviation then when n becomes very large that is the sample size is very large it starts behaving like a standard normal distribution.

So when you take large samples you assume their population is behaving like a standard normal population. Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 06
Conditional Expectation-I

Welcome to the first lecture of the second week of this course. Today we are going to discuss something called conditional expectation. Conditional expectation is the first most important thing, that I need to discuss with you. The remaining part, that I was discussing with you is something I was, expecting that you had almost some idea about. But here we start with something very new usually not done in standard courses of probability. Standard courses mean courses which engineers usually take or economists take, etc.

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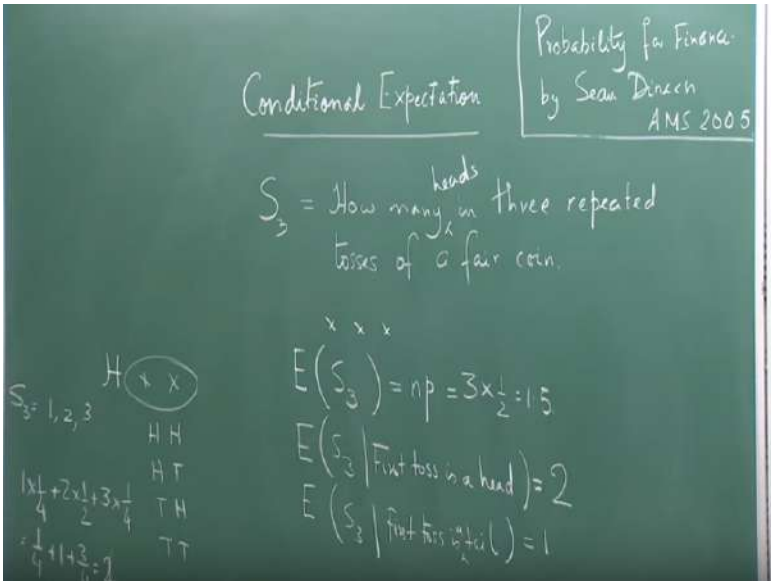

So, the exposition that I found most interesting was from a book called Probability for Finance by Sean Dineen published by American Math Society in 2005 under the series Graduate Texts in Mathematics. So, what does conditional expectation mean? We will see, a very simple example. Suppose, I take this S is a random variable or S is a random variable, which tells me, how many times head appear, when I toss a coin three times. So, how many heads in three tosses of a fair

coin, repeated of course each, first toss, second toss, third toss, three repeated tosses of a fair coin.

Of course, you know this is a binomial random variable. We have already know, we know that, if N be the number of success and Np is the expectation, given that p is the probability of success, sorry, how many heads, so just make a little correction, how many heads in, three repeated tosses of a fair coin. So, it tells you that how many successes you have. So, by success, I am meaning the appearance of a head. So, I toss any of the three can come; head, head, head, head, tail, head and all these things.

So, you know that expectation of this because, this is a binomial random variable, is nothing but $n \times p$ where probability of success is $\frac{1}{2} \wedge nis3so3 \times \frac{1}{2} whichis1.5$. What happens if I now ask you this question, find the expectation of S_3? find the expected number of heads that you expect if you make three repeated tosses of a coin, but with the knowledge now, that the first toss is a head?

So, now we are going to see how many tosses, you have three repeated tosses, so how many heads would appear if you already know, how many heads you are expecting rather, if you already know that the first toss is a head. What is the answer. This is interesting.

You will see now I know that the first toss is a head. I really have to look into the second two positions, this is given. So, here I can have head, head; head, tail; tail, head; and tail, tail. So, means, now I can have only 1 head appearing remaining two are tail. I can have only two heads appearing there is no tail at all so this one comes all here head comes and tail comes then tail comes and head comes.

So, if S_3 is 1, if S_3 can take value 1 it can take value 2 it can take value 3. It can never take value 0 because, I know the first one is 0 right. So, when I am talking about 1 means these two tail has come. So here among this four, now I do not bother about it, it is already a known fact so I just

have to look into this diagram. So, when tail comes both are tail so among these four choices I have only one choice, so there my probability is $\frac{1}{4}$. So,$1 \times \frac{1}{4}$, right.

Now if, I have the situation, where I have 2 heads which has come in three so among these two, one has been head; so, either head, tail; tail, head. So, any one of the two should appear with probability of the 4, this will appear two times, so the probability would be 1/2. You see the whole working changes. Now 3 means I have got this. So, among all the 4, I have one chance. So, it is $\frac{1}{4} + 1 + \frac{3}{4}$. So, it gives me 1, so does this answer sorry 2, $1 + 1 = 2$, $1 + 1 = 2$, sorry this answer is 2.

So, I leave you with the to show that, leave you with this question, that if the first toss is tail then what is my expected occurrence of heads? So, that would decrease. So, here my expectation increases. When the first one is head, I am expecting that among the 2, 3 at least 2 would be head. Here because the first one is tail I am expecting at least among the 1 there will be 1 which will be head. So, you see once more information is available my expectation changes. This is called conditional expectation essentially.

Now, how can I put this idea into a more rigorous form. See, this idea is being developed on a scenario, where your Ω is finite, your sample space has finite elements. So, your Sigma-algebra or the set of all events, is the power set of that finite set. So, let us look into this scenario.

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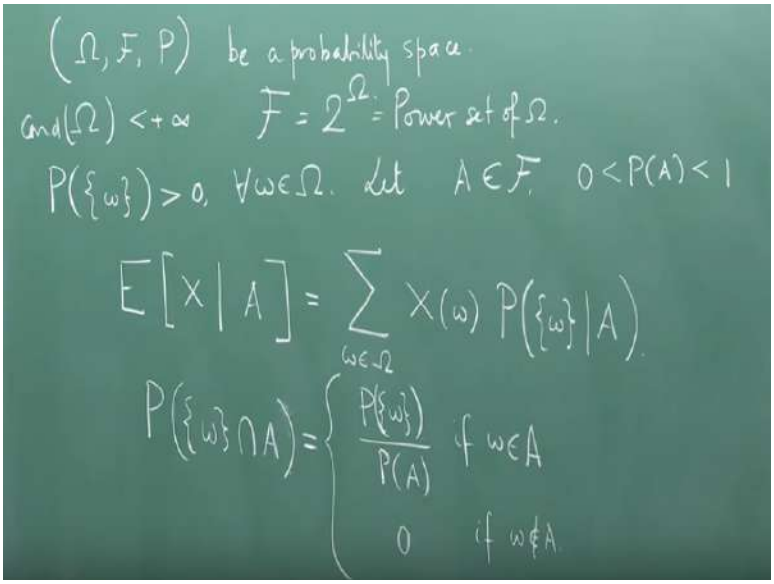

So, we have, instead of u I am writing F because, that is what is usually written in finance books. So, let this be a probability space. So, what is more interesting to know here, is that I take that this is finite. The cardinality, when total number of elements of Ω is finite, that is so my F is nothing but the power set. So, F is the power set of omega.

Now, assume that $P(\{\Omega\}) > 0$ for all Ω in the sample space. So, every sample point, every outcome of random experiment the probability is strictly greater than 0, just like a head or tail scenario. Let A be an event, such that $P(A)$ is strictly bigger than and we can take it less than 1 or strictly less than or equal to 1 does not matter much. So, this is what I have assumed and under this assumption I have to define what is called a conditional expectation.

So, I am looking for the conditional expectation $E[X|A]$. This is in some sense conditioning using a Sigma-algebra because A belongs to a Sigma-algebra and soon we will come to that, how we actually condition over a Sigma-algebra. This is nothing but the very definition, it will be

$\sum_{\omega\in\Omega} X(\omega)P(\{\omega\}|A)$. This is exactly what will happen. This is exactly the definition.

So, what is $P(\{\omega\}A)$=$\{\omega\}\ P(A) if \omega \in A\ P$. This is quite simple to understand because, you see what happens. When $\omega \in$ A,$(\{\omega\}A)$ is ω, so you have just $\{\omega\}\ P$ and when $\omega \notin A$.

$(\{\omega\}A)$ is the null event, empty, impossible event, so that time this will become 0 because, this will be the null event.

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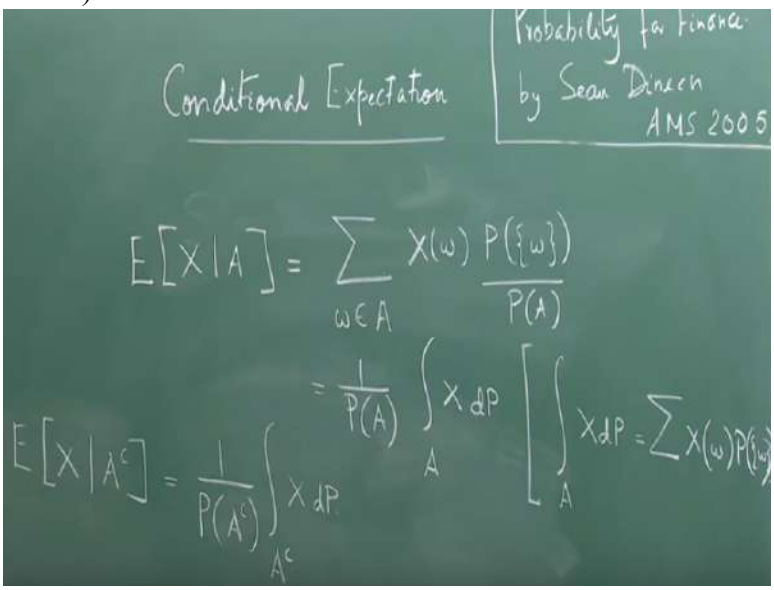

So now coming to this board again. I am rubbing off what is in here. That is why sliding boards are so important and this place has space for sliding boards actually. So, now I can write down the definition by putting in this in this place. So, what I finally get is $E[X|A] = \sum_{\omega \in A} X(\omega) \frac{P(\{\omega\})}{P(A)}$ and this I can write as $\frac{1}{P(A)}$ and symbolically, you know that this Σ is actually an integral if you were not considering, the discrete distribution or discrete setup, then this would be an integral.

So, essentially a sum, summation $X(\omega), P(\{\omega\}), \omega \in A$ but this we will always symbolically even if we have sum we would symbolically write this as $\frac{1}{P(A)} \int_A X.dP$. So, $\int_A X.dP$ is just a symbolical writing for the expression sorry is just a symbolical writing for the expression $\sum X(\omega) P(\{\omega\})$,symbolical writing.

Of course, you can move to non-finite spaces, which we will soon do, but this is just a useful trick to understand it. So, if you want to write about, any other set, what about the compliment, if I know the compliment of A has occurred. Then also as $E\left[X|A_c\right]= \frac{1}{P(A_c)}\int_{A_c} X.dP$ this is nothing but, the same you can actually prove that, they are you can just replace A with A_c.

Now let us look at the Sigma-algebra which the event A can generate, that is the smallest Sigma-algebra which contains the set A, it should contain A, it should contain the empty set, it should contain the whole set, it should contain its compliment. So, Sigma-algebra generated by A the empty set, the whole set A and A compliment.

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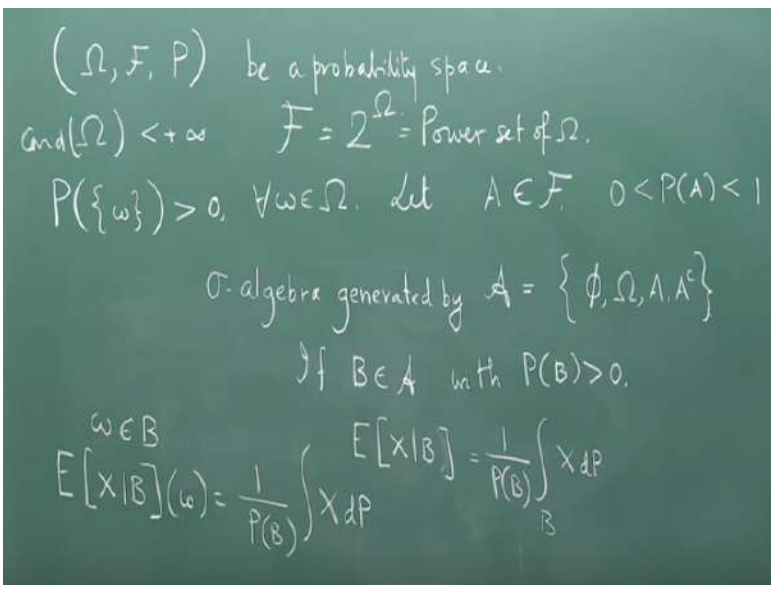

So, if B, so we are just writing it B is element of A so some sigma element B from this Sigma-algebra with element $P(B) > 0$, so we can write $E[X|B]= \frac{1}{P(B)}\int_B X.dP$. So now observe that, whenever B occurs, what does this means B occurs, some omega belonged to B has occurred. When you have random experiment, what is the meaning of B has occurred. When I say, I throw a dice, odd number has occurred. If 1 occurs odd comes over the face, then the event that odd number has occurred has occurred.

So, I can actually view this as a function. So, for any $\omega \in B$, I can define a function like this, rather a random variable like this. So, for every omega in B, that is over the whole set B, whole event B, $E[X|B](\omega) = \frac{1}{P(B)}\int_B X.dP$this is the value. So, it remains constant over the whole event B, so if I look at it like that, that I am looking at a function which is constant over various events, then the conditional expectation itself can be viewed as a random variable.

So, you see because it does not matter whatever, whenever any omega occurs, B has occurred actually. So, for any omega in B this should be the story.

So, whatever once you know that, B has occurred so for whatever omega you are taking this should be the story, this should be the answer. So, this idea, that you can actually view, the conditional expectation itself as a random variable is a very-very fundamental idea and is a very helpful idea. So, this is itself a random variable r.v, where r.v. is just a short shorthand for random variables. So, this will allow us to make shift from the standard probability space where we do not bother about the finiteness of omegas and all those sort of things.

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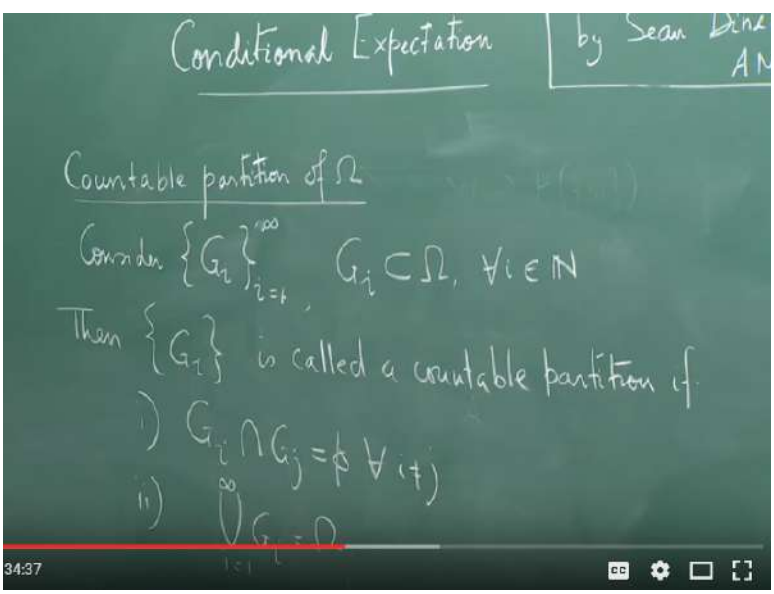

So, first we will talk about a countable partition. Now, what is the meaning of a countable partition of Ω. So, consider a sequence of sets, consider $\{G_i\}_{i=1}^{\infty}$ right, 1 2 3 by $G_1, \ldots.. G_n$ so consider G_i where $G_i \subset \Omega, \forall_i \epsilon N$, set of natural numbers. Then this $\{G_i\}$is called a countable partition, if number 1, $G_i \cap G_j = \phi, \forall i \neq j$, and number 2, $i = 1 \infty G_i$, should give me back Ω the whole sample space.

So, this is called a countable partition and what we would now require, is a Sigma-algebra generated by that countable partition that you take elements of that countable partition, when I basically, take, do not take intersections, because intersections would ultimately generate either the, if you take intersection of the same set it will generate the same set itself or it will generate the empty set.

So basically, if you want to take any non-empty set of a Sigma-algebra generated by the set, you just have to construct unions of this set using some subset of n could be finite could be infinite whatever right.

So, what we will be interested is, in a Sigma-algebra generated by a countable partition, which will be a sub Sigma-algebra of F, naturally because, we are taking a subset, when taking a subset of A and generating a Sigma-algebra.

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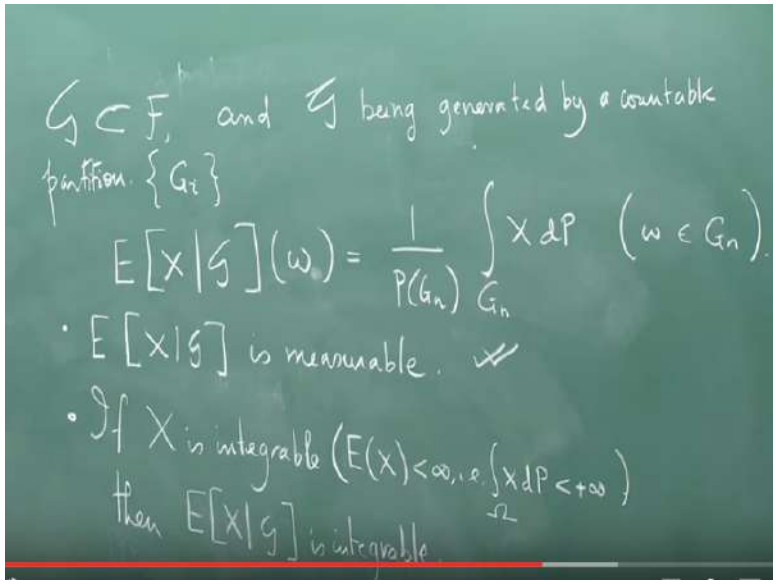

So, consider a Sigma-algebra $G \subset F$, and G, G being generated by a countable partition. Then, we define it like this and the conditional expectation of the random variable X, conditioned on the Sigma-algebra G, is defined as like this $E[X|G](\omega_n)$, right, say omega n or whatever, any omega is $\frac{1}{P(G_n)}\int_{G_n} X.dP$ because this is a countable because this is a generated by a countable partition say $\{G_i\}$.

So, given any omega it must lie in one of the $\{G_i\}$ because omega is the element of omega and this is a countable partition of omega so it must lie in one of the$\{G_i\}$. So, suppose take any omega then this will happen. This is the way I define how you compute this random variable at the point omega if $\omega \in G_n$, good.

I will now, been a random variable, it is important to know that because I am claiming it to be a random variable, you should be able to prove that, I leave this as an exercise to you, in your assignments, so of course and this is true. You are going to prove this fact that $E[X|G]$is measurable.

An interesting thing is that if x is integrable random variable, that is it has a finite expectation then this random variable also has a finite expectation that is the, conditional expectation,

conditioned on a sub Sigma-algebra of F which is generated by a countable partition, has also got to be integrable. That can be proved, but I would not rather prove this fact now.

So, this would be a part of the exercise. If X is integrable, I am expecting that, you know what is integrability though we have given some basic definitions. So, if X is integrable then, integrable means it has a finite expectation because expectation is expressed in terms of an integral,that is $\int_{\Omega} X.dP$ this is finite, then $E[X|G]$is integrable. That is this random variable also has a finite expectation. This is important.

Now, once we know this, we are now going to state a very important property, of this random variable. This property essentially characterizes and this idea actually allows us to move beyond Sigma-algebras, which are Sigma-algebras of this type, that is you will now, once you know this you can, once we know this result what we are going to state, using this idea, we can give a very general definition of a random, conditional expectation as a random variable that is we can define the conditional expectation, conditioned not just by a Sigma-algebra generated by a countable partition, but by any Sigma-algebra which is a sub Sigma-algebra of F.

So, you are writing this important property, give a very small proof of this and then we will wind up today's discussion. So, this section is the first part of conditional expectation and tomorrow we are going to discuss the general case and the properties of conditional expectation that is it.

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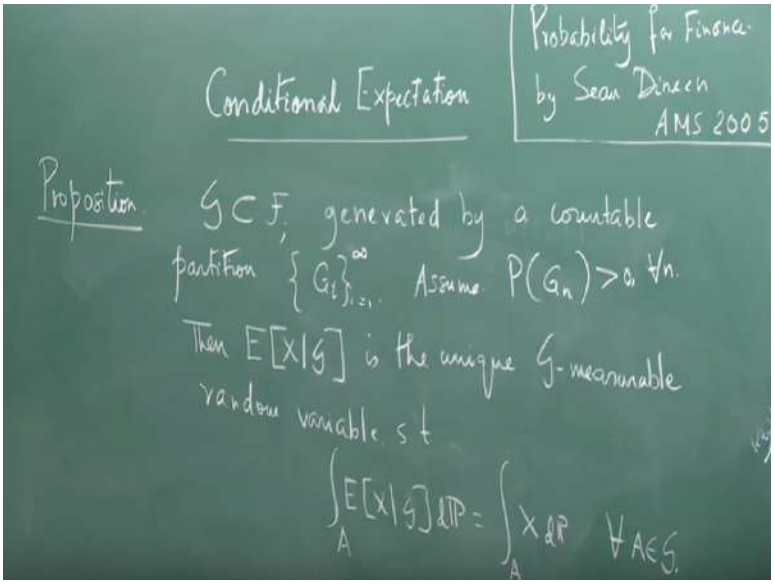

So, what does this proposition say? So again, $G \subset F$, generated by a countable partition, $\left\{G_i\right\}_{i=1}^{\infty}$. Assume, of course this definition when you are writing, you are assuming that $P\left(G_n\right)$ is strictly bigger than 0 for all N because without that you cannot write this.

Assume that $P\left(G_n\right)$ that is probability of each of these pieces of the partition is greater than equal to 0 for all N. Then $E[G]$conditional expectation is the unique G measurable, of course when I am talking about measurability here, I am essentially talking about G measurability, is the unique G measurable random variable such that, random variable on the probability space of course, such that $\int_A E[G]dP = \int_A X.dP \forall A \in G$.

So, on G, integrating over a set of G, subset of G, integrating over any element in G, any set which belongs to G, over X is same as, integrating over the conditional expectation. So, it is essentially it says that, over G conditional expectation and X are almost the same thing. Over such as Sigma-algebra, X and conditional expectations, they behave in a almost similar fashion. The integrals are same and this idea is actually used to extend, to a higher basically, when you go to a case when you do not take this countable partition you take this as a definition of condition expectation.

You see this is how mathematics develops because, you do it for some simple case and then you know that, it is not so easy to talk about the general case, so you pull it up by taking the, what you know for the simple case as a general definition and that definition will always work if the case is simple. So, we will come to that later on. It has lot of links with certain things called Radon-Nikodym theorem. So, let us give a proof of why this is happening.

I am not going to prove, the case for uniqueness because, that is very simple. So, to prove the uniqueness, we really have to use take any other Y, which for this the same thing happens, which will tell me that $\int_G E[G]$ variable - $\int_G Y.dP = 0$. So, which means that if this function is 0, what does that mean, function is 0 almost, that is exactly. So, that is the how you receive your uniqueness. So, we will just do the existence of this that okay if I define the random variable in this way the way which we have defined in this particular case then this must be equal to this. So, proof and with that we will end today's discussion.

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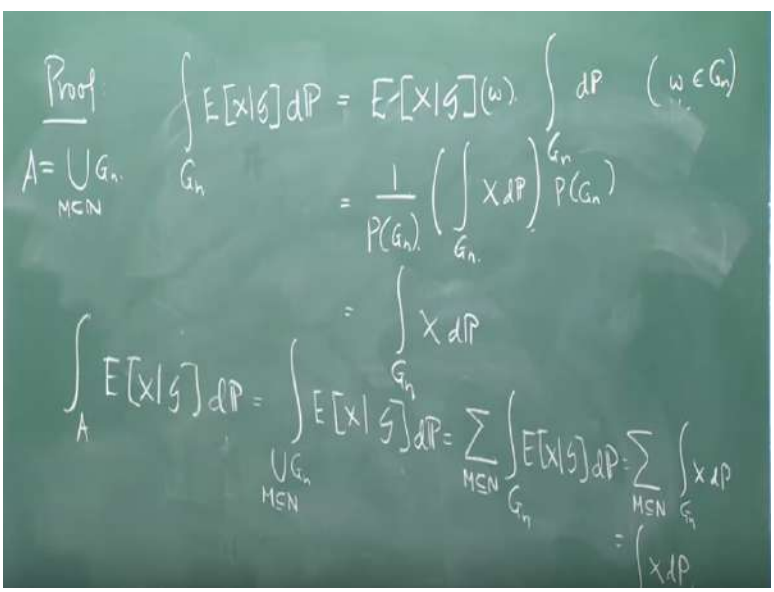

So, I am just taking with $\int_{G_n} E[G]$ first with the G_n, let us see why we are doing so. First, you have to understand why I am just taking first with G_n because, if A is any non-empty set in G, then $A = M \subset NG_n$ where M is a subset of N, could be finite could be infinite whatever.

So, take some G_n, from there and you are writing this G_n,, M is capital M is a set. It contains the indexes which is subset could be just 1 2, 3, 4, 5 so any subset of M. So, every A can be expressed like this right because the intersection you could take intersection of the empty right that is why you cannot. So, every non-empty set has to be expressed like this okay.

Let us see what happens on this G_n. Now you observe, that on this G_n, for every part, if you look at the definition this is a fixed quantity. You take any omega n element of Gn, whatever omega you take it will give you the same value. It is a constant on that Gn part. So, I can take to be some omega integral Gn dP, that omega is element of G_n.

Now what is this. This is nothing but probability of G_n but I can again use this, I will write down the definition here, $n\,G \quad P\,\frac{1}{}$ because this is nothing but $\int_{G_n} dP$ is $P\left(G_n\right)$. So, what I will get here is $\int_{G_n} X.\,dP$.

Now if I am talking about $\int_A E[X \vee A]dP$, so I can write this as $\int_{M \subseteq NG_n} E[X \vee A]dP$,M subset of N because, these are all disjoint, I can actually sum them up sum each of the integrals. So, it is $\sum_{M \subseteq N} \int_{G_n} E[X \vee A]dP$ and this summation, what is this, for each individual piece this is nothing but this. So, this is nothing but $\sum_{M \subseteq N} \int_{G_n} XdP$ which is nothing but, integral union of Gn, union of Gn is

nothing but A which is $\int_{G_n} XdP$ and that is the answer that is the proof.So, with this proof we end our discussion here today.Thank you very much

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 07
Conditional Expectation-II

So, now we are going to talk about how to get conditional expectation, when I do not have the Sigma-algebra G given by a countable partition of omega, so how you generalize. So, in the case where we had this issue of countable partition, you know that we proved that there would exist a G measureable random variable $E[X|G]$ such that $\int_A E[G]dP = \int_A X.dP \forall A \in G$.

Now this idea, cannot be so easily proved, if you consider that G is just a Sigma-algebra. So, in many books, especially books of finance, they define the conditional $E[G]$ as follows.

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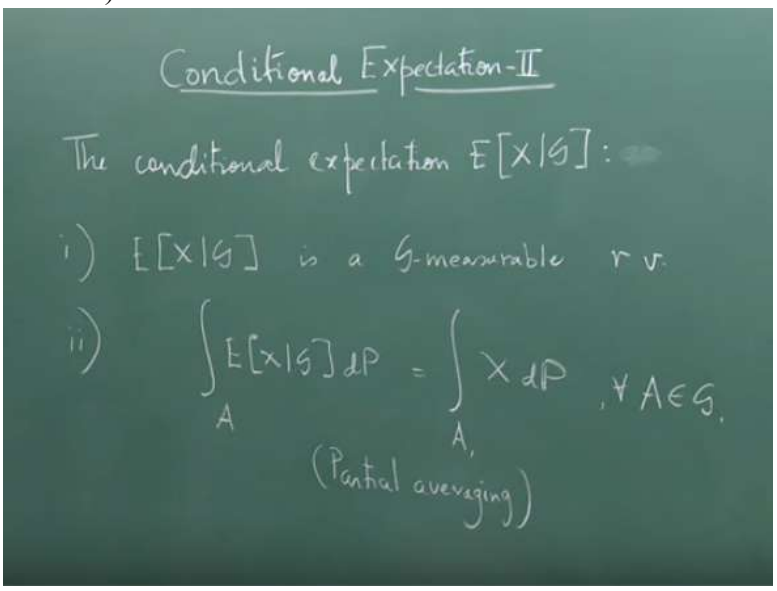

So, it is a random variable, is a G measurable random variable, because that is what we proved earlier, so we are assuming it okay, random variable r.v. And number 2, whatever we have proved in the last class, in the last result, we want to assume that such a thing actually happens, as integrating over this and integrating X are the same thing.

So, this variable and X does not have much of a difference, that is what one other crude. So, this property is sometimes called partial averaging. So, this is the definition. So, conditional expectation is a G measurable random variable which satisfies this, which shows that, this may not be unique, there cannot be any more than one G measurable random variable which can actually get you this. But the key question is, whether such a G measurable random variable would at all exist.

Okay, I am making a definition, but try to generalize my last result, of the last class, but how do I know, that this such a thing such a random variable would exist. If it does not exist, then such a thing is nonsense. To know that it exists, we need to go through some steps and get into some slightly deeper probability theory. We will not push you for proofs etc.

We will try to give you an information about how things are done what are the thought processes behind the steps what that is required or tools that is required to prove that such a random variable would exist which will exhibit this property, okay.

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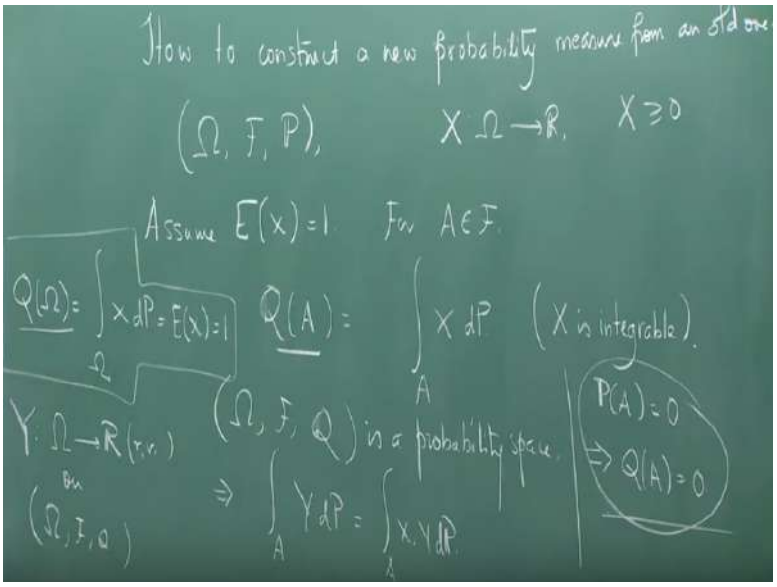

One of the, first steps is to know, in finance and in probability theory is, how to construct, a new probability measure from an old one, that is the key question. So, in that sense suppose you have a given probability space, so whatever handwritten notes I have, I will see whether, I mean, I finish delivering the lectures, I can hand it over to the Muk, support and staff, who can actually very good technical staff here, who can actually scan it and put it up on the

portal, so that you can see all these things. So, I cannot write down each and everything in detail which is written in the notes because I am just trying to explaining the real issues.

So, here we take a probability space and we take a random variable and assume that this random variable is non-negative. You can assume it, in the sense of almost everywhere but okay let us for the moment assume that, X omega is greater than 0 for every instance omega. Now, also assume, you have X is equal to 1.

You see, once I know this, then I construct for A in the Sigma-algebra, if A is a event then, construct the following function of A. This Q, which is actually taking an element of a Sigma-algebra and mapping to R given this is integrable. So, suppose that X is integrable actually. Then this Q generates a new probability measure. So, then Q okay maybe I will write it like this no maybe I will not do this because that will make it look like rational numbers. So, this Q is a probability space, that is absolutely fundamental. So, given a probability measure, I can construct a new probability measure in the following way.

So, another interesting property that we have is that, if Y is a random variable on and suppose I can integrate, then for any A that you have in F, so this is nothing but, so this is an additional result. So, this is the key step, where you have constructed the new probability measure.

Finance is very important, how to move from one probability measure to the other. An important result that can be shown is that, suppose $P(A) = 0$ okay. Then, in this particular case, it would imply that $Q(A) = 0$. This is called Q have been absolutely continuous with respect to P. So, if you have an A for which $P(A) = 0$, then $Q(A)$also equal to 0. This is fundamental which comes out from here, this definition.

So, here you see Q, now how do I know that Q is a probability measure; because you see, $Q(\Omega) = \int_{\Omega} X.\, dP$and this is nothing but $E(X)$, which I have taken to be 1, so probability of the sure event omega that whole sample space must be 1. This is something very important that, it is truly a probability measure and you can show that this will happen. So, the interesting question that we now can ask comes from here.

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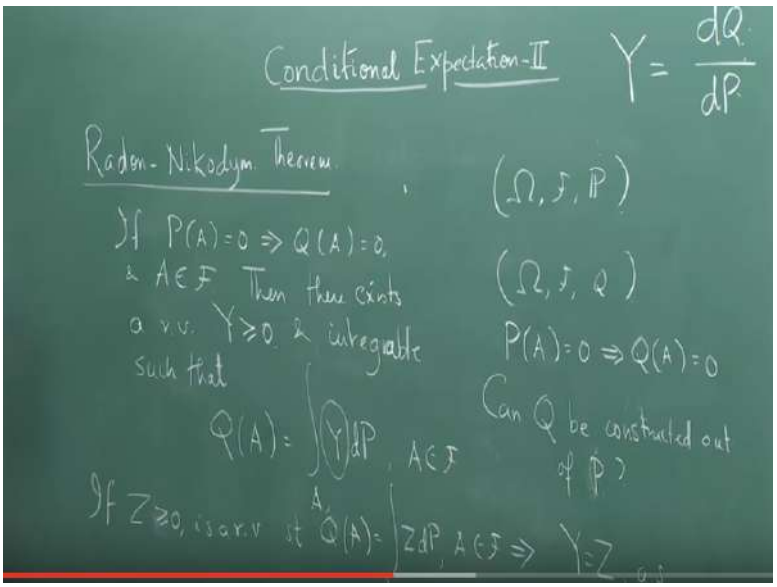

So, if I have a situation where P is the probability measure and Q is the probability measure on both on so P Q are both probability measures on omega F. So, we have this probability space and we have this probability space also right. So, this probability space and this probability space, but we also have this connection that $P(A) = 0$ would imply $Q(A) = 0$. Whenever $P(A) = 0, Q(A) = 0$.Then is Q, a probability measure constructed out of P. So, which means the question is can Q be constructed out of P?

Answering this question leads to a deep result called the Radon-Nikodym theorem and the random variable which will require to construct Q in this manner, the one which we have already shown in this result, that random variable would be called the Radon-Nikodym derivative and this is a very-very famous result in measured theory and in probability, also it is called the Radon-Nikodym theorem. Just take this for information do not get too worked up because you are not going to really use Radon-Nikodym theorem in such a way in your actual practice of finance, but it is good to have some additional information.

So, what does Radon-Nikodym theorem tell me? Radon-Nikodym theorem tell me the following, tells me that okay, very good, so you have this information. If $P(A) = 0 \Rightarrow Q(A) = 0$, where A is of course element of F and $A \epsilon F$, A must be an event of

course you do not compute the probability, then there exists a random variable Y which is nonnegative, an integrable.

This is a very deep result because we are proving the existence of something. Every existence here results in mathematics as deep results. It is not easy to prove that something actually exists whether you are able to really hand calculate it that is a different matter but the question of its existence is a very-very important issue in the way modern mathematical thinking goes.

So, you are telling such a thing would actually be there and so then there exists random variable this which is integrable such that, $Q(A)$can be constructed in the following manner. See, why you need to talk about a positive random variable not just any random variable taking any value.

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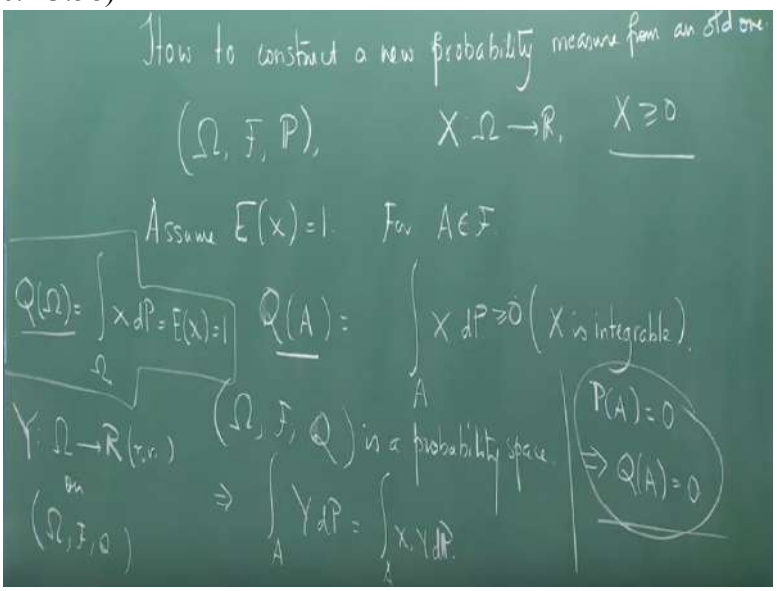

You need to talk about a positive random variable whether it is here or here because then you will always have $Q(A) \geq 0$. You put five here it would become 0. It will put omega here it is 1 so that is a true probability measure. $Q(A)$values are lying between 0 and 1 because X is nonnegative this will always be greater than equal to 0 and that is why you need to talk about our nonnegative random variable, okay.

So, you will always get this. There would always exist a nonnegative random variable such that this will happen. Of course, $E(Y) = 1$ because, you already now know that, Q is the probability measure so $Q(\Omega) = 1$ so expectation of Y is 1. So, let now interesting fact is, that this is unique up to a measure, means it says that if $Z \geq 0$is a random variable, such that $Q(A)$ can also be written like this, for every A element of F. Then it implies that Y and equal to Z almost surely that Y and Z agree at every omega except on a set of omega of measure 0.

So, this is essentially unique so this Y is usually written as $\frac{dQ}{dP}$ the Radon-Nikodym derivative of Q with respect to Q given P and Q are absolutely continuous measure means Q is absolutely continuous with respect to P.

So, this is the Radon-Nikodym theorem which tells you that okay if you can construct a probability measure if you have random variable like this you can always construct a probability measure, for which this property will hold. And the reverse says that if this property hold this will always get a nonnegative random variable, for which would be able to through which you are able to represent the measure Q, in terms of the probability measure Q, in terms of the probability measure P.

But these two things sum up to give you, there is no time to really write down proofs, but I am trying to tell you that this two results these two ideas sum up to prove the existence of a random variable, which will satisfy exactly what we have seen in the definition of conditional expectation in the general setup.

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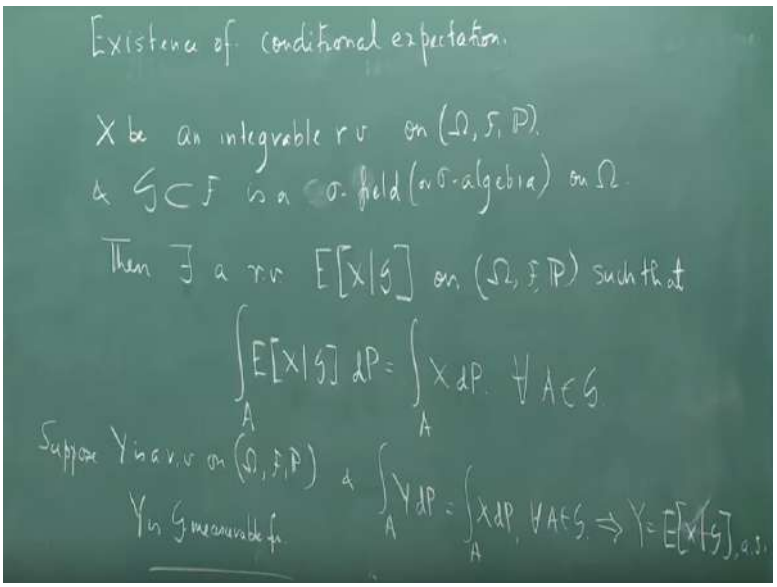

So, existence of conditional expectation. So, what does this result say. So, let us start with this fact that, X be an integrable random variable on the probability space, okay and $G \subset F$ is a sub sigma field okay otherwise we ll just say this sigma field or Sigma-algebra whatever term you want to use you can use, Sigma-algebra on, then there exists a random variable $E[G]$on.

So, this is an important result, which is integrable. So this sign is very common to mathematician, this means there exists. So, then there exists a random variable on this, on the probability space such that, the partial averaging property, which we had written down actually holds. So, it shows, that though this random variable may not be unique, but such a random variable do exists and exhibits this partial averaging property. So, this is true for all A you take in Sigma-algebra G with respect to which you are doing the condition.

Now the interesting part is the following. So, suppose Y is a random variable on this probability space, omega F P and $\int_A Y.dP$ equal to $\int_\Omega X.dP \forall A \in G$ then it implies that, $Y = E(X \vee G)$ G almost surely or almost everywhere. So, any random variable Y which satisfies this property must be same as the conditional expectation that is the fundamental idea.

So, that also shows that this is really not unique you can just find some Y which satisfies this, that is, which is G measurable. So of course, Y has to be G measurable. So, if you can find any G measurable function this will just work.

Now, I will state the properties of conditional expectation. I will not give the proof of the properties. The proof of the properties would be coming through exercises. You will be asked to give a proof of the properties and then the proof would be given in your homework. Maybe if there is time, time permits I will try to do for 1 and not each and everything. So, let me write down the properties of conditional expectation. So, these are important properties used very often in finance so it is important to know this.

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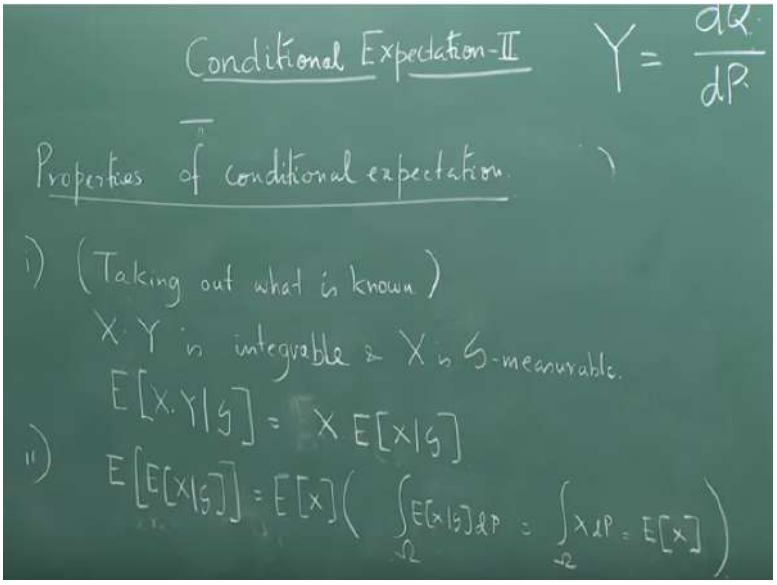

Properties of conditional expectation. So, what is the first property? First property is taking out what is known. Number 1 property, is called the taking out what is known property. So, what do I mean by this.

So, suppose I have 2 random variables X and Y and I say, that X and X into Y that is X omega into Y omega for the omegas, when you compute them is, integral, is integrable and X is G measurable, means, G contains all information about X that is the meaning. So, that is the meaning of something is known. So, X is essentially known by knowing G. So, the nature of X is completely revealed to G. So, this is nothing but X into expectation of, this is the thing.

Number 2, $E[E[X|G]]$ is $E[X]$. This is a very simple thing, this property because, if you use the partial averaging property, what it means. What does expectation of a thing mean? $\int_{\Omega} E[G]dP$, this means this, and this means $\int_{\Omega} X.dP$ and that is $E(X)$. So, this is the partial averaging property and this nothing, it gives me this because these 2 for any A in G and of course omega the whole space A is in G because G is a Sigma-algebra.

So, as I have told you in the very beginning that the whole omega has to be a null event and sure event has to be the whole sample space has to be in any sigma fields. So, that is it and this is nothing but the $E[E[X|G]]$. So, you have taken the expectation of this random variable which is this and the partial averaging do this which gives you this.

This is property two then we come to property three and property four.

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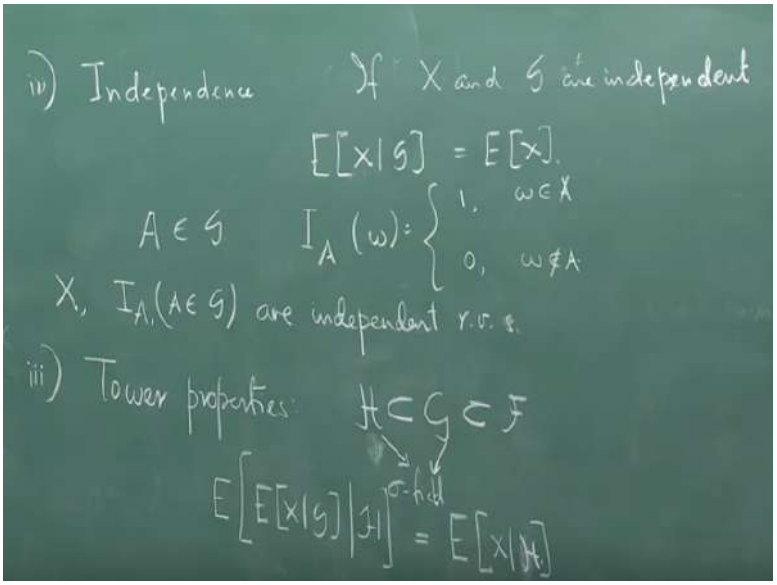

Property three is called independence. So, if X and G are independent, I will explain to you, what I mean, if X and G are independent, you will be very surprised that I am talking about independence of a random variable with respect to a Sigma-algebra. I will tell you what is that.

So, mathematicians hardly mix-up lot of language.

So, if X and G are independent then basically X does not depend on G so the conditional expectation of G is basically a constant random variable; for any omega that you take this gives you the values $E(X)$. What I mean by X and G been independent. By the term X and G been independent, I mean that if you take any $A \in G$, how do I identify the element A, how do I say that the event A has occurred. Event A, if I take, I look at the outcome of random experiments some omega has occurred in the sample space. If that omega remains is in A then I say that the event A has occurred.

For example, I throw a die and the event A is that an odd number occurs. If I say that the number 3 has occurred, then I say that the event A has occurred. So, which means, this event A, can be identified in terms of a random variable I_A the indicator function. So, where I_A is a function defined as follows where $I_A(\omega) = \{1 if \omega \in A \; 0 if \omega \notin A$.

So, which means that $X \wedge I_{A,} A \in G$ this are independent. So, this are for any A and G are independent random variables. So, this is a random variable, independent random variables. That is the meaning of the fact that X and G are independent. So, if these are independent, so X really does not depend upon the occurrence of any event in G then of course this is the case.

Third and the last property which is used pretty often is a tower property. So, it is very important for you to learn these properties. So, you will have exercises to work on. So, what do you do. You have a sigma field H, which is contained in the sigma field G, which is contained in the sigma field F, so this both sorry, so both G and H are sigma fields, which is contained in the sigma field H.

Then the property that we have is the following that, $E[E[G] \vee H]$, is same as expectation. So, conditional expectation of the random variable $E[E[G] \vee H]$ has occurred is same as sorry, $E[H]$.

So, this is something I would like you to remember. So, these are four properties that you learn about conditional expectation are just very crucial. So, lot of things you have to use repeatedly when you study finance and when you start studying Martingales.

Martingales are something used for when it comes term actually comes out with gambling, so you have gambling and then as time passes more information is revealed and if you have some information now, what is your expected gain or profit or expected income or expected, whatever you want to say, expected payoff at the next time. So, that is what Martingales talks about and that is what we are going to learn in the next class.

It is very interesting and hope because at the end stock market is gambling. At the end the stock market keeping money in the stock market is gambling, but people do keep money in the stock market because, the rate at which money grows in the stock market if everything goes on, is much faster than if you keep it in the money market which is fixed deposit like or invest in government bonds.

So, it is very important that you have to have an understanding of this. If you really want to know in the second part of the course how derivatives are priced, of course some derivative pricing will, of course come in here. We will talk about at the end when we learn it to calculus. We will learn to apply it to calculus. We will talk about how to compute derivative rates. We will talk about, what is the actual behavior of the stock prices, a pretty good model called the geometric Brownian motion.

All these things will come gradually, but these properties will become pivotal there and as a result, I would like you to remember and use these properties. Do these exercises which is associated with this conditional expectation in a pretty serious way. At least learn this part very well. Of course, the internet is there. You can keep on looking at it but here the course is basically, to give you some understanding. Maybe you will do slightly better than the internet, but anyway, you can use whatever resources you want but these things have to be known very well.
Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 08
Martingales

Okay so we are going to talk about Martingales today. So what are Martingales? We cannot immediately approach that Martingales are particular type of stochastic processes because stochastic process behaves in a certain way, we will call it Martingale.

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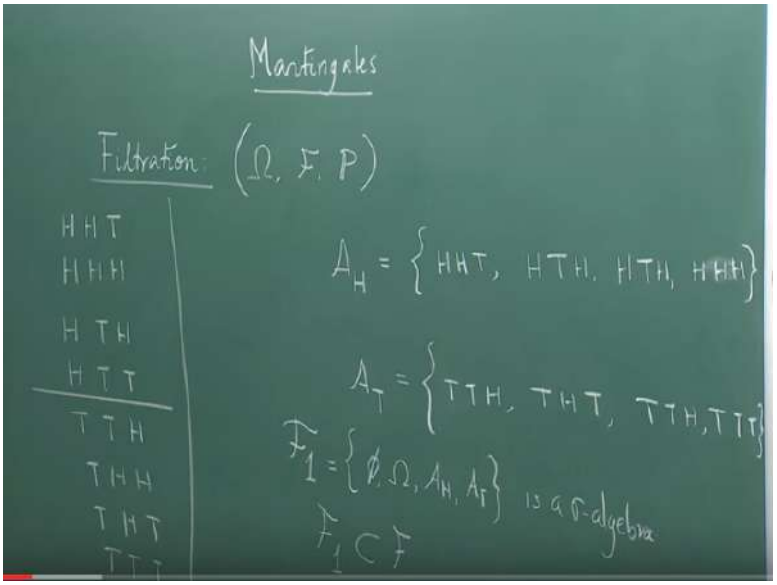

In order to understand martingales, we need to first talk about the notion of a filtration. Of course in any discussion that we begin with always keep in mind that even if I do not mention it underlying everything is the following probability space okay. Now what essentially is filtration? Filtration is the part what happens is that as a random experiment progresses and new information becomes available you know which part of the Sigma-algebra you already know.

So some part of the Sigma-algebra would be completely revealed. So some part of F or subset of F would be revealed once we start getting more information. For example I throw a fair coin. So if I throw a fair coin, there are 8 possibilities okay. So let me write down what are the 8 possibilities.

This is 1 or we can have all heads; head, tail, and then head; head, tail and tail; tail, tail, and head; tail, head, and head; tail, head, tail; and tail, tail, tail. These are the 8 possibilities if you throw 3 coins or 1 coin thrice in succession. So you are doing a repeated experiment. Now suppose I know that the first coin I tell you okay suppose I do not allow you to see the experiment, I am conducting the experiment and I tell you okay the first coin has turned out to be head. Then what are revealed to you?

What events are revealed to you? If the first coin is head is the situation is this one, means you know if the first coin is head the only possibilities now are the following sorry. So once I know that the first coin is head there are only these possibilities that can occur or maybe if the first coin is tail so if the first coin is head then these are the possibilities that can occur. If the first coin comes out to be tail then these are the possibilities.

So if the first coin, the outcome of the first toss is known to me, then given any Ω , if you give me any omega in any sequence of 3 tosses, I can tell you whether this will be revealed or this would not be revealed, this will be an outcome or this will be not an outcome. So what I tell you okay the first coin has come out to be head then you say okay can tail tail tail come, no. the tail tail tail come will come in the compliment of this H this is the compliment.

So what we have done? We have essentially segregated this thing, separated this out. Made a finer division of these 2. So given any Ω now, any omega does not belong to the empty set, every omega is belonging to the whole sample space which means the sample space and the empty set is always revealed. You know that either there will be nothing or there will be everything basically.

But the interesting part is that now given Ω I can tell you whether that event will now occur or will not occur. I have the information. If you say first toss is actually head, I can tell you what will actually happen, what are the next consequences any of those 4, right. If you say okay what about T H T can this consequence will be there no, it cannot be.

So if you look at it the following sets of the Sigma-algebra F is now revealed which I called F_1 which is$\{\phi, \Omega, A_H, A_T\}$, see if I know I have a knowledge about the first toss. These are the sets of the Sigma-algebra which is revealed and this itself F_1 itself is a Sigma-algebra

whether it follows all the rules of the Sigma-algebra whether if you take the union of these 2 it will become omega.

Now if you take the intersection it will become phi. If you take the compliment of H it is A_T compliment of A_T is H. So this is a Sigma-algebra. So this part of the Sigma-algebra, so this is of course you immediately see that F_1 is a subset of the Sigma-algebra F. So part of the Sigma-algebra gets revealed when some information is revealed and the next stage I said okay good I tell you what has happened in the second toss.

Then I will have finer revealing. I can again partition this into finer parts when basically I am breaking up the space F the Sigma-algebra F.

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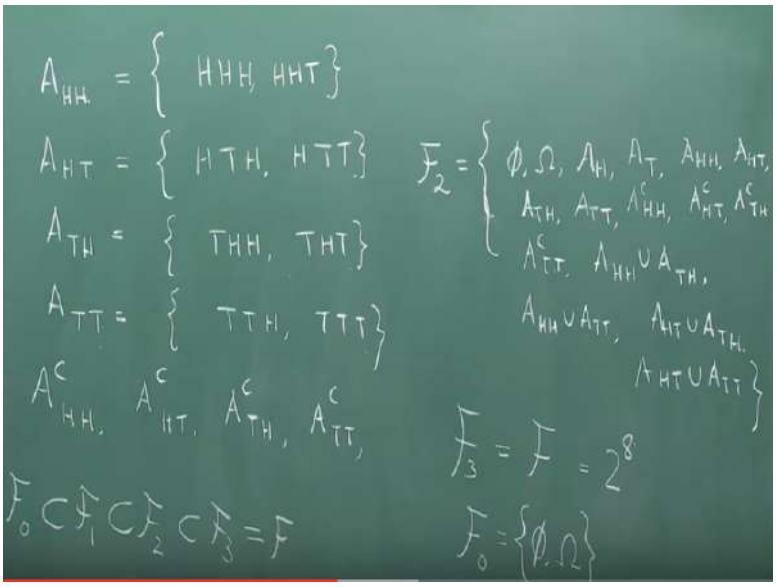

So you tell me that so what can be the second toss either both can be head first head and then can be tail or the first can be tail the second can be head, the first can be tail and the second can be tail. So if I tell you what are the 2 consecutive things you know what can occur now. If I have the knowledge of what is also the second outcome and also I know both the outcomes I know what are the occurrences.

So here it will be this, nothing but just augmenting with head or tail that is all. So you have made more finer divisions of this basically. So I know if you said that the first coin is head and the second coin is head if you said the second coin is head so these are the 2 things that can happen, anyone of the things can come. Second coin is head because the first coin is could be head could be tail. So any one of these 2 things can come.

Now you see can I now make some Sigma-algebra out of this information. So this sigma that sigma here for example this Sigma-algebra totally encodes the information that the first toss is known. If I know the second toss, can a Sigma-algebra be constructed which can encode all the information. So of course you should have once if you want to put construct a Sigma-algebra if they are all inside that Sigma-algebra then all this has to be also part of the Sigma-algebra because A_{HH}^{C} is not any one of them but this whole the union basically.

So you basically do not have to when you construct the set you do not have to write the union of this union of this union of this 3 or union of any of the 3 because that is the compliment of the remaining. So you can take the compliment $A_{HH}^{C}, A_{HT}^{C}, A_{TH}^{C}, A_{TT}^{C}$. So that is it. So you take the compliment and that is so you take the compliment. Of course you have to take some unions also. For example if I take this union such a union does not such a union for example does not appear in any one of these sets already known.

For example if you take this set and this set and take their union there is no where I can find anything right. But if you take this set and this set and take their union and this set is A_T. So I do not need to bother about the union of these set but I can I have to bother about the union of these 2 sets, union of these 2 sets which were not there to create the Sigma-algebra.

You see H and $A_T$is anyway revealed even if I know the second choices. The first choice must be either head or tail so these are already there. So whatever is known at the first stage will always be carried on to the second stage because they are anyway revealed. For example if I take the union of these 2 and take the compliment of that that would anyway give me A_H. So $A_H \wedge A_T$anyway will continue to be revealed right.

So essentially if I want to construct F_2you see how far the size of the cardinality of this set will grow. So $A_H$andA_T will anyway be there okay. You will also have these sets $A_{HH}, A_{HT}, A_{TH}, A_{TT}$. Of course then you have to line up their compliments$A_{HH}^{C}, A_{HT}^{C}, A_{TH}^{C}, A_{TT}^{C}$

okay. Now you can combine this with this. Do not combine this with this because this with this will give you A_H so this is already there so you do not have to show that combination.

So $A_{HH} \cup A_{TH}$ must be there must be considered. $A_{HH} \cup A_{TT}$ must be considered because they would not form anything which is already given. $A_{HT} \cup A_{TH}$ must be A_{TH} must be considered these 2 and $A_{HT} \cup A_{TT}$ must be considered means those things which do not appear at all. You see your from just 4 elements here we have increased to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 sorry 1, 2, 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17, 18 just blowed up actually.

Now it comes F_3, all is known. All is known, means F_3 means I know all possibilities so I basically know the whole Sigma-algebra F. So F_3 and the third if I know all the coin tosses are known and F_3 is nothing but F and that will have 256 possibilities, 2^8 combinations because there are 8 possible elements in the Sigma-algebra.

So what I have, now let us talk about what is F_0 then. F_0 means when nothing is revealed to me. Then I have only 2 choices. Either nothing will happen or every possible choice might happen right. So F_0consist of just so this is under no information. This is under 1 information that first toss is something. This is under more information the second toss is revealed and this is under the third we have revealed everything.

So everything is revealed so which means if you look at them you have F_0belonging to F_1 belonging to F_2 belonging to F_3belonging to F where $F_3 = F$ you do not write have to write F_3 subset of F or equal to F. So what I have done is a chain of Sigma-algebra each containing the other. So as time evolves more information is revealed, you know more about the structure of the Sigma-algebra.

So such a sequence of Sigma-algebras which each of them are subset of the original Sigma-algebra and each of them are bigger than the previous one, such a thing is called a filtration which we can obviously formally define.

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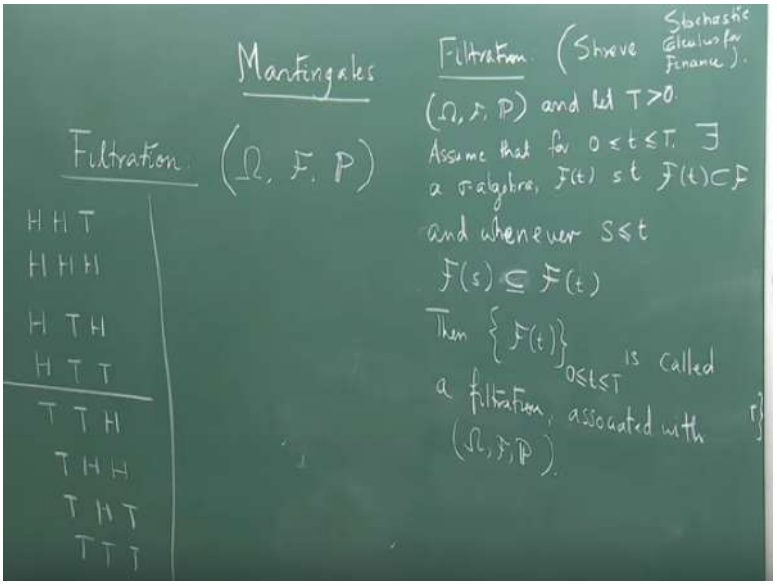

So this is in a very discrete setting. You can define a filtration in the following way. So this definition I have given from the book of Shreve, Steven Shreve's book called Stochastic Calculus for Finance okay. Now observe what I want to say. So if you have (Ω, F, P) let $T > 0$. Assume that for $0 \leq t \leq T$ there exists, this is the symbol of there exists, a Sigma-algebra $F(t)$ T.

So when we are going to write continuous things we will put it $F(t)$ instead of this t just to differentiate between the discrete thing and the continuous event. When we write the discrete thing we will write it as a lowered index and when we write the continuous thing it is as if it is a function.

So there exists a Sigma-algebra $F(t)$such that whenever I have s less than equal to T, $F(s)$ must be contained $F(t)$such that $F(t)$is contained in F and whenever $s \leq t$ will $F(s) \subseteq F(t)$ say $F(t)$will contain in $F(s)$. So this condition has to satisfy it. Then if these are all satisfied, then the family of Sigma-algebras $\{F(t)\}_{0 \leq t \leq T}$is called a filtration associated with the probability space.

So it might, the definition might look tricky, but I think you can go and read it up in books or just think about it a little bit just think about the definition you can if you want rerun the whole lecture. So you can see it how many times you want so that you get your concepts cleared once for all. So then we are going to talk about something called a stochastic process adapted to the filtration.

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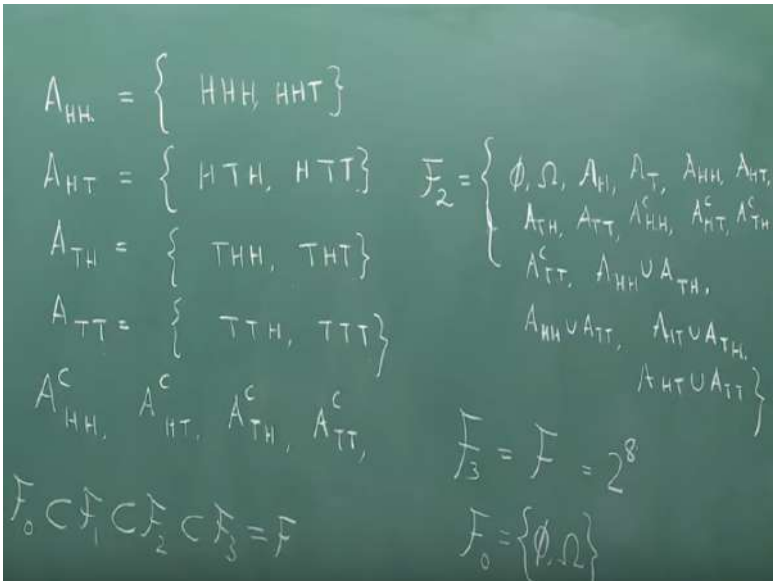

So given a stochastic process $\{X(t)\}_{0 \le t \le T}$ say is of this form, so this is the stochastic process given to you. We say that this $\{X(t)\}_{0 \le t \le T}$is adapted to the filtration $\{F(t)\}_{0 \le t \le T}$ if $X(t)$ is $F(t)$ measurable so all these are these are all random variables. For every t between 0 to t, $X(t)$ is some random variable; adapted to the filtration if $X(t)$ is F t measurable and $X(t)$ measurable that is $X(t)^{-1}(B) \in F(t)$ for all Borel set B in R okay. This is the meaning of an adapted stochastic process.

Martingales are special type of adapted stochastic process. So first we will talk about discrete Martingales and then we will talk about continuous Martingales. Do not get too much bothered about their properties right now except the one which we will require.

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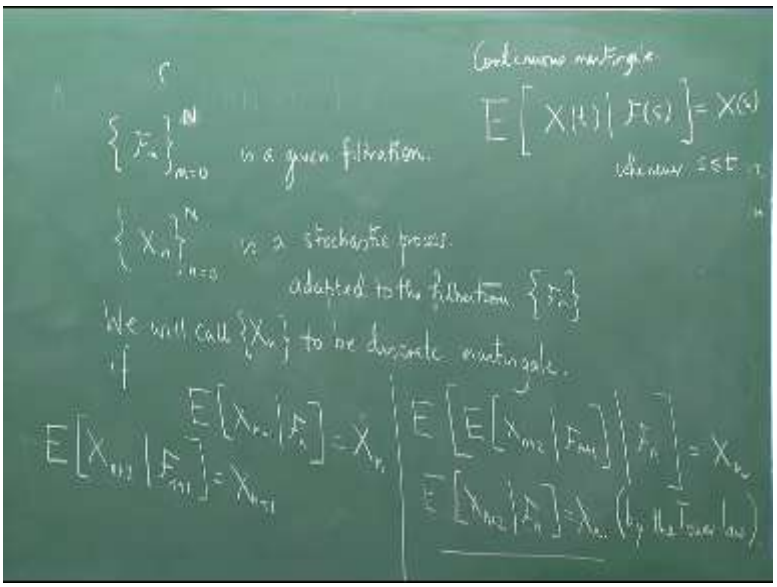

So suppose you have a discrete filtration say $\{F_n\}_{m=0}^{N}$ why I am using a finite class because I am essentially talking about finance. In finance you have a trading horizon, trading time, so trading starts at 0 and ends at T so ends at time N. So that is why I am taking the simplest definition. You can possibly take N equal to 0 to infinity it does not matter and you can have a stochastic process which is infinite.

Stochastic process are nothing but sequences of random variables where the indexing is actually overtime. $\{F_n\}_{m=0}^{N}$ is a given filtration. And then you have $\{X_n\}_{m=0}^{N}$, a stochastic process adapted to this filtration. So you will hear this term adapted to filtration repeatedly after this. Stochastic process adapted to the filtration $\{F_n\}$. So now we will call $\{X_n\}$to be a distinct Martingale if a $E[F_n]$ conditional expectation of the random variable at n+1 there is X_{n+1} having the information till time n is X_n.

So how it is related to gambling possibly? It says that a stochastic process is a Martingale if your expected payoff at the n+1 th move is same as whatever payoff you have received at the nth time provided you have only information up to nth time which is that fact. You will not know what will happen at n+1 it is uncertain. Up to nth time you have information, you know that this is X_n.is what I have got is a random variable or the random process which was revealed.

When I had done the gambling at nth instant but when I am going to do at the n+1th instant conditioning upon the fact that this is known then this is my answer, when I cannot expect to improve. Once I improve then I call it something like a super Martingale if I do bad then I call it like a sub Martingale.

So this is what one should expect that I can only expect I will do as much as I did last time. My payoff would be as much as I had last time, nothing else. Of course you can ask that, this will go on basically, $E\left[F_{n+1}\right]$ this will again become X_{n+1}. So this is what is known to me. Now I will put this fact here. Let me see what I get out of this. Some little game with conditional expectation.

So this itself is again this. So if a scenario has revealed at the n+1 th time then what I expect to get for a given scenario is same as what I would get at the previous time. That is the meaning of, I cannot expect something more. I have to remain status quo. That is what I can expect. See if I am putting this fact here then by using the tower law of conditional expectation because F_{n+1} is the largest Sigma-algebra than F_n. I can write this is nothing but the $E\left[F_n\right]$.

So it does not matter whether it is now n+2 or n+5 or n+ 6 you can so this is an application of the tower law. So it does not matter what is your X_n, after X_n however large n you put here put any n does not matter n is n+5 or n+100 or n+10. If you just have information up to n level move whatever X_n you put n is any number bigger than n, conditional expectation of X_n conditioned on the fact that I know only things up to n is nothing but X_n.

That is the meaning of the Martingale and this idea this fact that does not matter whatever you put here. If you just know up to F_n you know you can only expect what you have got at the nth level and this fact this beautiful fact has some beautiful properties and so this idea actually allows you to also look into the continuous case. If you have a continuous Martingale you just have to write this.

So a continuous stochastic process $E[F(s)] = X(s)$ whenever s is less than equal to t. So this is the meaning of a, this is just the definition of a continuous Martingale okay.

Now I will just tell you one property is that it does not matter whatever be your position. If you take the expected value just calculate this expectation of any one of the random variables that is the expectation for each of the random variables. So you have the stochastic process X_1, X_2, X_Nlike this so $X_1, X_2 \ldots\ldots X_N$, then if you take each of them as separate random variables and take their expectation provided that this sequence is a Martingale then every random variable will have the same expectation.

So this calculation can be done very simply, I will just do it on the top and end our talk today and tomorrow we will start some lovely thing called Brownian motion and that is the hardcore stuff because Brownian motion is to be known because that will allow us to model the behavior of stock prices in a financial market in a stock exchange possibly.

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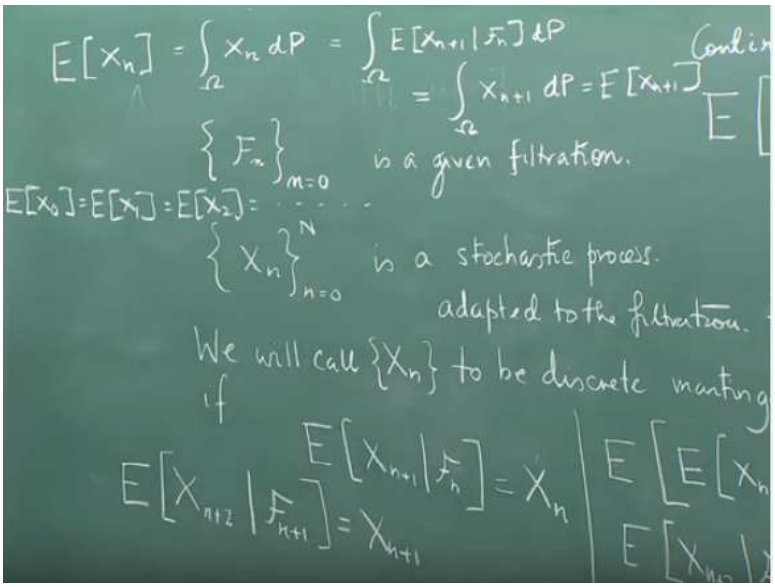

So here we just do it in a very simple way. So take $E\left[X_n\right] = \int_{\Omega} X_n . dP = \int_{\Omega}\left[E\left[X_n\right] \vee F_n\right] . dP$ so this okay I am just writing maybe I should write in a much more different way but okay I am just writing in a standard symbolic way. So this is nothing but integral but by the very definition of conditional expectation when you do it for the general case general Sigma-algebra F n not the one generated by a partition you know you have to use this partial averaging idea that is essentially you define it in that way. This is nothing this one is nothing

but $\int_{\Omega} X_{n+1} . dP$. This is the definition of conditional expectation for the general Sigma-algebra.

This is called the partial averaging and this is nothing but expectation of X n+1. So does not matter. So which means if $X_0 = X_1 = E[X_2]$ so what you finally get is $E[X_0] = E[X_1] = E[X_2]$ ..... So with this lovely property of the Martingale we will stop our discussion here and tomorrow we are going into this wonderful stuff called Brownian motion and we will take our discussion off from there. Thank you.

Probability and Stochastics for finance
Professor Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 09
Brownian Motion-I

So if you go to the stock market and look at the price of say a favourite company's share. So what you would observe is the following.

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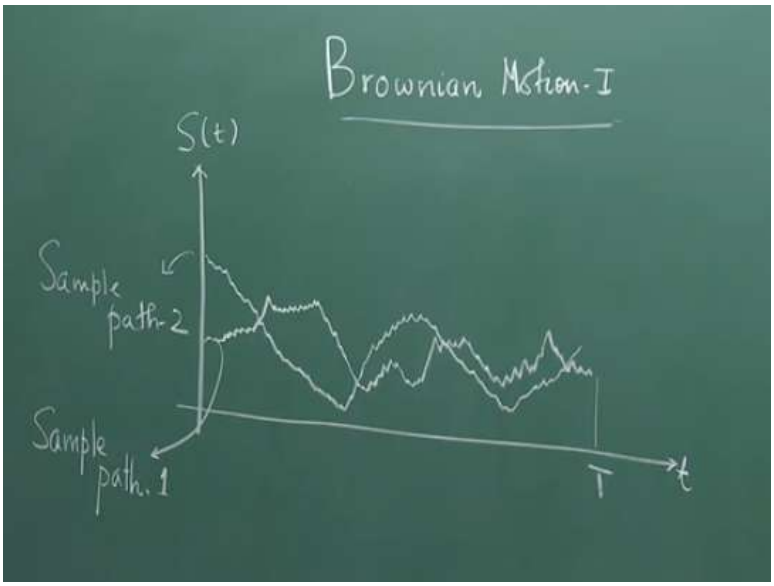

So if the horizontal axis is the time axis and if this is the price axis it tells you what is the price then you will see starting from a certain time price you will see some zigzagging motions like this. So you observe it up to time T and you observe this zigzagging motion.

This is of course random. Nobody knows what is the next price is. So if a particular scenario evolves, you have a particular path. This is called a sample path. So if another scenario evolves, if another scenario evolves there would be another path, for example it could be like this. The stock price is going down down down down and you are in a bad shape and then it again climbs up and again it falls down down down and again then again climbs up.

So under a different scenario it has a different path. So it is sample path 1 it is sample path 2. So it is what type of scenario one evolves. Now of course you can ask me what is this term scenario

that you are talking about, what is the meaning of this goddamn scenario? We will come to this very soon. But how do I model such zigzagging paths, what way to model it. Is there any mathematical way to say that us or can I construct the stochastic process whose sample paths are represented in this form?

Let us do, to do that we need to study what is called Brownian motion. Brownian motion is a type of stochastic process which will help us to model stock prices at the end. So the whole term Brownian motion comes from the name of Robert Brown who first studied the movement of pollen grains in water and he found that they were having a zigzag haphazard movements. But it is not so immediately apparent that you can just start writing about this particular stochastic process.

We need to have some more idea and built upon some simpler stochastic process. So we will begin by introducing what is called symmetric random works.

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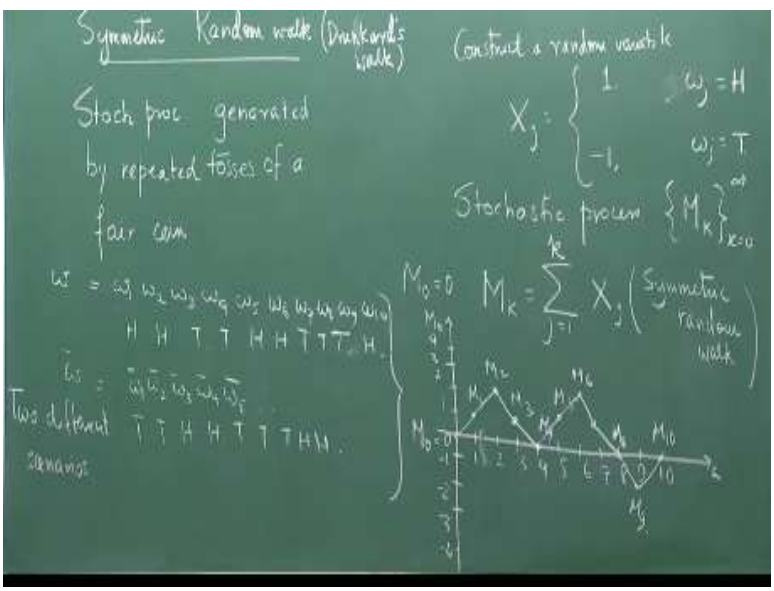

Where there are only 2 possibilities you can either go up or down that is like a coin toss head or tail and that here I can have infinite such possibilities, infinite such sample paths and there are infinite possibilities also. Here also we will have infinite possibilities but generated out of only 2 possibilities. So symmetric random walk. So when you take a fair coin and then you keep on repeatedly tossing it.

So it is a so this is a stochastic process which I will write in short form now as stochastic process generated by repeated tosses of a fair coin okay and if you look at it very carefully what I mean by this? So you start tossing the coin so repeatedly you are tossing omega 1 omega 2; so omega 1 is either head or tail; $\omega_1, \omega_2, \omega_3, \omega_4, \omega_{5,\ldots\ldots}$ and so and so forth. Suppose you have here head, head, tail, tail, head, head, head, tail, tail, tail and it goes on.

So this is one particular scenario that has evolved. You could have another scenario say ω′which is consisting of say $\omega_1', \omega_2', \omega_3', \omega_4', \omega_5'$, it could be something like this; tail, tail, head, head, head, tail, tail, tail, head, head, head and so on. So these 2 are different scenarios and these 2 each would generate 2 different sample paths. So how do we generate this symmetric random walk?

So these are 2 different scenarios, 2 different scenarios. Now construct a random variable X_j which takes the value 1 if j is equal sorry if ω_j is equal to head and takes the value -1 if ω_j is equal to tail if tail appears and 1 if head appears. So now you define a stochastic process, define a new stochastic process $\{M_k\}_{k=0}^{\infty}$ or let us we can need not bother we can also fix it after some time.

It could be some time capital say K is say 25 something here 25 or 30 whatever. But in general it is alright to take plus infinity just a sequence where M_k is given as follows. Each of these M ks are calculated by starting from $M_0 = 0$. $M_k = \sum_{j=1}^{k} X_j$ So let us see what would happen if one particular scenario like this evolves. Let us see then what is the sample path of this. This symmetric random walk is also called a drunkards walk.

So somebody has had a good drink and he has become drunk and if you look at his walk so a drunkard would walk like this if I am here so I start from here then I can just go like this and he goes like this just it is just or like this you know I am coming here and then I am going there

something like this. So this sort of thing you will immediately observe as I start say checking out with this scenario.

So here is my k and here is my M k value. Now the first one here has turned out to be head. So $M_0 = 0$. Let me write -1, -2, -3 and so on -4 here 1 2 3 4 and so on and of course here also you have to have k values which is 1 or maybe 1 2 3 4 5 6 7 8 9 10 and so and so forth. So $M_0 = 0$ this is 0 is 0. Now you toss a coin and you have head.

Omega 1 is head so you go up by +1 because X j will take +1 because M_1 is just X 1 so here is the value of M_1 this is your M_1 so you join the M_0 and 1 by line so 0 is M_0 and then you again had head so M_2 is again 1. I am looking at this scenario M_2 is you go by 1 so it is 1+1 now 2. So M_2 is 2. So you join again by this line. But $M_3$is tail so you will drop by 1 so it will be -1 so it will again drop back to the point 1.

So this is your M_2 and this is M_3 and then omega 4 is again tail so it drops back to 0 again you -1 subtract. So this is your $M_4 = 0$. Again then you have head for omega 5 say so here you have omega 6, omega 7, omega 8, omega 9, so for M_4 you have tail you have come to 0 again then it goes up again for M it goes to plus 1 again.

So this is your M_5. Again, it goes up to 2 M_6 but then you have tail again so M_7 comes down to +1. Again, you have tail so M_8 comes down to 0 because you are adding up everything. At every step you are going up or down. So you add up in this fashion and you move like this. So M_8 I have here omega 8 it is again omega 9 is tail so again I have to go down so I go down by so from 0, I will have to go down by 1 so I will come to -1.

So this will be your M_9 and if suppose M_{10} omega 10 is head then it will again go up to 0 at the 10th place because you will again add 1 so it will become your M_{10}. So what you see that the

symmetric random walk is providing me some zigzag looking curve which might tempt you to think that possibly these 2 have some relationships. They are they do have some relationships and we will talk about that slightly down the talk.

But let me tell you some more properties of this symmetric random walk M_k. So here is my stochastic process and this stochastic process is called the symmetric random walk. Of course we are not mentioning but underlying we are always taking some probability space and all of those things.

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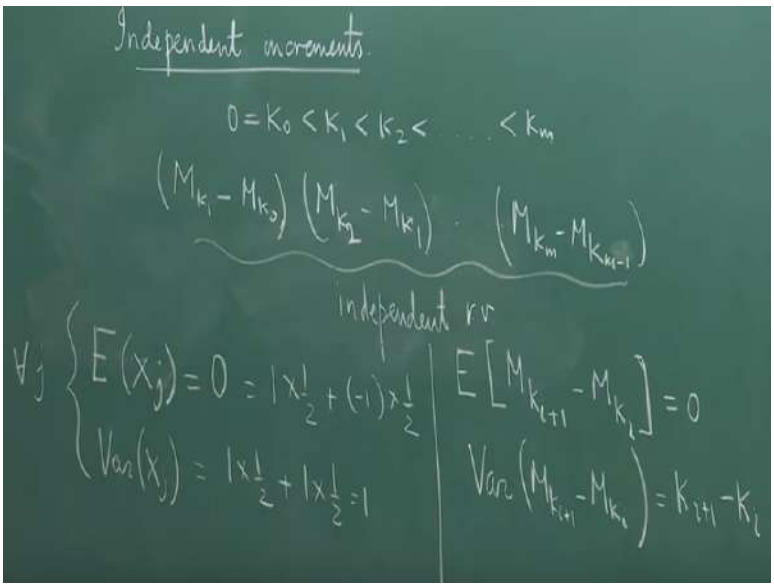

So now we will these random this particular random process of stochastic process has independent increments. What do I mean by the fact that they have independent increments? What I mean is the following. So if you have you take certain numbers sometime say some K m then you have the following.

You have M_{k_1} that is once you consider non-overlapping intervals then this difference is independent because they depend on independent coin tosses because coin tosses are independent when the coin tossed at the second level really does not the second outcome does not really depend on the first outcome right when you do a repeated coin toss sorry $M\left(k_2 - M_{k_1}\right), \ldots\ldots, \left(M_{k_m} - M_{k_{m-1}}\right)$

So these random variables are independent. All of these random variables these form a set of independent random variables. So that is once this happens this is when we say that it has independent increments and this actually has independent increment. These are independent because their difference which really does not depend on the coin tosses here does not depend on the coin tosses here and here and so you have independent increments.

So this change that you see here does not depend on the change that you see in this interval or the change that you see in this interval right. So you can still observe that it is like a drunkard walk so drunkard walks like this goes down goes up. In George Gamow's famous book One, Two, Three Infinity this has been described in a very very nice way.

How do you, see here, my success probability this occurs with probability half and this also occurs with probability half so if you observe that exponential of X j sorry $E\left(X_j\right) = 0$, because this is one into half plus minus one into half. $Var\left(X_j\right)$ so what is variance of X j exponential $\left(X - X_j\right)^2$ which is 0 so it is one into half plus minus 1 minus 0 whole square plus 1 into half which is 1.

Once you have this information this is true for all j. It is immediate that $M\,E\left[k_{i+1} - M_{k_i}\right] = 0$ and the $M\,Var\left[k_{i+1} - M_{k_i}\right] = k_{i+1} - k_i$ to I leave it to you to calculate these stuffs. Our second property about this random walk is to show that this is also a Martingale. You see Martingale thing comes up. So symmetric random walk is a discrete Martingale.

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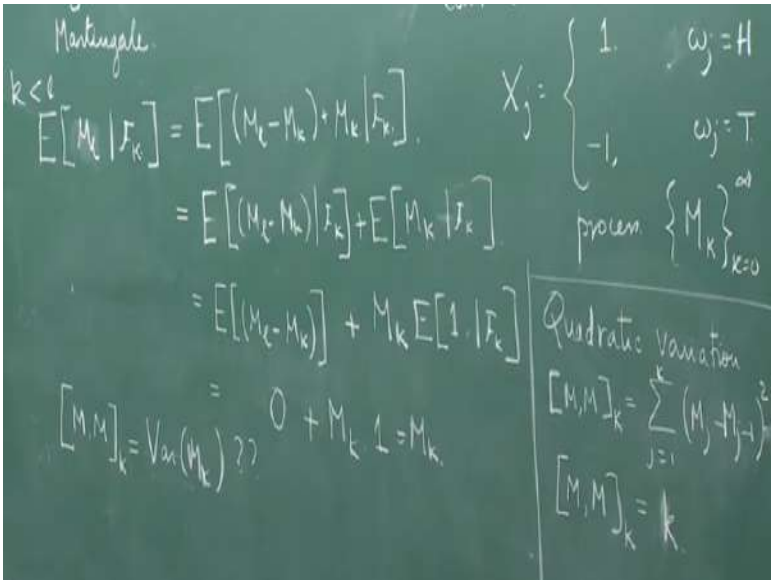

So you can easily prove that it is a Martingale. So you take any $k < l$ and look at the $M\ E\left[1 \vee F_k\right]$, the Sigma-algebra F_k which is the part of the filtration okay.

So you can write this as $M\ E\left[\left(1 - M_k\right) + M_k \vee F_k\right]$. So these can be summed up just like expectation can be some conditional expectation this random variable can be decomposed into 2 parts which you can actually prove which will be a part of your exercise but we are just using this fact here so, anyway I should rub the board a bit.

Now let us look at the first part. Since l is strictly bigger than k this increment $M\left(1 - M_k\right)$ is independent ofF_k. F_k does not have the information of anything which is beyond the time k. So here by one of our rules for conditional expectation this is nothing but $M\left(1 - M_k\right)$ and here at time k everything about M_k is known. So $F_k$contains all information about M_k. So the first law was taking out what is known, I can write this M_k as M_k.1where 1 is the constant random variable 1. So whatever be the scenario it will just give you the value 1.

So I can write this as $M_k$so I will write this as M_k dot 1 so I can take out what is known $1. \vee\ F\ M_k E[k]$. Of course, 1 is a constant random variable. It does not really depend on is independent of F k so it will be E of 1 which is a constant which will be just 1. So everything

will be 1 so the sum of the probabilities will sum up to 1. So the expectation will be just the number.

So this $M\left(1 - M_k\right)$ again is 0 which we already know plus M_k. 1 which is M_k and so this shows that M_1 is a this symmetric random walk this thing forms a discrete Martingale. Of course, F_k is M_k has to be adapted to this filtration that is the basic definition of Martingale. There is another notion which crops up in the study of these sort of processes is called the quadratic variation. So you essentially look at path by path.

You look at how much the random variable values are varying between one end of the path to other end of the path that is between k_1 and k_2 say how much it is varying but do not take just the sum of those variations they might just be 0 so you have would not get any information but take the square of the variation. It is like a mean square error type thing so we again take here and introduce the notion of a quadratic variation.

So the quadratic variation is expressed in the following way. $M \sum_{j=1}^{k} \left(j - M_{j-1}\right)^2 [M, M]_k =$ is equal to and this if you look $M\left(j - M_{j-1}\right)$ whole square this value is always 1 . If you sum them up what will be left here, X_j would be left here, the X_j. If you take the difference between M_j and M_{j-1} you will have the value X_j left. X_j is either plus 1 or minus 1 so the squaring will always give you 1.

So this expression of the square errors basically or the square changes around every path of a given sample I can take the changes but whatever be the path independent of the path it turns out to be k. If you do up to the kth level it turns out to be the k independent of the path that you have taken which is very very interesting.

It does not happen for suppose you want to compute the variance so $[M, M]_k$ is actually $Var\left(M_k\right)$ think about it how it is possible. But you see to compute this I really do not need to bother about

the path but to take variance of M_k we are essentially averaging over all the paths. So this is a difference.

Now how do I can I do something with this process. Can I increase the jiggling of this process a bit this symmetric random walk a bit and generate some sort of an approximation of a Brownian motion. Generate this sort of zigzagging that we had just seen in the beginning when I had drawn the picture of the stock price that this sort of zigzagging can we generate this sort of zigzagging this sort of zigzagging can be generated by using the symmetric random walk and that leads to what is called a scaled symmetric random walk.

We will not go too much of details into it because that might you know take you off track and you might feel a little bit of discomfort for those who are not so very comfortable with very complicated analysis. So what we are going to now show by this scaled random walk is that what we are going to show by scaled random walk is that we can construct an nth level approximation for the Brownian motion.

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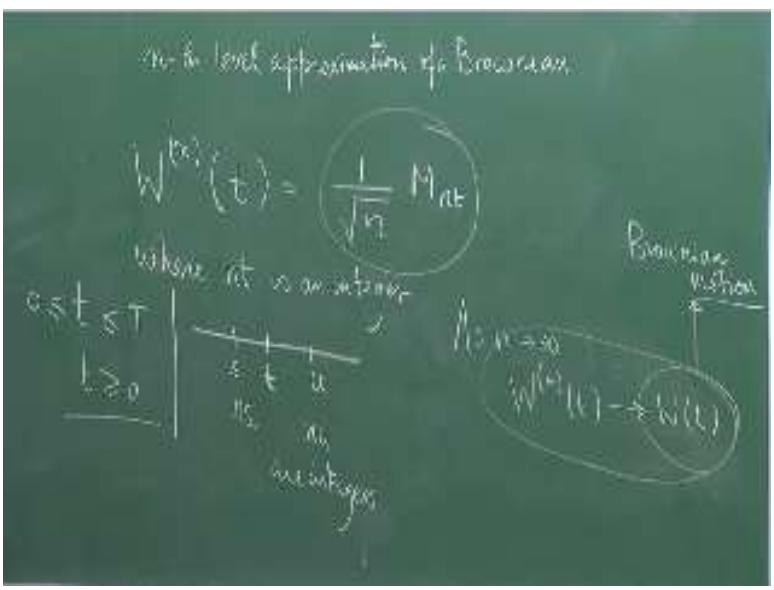

Let us construct an nth level approximation. So if you are zigzagging by say +1 and -1, I might zigzag by 1 by 10th and minus 1 by 10th. So I will decrease my zigzagging steps but increases my time size right so we construct the scaled symmetric random walk which is the nth level approximation of a Brownian motion. So all these are stochastic processes so this is a discrete

stochastic process from which I am trying to go to a continuous stochastic process. I define it like $W^n(t) = \frac{1}{\sqrt{n}} M_{nt}$.

You see if I do not have nt to be an integer I cannot define this. So here my t is a t say $0 \leq t \leq T$ or even t goes to infinity so basically for me here this t is just greater than equal to 0. So using the discrete thing I am trying to construct a continuous stochastic process but I have to be aware that if I really want to use it so at the nth level approximation this m and t this nt has to be integer if I want to actually compute this.

Otherwise m is a discrete thing it is computed only at integer points you cannot compute it at non integer points. So what happens if it is not computed, if nt does not turn out to be an integer? So basically what you are considering for n very large at various time points nt would be an integer and you are actually computing out of nt. So if nt is not an integer take the t for which is not an integer then take some u and take some s which is nearest to t such that ns and nu.

These are integers and then compute the value of $W^n(u)$ and $W^n(s)$and then make an interpolation linear interpolation to approximate the value of Wnt and that is how you can actually do you can generate it in a machine by taking a sample so you can take a sample of say so you can do the coin tossing 400 times with one by tenth you can toss the coin 400 times with probability of half of going you go one by tenth if it is h and you go minus one by tenth if it is tail.

So you decrease your movement so you actually increase the zigzag by decreasing the movement and at every time you have to observe that your Mnt nt has to be an integer. Once you do that you will find all the properties that you had for here is in here provided that this M n into t is an integer. So you first do it only for n into t is a integer. Whatever is left you do the interpolation and you will see you will start getting a zigzagging curve much zigzagged than the symmetric random walk itself.

Actually it can be shown that as n tends to infinity as n tends to infinity this $W^n(t)$ converges this random variable converges $W^n(t)$ almost surely sorry not almost surely I made a mistake converges in distribution rather converges in distribution.

Okay these are terms which I have not mentioned. Just forget them for a while, converges. In some sense $W^n(t)$as n becomes infinity. This this stochastic processes gets changed into what is called a Brownian motion. So this is what we are going to talk about in the next class. So tomorrow we are going to study the properties of Brownian motion for the next 2 classes.

So tomorrow's class would be the last for the second week of the course. In the third week we continue our discussion on Brownian motion and then go to understand stochastic integrals or Ito integrals and doing Ito calculus which is the foundation of any financial mathematics that you do. Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Instutute of Technology Kanpur

Lecture - 10
Brownian Motion-II

As, we have discussed in the last class, we discussed symmetric random walk and how we can scale it slightly up and down and so that we can make it more zigzag and go towards what is called a Brownian motion.

So, the Brownian motion is a continuous stochastic process which exhibits the property of a symmetric random walk. So, that is the idea of a Brownian motion. So, for example, if you look at the motion of a pollen grain in water, it would be something like this, more zigzagging than this than, I can draw. So, this is something a motion of a particle in a gas chamber a motion of a molecule.

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So, Brownian motion encapsulates lot of phenomenon. So, what did I say, the best way to remember about Brownian motion is that, Brownian motion is a continuous analog of symmetric random walk. In a continuous setting, it behaves in the way a symmetric random walk behaves in the discrete setting.

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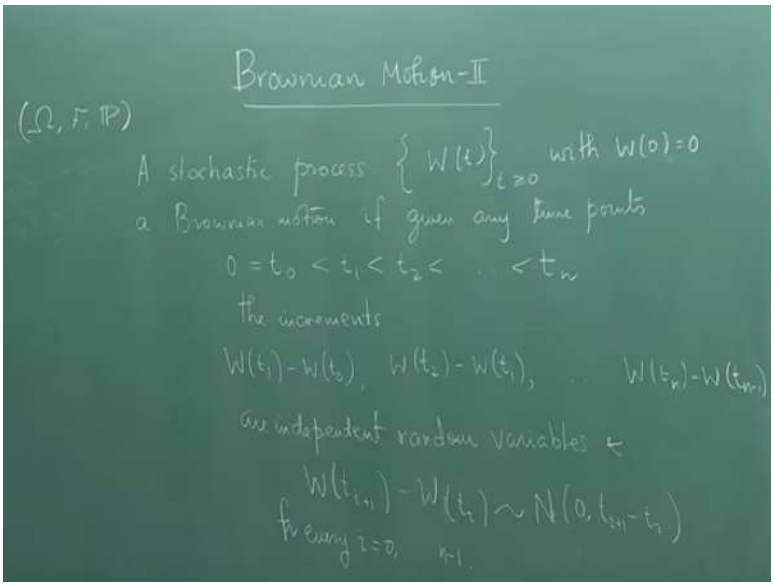

So as usual, I will have the probability space which is written down here. So, Brownian motion, so, your stochastic process, $\{W(t)\}$some people write W lower index t. it is up to you. it does not matter, is called a Brownian motion, if given any time points, t is time actually, spreading time points say tm tn whatever. Given any time points like this the increments are independent.

The increments mean just like the way you have done increments in the symmetric random walk, the increments $W(t_1) - W(t_0)$. $W(t_0)$ is actually 0, $W(0) = 0$ just a minute, I think, I should just change it a bit.

A stochastic process with $W(0)$ is equal to 0. So, it does not matter. Whatever be our scenario omega, $W(0)$ is always 0. So, it is identically a zero function. So, $W(0)$ itself is a function. Please remember it is a random variable. So, whatever be the form of that random or whatever be this random variable, whatever be the scenario omega, W of that would always be 0. That is the meaning of the whole thing.

So, this $W(t_2) - W(t_1)$ this random variable, these increments in the random variables. Hence, how much change you are having how basically, you are telling that how much zigzagging has taken place in the interval $t_2t_1, t_0t_1, t_1t_2, t_2t_3$and so forth. So, it is some sort of a broad measure

of the zigzagging that has taken place. How much the function value has changed at the 2 ends. These are independent random variables.

Now the independent random variables. This is one property you know from the property of the symmetric random walk. Also, you know the symmetric random walk, the expectation of increments are 0 and the variance of the increments are the difference between the 2 time end points.

Here, actually one can prove through the Central Limit Theorem that, these increments actually follow normal distribution, which we do not prove because, that would take too much of time, it is a very short and compact course. So, these are independent random variables and $W\left(t_{i+1}\right) - W\left(t_i\right)$. this random variable follows normal distribution, with mean 0 and variance t_{i+1}-t_i.

So, if all these properties, 2 properties are followed, first that this has to be maintained whatever be the scenario, whatever be your time points increasing, if you take an increasing set of time points finite set of time points, then these things are independent and these increments themselves follow normal distribution, for every i. Of course, this has to be for every i, now start $i = 0 i = n, i = n - 1$ basically.

So, each of these, is following a normal distribution. So, this is what is called Brownian motion. Does stock price follow Brownian motion? The stock price truly does not follow Brownian motion, though it looks like one. Because Brownian motions can take negative values, because the symmetric random walks can also take negative values, but a stock price can never take negative value. Once a stock price is 0, that stock has to get out of the market now it is a 0-value stock.

So, stock prices are not really a model using Brownian motion, but actually these stock prices actually were modeled by Bachelier in 1900, who was the student of Henry Poincare using Brownian motion and showing that the pricing of such commodities in the stock market, of various instruments in the stock market can be achieved by solving the heat equation. So, he

linked the processing the financial markets to partial differential equations. And of course, his idea was lost and now later on he has become a very famous name.

But the interesting part is that, the way stock market or movements of stock prices are modeled that is called a geometric Brownian motion and then that is built using, that will always give you a nonnegative thing, which is built using a Brownian motion and you can understand and taking the trick that, if you want to get non-negativity always use the exponential function. So, that trick is being played in this area pretty often.

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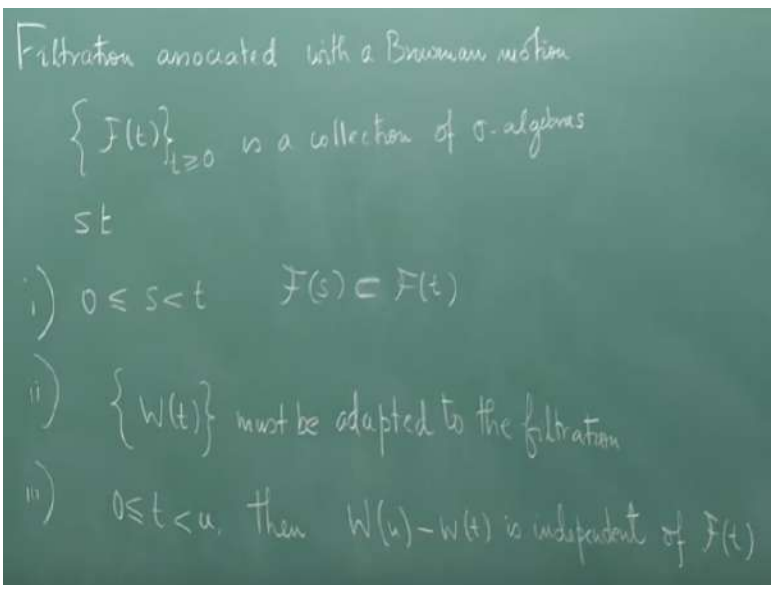

So, now we will introduce what is called, if I want to define a filtration associated with a Brownian motion, how do I define a filtration associated with a Brownian motion? So, what I will do in this filtration associated with the Brownian motion.

So, this filtration $\{F(t)\}_{t\geq 0}$ is a collection of Sigma-algebras, this is a collection such that, number one, whenever t is strictly bigger than s. $F(s)$ must always be contained in $F(t)$, but you have information as time evolves. So, this should always be contained in $F(t)$ okay. Now number two, is that $\{W(t)\}$ must be adapted to the filtration. The Brownian motion must be

adapted to the filtration and the third point is, so if you have this situation then, $W(u) - W(t)$is independent of $F(t)$.

That is, once you move beyond t, $F(t)$ does not have any information about this. This is independent of what information you have in Ft. Ft can also be viewed in some sense as a smallest Sigma-algebra generated by the stochastic process till a given time.

So, take all the values of the random variables omega t, up to a given time and take the Sigma-algebra generated by them, that can be also viewed as one of the filtrations right. You could have larger filtrations also but that is essentially it is. So, once we know that that we can have some filtration defined like this, we can prove that the Brownian motion is a Martingale.

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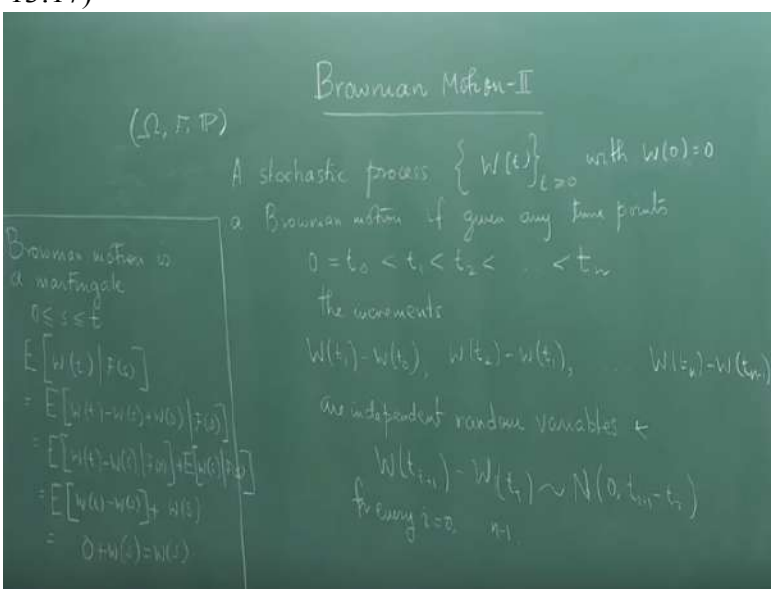

So, again, just do like this, take $t \leq s$, so you will find that the tricks are almost similar. So, what I do is, I, and now I breakup the whole thing. You know, $W(s)$ is completely determined by $F(s)$ is known. So, you have to take out what is known. So, $W(s)$ is equal to $W(s)$ into 1 you take out the $W(s)$ expectation of 1 of constant random variable is the same so.

Now here, $W(d)$-$W(s)$ as the third definition, does not depend on $F(s)$. So, it is nothing but it is so there is no conditional expectation here, so is just the expectation, so this random variable is just this constant random variable plus $W(s)$. You take the $W(s)$ out, expectation of 1 given $F(s)$ expectation of 1 is 1 basically that is it. Because 1, just the constant random variable does not depend on $F(s)$ but this you know is 0 because of this fact, $0 + W(s)$ which is equal to $W(s)$ and that proves that it is a Martingale.

There are several other Martingales, which are associated with the Brownian motion. A Martingale that we are going to write now, is very-very important in finance specifically in calculation of risk neutral probabilities and all those things. So, this is a very important Martingale called the exponential Martingale.

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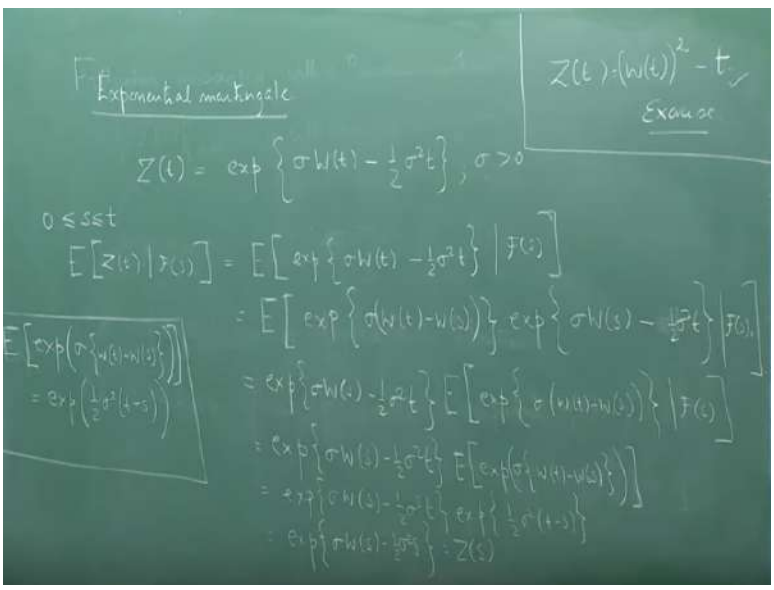

So, the exponential Martingale is defined like this. So, exportation means, e to the power, so $Z(t) = exp\{\sigma W(t) - \frac{1}{2}\sigma^2 t\}$, where σ is positive number, exponential means e to the power of this. Of course, this itself is a random variable because you are taking exponential to the power of a random variable.

So, it is, when you are taking the exponential, exponentiated by a random variable. So, this itself is a random variable. Now the idea is that if you have a filtration associated with respect to $W(t)$ then that same filtration will be associated with $Z(t)$, because if $W(t)$is adapted to a given filtration $Z(t)$will also be adapted to the same filtration, because knowledge of $Z(t)$ singularly depends on the knowledge of $W(t)$ right. The question is whether it is a Martingale. So, one has to prove that this is also a Martingale.

So, we start in the similar fashion. This proof is not as straightforward, as this proof, though similar type of approaches would be used but let us just go and do it. This omega, the σ that you see, this we will finally talk about as volatility of the stock price movement. It captures, how the randomness, captures the massive movement. It really gives you a feel of the zigzagging of the path how the prices are how fast they are going and coming down so that sort of so this captures that idea basically.

So, F is a filtration associated with the Brownian motion $W(t)$. Say I have added $\sigma W(s)$ and subtracted $\sigma W(s)$from here. Observe that $\sigma^2 t$ is a nonrandom part. It is not a random variable right. So, here is the product x into y okay. So, the question is, in this sort of situations right, there are some deeper questions. If you go back to your original, where we had written down the laws of conditional expectation, the rules then, we expected this and this, this product has to be integrable because if you have to maintain the definition of conditional expectation.

Of course, product has to be integrable means you need to define, what is the meaning of integration of these two random variables okay. Is such random variables are integrable? That is the question we are not going to answer right now. We will later on show that these are actually integral, let alone discuss the integrability of this, that is they are actually integrable random variables, because they will come out to be solution of certain equations.

So, and if you have a Brownian motion, how can you integrate it. Can you integrate it just like any other random variable, can you find the expectation of a Brownian motion? The answer is yes. I can find the expectation of this Brownian motion because this is 0. Can you find the

expectation of this Brownian motion? The answer is expectation of Brownian this Brownian motion is this, minus $\frac{1}{2}\sigma^2 t$.

Now it will be left to the reader to decide whether. Of course, there are certain little technical issues, I am not getting into, but it is clear that expect. What is meant by the meaning of integrability of a random variable? Integrability means that the expectation is finite. The expectation of this is finite, this is finite.

So, this exponential function will have the integral will be integrable and this exponential function would be also integrable. So, at the end if you look at this part. So, this is nothing but this part, but this part this is because, this has mean minus half sigma square t because, this mean is 0 and, this exponential is nothing but a constant thing, so mean of this is nothing but minus half sigma square t so this is a thing with finite mean.

Now if you take a you are taking exponentiation so basically you are taking expectation of the exponential function. If the random variable itself has a finite mean, then if you take the exponential function, that exponential function will also have a finite mean it will have finite expectation. That is why it is meaningful to apply the fact that I can take out what is known.

You see this part is known to me when, at time s, this whole part is known to me because this only depend this is just the evolution of W the random variable up to the time s. So, I will take this part out. Taking out what is known I will have exponential. So, it is very important to get certain technicalities clear, before you move, where you are actually applying the results correctly. That is an important thing, that one needs to learn as one goes on doing more mathematics. Sometimes we can do some hand waving but not always.

Now once, I have this, remember that this thing is independent of Fs, which means, I just need to calculate this. Now what is this, sorry, I would not have the Fs here sorry it is independent. Now if those, who know some probability, they will understand that, I know that $W(t) - W(s)$ is a normal distribution with mean 0 and variance $t - s$and this is nothing but the moment generating function of a normal distribution.

I have not spoken about moment generating function in this discussion, but if you forget about this term moment generating function, you can directly compute this expectation. I am not going to compute for you, this will come as a homework and this will appear in your assignments.

So, what I am just writing down the answer. Even this is simple integration, so I am just not going to do that. This is equal to $exp\{\frac{1}{2}\sigma^2(t - s)\}$ okay. So, this is what you will have that is the answer.

So, once you write that down then, the final answer would be the following, that if you write this as exponential so you write this as$exp\{\sigma W(s) - \frac{1}{2}\sigma^2 t\}$ and then you write this as $exp\{\frac{1}{2}\sigma^2(t - s)\}$, so this is nothing but $exp\{\sigma W(s) - \frac{1}{2}\sigma^2 s\}$ so this is nothing but $Z(s)$.

There is another Martingale which is helpful in finance is the following. It is $Z(t)$ $(W(t))^2 - t$. So, this will also go as an exercise in your homework assignments, to prove that this is a Martingale.

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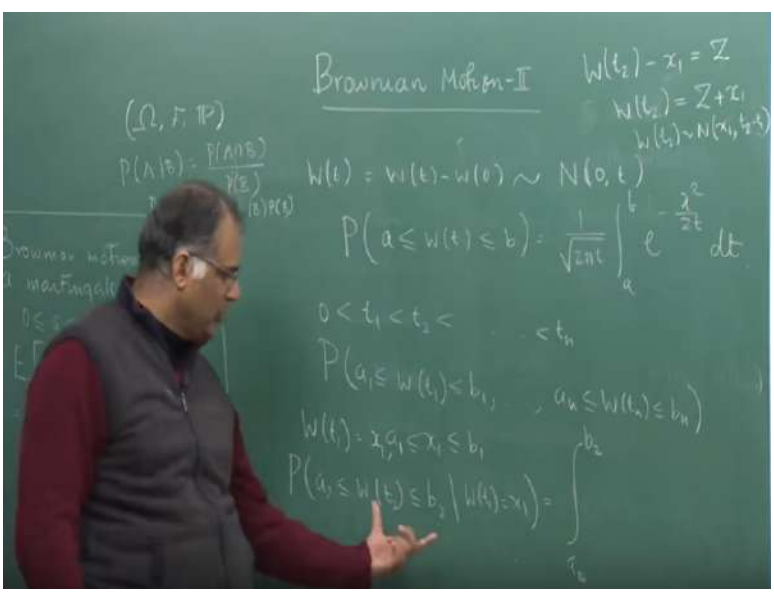

Now we will talk about, so what is happening. We will talk about how to compute the joint probability of a Brownian motion at certain given time points. You take $W(t)$ at any t, what is the distribution of this random variable. $W(t)$ can be always written as, $W(t) - W(0)$ and this has normal (0, t). So, if you want to know whether, at a given time t your $W(t)$is lying between some points a and b, it is obvious you can just use the basic idea that is.

The question is, suppose I have now n time points which are greater than 0, 0 I know where it is, so with n time points and we ask you the question, how to find sorry, say $W(t_1)$, how do I do it. So, let me look at the case at the very first, this is how you compute what are called the transition probability densities let us see.

So, now suppose, you are given the information that, under the given scenario $W(t_1)$, under the given scenario omega $W(t_1)$ omega was x okay. So, suppose I know that it is known to me that, at t1 for the given scenario W, at time t1 it was x where say x_1, x_1 lying between b_1 and a_1. So, what I now want to do is, I want to calculate the probability at $W(t_2)$ lies between a_2 and b_2 given that $W(t_1)$ is x_1. That is what I want to do .

But, if you see, if I fixed $W(t_1)$ asx_1 then, what do I have. I have $W(t_2)$- x_1 as my random variable, x_1 is known to me that, at time t_1 x_1 is what happened. So, now what is the probability that $W(t_2)$ would lie between the value of $W(t_2)$ for the given scenario omega will lie between a_2 and b_2. So, for the scenario omega t_1 I know what will happen. It was at x_1. Now find so it is in the state x_1 so what is the probability that, the next state would lie between a_2 and b_2. It is like sort of a computing transition probabilities.

So, here if you look at this thing, the expectation of this, this is also normal random variable, so this also follows normal random variable, but the expectation of this, is what, expectation is, so it is a, see here, we had, so this new random variable z, so $W(t_2)$ so, is equal to z plus x_1. So, under this given information the expectation of $W(t_2)$. So, $W(t_2)$ under this information follows

normal x1 with, because this is the time t_1, so this z follows normal 0 at time t_2 t_1, so $W(t_2)$ is nothing but x_1 plus z. So, $W(t_2)$ follows this one, because this expectation is 0. So, once you know this fact then what would happen.

This means now I can compute, sorry, I should be having, so we will be applying the same idea of conditional probability, that $P(A \cap B)$ is probability of A into probability of B sorry, $P(A \vee B) = \frac{P(A \cap B)}{P(B)}$. This is what we had learnt. So, you can always write $P(A \cap B) = P(A \vee B).P(B)$. That is exactly what we are going to do here.

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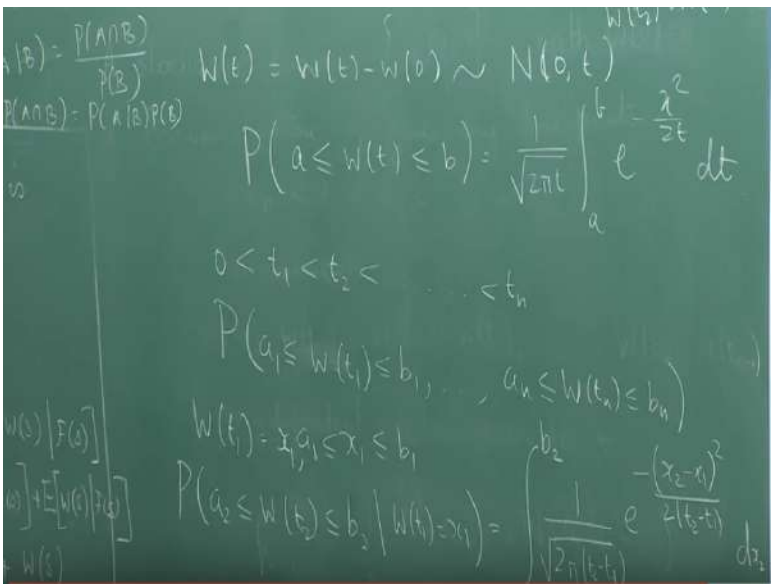

So first we are computing the conditional expectation and then using this we will write the joint probability. So, this conditional expectation is, 1 by the conditional probability is, $\frac{1}{x_2}$ is the variable that I am using for the variable x_2, x_1 is what is the mean.

Now, if I want to write the joint probability so this is my conditional density. This is my conditional density function and thus I can write the conditional density function in a more

simpler way. So, I can write the conditional density function in a compact form but let us do a general form and then we will tell you how to write the conditional density function.

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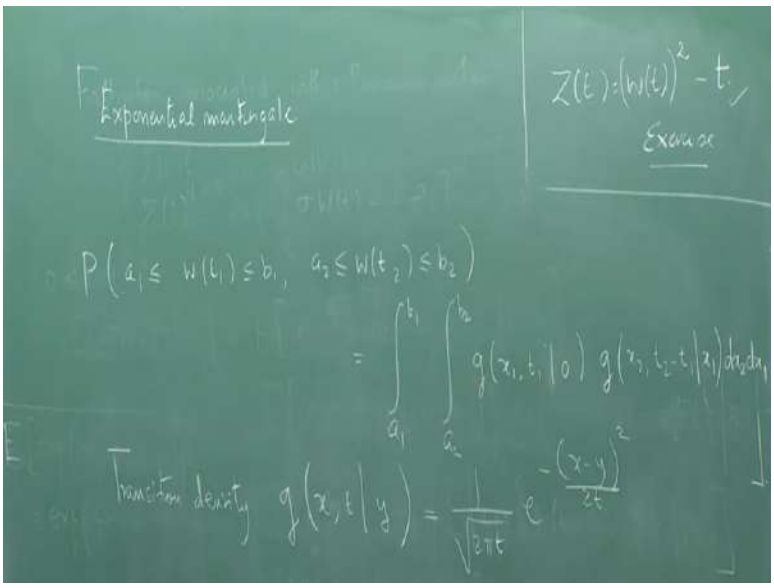

If you look at this, now I am writing this one, is a joint probability okay, and this is equal to, so first I will write, I will tell you, what is the meaning of this. So, these are the marginal probabilities, a marginal densities. So, the marginal density, in general is given like this.

So, what is the density function associated with the fact my current state is y and then in time t, I will move to the state x, at time 0 say if my current stated time 0 is y in time t, I will move to the state x, what is the density function associated with that particular transition. So, this is called the transitional density function or the conditional density, transition density and that is given as, $\frac{1}{\sqrt{\pi t}} e^{\frac{-(x-y)^2}{2t}}$.

So, this is my conditional density function. So, I would keep it as an exercise for you to write down for the case till end. How can you write it? So, here you see, from 0 my state was 0, it is

always 0, W0, 0. I have come to the state x_1 at t_1. Given that, I am in state x_1 at t_1, within the time t_1 tot_2, I have, within the time span $t_2 - t_1$, I have come to the state x_2.

So, essential a marco process. Brownian motion is also a marco process those who know about marco process. So, we are just trying to compute, this is nothing but a conditional, this is a transitional probability actually. It tells you, given the state is at x_1, what is the probability that the state is between a_2, and b_2, so next state. Because it is a continuous thing, you cannot say that probability $t_2 W$) is x_2 because that will become 0, so you just have to say what is in between. So, that is the whole thing.

So, I know what is at 0 I will come to the state x_1. So, the conditional density at x_1 is the conditional density at x_2. I am at time t_1 and I am at x_1 and at time t_2 so within the time spend $t_2 - t_1$ I have come to x_2 then what is the conditional density. So, the joint integration of this, would give me the conditional probability, which can be motivated, is motivated from this very basic idea.

Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 11
Brownian Motion-3

Today we are going to do the last part of the Brownian motion, which I am writing as Brownian motion-III. And today we are going to do a very important part, so we will be today talking about quadratic variation of a Brownian motion, what does it mean?

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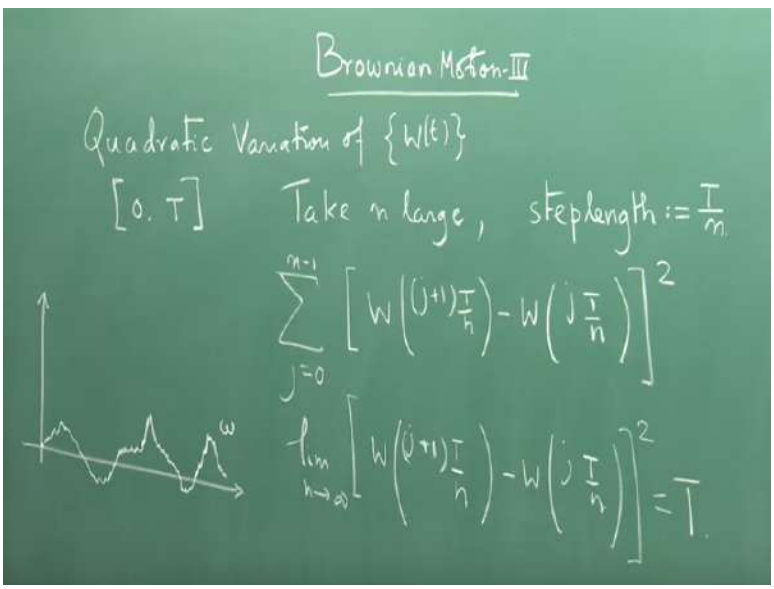

Variation of this stochastic process $\{t\ W\ )\}$. W actually stands for Wiener, Norbert Wiener who actually formally studied the process in detail and made lot of applications.

So, a natural thing to do is, that if you have given interval $[0, T]$ there is no natural way to break the steps, like for the symmetric case symmetric random walk here a step every step 0, 1, 2, 3, 4. So, here you do not have any selected step length, so what you do is take n very large and consider step lengths of the size $\frac{T}{n}$ and form the following sum $\sum_{j=0}^{n-1}\left[W\left((j+1)\frac{T}{n}\right) - W\left(j\frac{T}{n}\right)\right]^2$, it is not so apparent what would be the limit, if I take the limit of this quantity. So, I take limit of

this quantity as n tends to infinity, what is the idea? how to compute it? It is not so apparent that the $\left[W\left((j+1)\frac{T}{n}\right) - W\left(j\frac{T}{n}\right)\right]^2 = T$.

So, before really getting an idea, what it means, let us talk about the first variation and second variation of a continuously differentiable function.

If you look at the sample path of a Brownian motion, then for a fixed ω, so once you fix an omega, it starts from 0, this is too much zigzagging basically, I am unable to draw, I am putting straight lines at some place. You magnify them there will be zigzagging. For every small zigzagging, there are further many infinitely, many zigzagging's.

So, if you look at this it essentially tells you that the curve is continuous, as a function of T given a sample path omega, so this for a sample path say ω, the curve is continuous in T, but is not differentiable at any point that you want. And so, second variation will take the value T. It will accumulate, some second variation, while will show that, if you have a continuously differential function second variation will always be 0.

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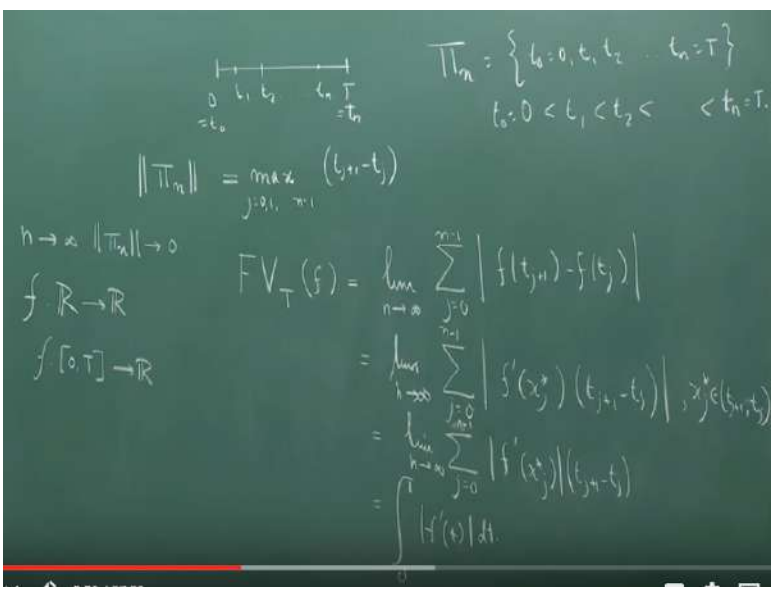

But let us just talk about what is first variation. So, if you take the interval 0 to T, the key point is to, break it up into partitioning points say n partitions. So, the nth partition will be given as

points

$\pi_n = \{t_0 = 0, t_1, t_2, ..., t_n = T\} where you of course have t_0 = 0 < t_1 < t_2 <....., t_n = T.$

So, this is the basic partitioning of an interval, that you do to compute the integral. So, if you write the first variation. So, you define some quantity of $\vee \pi_n \vee$ form and max of, so for at the nth partition, this is what is called the maximum size of an interval, because $(t_{j+1} - t_j)$ intervals need not be of equal size. One thing is very clear that, as you keep on increasing the number of points, this one will go down to 0, this will come down, whereas this is the maximum of all possible gaps but if you put in more points this value will come down right.

So, once you understand this, you can immediately conclude for the time being, that if $n \to \infty, ||\pi_n|| = 0$ this is the first observation. Once you make this first observation, you see how simple it is to calculate the first variation. The first variation of a function over the time interval 0 to T of the function F, which is continuously differentiable. So, you will take a function $f. R \to R$ does not matter, you can take Rn into R, we are just taking R to R because, this W as a$f. [0, T] \to R$ function of t is a function from R to R for each sample path.

So, let us look at it, R to R or 0 t to R does not matter. If you are more comfortable, let us just put this is what. So, I am just bothered about this path. I am not bothered about the function elsewhere because then I can simply write this as, $\sum_{j=0}^{n-1} f(t_{j+1}) - f(t_j) \vee$, which is same as telling that $||\pi_n|| = 0$. This one now how do I look at it, because once you have some information that this has a derivative and you want to have some information about the function always use the mean value theorem that is the biggest trick and a simplest trick in analysis.

So, what you are doing. When you are taking the difference of this so by mean value theorem there would be some $\sum_{j=0}^{n-1} |f'(x_j)(t_{j+1} - t_j)|, where x_j \in (t_j, t_{j+1})$ in open interval not the closed interval. So, this is nothing but $\sum_{j=0}^{n-1} |f'(x_j)(t_{j+1} - t_j)|$ but if you look at it this is nothing but, a

reimann sum for the function mod fj mod f sorry. So, it is a reimann sum for the function mod f and this is nothing but the integral from 0 to t, sorry not infinity, n-1, so it is nothing but the integral by very definition of $\int_0^T \left|f^{'}(t)\right| dt$.

So, this is what is the meaning of first variation which is pretty natural. So, let us come to the definition of the second variation. What is the meaning of the second variation of a function? Of course, you will do a little bit of jugglery I am just removing this part. So let us look at the second variation or the quadratic variation.

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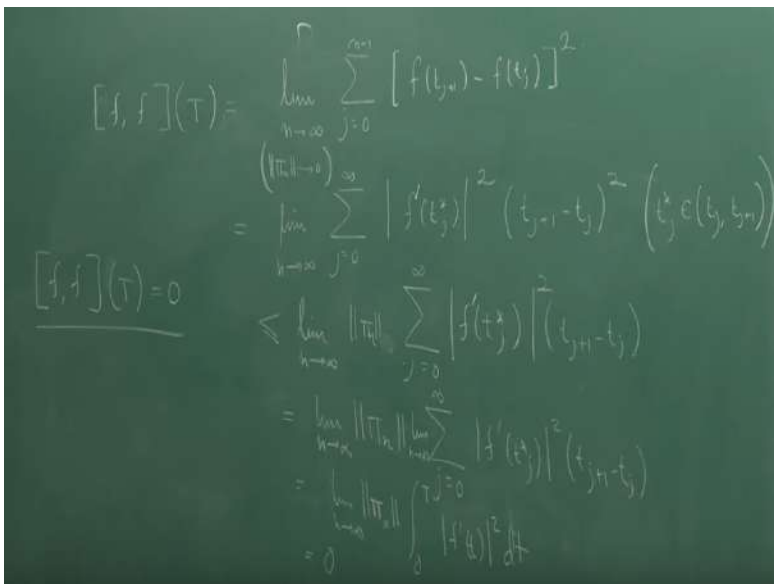

So, quadratic variation is usually symbolized like $[f, f](T)$ and this is written as Limit n tends to infinity. Limit n tends to infinity is same as writing $\left|\left|\pi_n\right|\right| = 0$ okay the same thing, either we write that or we write that. So, as n goes to infinity, so j is equal to 0 to n-1. Instead of having the modulus, the absolute value of the difference.

So, what does the first variation do. It counts the total number of times the function goes up and down up and down up and down. That is what it is counting, it is counting the variation. From

here say if this is positive and this function value is say lesser than this it goes down and then possibly it can go up so it is counting the up and down total up and down the crossings.

So, here I will write the same $\sum_{j=0}^{\infty}\left[f\left(t_{j+1}\right)-f\left(t_{j}\right)\right]^{2}$ but instead of looking at the variation in terms of the absolute value, I am doing in terms of the square, something like the mean square error type of thing that you must have heard if you have learnt some statistics. So, this is summation. Now I can here write the same thing, but I have written x-star j where x-star j is here anyway, I will not write x star as n goes to infinity, so j is equal to 0 to n-1j, I will write say $\sum_{j=0}^{\infty}\left|f^{'}\left(t_{j}\right)\right|^{2} \vee \left(t_{j+1}-t_{j}\right)^{2}$ okay.

Now if I write this $\left(t_{j+1}-t_{j}\right)$ into $\left(t_{j+1}-t_{j}\right)$. This breaks the square into two parts. The product of the two similar parts. Then one of them is obviously less than equal to this π_{n}. This π_{n} n is the maximum of the interval size. So I can write this thing of course as, less than equal to limit n tends to infinity, $\left|\left|\pi_{n}\right|\right|$that is, into summation j is equal to 1 to infinity

So, here again I have applied the mean value theorem if you have not observed that I have done the same thing only instead of x-star j I have put t star j x t star j is an element between t_{j}sorry, I made a mistake here, it should be$t_{j} \wedge t_{j+1}$, $t_{j+1}$is bigger than t_{j}, please correct that t_{j} to t_{j+1}.

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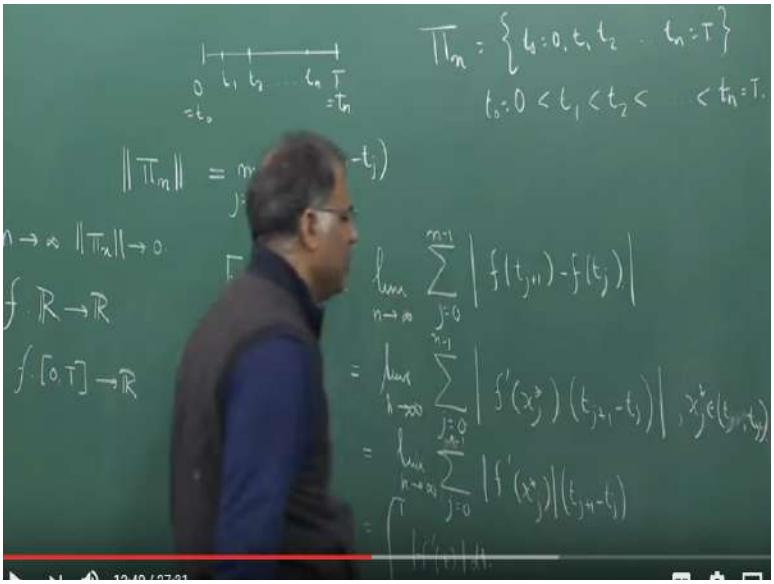

So, it is lying in the open interval t_j and t_{j+1}. So, basic mean value theorem Lagrange's mean value theorem basically. Now, you will write f of f, sorry f dash, so here we observe this f dash. So, we will have $\left|\left|\pi_n\right|\right| . \sum_{j=0}^{\infty} \left|f^{'}\left(t_j\right)\right|^2 \left(t_{j+1} - t_j\right)$. Now limit of this exist, limit of pi n. Limit of $\left|\left|\pi_n\right|\right|$ exist this is 0, and limit of this also exists, because these are nothing but the integral of the mod square. So, what you will have.

Of course, because I have taken f to be a continuously differential function its square is also continuously differentially function and hence it is integrable, basically in the Newtonian sense and hence in the riemann sense.

So, this means, this is nothing but, limit of both of them exists. Limit n tends to infinity so this is exactly equal to norm-pi n, so here also I have to write $\left|\left|\pi_n\right|\right| . \sum_{j=0}^{\infty} \left|f^{'}\left(t_j\right)\right|^2 \left(t_{j+1} - t_j\right)$. So, this is nothing but limit of n tends to infinity $\left|\left|\pi_n\right|\right|$and $\left|\left|\pi_n\right|\right|$and this one is nothing but $\int_0^T \left|f^{'}(t)\right|^2 dt$.

So, this is a finite number because $\left|\left|\pi_n\right|\right| \int_0^T \left|f'(t)\right| dt$ is a integrable function. So, this is nothing but, that has to be one has to make sure that this is finite. So, because of our assumption this is finite this limit is going to 0 so this whole limit goes to 0. So, what have you proved. We have proved the following that for a continuously differential function f defined from on the interval 0 to T ,it does not accumulate any quadratic variation.

The square of the ups and downs the total accumulation of the ups and downs but the square of the values is added up and that variation is 0. That is very very interesting because that does not happen with this one but it does not happen with this but you will see immediately that the behavior of the, you can immediately understand also the behavior of the Brownian motion is so different because it accumulates quadratic variation while a smooth nice function does not accumulate any quadratic variation.

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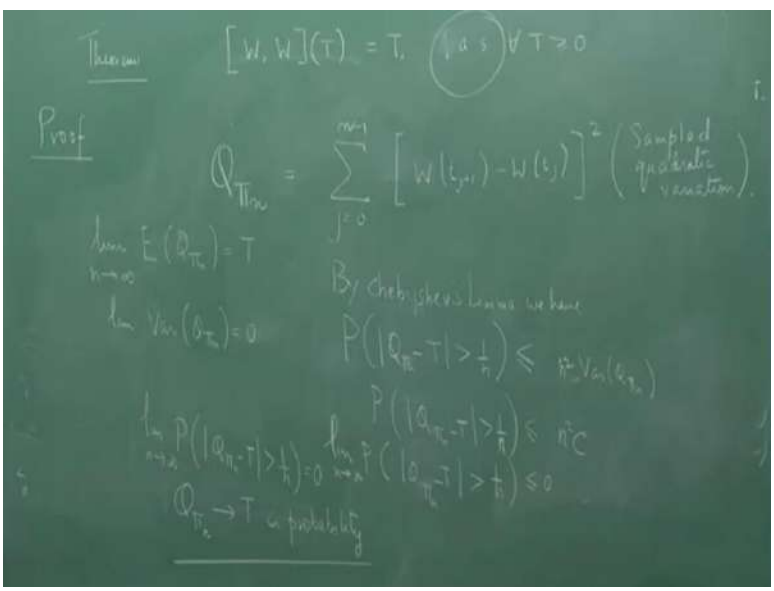

So how do I go about proving that the Brownian motion actually accommodates some quadratic variation. So, we will just prove this fact about Brownian motion. So, the theorem $[W, W](T) = T$almost surely for T greater than equal to 0. Now it is very important to note that, this quadratic variation here is actually a random variable, because W itself is a random variable,

is a function but is a random variable so when you make these sorts of variations you are actually getting random variables.

So, your convergence is in the almost surely sense and that is we are going to show and you will see a fantastic application of Borel-Cantelli lemma and a fantastic application of the Chebyshev's lemma. That is what makes mathematics beautiful actually.

So, when you deal with random stuff always be sure that, this guy would be always around you, almost surely this term, which means same as almost everywhere, this term would be always around you. So, let us try to prove this fact. This is known, this is very-very much a useful thing in finance. We also at the end write down some shorthand, which will be used in finance when we do the Ito calculus.

So again, you do the same sort of partition. Partition in the interval, portioning is something I am not going to write, because I have already written how I am partitioning the interval, what is your big pi n, what is the meaning of norm, $||\pi_n||$ is going to 0 as it goes to infinity.

So, I am going to define a variation, total quadratic variation on a given partition the nth partition, which as before, would be, $Q_{\pi_n} = \sum_{j=0}^{n} \left[W\left(t_{j+1}\right) - W\left(t_j\right)\right]^2$ This is sometime called sample quadratic variation.

So, when you are taking about this sample quadratic variation, you have to observe that, this Q_{π_n} is actually a random variable, because this is random variable, I can put a particular scenario omega and so you will have a Qn omega, that will give you a particular value of this, for the same partition. Now if you change the scenario omega to another omega 1, then for the same partition it will give you a different value.

Now, what we are going to prove first is an amazing thing. Let me first tell you, what we are going to prove. We will first going to prove that, $E\left(Q_{\pi_n}\right) = T$. This thing will allow us to prove that, and we will prove that, $Var\left(Q_{\pi_n}\right) = 0$.

Suppose, I have proved this, basically what I am telling, that if you average over all the possible parts omega, then you essentially get the value of the quadratic variation, that is the meaning. You take this and average over all the possible parts then you get the value of the quadratic variation, which I claim to be T.

Now how do I actually say that, this Q_{π_n} goes to T, almost surely and here you will see the use of Chebyshev's lemma. We will prove this later on, if we have time, but let me, once I know this how will I use this, Chebyshev's lemma and the Borel-Cantelli lemma. Actually, I will use this idea.

So, now my Chebyshev's lemma what will we have, if we have this information then by Chebyshev's lemma we have, this minus T sorry, Q_{π_n} I should write, $P\left(\left|Q_{\pi_n} - T\right| > n\right) \leq \frac{1}{n^2} Var\left(Q_{\pi_n}\right)$. So, $Var\left(Q_{\pi_n}\right)$ that been bounded and that that goes to 0 and hence it is bounded. I can still write this to be, sorry, just do it 1 by n. So, this will become $P\left(\left|Q_{\pi_n} - T\right| > \frac{1}{n}\right) \leq n^2 Var\left(Q_{\pi_n}\right)$.. That will be $n^2 C$.

So, what does this tell me. It tells me that, , $P\left(\left|Q_{\pi_n} - T\right| > \frac{1}{n}\right) \leq 0$, but probability is never a negative quantity, so limit n tends to infinity bigger than 1 by n.

So, it tells me that as n becomes very large, then the set of all omegas for, which this is true is, as measured 0, which says that $Q_{\pi_n} \to T$ in probability. So, if it converges to T in probability, does it converges to T almost surely. Let us look back at the literature. So, if you go back to our very old

lectures on the Borel-Cantelli lemma section, then you can see that, we have spoken about this, so if xk converges to x almost surely, we have proved that it is Q pi n goes to T in probability okay.

Now does it mean that, Q_{π_n} will go to T almost surely. This is not a clear thing that it will go to Q n, will go to T almost surely, but that is what essentially you have to prove. So, we have come to this conclusion immediately by applying Chebyshev's lemma and this information. So, what we have that Q_{π_n} goes to T.

Now we can use some other techniques some more involved techniques to actually show, that Q_{π_n} also goes to T almost surely, which we will put up in the notes, in your, what is that called in our forum because that will take too much of time to describe. So, just by applying the Chebyshev's lemma by noting these notion we have proved that, it goes to Q n converges to T in probability. So, what we have to show that Q n converges to T almost surely.

So, this is something which is quite involved which we will not do in the lecture, but we will describe it in the class, so hence we have come to the following conclusion. That this is equal to t. You are obviously showed that this is equal to T in probability, but this is equal to T also almost surely. That thing needs to be proved. That needs a little rewriting of this Q_{π_n} and needs a little bit of, you know tweeting the whole thing so that we can apply Chebyshev's lemma and Borel-Cantelli lemma both to come to this conclusion.

So, we will take a little bit of, it will be a little more complex thing, so we will not do it now. So, what we have proved here that if I can this is easy to prove, this will take some time to prove which we will not again prove it. So, we can assume that this if you agree to these two things this is of course can be proved very easily you can prove it yourself and this is take time then by applying Chebyshev's lemma.

You can at least prove that Q_{π_n} converges to T in probability but proving that it also converges to T almost surely needs a machinery needs some more machinery which we will not get into the class but we will put it on the forum.

Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 12
Ito Integrals-1

So, here we are, on our third week run and today we are approaching or rather entering the fascinating land of the Ito calculus or the Ito integral. Ito is the name of a Japanese scientist who introduced this and with that, he changed many things in physics, his things changed many things in physics and also in probability and Ito integrals and the calculus associated with it, the Ito calculus, is the key to doing a lot of computing or solving lot of models in finance.

So, this is the holy grail of stochastic calculus and we are going to do it step by step. We are not going to be in a hurry, we are not going to push ourselves into too much of rigor. Of course, rigor is a necessary tool of a mathematician, but looking at the diverse audience that I have, you cannot push rigor too much, but I will try to get you some understanding and in this whole the remaining part of the course, the book, I will follow a single book, which is one of my favorite books in mathematical finance. It is called stochastic calculus for finance.

So, this book is written by, Steven Shreve a very famous name in mathematical finance from Carnegie Mellon University. I have taught from this book earlier and this is my favorite book in this subject. I would rather like to show it, this is the book and the interesting part is that, this book now is an Indian edition, both the volume.

This is volume 2 from which I am teaching. So, volume 1 and 2 there are 2 volumes. Volume 1 talks about discrete time. Here we are talking about continuous time and both these books are very a lovely read and those who are serious about mathematical finance should actually procure the copy of these 2 books. This will be helpful in many-many ways and this book takes you from a very basic to pretty advanced level.

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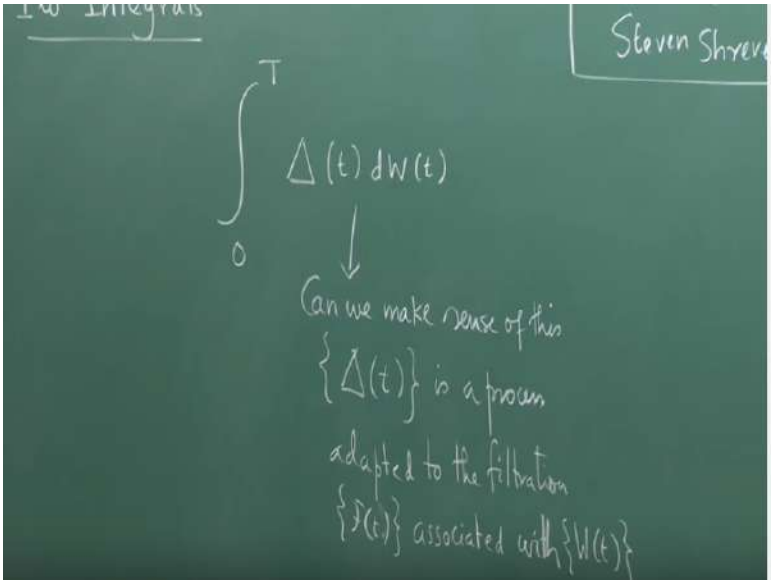

So, now what we are going to do. In this section, we are trying to make sense of the term, sense of this $\int_0^T \Delta(t)dW(t)$. Of course, you might ask what is $dW(t)$. We just said in the last class that, this Brownian motion is not differentiable anywhere.

Yes, but assume that this is like writing the difference. This is the increment of the Brownian motion we and we had been talking about increments of this Brownian motion. So, we want to so the question is, can we make sense of this and what is this $\{\Delta(t)\}$?

$\{\Delta(t)\}$is a process adapted to the filtration $\{F(t)\}$ associated with the Brownian motion. So, just like in standard integration theory, we will start by trying to look into the meaning of this, when this process is nothing but step functions. At every t it is some step function and on a particular interval it will hold a particular value fixed value.

So, our job would be now to make you, will see that, finally when we will compute things, you will see that this will give us very different answer than standard calculus. It is the quadratic variation of the Brownian motion that would be the culprit. So, here again, we are going to partition 0 T into intervals. So here we are partitioning where you have this.

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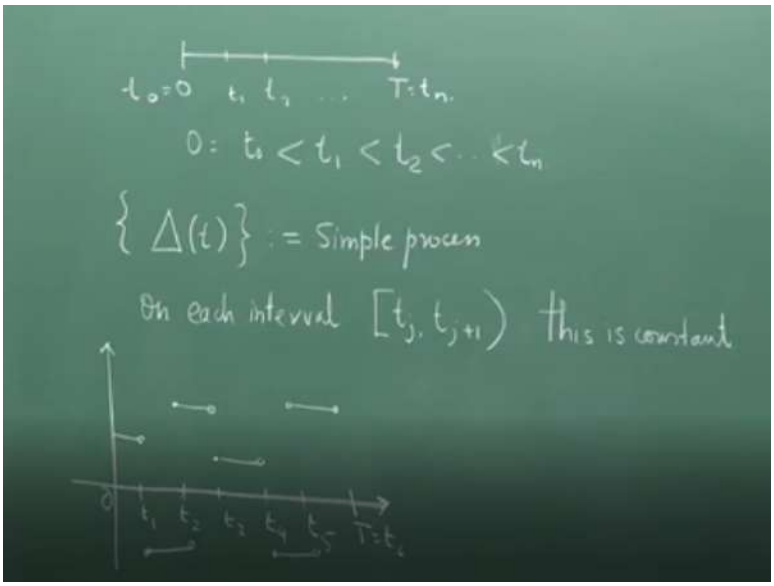

Now, we will talk about a simple process or a simple function basically. We will take this to be a simple process.$\{\Delta(t)\}$ = Simple process means on each interval $t_j t_{j+1}$ which includes t_j, not t_{j+1}, this is constant.

So basically, you are looking at a function of this form. So here is your 0 to T, here is t_1, here is t_2, here is t_3 say I will just do it 5, t_4, t_5 and this is t_6. So, what you are expecting that from 0 suppose this, to here, you have some particular value. Then t_1, so here you do not have the value, this value is not valid here it starts again from here, t_1 to t_2 that is why you are giving this round.

This is one particular value and t_2 to t_3 it could be like this, t_3 to t_4 it could be like this, t_4 to t_5 it could be this one, and t_5 tot_6 it could be like this just. This is a step function basically and this is an example of a simple process.

So, this is one sample path of that simple process right. So, in this thing, if this is the way you have broken it up the partition, on each partition we will consider that it is constant. Of course, if you take more smaller partitions, the same idea will remain fixed. Okay, there will be 2 intervals where you will have the same value.

Now given this partition on which it is constant, so we will look at only those intervals on which it is constant and that will partition the interval. We will try to calculate it. Now by $\Delta(t)$ let us work like a financial guy.

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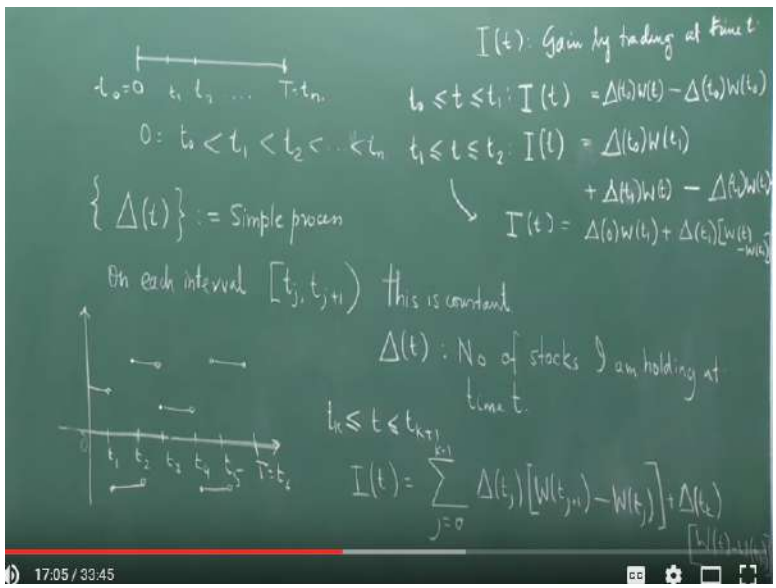

Let us in $\Delta(t)$ is the position I am taking in an asset at time t that is delta like a stock so $\Delta(t)$ is the number of stocks I am holding at time t of some company for example. So, number of stocks I am holding at time t. Now let us, now assume that $W(t)$ the Brownian motion, just like as Bachelier thought, describes the price process of the stock. But you know that, this is really not true because, price process of stocks never take negative values while Brownian motion can.

But for the moment, just for the heck of it, think that we are in Bachelier's days, we are not bothering of those things and we are talking about the fact that that $W(t)$ is the price. So, what I

have done so at time t, what is my gain in trading at time t. So, $I(t)$ is the gain is the process is actually a stochastic process, describing the gain by trading at time t.

Now consider the inter t, which is lying between t_1 and t_0 right. Now I have actually made, I have actually, bought say $\Delta(t_0)$ quantity of objects at time t_0, and for which I had paid $W(t_0)$ as the price. So, $W(t_0)$ is the price per unit of course. So, my total money I paid, so let me write I t so what is $I(t)$ in this case.

So, total money I paid, to get this $\Delta(t_0)$ amount of stocks at t_0 is $W(t_0)$ and total money now I take hold on to this stock at time t at a time k to sell that number of stocks and at a price $W(t)$, because price is changing at every time. So, by selling those stocks, I make this amount of money. So, what is my gain? Gain is nothing but the difference of this two. Amount of money I spent and then amount of money I actually bought by selling, so I will just subtract these two to know whether I am at profit or loss, gain could be negative also.

Now think that what would happen if my t, is between $t_1 \wedge t_2$ what would happen to this, what would happen to this scenario right**.** So, what would be $I(t)$ when I am in that particular scenario. In that case, of course t_0 is in this case 0, so we can write delta 0 also here, does not matter. So, when I am writing this t here, one has to be very careful. I am meaning that, I am ending the trading at this time.

So, what happens when I am telling that t is between t_1 and t_2 because which means I have held to the stock $W(t_0)$ up to the time t_1. After that, I will change it, so at time t_1, I sell my stock which I bought, this amount of stock at time 0 with the price $W(t_1)$.

Then I bought, a stock with this money that I have got, I have bought a stock $W(t_1), \Delta(t_1)$ at time t_1, paying price $W(t_1)$ then at t_2, at time capital T which is lying between t_1 and t_2 I sell this this $W(t_1)$ holding, a $\Delta(t_1)$ holding so $\Delta(t_1).W(t_1)$Wt. So, by selling I totally gained this

amount and this is the amount I have spent to actually buy the $\Delta(t_1)$ stocks. So, it will be this minus this. So, which I can also write nicely as $I(t)$ in this particular case $I(t) = \Delta(0).W(t_1) + \Delta(t_1).W(t_1) - W(t_1)$.

Again, if you are slightly not comfortable, what I did, I know that my maximum time I can hold the stock, I bought at time 0 is up to time t1, then I have to sell it. That is the period of holding. Then at time t_1, I know what is the stock price I may sell it with this price. Now with that price I buy, stocks $W(t_1)$stock of the amount $\Delta(t_1)$ with the price $W(t_1)$. So, this amount now has to be subtracted from this one, because I sold and got some money, from that I spent some money to buy some new stocks.

Again, I sold out stocks at time capital T and I finish my trading, time small t and I finish my trading. I got this amount of money. So, this minus this plus this is the amount of money I have now.

So, this is a very simple thing. Now this can be written in a more general way, that if t, so if you have a t lying between t_k and t_{k+1} then your I t, so this is what does it say. Say at every time point, before the time, till time t_{k-1}, what did he do? He bought something, he hold something in 1 interval, then sold that thing and bought the new stock by using the money from the same thing that he had gained.

When he sold something whatever money he had got he spent a part of it to buy something. So, he held at this stock, up to time $t_{j+1}$and sold this at this amount, but at time t_j, he had at, sorry at time t_j he had bought the same amount by spending the money $\Delta(t_j)$into $W(t_j)$. And then of course, we will look at the last part here this into this whole thing into $W(t) - W(t_k)$.

So, $I(t) = \sum_{j=0}^{k-1} \Delta(t_j)\left[W(t_{j+1}) - W(t_j)\right] + \Delta(t_k)\left[W(t) - W(t_k)\right]$ is the general form of writing $I(t)$. if you are not comfortable with, the cannot see it I write it again here for your convenience.

This is very simple profit and loss thing. I trade make money use that money to buy something, again I sell that and buy some new thing. So, that is the whole thing that is whole idea that is the way the market actually operates.

This $I(t)$, so this $I(t)$ is often written as $\int_0^t \Delta(t)dW(t)$. So, this whole thing is written in a shorthand like this. That is how you give meaning to this thing. So, this is what we want to write as $dW(t)$. $I(t)$ is a shorthand writing$I(t)$ is not exactly the differential. There is nothing like a differential of Brownian motion. It is a shorthand writing. Now $I(t)$ then itself is a stochastic process.

So, $I(t)$ is called the Ito integral of the simple process $\Delta(t)$. So, you are taking integral of a stochastic process that is very important and it is not like integrating a random variable. Integrating random variable is just like integrating any function right that is it.

So, here what you create is also a stochastic process. Your integral itself is not a number but your integral itself is also a stochastic process. $I(t)$ is a stochastic process. Son you see there is a huge leap or huge difference in the thought process. Son when you come to this stochastic wall your life completely changes.

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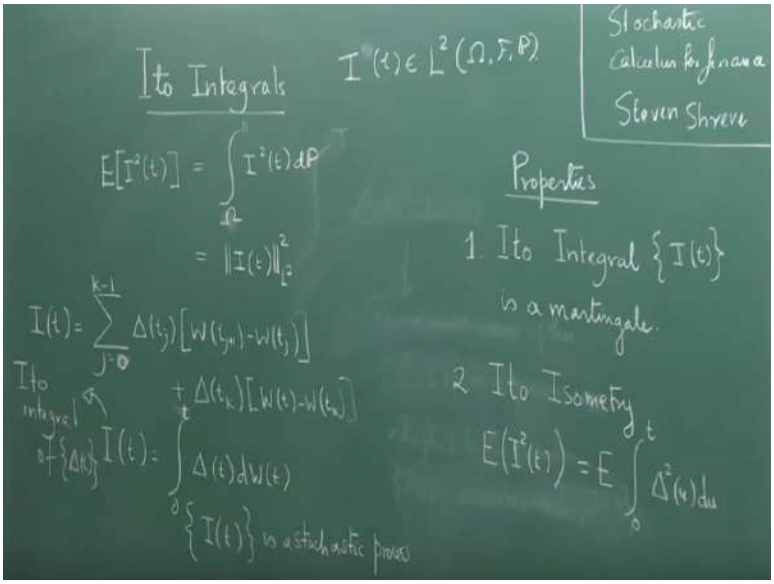

Now what are the key properties of this stochastic process. The key properties are following. The number 1 property is that, the Ito integral properties, if we have time we will prove any one of them, the number 1 property is that, Ito integral is a Martingale with respect to the filtration Ft adapted to the Brownian motion W. The Ito integral is a Martingale.

Ito integral, so when I am taking integral, do not think it is an integration, it is a number. It is not definite integral, Ito integral $I(t)$ means, Ito integral is a stochastic process. Ito integral $I(t)$, is a Martingale. Martingale are essentially processes which possibly are trying to maintain status quo at every time.

Number 2, proving this is not so simple. It needs little bit of work nothing else, it is just the computation is complex. Another thing is called Ito isometry. We will see, what is the meaning of Ito isometry. Ito isometry says the following. Expectation of I square t, so expectation of the random variable $I\ E(2(t)) = \int_0^t \Delta^2(u)du.$

Now why it is called isometry, would be clear, if you only know something about L2 norms. Say look at this quantity, $I\ E(2(t))$. So, if I write here as $I\ E(2(t))$ this simply means integral right,

whatever is 0 to t, I am sorry, integral over, sample space whatever the sample space is $I\ I\ \ 2(t)\ \ E(2(t)) = \int_{\Omega}$.

So, this because, we are assuming that this is finite. So, once you assume that this is finite you are assuming that $I^2(t)$actually belongs with a capital L2 of, L2 space of, this is only for those who know this stuff, this function analysis. Those who do not do not bother. Think that Ito divides this formula.

Isometry means they are similar distances. So, distance in the original space between 2 points and distance in the range space between the 2 functional values are similar, then we call it is isometry. So, this actually means if you look at it very carefully, it is, if you take I, no sorry no, not I, $I(t)$ is in L2 f not $I^2(t)$. So, $I(t)$ this is a square of the L2 norm. So, this is nothing but the $||I(t)||_{L^2}{}^2$.

So, this is I 2 is in infinite dimensional space in this when $I(t)$itself is so the Ito integral itself is an element of L2. So, it is in the infinite dimensional space is an infinite dimensional object. So, it is so interesting things are coming in and if you look at this.

So, they are telling that this is expectation of $\int_0^t \Delta^2(u)du$. So, you are telling that the distance, the norm of $||I(t)||_{L^2}{}^2$ is computed, is nothing but the expected value of this integral, because this integral, if this the expected value of this integral means what? This is a normal integral. $\int_0^t \Delta^2(u)du$ is a standard level integration right.

Here, I am not treating Δ^2 as the function of omega basically, keeping the u fixed, but I am treating it as a function of u itself. So, for every sample path you can compute this as t changes fix up the sample path, the scenario omega and you can compute this and then you take

expectation over all the sample path, average over all the sample path. That is exactly same as the L2 norm of $||I(t)||_{L^2}{}^2$. That is the idea.

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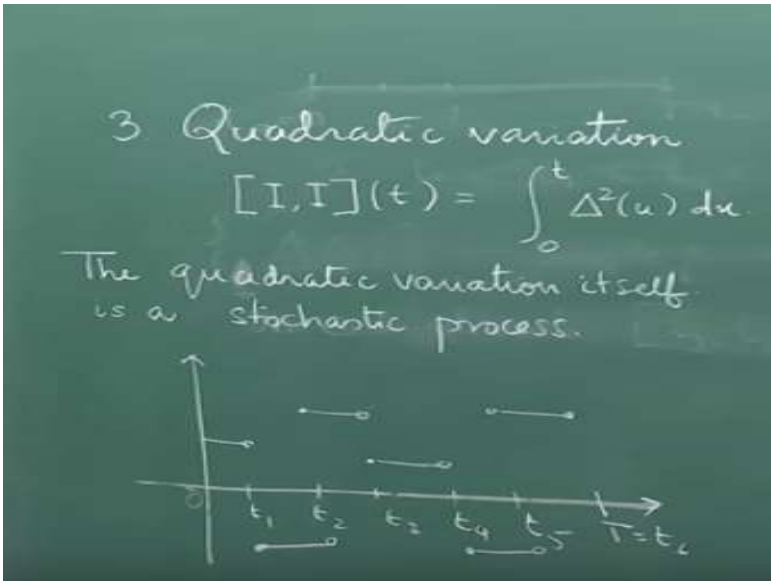

Now comes a very-very important property. So here, all these things are being done for, the Ito integral for the simple process delta t. We have not spoken about Ito integral for a general case and that we will come tomorrow, in the next rather in the next class. The very important thing, that we should now understand, that just like Brownian motion, the Ito integral itself has a quadratic variation and what is that quadratic variation.

So, Ito integral does not have a 0-quadratic variation. Ito integral itself has a quadratic variation. The Ito integral states, the quadratic variation of the Ito interval is being given by $[I, I](t) = \int_0^t \Delta^2(u)du$ this simple thing it is amazing the answer. So, the Ito isometry says the following, that the L2 norm of I the integral, is nothing but the expectation of the quadratic variation. See the quadratic variation itself, whether it is of a Brownian motion or whether of this integral itself is a stochastic process. That is the thing that you have to take into account.

The quadratic variation, but it is very interesting, the quadratic variation has a stochastic process, but for every t it gives you a number, every t it is a fixed number, if for a given sample path, for every omega, it will calculate a number very nicely. So, and for every omega whatever be your omega whatever be your path for a given t, therefore whatever be the omega, this will be the answer.

So, it is path independent essentially. So, the quadratic variation itself is a stochastic process. So, what we have now learnt, is completely for the case, where $\Delta(t)$ is that simple function, that is all. $\Delta(t)$ is this so, I know that delta t given the t $\Delta(t)$varies like this. So, given any omega if it is in the time interval, this $\Delta(t)$will have this values that is all. It does not depend on what omega and so delta t is the simplest simplest function.

Delta t sorry, I made a mistake. So, given an omega $\Delta(t)$ will take this this this this values in this interval. If you change the omega, it will take some other values in this fixed intervals. So, the intervals are fixed and in those intervals, it is taking constant value but it will depend on the omega. If you change the omega you will have different values and that omega will change the value of $I(t)$. So, that making $I(t)$ a stochastic process. Similarly, this quadratic variation itself is a stochastic process.

So, this is a very-very important understanding that we are essentially generating not numbers, we are at every step generating a stochastic process. So, this is something of very important thing that we have to keep in mind. So, when we are going to talk about general integral, this is some adapted process. $\Delta(t)$ is an adapted process, not a simple process like this, just not say function type thing then, we proceed as we do in integration theory.

We write the, we approximate the given function $\Delta(t)$, in as a, we approximate it by a family of simple processes, simple functions, and then you take the integral over those simple functions and then take the limit.

That is the idea, we are going to apply and then we will see for one particular calculation what will happen and there is a so, we had learnt in the last class that a quadratic variation

$[W, W](t) = t$ of course, we should ask what is the, what is when we take the limits of those Q_{π} what was what sort of limit it was.

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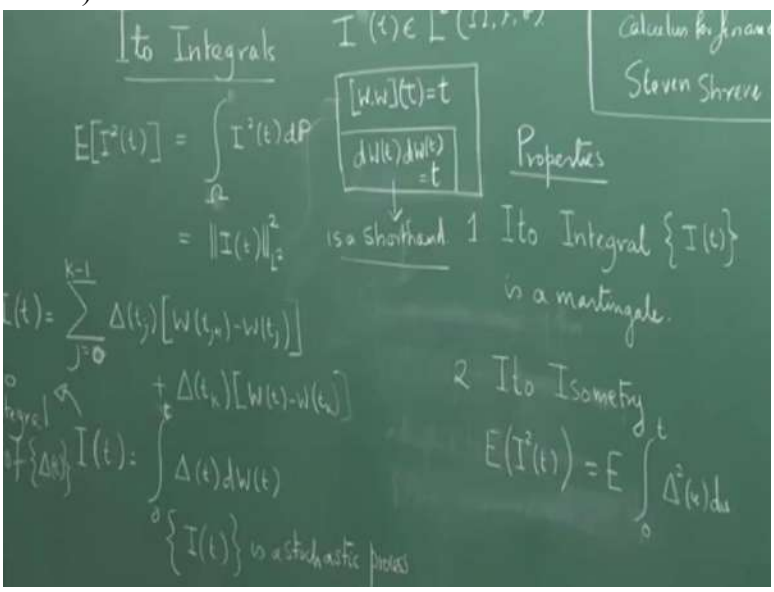

We proved that what the limit we get is a limit in or the convergence that we get that we can always have convergence in probability. But under slight, and also if we just bother about the fact the variance is 0 and the expectation is t then we have something called L2 convergence, which I am not going to bother you with. Or else by slightly modifying the definition, slightly modifying the requirements, of the behavior of the partition, we can show that it can be made to be convergent almost surely. So, you just have to remember this. But remember this is also a process stochastic process.

Sometimes, now because we have started using this $dW(t)dW(t) = t$ symbol, we will use this shorthand. $[W, W](t)$ actually means quadratic variation. This is this is the meaning of this. We will very, this is nothing but these two difference right the difference square basically. So, we will use this shorthand. Please note that this thing, whole thing, is a shorthand, shorthand to write quadratic variation. But this shorthand would be very helpful when you do finance. They are extremely useful to use this shorthand.

Thank you and tomorrow we are going to talk about, some more interesting facts doing, Ito integral for a general integral, and all these properties that you have learnt here, would be actually translated to that general integral and then we will do an example.

Probability and Stochastics for finance
Professor Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 13
Ito Integrals-II

So we are gradually getting deeper and deeper into finance and looking at the type of people who have really enrolled for the course getting some 1 or 2 mails from them I have decided to if you look at the course format I have decided to change it a little bit and take you slightly more into finance so that you get the finance flavor.

So here we are going to start getting more deeper into the tools for mathematical finance and then we will end our show with 2 last lectures which would be on the Black Scholes formula for pricing a European call option. Of course, then we will try to describe what is a call option and etc., etc. and what are the things, how to go ahead with the things. So we are now going to talk about a Ito integral for a general integral $\int_0^T \Delta(t)dW(t)$.

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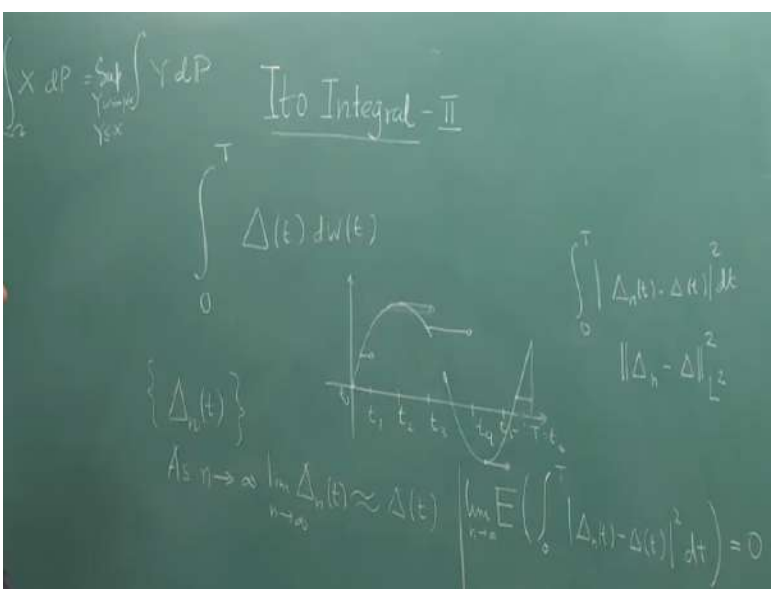

So this is no longer a simple process right. It is a it is essentially a general one. If you look back and go back to the sections on expectation, then you will see we have tried to define an integral

in the following way that if you want to define an integral of a random variable defined over a probability space $\int_{\Omega} XdP = \int YdP$ then this has been defined as a limit okay of these integrals.

Limit Y is simple and $Y \leq X$. So it is a limit but sorry it is essentially not the limit but supremo because you are taking $Y \leq X$. so the integral of Y is less than integral of x and then you are taking the supremo but it is a limit in some sense you can also put it in a limit form. So this is what you have learnt. So the same idea should come here.

So as you keep on changing the partition of the interval 0 to t for each partition you can define 1 particular simple process and each as you make the partition smaller and smaller the simple process will actually go closer and closer to the suppose this is your actual process up to T. Now you have divided into say this $t_0 = 0, t_1, t_2, t_3, t_4, t_5$ ...and so on and this is t_6.

So what you do is you define 1 value here maybe then you define another value here in this way sorry from t_2 maybe like this, from t_3 might be like this t_3 to t_4 something like this, from t_4 it might be just like this and from t_5 it might be just like this. So this is very crude approximations very bad approximations but as you make the thing smaller and smaller and smaller these partitions your approximations would start becoming better and better and better.

So what do I mean by that that a simple process delta n t so I will have a sequence of simple process $\{\Delta_n(t)\}$ each process is simple. So if I take this sequence of simple processes as $n \to \infty$ we expect that this$\Delta_n(t) = \Delta(t)$ simple process simple process should somehow be same. Of course you have to define what is the meaning of this sameness, same as this continuously varying process $\Delta(t)$.

So these are discrete processes whose limiting form is this continuous process $\Delta(t)$. Continuous not a continuous process this continuous process here but okay continuous more continuous than the things that we are talking about the parts are continuous as they are continuous parts. So there

is nothing like every step is discrete. So it may not be just a constant function on the interval, it could be just varying.

So we expect that this is somehow what we will get but what is the meaning of this sameness how do we talk about this sameness. In probability theory, the sameness is spoken in a slightly different way. It tells you I expect that the distance between these can be made smaller and smaller and smaller. So it tells me, now each of these so what we expect is the following. $\Delta_n(t) - \Delta(t) \vee \quad ^2 dt \quad \int_0^T E$ I will tell you what it means.

So Shreve tells us that if by these functions simple functions coming to $\Delta(t)$it means that it what it means that over all the sample paths if you average the square of the distance. So this is nothing but the L2 norm. If you look at $\Delta_n(t) - \Delta(t) \vee \quad ^2 dt \quad \int_0^T$ integral. So in the terms of the function this is nothing but $\left|\Delta_n(t) - \Delta(t)\right|^2$ and those who know about the L2 spaces the space of all square integrable measurable square integrable functions.

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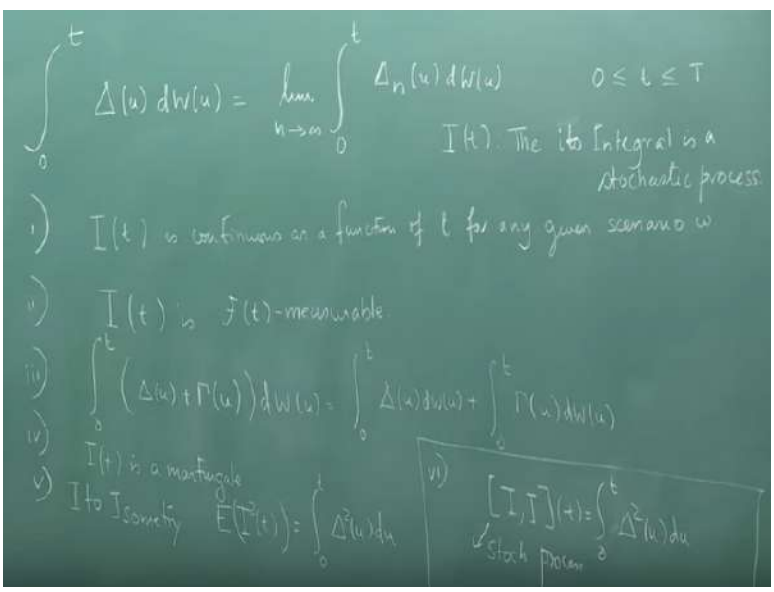

So once you have and you expect so let us forget this thing and let us not bother because people might be not so comfortable with it. Let us just look into this. Shreve tells us that when I am

expecting the fact that I expect that when n grows this simple processes must somehow manage to imitate or will be almost same as the given process.

By this statement we are actually meaning this statement, we are meaning that if we take over all the paths we take the distance between these 2 by the square of the distance then the expected value of the distance between these two points should be 0 because remember in a random setting we cannot be telling that it is exactly equal to 0, this will be exactly 0. We have to talk in terms of expectation because we do not know the outcome. So we have to only speak in terms of expectation.

So the expected distance so this is this distance so as n grows the expected distance should be going to 0 because the expected distance itself is a random variable so the sorry the distance itself is a random variable, so the expected distance between these two has to go to 0 that is the idea. So by this we actually mean this.

So once we mean this $\int_0^t \Delta_n(u)dW(u) = \int_0^t \Delta_n(u)dW(u)$ the definition of a standard integral standard Ito integral from any point so would be limit. So it can happen when you take any t between 0 to T this will happen. This is the way I define. Now why such a limit would exist. Remember that this is a stochastic process. I am taking a limit of another stochastic process. So why should such a limit exist right. For every given t why should such a limit exist.

That can be answered by using the Ito isometric theorem and showing that this actually forms a Cauchy sequence but we are not going to get into that sort of argument at all though I assume that you must be understanding this what Cauchy sequence and those who do not just forget it for the time being. Just assume that okay I can calculate it in some procedure.

Now what sort of properties would such an integral satisfy. You will be amused that all the properties that we learnt about integrating the simple process would be applicable here. For example that $I(t)$ is continuous as a function of t for a given sample path t for a given scenario omega t for any given scenario omega. Number 2. $I(t)$ is $F(t)$ measurable that an $I(t)$ is a

stochastic process adapted to the filtration F t which is the filtration associated with the Brownian motion.

From now on whatever filtration we are talking about the filtration would be associated with the Brownian motion. So $I(t)$ is $F(t)$ measurable. The third property is linearity. So if we take any 2 processes gamma and u, sorry delta and gamma it can be proved that $\int_0^t (\Delta(u) + \Gamma(u))dW(u) = \int_0^t \Delta(u)dW(u) + \int_0^t \Gamma(u)dW(u)$is same as we are not going to prove these things just to announce you that these are the important properties. Another fourth most important property is that $I(t)$ is a Martingale.

Then of course we should talk about the Ito isometry property. So Ito isometry you know what is Ito isometry, it is exactly the same. You basically copy what you know for $I\,E(2(t)) = \int_0^t \Delta^2(u)du.$ to this part. The Ito isometry says a second movement of this and the last property which is one of the most seminal property so I separate it, the quadratic variation of the Ito integral $I(t)$.

Remember Ito integral $I(t)$ itself $I(t)$ the Ito integral again I this is the point I want to state again and again the Ito integral the Ito integral is a stochastic process. So the major property that I want to state is that a quadratic variation$[I, I](t) = \int_0^t \Delta^2(u)du$ accumulated up to time t is t. That is independent of the path quite a Brownian motion type behaviour sorry not t ,I am making a mistake here.

This is not Brownian motion type behaviour sorry it is the same as same as the isometry. So here you see so it gives you the standard. This is the normal integration in terms of the variable u. So this is a stochastic process. For every omega, you take you will get something. So as omega changes this will change sorry I made a mistake in my exit, it is path dependent actually.

So this is itself this quadratic variation itself is a stochastic process just writing in short and this is path dependent of course not path independent I made a mistake sorry. So these are the important properties that you learn about the Ito integral. So these are the properties that you have to keep in mind. Of course, because of the time we cannot be proving each and every step. Now what we will do we will show you a simple example of how to compute the Ito integral and we will make a difference between computing the stochastic integral and the standard form of computing this integral.

We will just see. So the next part remaining part of the class would actually consist of computing this integral. Please understand quadratic variation is a very very fundamental thing here. So our goal would be to compute.

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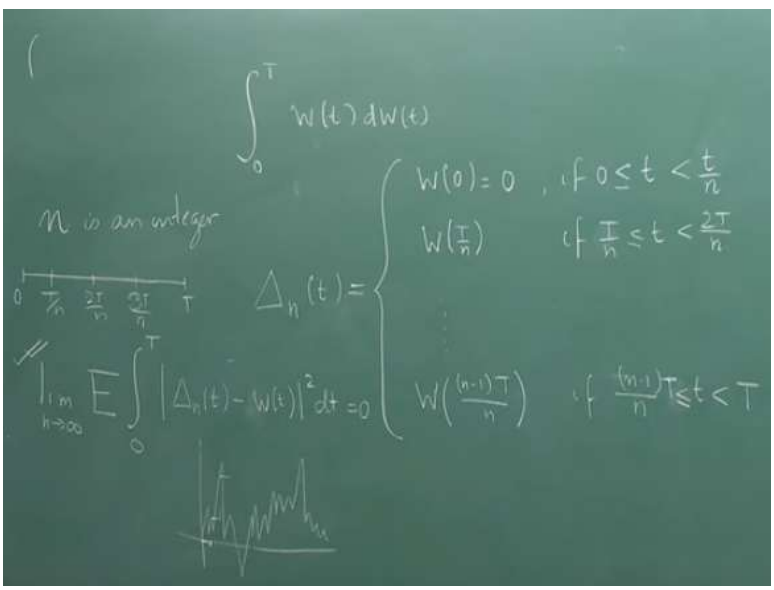

So $\Delta(t)$ we have replaced by $W(t)$. So what would be this? That is basically $\int_0^T x.\,dx$ that will be $\frac{x^2}{2}$ that will be $\frac{T^2}{2}$would be the answer if I just take it as a standard calculus integral but you will soon see that $\frac{T^2}{2}$ is not the answer right. We are just putting x here and dx will be $\frac{x^2}{2}$ would be the answer. $\frac{x^2}{2}$, 0 to t so $\frac{t^2}{2}$.

But here you will see is the quadratic variation which will get you an additional term with a term of that type and that makes stochastic integrals very different from the standard integrals of calculus. So here we do it start doing it step by step. So we will construct for this particular we will make a construction of a sequence of simple processes. Now we will choose n to be a very large integer so n is an integer so we will keep on increasing n, n is an integer.

So take an n and have for every step so you divide every interval length is of the length $\frac{T}{n}, 0, \frac{T}{n}, \frac{2T}{n}, \ldots\ldots \frac{.3T}{n}$ that is all that is all. Now as n becomes larger and larger these intervals become smaller and smaller. So we will now define that for every interval how would we define the $\Delta_n(t)$ because when interval these intervals become smaller and smaller and the length of this interval that this will be larger these lengths would be smaller and smaller and smaller.

There will be more and more intervals of those given lengths. So we will put $W(0)$, it will be same = 0, so it will be 0 if $0 \leq t \leq \frac{t}{n}$, $W\left(\frac{T}{n}\right)$ it is from here $W\left(\frac{T}{n}\right)$so W let me just put this is $W\left(\frac{(n-1)T}{n}\right)$. So, you are just taking these values itself. So that is that is what it is, sorry that is T here. So that is how I have constructed it. Now the next question would be to know whether such a construction can prove this thing sorry this will be W t that would become 0.

So what is the expected value of $E\int_0^t \left|\Delta_n(t) - W(t)\right|^2 dt$? That is an important question. So you can understand if you look at the picture very well I would advise you to draw the picture yourself then that is the way you can learn. If you draw the picture is a Brownian motion and here you have kept say 0 and then you have taken the t_1 value is here you have kept it here till say t_2 and t_2 value say is here started like this.

So in that way you will see that as n will become larger and larger these straight lines will become smaller and smaller and smaller become smaller and smaller and smaller and so you will have many of these straight lines as n becomes large and large there will be infinitely many

straight lines which would very hardly vary from the actual value that you will see. So this distance would any way become smaller and smaller.

So if you take their difference for a given sample path and integrate over them then basically you are calculating of the distance between the areas under the curves for a given sample path square of the distance basically not really area but square of the distance then this would turn out to be 0. Of course you can make a very rigorous calculation right of actually putting in the values breaking up 0 to t into these intervals and then actually looking into the difference $W(0) - W(t)$.

When t is lying between this so the square of the difference you can write them down and then you can separately come and do the job so doing this will take a little bit of time. It looks slightly obvious that you can easily do this but this thing actually getting this thing done of course one might think that how I should be able to change with the expectation inside the integral.

Yeah, you can do under certain circumstances you can switch expectations and integrals. So if you can switch the expectation integral then the things would be much more simpler then you would know that okay their difference would be 0 their mean would be 0 etc., etc. and sorry the variance would be not the mean that will be the variance so variance would be basically $\frac{t}{n}$ and so and so forth.

So you would add those quantities up right so you would add those quantities up and then you integrate with n becoming larger and larger so the area would actually keep on falling. So that is 1 way of thinking about it but let us not think about it at this moment. Let us assume that we can show this that this will actually work. So this is a very good approximation which it is if you look at it pictorially which it is.

You see in actual finance you really have do not have to bother too much about these sorts of approximation nobody is going to ask you to do such calculations but okay if we find time we will really put up these sorts of things up in the website.

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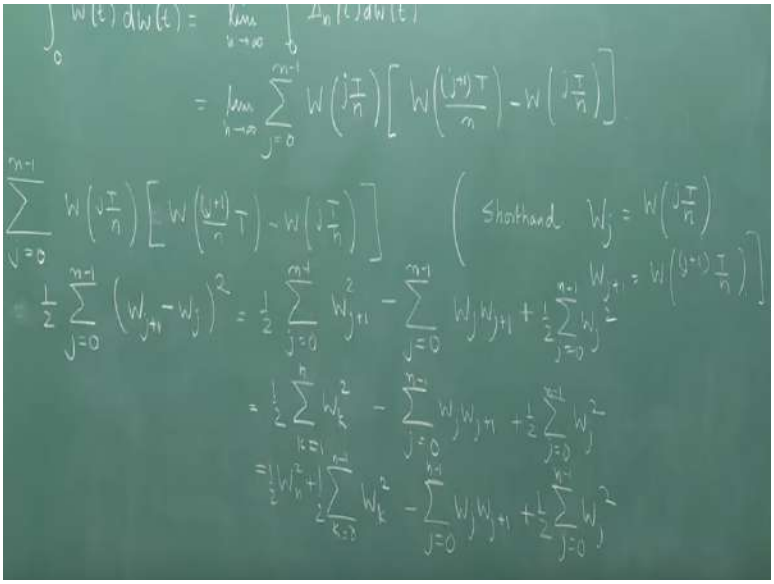

So now what we have is the following $\int_0^T W(t)dW(t) = \int_0^T \Delta_n(t)dW(t)$ is my definition. So how do I define this delta you already know. So, it depends on so it is from j = n so it is just using the values there. At every step you have to calculate $W\left(j\frac{T}{n}\right)$. and j it is 1 it is j is 0 it is 0, j it is 1 it is $W\left(\frac{T}{n}\right)$, j is basically you have to do this because these are the function points where these are the function values. You can easily understand these things.

So what have I done. So when j is 0 it is 0 done finished. When j is 1 it is $W\left(\frac{T}{n}\right)$. $\left(W\left(\frac{2T}{n}\right) - W\left(\frac{T}{n}\right)\right)$- $W\left(\frac{T}{n}\right)$of course because over that particular interval it will take the value j T by n. Over this interval it will take the value j t j W this will take the value $W\left(\frac{T}{n}\right)$. So that is exactly the interval that we are considering. So and so forth. So this is the sum.

So essentially now we have to compute this sum and then take the limit. So our job now is to compute this $\sum_{j=0}^{n-1} W\left(\frac{T}{n}\right)\left[W\left(\frac{2T}{n}\right) - W\left(\frac{T}{n}\right)\right]$ sum. Now because there is so much of clumsy things we will consider some shorthands. We will write W j shorthand let us do shorthand some

shorthand notations. In the shorthand notations let us put W j to stand for $W\left(\frac{jT}{n}\right)$so which means W j+1 would stand for $W\left(\frac{(j+1)T}{n}\right)$.

So with this$W\left(j + 1 - W_j\right)^2 \frac{1}{2}\sum_{j=0}^{n-1}$ shorthand notation we will start the computation. Now let me do the calculation. We have to check up this but let me start by doing the following calculation. You will see why I am doing this. I will just go ahead and just tell you a bit more about this. Suddenly I remembered it is a good way to talk about it. You see when n is becoming large see the first part is very simple because when you put any t here it does not matter.

Here W it will be 0 and when n is becoming very very large because W_T is continuous and when n is becoming very very large this thing becomes very small this interval. So there is hardly a difference between t and delta and t these values, very small difference and so this square values would actually come down further and that is why their values would basically come down and the expectation would go towards 0.

So I want to calculate this. You will see why we are calculating this. This would allow us to get a fairly good this will allow us to calculate this. So it is just an opening of the brackets nothing else. Once I do open the brackets because $W_0$is 0 then so I just very important note that W_0 is always 0. So here because I am putting when I am putting j = 0, I get 1 and I am putting j = n - 1 I am getting n.

So I can as well as write these expression as this particular expression as $W_k^2 - \sum_{j=0}^{n-1} W_j . W_{j+1} + \frac{1}{2}\sum_{j=0}^{n-1} W_j^2 \sum_{k=1}^{n}$. So now I will break this up into 2 parts. I will break this up into W half W n square plus summation. Now I will use the fact that W 0 is 0. So what I will have? If I break it up I will have summation $\sum_{k=1}^{n} W_k^2$ but noting the fact that W 0 is 0, I can add W_0^2 to it and write here this break it as $\sum_{k=0}^{n-1} W_k^2$. It does not matter because W k is anyway 0.

So here is what we have got. But now we will go and transfer our calculation to this part. So we will do our calculation in this part. So I will have $W\left(j+1-W_j\right)^2 \frac{1}{2}\sum_{j=0}^{n-1}$ and again I can begin up from this part. So what I will do I will add sorry there is a half here. I will add this half and this half because these j's and k's are dummy indices.

So I will have $\frac{1}{2}W_n^2$ plus j = 0 so half plus half is added so it is 1. $\frac{1}{2}\sum_{j=0}^{n-1} W_j^2 - \sum_{j=0}^{n-1} W_j W_j + 1$.

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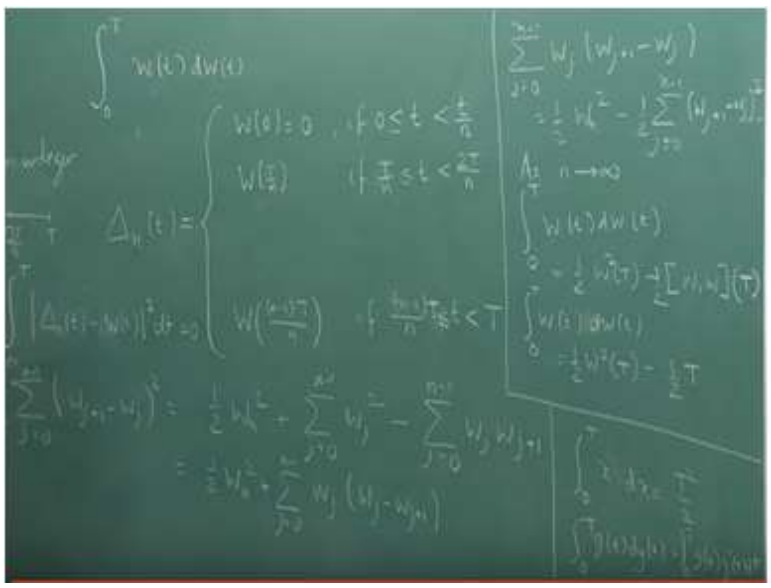

So basically I am getting $\frac{1}{2}W_n^2 + \sum_{j=0}^{n-1} W_j.\left(W_j - W_{j+1}\right)$. You see I have almost got this expression but with a minus sign. So what I will do is the following. But what is this? This is a quadratic variation term. So I will take this part to this side and bring this thing to this side. So my next job is the following.

So I will now write $\sum_{j=0}^{n-1} W_j.\left(W_{j+1} - W_j\right) = \frac{1}{2}W_n^2 - \sum_{j=0}^{n-1}\left(W_{j+1} - W_j\right)^2$. This is actually the form of the quadratic variation of the Brownian motion you see so quadratic variation has come.

So basically, now if I take the limit on both sides what is this? W n square W n square. So what will I get? W n t. So if I put n here W j W n is W t. So here it will be nothing but $\frac{1}{2}W_t^2$. So if I take as n tends to infinity this limit is nothing but the integral. So I will get $\int_0^T W(t)dW(t) = \frac{1}{2}W_t^2$ - this limit, assuming that all the limits that I am taking here I am assuming that okay there are in almost surely sense. You can even take them in the sense of limits in the sense of probability convergence in P that will also do.

So I am writing so this equality is a convergence in probability. Essentially this is because we can actually show the quadratic variation goes in or can be shown in almost surely sense. So these are actually in almost this equality is always in almost surely sense because this is a stochastic process random variable at time t so this equality holds only everywhere except a set of measured 0.

So this is very very important to understand when a set of a null event if you throw away null event it holds everywhere. So what do you get? So you get 0 sorry this is half here. So what you get 0 to t W t d t sorry $\int_0^T W(t)dW(t) = \frac{1}{2}W_t^2 - \frac{1}{2}t$. You see this $\frac{1}{2}t$ is an additional term because if I take a standard integral $\int_0^T x.dx$0 to t x square dx sorry x dx. This is nothing but t square by 2.

This is nothing but $\frac{1}{2}t^2$. Or if you take any integral as $\int_0^T g(t)dg(t)$. $dg(t)$is nothing but $g'(t)$, $\int_0^T g(t)g'(t)$ because this is the differential of t d t and this will simply give you $\frac{1}{2}g^2(t)$. So you see this is something very different and that is the crux of stochastic calculus that stochastic calculus is a quadratic variation actually makes the difference the quadratic variation.

See essentially, we were trying to compute the quadratic variation so these are some sort of a trick. The quadratic variation cannot be forgotten when Brownian motion is in the game. So we stop here and we will talk about Ito's calculus in the next we will have a Taylor theorem type things that you learn in standard calculus. What is the meaning of Taylor's theorem in stochastic calculus? So we will learn about those things.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 14
Ito Calculus-1

Today we are going to learn something about Ito calculus, that is how do you write, a Taylor expansion for Ito's integral. Now you see quadratic variation will always add some extra term.

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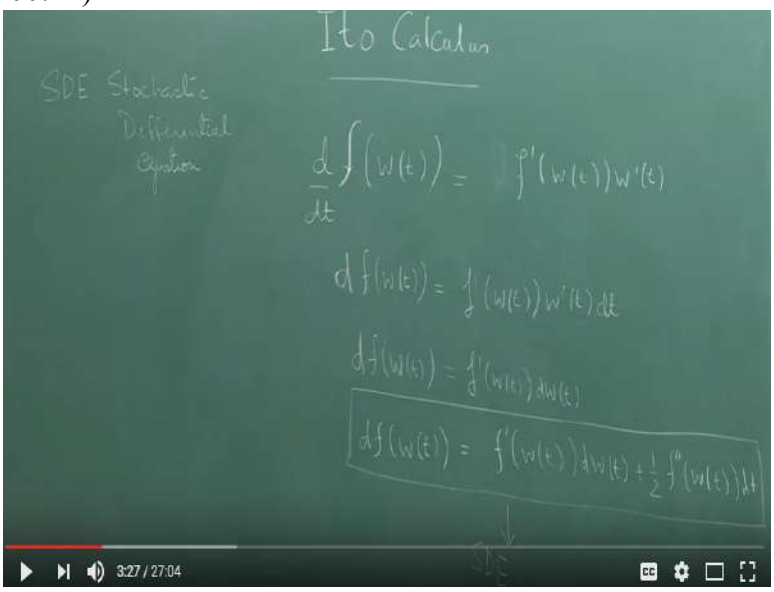

For example, for the moment, just think, that you have function $f(W(t))$ and assume that $W(t)$ is differentiable, which you know that it is not, but just for the heck of it, so just to gain some insight. So, if you take a derivative $\frac{d}{dt}f(W(t))$ it is exactly equal to, if you use the chain rule then it will be $f^{'}(W(t))W^{'}(t)$okay.

Now you know that, this is not really true, you cannot have W dash t. So, you can write $df(W(t)) = f^{'}(W(t))W^{'}(t)dt$ if you go by the standard way of doing differential calculus, this actually symbolizes the differential of t and so you will write. Though we know that we have been able to give some meaning to $dW(t)$, but we really have not, this is not really the job.

But as we will show that, if we can give meaning to this, because we have given some meaning to this $dW(t)$,, in actual case this expression does not end here, but a term is added because of the quadratic variation. So, in the Ito sense, this actually means, assuming that f is nice enough to be twice continuously differentiable, this is what you will get. Instead of getting this, which you get, if you are thinking in terms of standard calculus and $df(W(t)) = f^{'}(W(t))dW(t) + \frac{1}{2}f^{''}(W(t))dt$ is what you get as a stochastic calculus.

This equation is often called a stochastic differential equation or SDE. So, I will write down what is the full form of SDE. It is called the stochastic differential equation. Now I would just want you to recollect a few facts.

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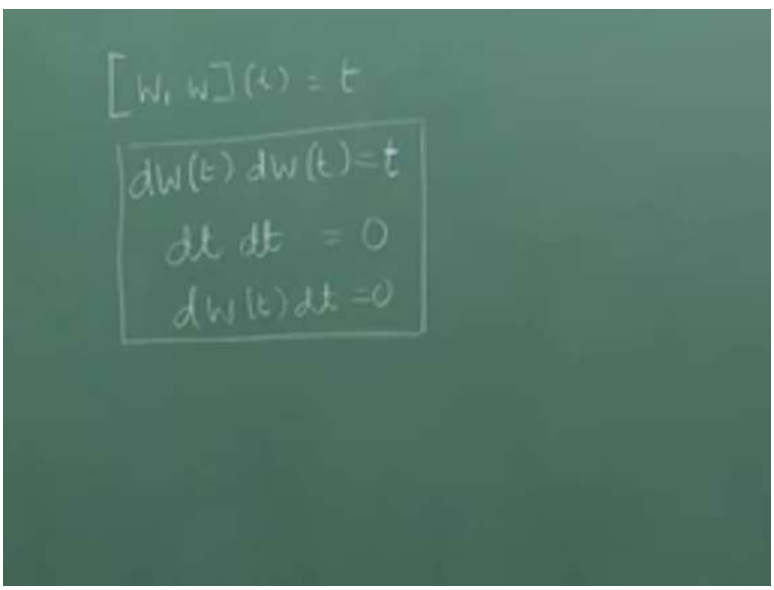

One of them you know that, the quadratic variation of Brownian motion is path independent. A shorthand short form of writing this is$[W, W](t) = t$, so this is the actual process, but we are writing it in a short manner this $dW(t).dW(t) = t$ is a shorthand. Do not think that do not immediately try to attach meaning to this. So, this is just a way of doing the math. It is not.

Similarly, if you take $dt.dt$ quadratic variation of t it been a normal function you know that will be 0 and then if you take a cross variation between $W(t)$ and dt that is $dW(t).dt = 0$. So, this is a basic shorthand that you should remember when you are going to study finance or doing do derivative pricing. This is a very basic shorthand, very basic shorthand.

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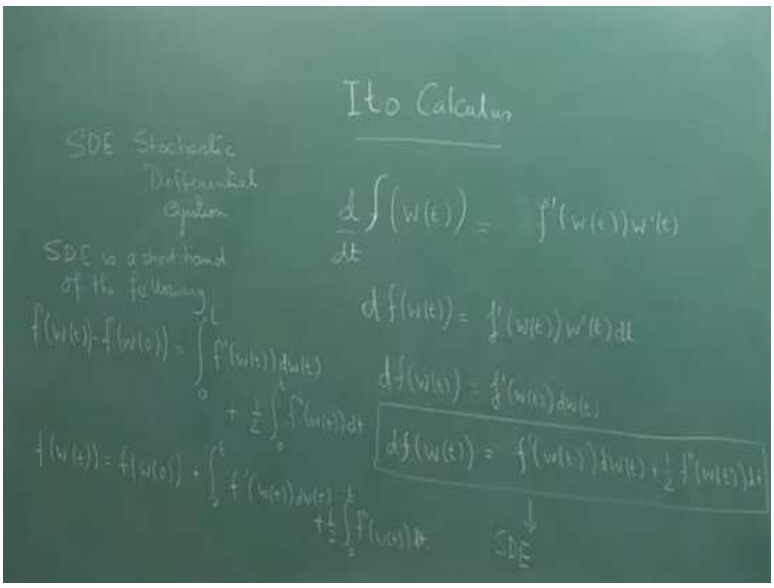

Now this SDE, what you see here, is actually a shorthand. This SDE is a shorthand of the following. Suppose, I integrate from 0 to capital T, I can integrate it out, if you think in terms of standard integration, then it is a shorthand of the following, or 0 to small t, so this one this SDE is a shorthand of $f(W(t)) - f(W(0))$ which is essentially giving you this, is equal to $\int_0^t f^{'}(W(t))dW(t)$ which an Ito integral for which we have clearly shown how to think about it, plus this additional normal integral. So, this becomes an Ito's expansion, Ito's calculus.

So basically, you can rewrite it as, $f(W(t)) = f(W(0)) + \int_0^t f^{'}(W(t))dW(t) + \frac{1}{2}\int_0^t f^{''}(W(t))dt$,Now a, Ito's formula which is also called the Ito Doeblin formula in Shreve's book, can be done at a slightly general level, where you take a function of two variables.

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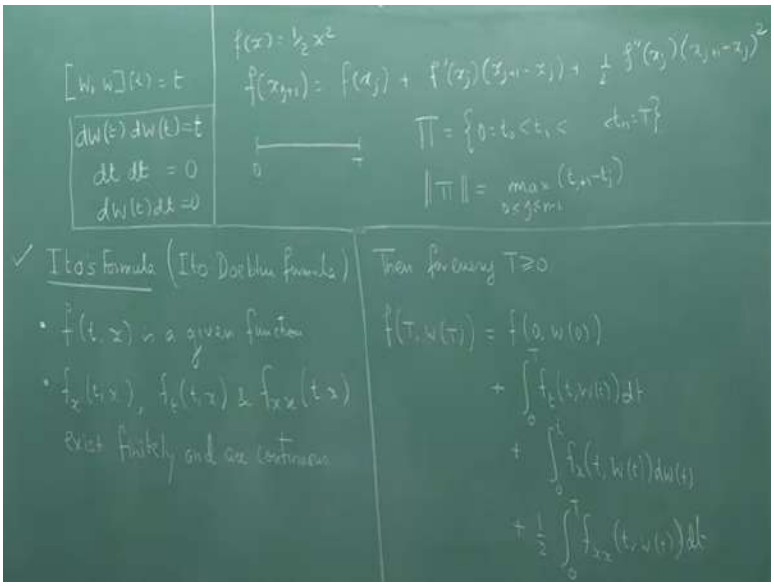

So, I will now write down Ito's formula for two variables. So, Shreve also calls it Ito Doeblin formula in the name of another young researcher, who possibly died in the world war, in the World War II, but he also had almost got the same, came to the same idea. So, whichever you want to call it. I think I found this Ito Doeblin thing only in Shreve's book, everywhere it is mean called Ito's formula. So, we will just go by the traditional name Ito's formula. So, let us see what does Ito's formula tell us.

Ito's formula tells us the following. It considers not a function just of t just of $W(t)$, f of some function of a process, but t and $W(t)$ where t and a process. So$f(t, x)$is a given function. X is in r, t is in r right, is a given function okay. Then to $f_x(t, x)$ that is the partial derivative with respect to x $f_t(t, x)$, that is partial derivative with respect to t, and partial derivative with respect to, everything else, $f_{xx}(t, x)$partial derivative with respect to xx, second order partial derivative. These exists and are continuous. Exists means, exists finitely.

Now, once these things are done, what formula we have. Once these things are given, we get the following formula. So, Ito says then, for every $T \geq 0$, we have the formula $f(T, W(T)) = f(0, W(0)) + \int_0^t f_t(t, W(t))dt$.

See what happens, only when I take the partial derivative with respect to x I will consider Ito integral because the random process is at the x in the place of the x variable, plus $\int_0^t f_x(t, W(t))dW(t)$. Then there has to be the correction term, to be a quadratic variation okay. So, that will be $\frac{1}{2}\int_0^t f_{xx}(t, W(t))dt$.So, it will not this will not be an Ito integral. This will be an Ito integral. So finally, you are talking about a stochastic process.

Now, what we are going to do, right now is to, instead of trying to prove this whole thing in a very general setup, we are going to look into the very simple case of f in half x square. We will forget the fact that the t variable is also included.

We will not take the t variable. We will just leave the t variable and try to deduce a formula like this when I have something f is half xx square. So, our job now would be to consider a function $f(x) = \frac{1}{2}x^2$ and show that this actually holds.

So, that will give you a fairly good idea about how to proceed. So, let us fix t greater than 0 and then do a partition and proceed. But before that you realize, that for a quadratic case, you have an exact second order Taylor expansion. So, if you have $x f''(x_j)(j + 1 - x_j)$ $f(x_{j+1}) = f(x_j) + f'(x_j)(x_{j+1} - x_j) + \frac{1}{2}$. The higher order terms will all vanish. They will be 0 higher order derivatives. That will be the Taylor's expansion. So, can we use this idea now to proceed, that is the idea.

So again, you will have we will start with the interval 0 to t and I will consider a partition, given partition, where 0 equal to t0 less than t1 less than, strictly less than, tn equal to T. And as before we will consider the norm of the partition, is the maximum value of the length of the intervals.

Now, it is nth interval, you could have large right, I am just writing like this. I am not giving n possibly should have given n. Okay I will just take the partition norm $\vee\ \pi\ \vee$ goes to 0. So, this partition norm is max $t_{j+1} - t_j$ as j goes from 0 to n minus 1.

So, this is the basic preparation that you take and then once you take this preparation, you start working on it, trying to prove this formula.

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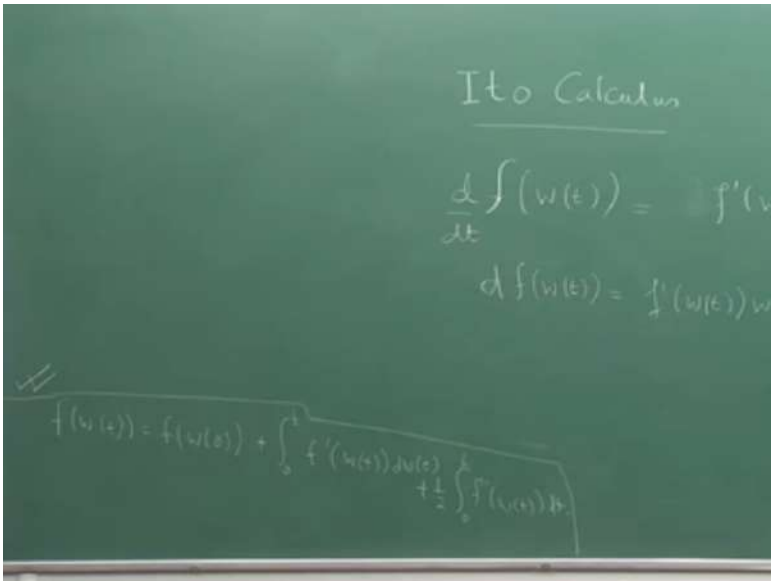

So, let us keep this formula on the side and try to see how do we arrive at it. But then so let me just remove everything and then start to prove this. So, once I prove this formula you just accept this formula as correct because it will be too complex to go and work it out. It is better to get a feel of it because this is many people for this is the first course in stochastic calculus rather than going into every detailed rigorous step.

Math can be sometimes understood with too much of rigor built in for the first time and then you increase your levels.

So now we take the partition and let me look at this. This is nothing but the collapsing sum W0 is 0 anyway. The collapsing sum, so I have forgotten the t, I am not taken t, f is half x square the $f(W(T)) - f(W(0)) = \sum_{j=0}^{n-1}\left[f\left(W\left(t_{j+1}\right)\right) - f\left(W\left(t_j\right)\right)\right]$. This is what you have. Now if I think that this is just this expression $f\left(x_{j+1}\right) - f\left(x_j\right)$ is a quadratic expression half x square. So, $f(W(t)) = \frac{1}{2}W(t)^2$.

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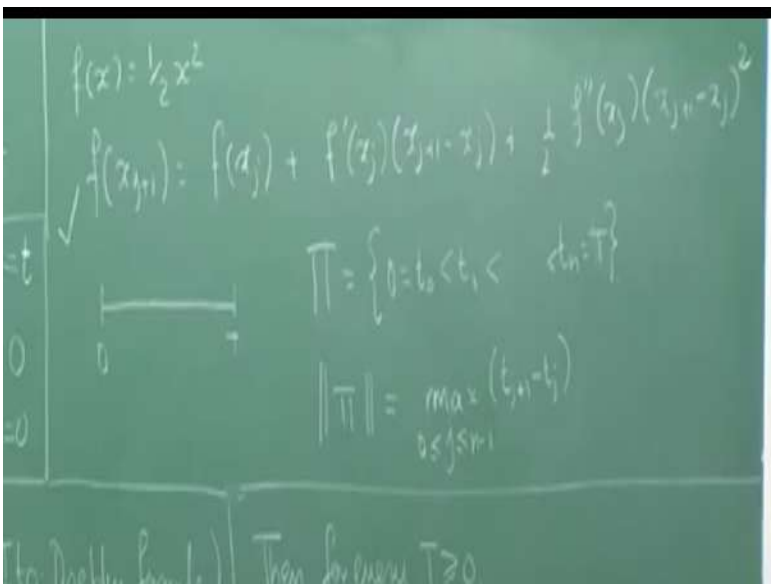

So, I am just using the Taylor's expansion at this point nothing else, the Taylor's expansion which you have known here. So, you know because of the higher derivatives going to 0 this is the exact Taylor's form Taylor's expansion.

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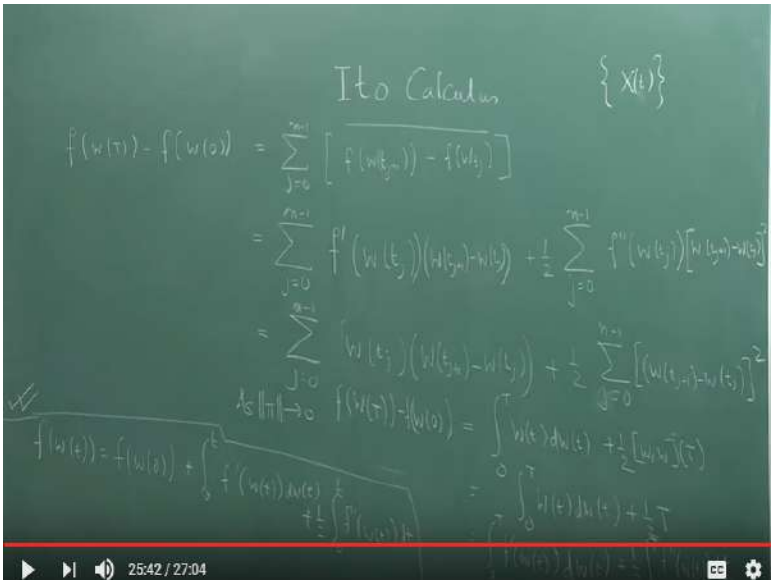

Thus, I can write just for each of them, I am writing so, I am finally writing the sum also. So, f is a continuous function. Now if I go by $f^{'}(x)$, which is x, so this is nothing but $W\left(t_j\right)$ sorry I made a mistake because this $W\left(t_j\right)$is the argument of the function f now. So, that is playing the role of xj plus 1, $W\left(t_{j+1}\right)$plus 1, please correct it.

Now what is $f^{'}(x)$? $f^{'}(x)$ is x. So, $f^{'}W\left(t_j\right)$is $W\left(t_j\right)$where $W\left(t_j\right)$is a real number so does not matter. This $W\left(t_j\right)\left(W\left(t_{j+1}\right) - W\left(t_j\right)\right)$and when you take the second derivative it is 1 so it is half of summation sorry j is equal to 0 to n minus 1, $f^{'}W\left(t_j\right)$ okay. Sorry $f^{'}W\left(t_j\right)$ is just 1, $f^{''}W\left(t_j\right)$ because that is the expression of th that is the value of the second derivative.

It is 1, so 1 into W you can immediately realize, what has happened see, if I go and take the limit as j as the ∨ π ∨ goes to 0, this would represent an Ito integral of W and this would represent the quadratic variation right and so what does this represents so as ∨ π ∨ goes to 0, f of Wt because this is not depending on that partition this is independent of it this is the two end points. This will

be nothing but $\int_0^t W(t)dW(t)$ the Ito integral, plus half of the quadratic variation. So, it will be W W of t. So basically, it will be $\int_0^t W(t)dW(t) + \frac{1}{2}t$ okay.

Now how do I semblance with this? What is $W(t)$? $W(t)$ is nothing x, which is $f^{'}(W(t))$. So, it is $\int_0^t f'(W(t))dW(t)$. This is one thing and then you have to realize that, what I can write? Half of 1 into t. What is 1? It is $W(t)\, f''$. So, I can write this as, half of 0 to capital T, $W(t)dt\, f''$, sorry dt small t.

So, you see finally I have got this formula, by using the $f(x)$. So, if I choose a particular f I can get a particular formula. So, I chose f equ al to something and then you see it matches this formula,$f(x) = x^2$ and it matches this formula. So, under this $f(x) = x^2$we have proved that I can write it like this. Or else in general when you take a function $f(t, x)$you write you have this formula.

You can ask what about $f(x, t), f, f(t, x)$. So because you see when you do $f(x, t)$and $f(t, x)$ you are doing this cross variation. That will become 0 when you take the limit when you actually write for this particular case which will be pretty complex, which I plan not really to get along with.

Secondly is that, if you consider the case, where you want to, consider the case f_{tt}but when you have f_{tt} that will be corresponding to dt gt, tt when I quadratic variation of this function t and that will be equal to 0 so that is why those terms really do not appear in this Ito's formula.

So, here the idea is not to give you something very, you know some very tricky thing, but really to give you some, very basic idea what Ito's calculus is like and see these formulas would be very-very useful and please keep in mind, this set of shorthand rules. A very important idea is to

understand that this $dW(t)$, what I can write from here I can take it to this side and write it as $dW(t) = f^{'}W(t)dW(t) + \frac{1}{2}f''W(t)dt$.

So, that thing is just a shorthand of this. You do not really have stochastic differentials. What you have is stochastic integrals. It is of fundamental importance to remember, that there is nothing like a stochastic differential. There is nothing called a differential of a Brownian motion because Brownian motion is not a differentiable function, so you cannot define the differential of a Brownian motion. So, since like $W(t)$ into $dW(t)$. So, you cannot define that.

What you can define is a stochastic integral and that is the key. We are essentially talking about stochastic integral. So, Ito's formula what you see in this Taylor's expansion they are not written really in terms of derivatives but in terms of integrals and that is the key idea that you really have to remember, if you want to make a progress.

In the next class, what we are going to talk about? We are essentially going to talk about, a very general process. So, we are going to talk about a stochastic process, a very general process. So, we are going to talk about a stochastic process, a very general stochastic process called an Ito process. So here we end this class by describing you what a Ito process is.

So, Ito process is a process, which consists of the constant x so there is a random variable x and $x_t$xt a stochastic process x_t and then this x_t sorry, $x(t)$ write $x(t)$so this xt can be decomposed into $x(0)$ plus a Ito integral plus a normal integral.

So any Ito process is Ito process is a stochastic process. So any stochastic process is called an Ito process. It can be decomposed into three different parts. One is $x(0)$, the 0th value of the process plus an Ito integral plus a ordinary integral. If we can do that then essentially, we are talking about something called Ito process, and what should be the Ito's formula for an Ito process. What should be the quadratic variation of an Ito process. All these things will come in the next class which you will see very soon.
Thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 15
Ito Calculus-II

So this is Ito calculus 2 the last version of Ito calculus that is needed for finance and that ends your third week's lecture. So we have 1 more week to go, 5 more lectures to go okay and then first let us just go back and let us see how Ito's formula can help you to compute the Ito integral. So we have computed the Ito integral if you remember in our last to last class we have actually computed $\int_0^T W(t)dW(t) = \frac{1}{2}W^2(T) - \frac{1}{2}T$ integral and we have written this as half.

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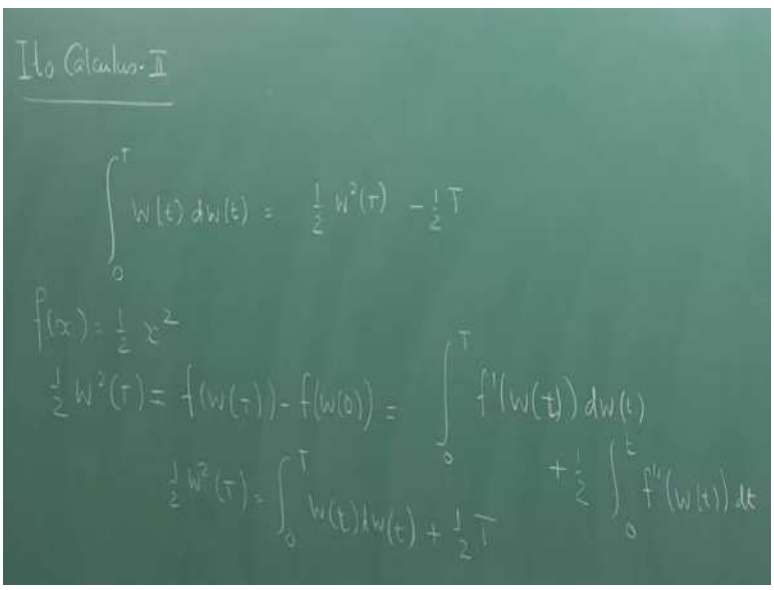

But you know this can come out just right out of the Ito's formula because if you take $f(x) = \frac{1}{2}x^2$ then $\frac{1}{2}W^2(T) = f(W(T)) - f(W(0))$ which is 0 and $f(0)$ is 0 so when a here x square is and this by our formula should be $\int_0^T f'(W(t))dW(t) + \frac{1}{2}\int_0^t f''(W(t))d(t)$. So what is happening here. So we will just we know that this is nothing but W T and we know this is 1.

So we will have $\frac{1}{2}W^2(T) = \int_0^T W(t)dW(t) + \frac{1}{2}T$ .So you immediately know that if I take bring it this side I get this formula. So it comes out of Ito's formula that we have studied. So now we are going to define something called a Ito process. We have just mentioned in the last class what an Ito process would look like.

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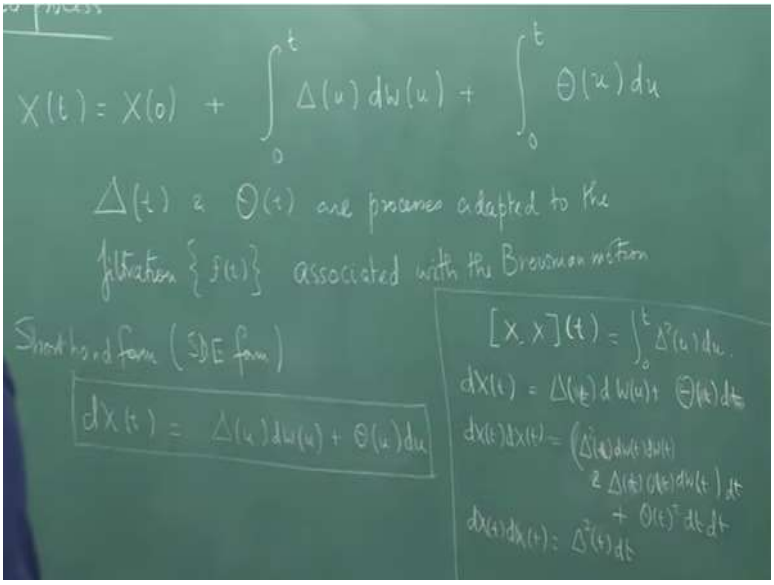

So an Ito process would like the following $X(t) = X(0) + \int_0^t \Delta(u)dW(u) + \int_0^t \theta(u)du$.So an Ito process $X(t)$ a stochastic process consists of $X(0)$ which is the 0th value of this which is non random plus an Ito integral so you have a stochastic integral and you have an ordinary integral. So this will be non random a number. This will be 0 I mean if you have W. This will be a number plus a random variable so basically it represents a stochastic process so $X(t)$ is a stochastic process.

Where $\theta(t)$ and sorry $\Delta(t)$ and $\theta(t)$ are processes of stochastic we are not writing the term stochastic process every time just write process are processes adapted to the $\{f(t)\}$ associated with the Brownian motion $W(t)$. They are adapted to the filtration $\{f(t)\}$ associated with the Brownian motion okay. Now you know you can write this in a shorthand form. The shorthand form is let me write shorthand form or the stochastic differential equation form.

SDE form a shorthand form or the SDE form would be of this type. You can write this as $dX(t) = \Delta(u)dW(u) + \theta(u)du$ Most of the models that you will use in finance would be represented through these sorts of shorthands. But please understand that this is nothing but a shorthand to this because we essentially talk about Ito integrals. We do not talk about differentials of Brownian motion okay.

One of the key formula that one now needs to talk about what is the quadratic variation of this process? So in this Ito world or the stochastic world quadratic variation is the major player. So, you really have to talk about quadratic variation. What is the quadratic variation of this process? That is something that we should know. So we will take 2 approaches to know about this quadratic process.

First, I will give you the easy approach. Then I will give you the actual approach. So let me give you the easy approach first. So we claimed this so we write $[X, X](t)$ and let us see what we do. Let us put the magic. The magic is we write the short form $dX(t) = \Delta(t)dW(t) + \theta(t)dt$. Then you do $dX(t)$ and you use these things that we have written earlier the last class. So what do you have if you do this?

You have $dX(t)dX(t) = \Delta^2(t)dW(t)dW(t) + 2\Delta(t)\theta(t)dW(t)dt + \theta^2(t)dtdt$ just you are doing squaring $(a + b)^2$ and then you are writing this as $dX(t)dX(t) = \Delta^2(t)dt$. So this $dW(t)$,$dW(t)$this this this and you know$dW(t)dt$ is 0 cross variation is 0.

Second variation of an ordinary function is 0 and but a second variation of quadratic function is t. So you have $dX(t)dX(t) = \Delta^2(t)dt$ and this is nothing but 0 to t so this is immediately so this is nothing but the short form of $\int_0^t \Delta^2(u)du$. Okay I could possibly go by the more deeper thing.

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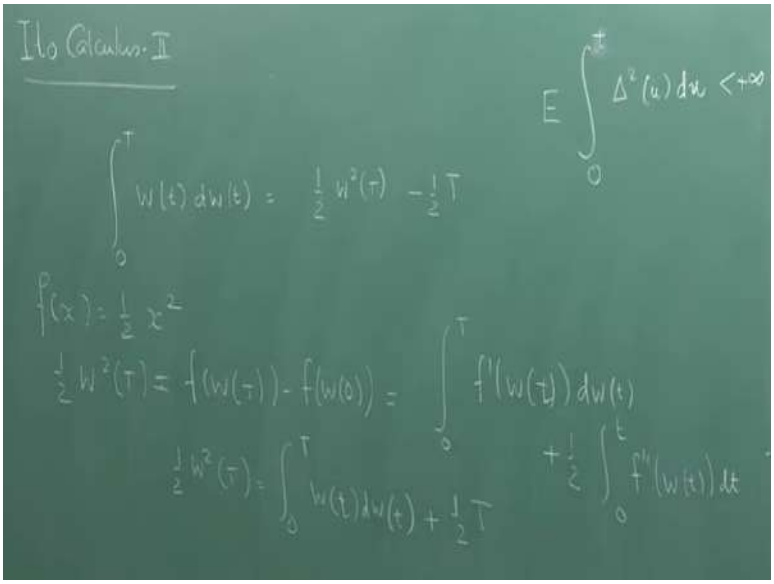

Always I know that when you are talking about Ito integral of a general function $E\int_0^t \Delta^2(u)du$ you should always assume sorry $E\int_0^t \Delta^2(u)du$ is always little $+\ \infty$ because the expected value of the you know of this integral of the basically L2 norm of this integral this is a pro cess so it will depend on t so this will also become a process. So the expected value of that random variable depending on t has to be strictly less than infinity.

That is the requirement because that requirement will actually help us write that the difference between the simple process and the continuous process when a delta n minus delta t whole square 0 to the expectation that limit would go to 0. So that sort of thing is there. So we do not get into all these things. We will assume that all these requirements are taken care when we are talking about this Ito business.

Suppose we are now going to write down formally this process. This is the shorthand process. You might say this is some sort of a backhand stuff. So maybe you will enjoy doing this backhand stuff lot more later on but we will do this backhand stuff later on. We will not get into writing too many things but if I do the whole thing then what I should do. Let me write down $I(t) = \int_0^t \Delta(u)dW(u)$ the Ito integral as and $R(t) = \int_0^t \theta(u)du$.

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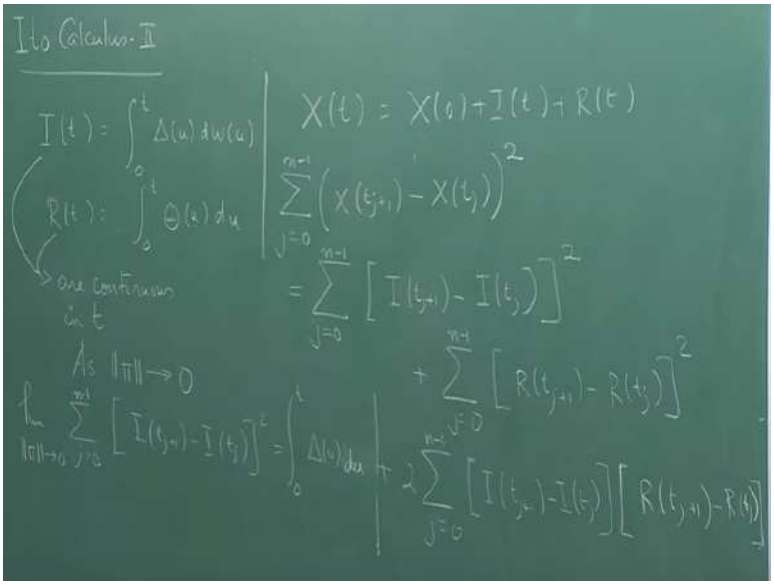

So if this processes are continuous in terms of u then from simple calculus from basic calculus you know both of these not these this is already separately known to be continuous but this you know would be continuous. So $I(t)$ and $R(t)$ are continuous functions in t. So these both are continuous and then you can write the Ito process $X(t) = X(0) + I(t) + R(t)$. So if you are going to write about the quadratic variation of this thing then you are going to actually write I will write in more detail possibly.

You are essentially going to compute the following. The sample quadratic differences or which I should I do not know I think this is called just give me a second I will just tell you the name I tend to forget this name. It is called the sample quadratic variation depending on the path. So this I will write as $\sum_{j=0}^{n-1}\left(X\left(t_{j+1}\right) - X\left(t_j\right)\right)^2$ where the partition is as before. I am not writing the partition every time because that is boring to write down the same thing over and over and over again so now you know what does this mean.

This means $X(0) +$ I t,$I\left(t_{j+1}\right) + R\left(t_{j+1}\right)$. So X0 gets cancelled. So essentially when you expand it a whole square you essentially get $\sum_{j=0}^{n-1}\left(I\left(t_{j+1}\right) - I\left(t_j\right)\right)^2 + \sum_{j=0}^{n-1}\left(R\left(t_{j+1}\right) - R\left(t_j\right)\right)^2 + 2\sum_{j=0}^{n-1}\left[I\left(t_{j+1}\right) - I\left(t_j\right)\right]\left[R\left(t_{j+1}\right) - R\left(t_j\right)\right]$and

so now this is just this one. Now if I want to do something here what would I get? That is the whole point. See this if I as and this will be the quadratic variation of the Ito integral and that would give you this formula. So this thing so $\sum_{j=0}^{n-1}\left(I\left(t_{j+1}\right)-I\left(t_{j}\right)\right)^{2}$ actually gives you.

The so the Ito quadratic variation of the Ito process is nothing but the quadratic variation associated with the Ito integral. That is that is the key idea. So sorry $\int_{0}^{t}\Delta(u)du$. So that that is exactly so this is nothing but the quadratic variation. So what would be these parts. These parts would actually be 0. That is something you really have to just I will prove for this part and then you can prove it similarly for this part by looking at the continuity of the processes I t and R t.

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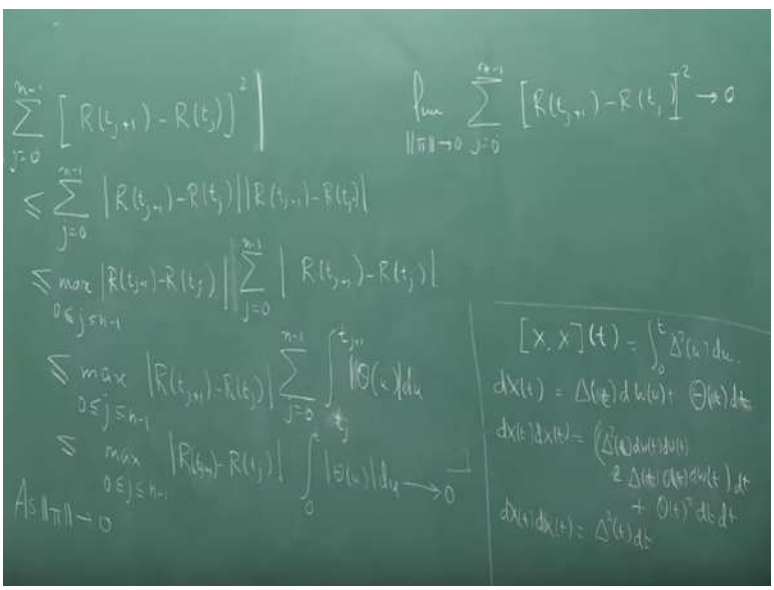

So let me just prove for this part. So what you can do that this value rather than the absolute value of this term $|\sum_{j=0}^{n-1}\left(R\left(t_{j+1}\right)-R\left(t_{j}\right)\right)^{2}$ ∨ I think there is no need of talking about absolute value because this itself okay does not matter. So I can write that this is $R\left(t_{j+1}\right)-R\left(t_{j}\right)$ ∨ $R\left(t_{j+1}\right)-R\left(t_{j}\right)$ ∨ $\leq \sum_{j=0}^{n-1}$ because this is a continuous function so this difference would have a maximum value of what the possible j s.

There will be finite number of such differences and hence I can write this to be $R\left(t_{j+1}\right) - R\left(t_j\right) \vee \sum_{j=0}^{n-1} R\left(t_{j+1}\right) - R\left(t_j\right) \vee \quad \leq \max_{0 \leq j \leq n-1}$. Summation this actually means I will write the whole thing as this one the summation n-1 sum this sum goes out and you have sum of $R\left(t_{j+1}\right) - R\left(t_j\right)$ So then I can write this term as 0 to t j to j+1, t_j to t_{j+1}, theta d d t or $\theta(u)du$. So which I can write again as because mod of the integral is less than the integral of the mod.

Using that idea a very simple fact about integration theory I can write this as and let me look at this part so I will write this as $\max_{0 \leq j \leq n-1} R\left(t_{j+1}\right) - R\left(t_j\right) \vee \int_0^t \theta(u)du \sum_{j=0}^{n-1} \int_{t_j}^{t_{j+1}} \theta(u) \vee du$. So that is less than equal to max of and we have assumed that let us assume that this is always finite. Then as $\vee \ \pi \ \vee \rightarrow 0$ sorry here I am making a mistake as the norm of the partitions goes to 0 so the distance between $t_j$and t_{j+1} vanishes comes down very significantly.

This mod goes to 0 but this is the finite number so this will be finally as goes to 0. This quantity whole quantity will go to 0 which will immediately mean that the norm of this whole square goes to 0 which will mean that the limit of this as $\vee \ \pi \ \vee \rightarrow 0$is 0. So finally we will get that $\sum_{j=0}^{n-1} \left(R\left(t_{j+1}\right) - R\left(t_j\right)\right)^2 \rightarrow 0$

When you take the third term out you take the I t plus $I\left(t_{j+1}\right) - I\left(t_j\right)$ out and do the same thing. How do you integrate if suppose you are given an adapted process, process is adapted to a filtration. How will you integrate it with respect to a Ito process means we are asking?

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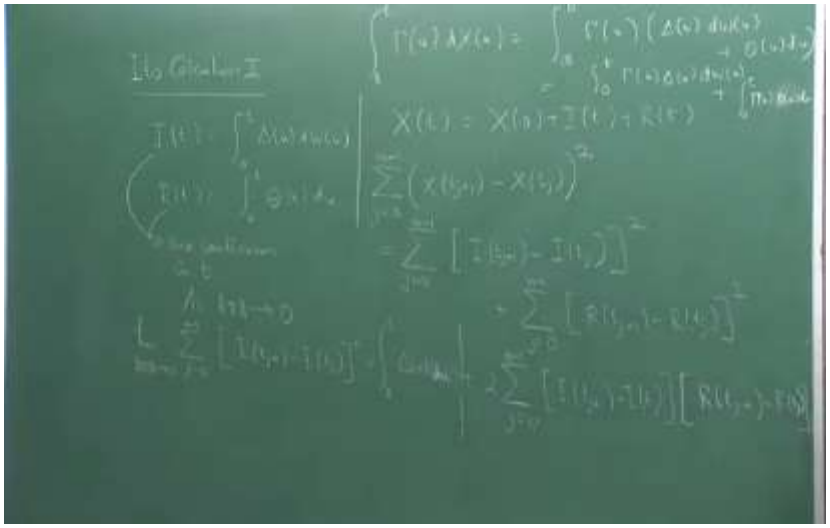

Suppose I give you a process what is the meaning of $\int_0^t \Gamma(u)dX(u)$? This simply means instead of $dX(u)$say 0 to t so instead of what you do is $\int_0^t \Gamma(u)(\Delta(u)dW(u) + \theta(u)du)$. That is it.

So if you want to take a process adapted to the filtration of the stochastic process W t there is a Brownian motion then the integration with respect to an Ito process is nothing but this $\int_0^t \Gamma(u)\Delta(u)dW(u) + \int_0^t \Gamma(u)\theta(u)du.$

Now of course the question comes what if I then start talking about the Ito formula for an Ito process. Let us see, what I can do. What is the Ito formula for an Ito process. That is what we so I am just going to write the formula not write the details. So Ito formula for an Ito process. With this we will end our discussion on the Ito calculus and then show you applications in the next coming classes and then end up with the Black Scholes formula. That which many people must be wanting to see.

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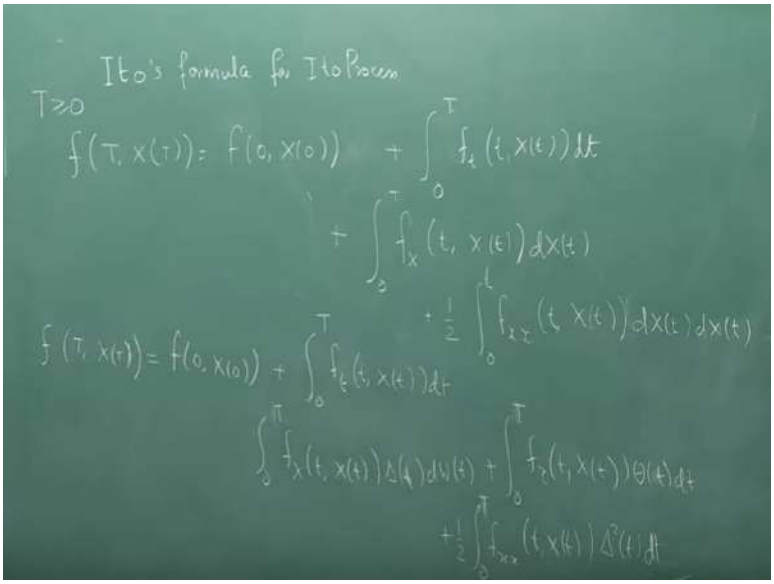

Ito process Ito's formula for the Ito process. So assume you take the f with the same sort of assumptions like the last class f_x and $f_t$are continuous and $f_{xx}$are all continuous functions the partial derivative and then Ito's formula for Ito process. Once we know this see this is just a repetition of what we have done.

The key formula is that you should know the whole thing is that the the key factor is that you just should remember $dX(t) = \Delta(t)dW(t) + \theta(t)dt$ and also remember that most models in finance are Ito processes Ito's formula for Ito process. So in the Ito's formula for Ito process it is just same thing. So for any $t \geq 0$ if t is greater than equal to 0 then same thing. Now instead of $W(t)$ it is $X(t)$here observe that.

In the same formula $f(T, X(T)) = f(0, X(0)) + \int_0^T f_t(t, X(t))dt + \int_0^T f_x(t, X(t))dX(t) + \frac{1}{2}\int_0^t f_{xx}(t, X(t))d[X, X](t)$ .I am just replacing. Instead of $dW(t)$, I will now have $dX(t)$. Now I had dtin the other case. My $dW(t)$the quadratic variation of $dW(t)$, $dW(t)$, $dW(t)$which became dtbut here I will have $d[X, X](t)$which is nothing but $dX(t)$, $dX(t)$okay. So basically I can write this instead of writing like this I can write this as$dX(t)\ dX(t)$

Now you know you can still simplify the formula by knowing how to putting the stuff of the Ito process. Do not worry, examples will now flow in as we will show that so many things in finance are actually Ito processes. You can you can model as Ito process sorry no d t here that is a mistake plus.

So here I will break it up into 2 parts in the same way. So I will have $f_x(t, X(t))\Delta(t)dW(t) + \int_0^T f_x(t, X(t))\theta(t)dt + \frac{1}{2}\int_0^t f_{xx}(t, X(t))\Delta^2(t)dt.\ f(T, X(T)) = f(0, X(0)) + \int_0^T f_t(t, X($

You see so this is the Ito's formula for the Ito process and with this we have covered a huge ground of stochastic calculus. Of course we have not been completely rigorous because if you need to be completely rigorous then you need lot of lectures to do it step by step and be very very rigorous at every step.

So we have not gone into rigour but just had some fun but at the end getting the formula that you would require to actually study finance. So with this I will end today's lecture. Thank you very much and we will start our next lecture very soon which is the last week of the course.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 16
Ito Integrals in Higher Dimension

So we are in the last week of our course. So we will today start about applications of the Ito's calculus. So you have learnt about Ito's calculus now you should learn about applications right? How can I do something with that Ito's calculus. One of the most important issues in finance is to express most financial processes as Ito processes. So one wants to express financial processes, all financial processes are obviously stochastic processes, processes as Ito processes.

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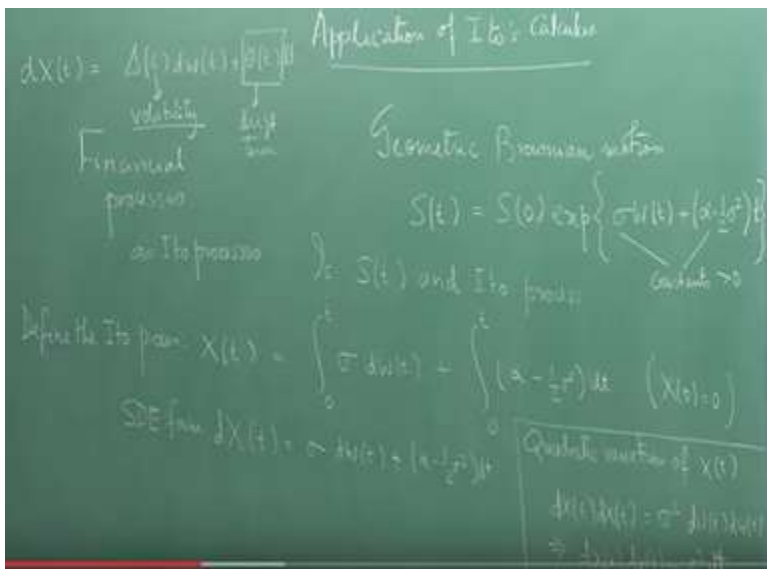

So Ito processes means you have to write down it everything in for more stochastic differential equation that is we have seen that the Ito process can be written down in this SDE shorthand as say what sort of symbol that we gave $dX(t) = \Delta(t)dW(t) + \theta(t)dt$. So θ term the coefficient of the dt term in the stochastic differential equation language is called the drift term essentially term similar to that in finance and this is called the volatility term which measures the unpredictability of the whole thing.

So the first question that one arises that can I model the movement of the prices of a stock as a Ito process. I have to model that. If I can model that then I can bring in the power of Ito calculus

to draw conclusions in finance. So in general you have to understand just having Brownian motion would not do. You cannot expect to model Brownian make Brownian model as the model for your stock prices though they may look similarly the jiggling type.

But the issue here is that you have to know that the stock price always has to remain non negative. So that brings us to the notion of geometric Brownian motion. So geometric Brownian motion is described as follows. Basically the stock price $S(t)$ at time t is the stock price $S(0)$at time 0 which is non random which is fixed which you know because you are at time 0.

$S(t) = S(0)exp\left\{\sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^2\right)t\right\}$E to the power of course the Brownian motion captures the Brownian motion captures the randomness. The question is this process $S(t)$ satisfying an Ito process. Is $S(t)$ an Ito process? So we will ask this question to ourselves. Is $S(t)$ and Ito process? How do we go about doing anything about it right. So let us define an Ito process of this form. X t so define an Ito process. So define the Ito process.

So we are constructing a Ito process. Our aim would be to show that this is the Ito process but let us construct the Ito process. Suppose $X(t)$ which is which we write as $X(t) = \int_0^t \sigma dW(t) + \int_0^t \left(\alpha - \frac{1}{2}\sigma^2\right)dt$, sigma these are constant this σ, α these are all positive constant okay. This is something we need to keep in mind. So this σ and α are positive constants. These are constants sigma and alpha. Sigma is called the volatility of course. So it is $\int_0^t \sigma dW(t) + \int_0^t \left(\alpha - \frac{1}{2}\sigma^2\right)dt$. This is what I have.

Let me define this Ito process. So if I write down this Ito process in the SDE form then we get d of $X(t)$ so here I have not taken $X(0)$, I have taken $X(0) = 0$ right. So basically, I am taking you must ask me what about the term $X(0)$ right because in Ito process there is a time term $X(0)$. So here we are basically assuming that $X(0) = 0$. So then I can write this as $dX(t) = \sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^2\right)dt$This is the SDE form.

Now if you look at the quadratic variation of this, so the quadratic variation of the process X the quadratic variation note that this process $S(t)$is equal to $S(0)$like this is called log Brownian motion or the geometric Brownian motion. You see because of the exponential term this of course we are guarantying that a price can never be negative.

So if you want to write the shorthand form then keeping in mind the formulas that we have given $dW(t)dW(t) = dt, dW(t)dt = 0, dtdt = 0$you can then write $dX(t)dX(t) = \sigma^2 dW(t)dW(t) \Longrightarrow dX(t)dX(t) = \sigma^2 dt$So basically this Ito process has the same same sort of it is also path independent quadratic variation. So $X(t).X(t) = \sigma^2 t$.

So this is the quadratic variation and if I look at it how my how do I look at my price right asset price. Now look at an asset price $S(t)$.

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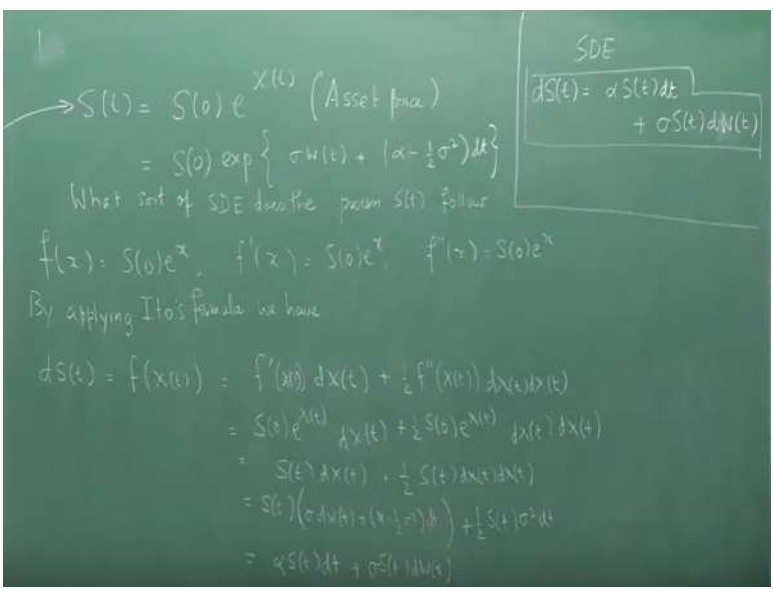

Suppose I define the asset price let the asset price we have written it in this form. Actually if you look at it I will define the asset price as now if I write like this we will come to the asset price. If I write like this what do I get? Let me write it down more carefully. I will tell you what do I get.

Let me define the asset price as $S(0).e^{X(t)}$ okay. So $S(0)$ see you might wonder why I am writing why I am not why I am writing in the integral form because this is nothing but if you look at it this $X(t) = \sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^2\right)t$.

Actually $X(t)$ is this process. Because if this sigma is constant of course you can start working with sigma and alpha and S where sigma and alpha not constant but since they are constant I can write it like this. So I can write this thing as $S(t) = S(0)exp\left\{\sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^2\right)t\right\}$.

So though I wrote a Ito process first just like the way I should write a Ito process but noting that these functions are constant I can now write from here this fact. See $\int_0^t dW(t)$ integral of 1 if you just go and use the, of course do not think that $\int_0^t dW(t)$is $W(t)$just by because this is a differential no. You have to apply the Ito's formula to come to this conclusion okay.
So that is the way you have written it. Now let us see what sort of stochastic differential equation this will follow $S(t)$will follow. So $S(t) = S(0).e^{X(t)}$ is the definition of an asset price. So I want to I know that there is a volatility term associated with the Brownian motion and some drift term. So we will talk about it when we explain things in finance in our next course in much more detail but assume that this is called the drift term.

And now my question is what sort of SDE does the what sort of SDE does the process $S(t)$ follow. Now to do this we will have to apply Ito's calculus. Now let us take $f(x) = S(0).e^{X}$. So we do not have the t here X only t is not separately taken or$f(X(t))$. So $f'(x) = S(0).e^{X}$ and so is $f^{''(x)} = S(0).e^{X}$by applying Ito's formula and if I apply Ito's formula what will have what will have we have the following.

Let us see what we have. We have $dS(t)$is nothing but $f(x)$is$S(0).e^{X}\ f(X(t))$ so it is $f(X(t))$t which by Ito's formula is $f^{'}(X(t))dX(t) + \frac{1}{2}f''(X(t))dX(t)dX(t)$. So what is this? This is $S(0).e^{X(t)}.dX(t) + \frac{1}{2}S(0).e^{X(t)}dX(t)dX(t)$.

So let us observe the next step. This $S(0).e^{X(t)}$thing is your $S(t)$. So $S(t)dX(t) + \frac{1}{2}S(t)dX(t)dX(t)$ So $dX(t).dX(t) = \sigma^{2}dt$. So we will write again $S(t)$. Now we will write down this whole thing. So I am writing now $dX(t)$. So $dX(t)$the SDE form of $X(t)$ is $\sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^{2}\right)dt$. Now here $dX(t).dX(t) = \sigma^{2}dt$ that we have already got. So we know that I can now replace this by $\sigma^{2}dt$.

So $S(t).\sigma^{2}dt\ S(t).\sigma^{2}dt$ cancels so to give you $\alpha S(t)dt\ +\ \sigma S(t)dW(t)$. So the price process of a asset so if you look at this this is exactly the geometric Brownian motion. So the geometric Brownian motion is following the price process of the stochastic differential equation just like so it is a Ito process basically. $\alpha S(t)dt$ is the drift term plus the volatility term sorry.

So this is the stochastic differential equation followed by a price process. So what so see the clever way we have applied the Ito calculus. We have seen this thing. So this function, this thing we have considered as $X(t)$ the process. So in general if you write a Ito process you have to write it down integral but we know that this is constant.

So this is so if I do the integration this is what I will get so basically this is what is $X(t)$. This is what I have taken as $X(t)$. So then so from here then from here we have found that $S(t)$ is expressed in this form and then by applying Ito's formula you come and show that the geometric Brownian motion actually satisfies the following stochastic differential equation and in all the books and papers etc.,

You would see in finance this is the form of the stock price movement which is usually taken if you do not add things like jumps and all those things to a model reality a bit more better in the

sense that there could be a sudden jump in the prices because of some problem in the market or sudden drop in the prices.

So this is 1 aspect. So we will go to some other application. I would expect you to work it out yourself at home when you take this as alpha t and this as sorry this as sigma t and this as t. This is of course $\sigma^2 t$ So when they are time dependent then what sort of thing a geometric Brownian motion what sort of a process would the price process look like.

If you have a random volatility and random drift term what sort of thing it would look like okay. So this is our first application of the Ito's calculus which shows that our stock price actually satisfies the Ito process satisfies this stochastic differential equation. Not let us look at some more 1 or 2 more applications. So this series of lectures the 5 lectures that we are supposed to give.

This week we will end with the computation of the or rather derivation of the partial differential equation whose solution would give us the Black Scholes price of a European call option which we will come very soon but now suppose I define an Ito integral $I(t) = \int_0^t \Delta(t)dW(t)$.

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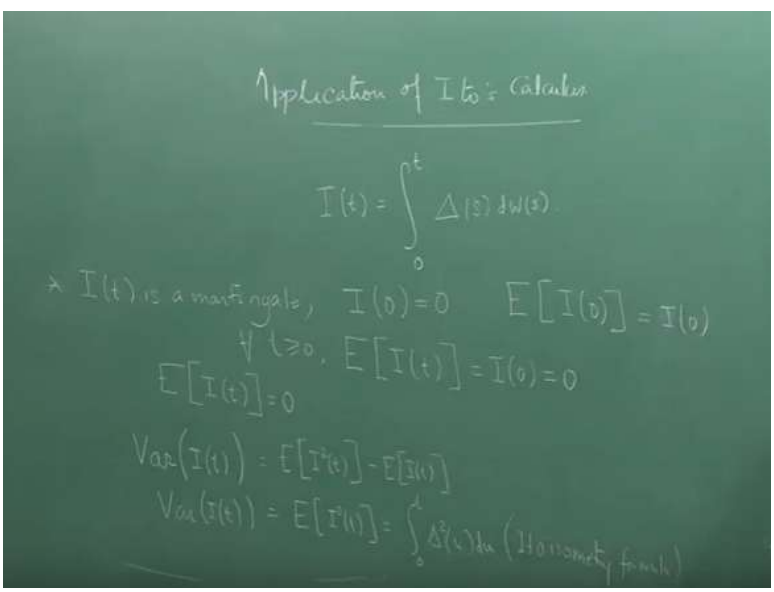

So once I define an Ito integral then what is my mean and variance? So these ideas would be required as we do more of our studies. So how do you recognize that what is the mean of a Ito integral. The key notion to remember is that the Ito integral $I(t)$is a Martingale. Martingale means the first random variable of the process or $X(0)$whatever be its mean that the same mean would be for all the other remaining random variables.

Now $I(0) = 0$ because you put equal to 0 it is $0.I(0) = 0$. So the $E[I(0)] = I(0)$ and we know for any t you have this and $E[I(t)] = I(0) = 0$ because $E[I(0)] = I(0)$ because this is constant it is 0 so is 0. So $E[I(t)]$of a Ito integral is 0. Now the question would be what is the variance of the Ito integral.

The $Var[I(t)]$is nothing but $E\left[I^2(t)\right] - E[I(t)]$ and $E[I(t)]$is 0 so basically $Var[I(t)] = E\left[I^2(t)\right]$ and $\int_0^t \Delta^2(u)du$ by the Ito isometry formula this is called this is by the Ito isometry formula. Now these ideas would be helpful as we do our next applications. We will talk about how to model interest rates in the next class.

So we will just introduce to you what are interest rates. We will not get into tremendous amount of details about finance because there is lot of issues when discussing very basic concepts of finance. So what is the interest rate?

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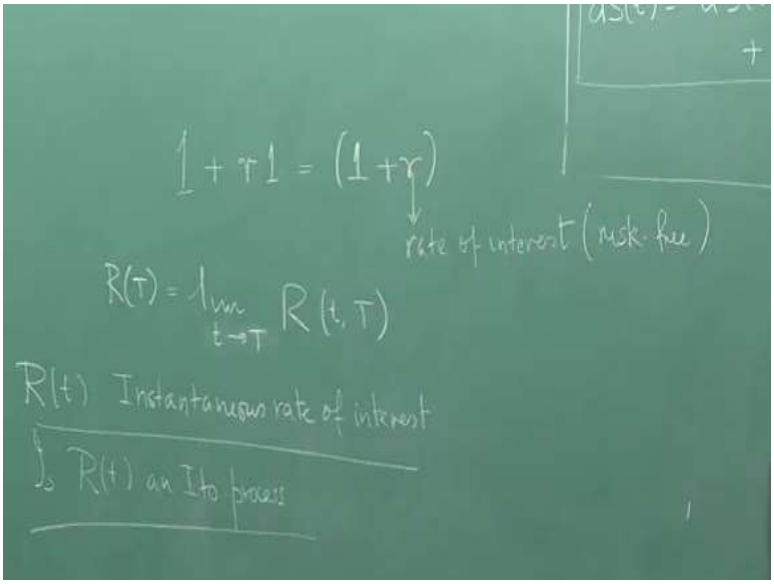

Suppose I have 1 rupee, I invest in a bank and the bank tells me that after time t you are going to after time small t you are going to get back or some money with an interest rate R. Which means he is going to pay me after time t not only rupees 1 but the interest rate which is the percentage rupees R into 1 so not only the principle but the fraction of my principle would be added to my original money.

So this is what I will get $1 + r$. So after given period I will get $1 + r$ money. So this r this fraction is called the rate of interest. Usually what happens is that banks suppose you have kept money in a bank in a fixed deposit or you have taken a loan. So you are paying back money to the bank. So usually for a given period say between time small t to capital T for a very small period the bank interest rates are usually kept fixed.

This is called the risk free period of interest. This rate of interest is called risk free because I guarantee you that if you take money from me now after say 1 month I will give you 1 plus if you give me 1 rupee I will give you 1 plus r rupee. So that is that is what is a for example if I say if you give me 100 rupees now I will give you back after say 1 month 110 rupees. So my interest rate is 1 by 10th.

So r is 1 by 10th. So that sort of idea. So the interest rate usually in a bank is kept at a hold between it is held for very short time. Usually the interest rate keeps on changing in the bank. So

in general just like in physics we want to know what is the instantaneous velocity is very difficult a concept. It is not so easy to understand that though we do lot of problems and do a lot of things with it.

But instantaneous velocity is one of the most deepest concepts. It is not so easy to fathom it immediately. Similarly we are talking about we will talk about instantaneous interest rates, what is the interest rate exactly at time t. So just like we did in physics that we look at the difference of the distance travelled by difference of the time taken and take the ratio and then take the limit as the time goes towards 0 and that is what we call instantaneous velocity in mechanics in physics.

Similarly, we call or rather I should say that $R(T) = R(t)$ is called the instantaneous rate of velocity at the time sorry instantaneous rate of interest at the time capital T. So here $R(t)$ or in general $R(t)$ is a instantaneous rate of interest or called sometimes called short-term interest, instantaneous rate of interest. So but this rate of interest is random. It changes.

This itself is random. You cannot say that today's interest rate is this next day the interest rate will be as per as you have decided it will it will be just based on the market, it will change. So $R(t)$ is called the instantaneous rate of interest. Our next question would be can this $R(t)$this instantaneous rate of interest which is now essentially a stochastic process. It is a stochastic process is we are asking the question is $R(t)$an Ito process? Is $R(t)$an Ito process.
So with this we end our first lecture of the last week and in the next class we are going to answer whether $R(t)$ is an Ito process or not and what sort of models had been there for $R(t)$ and how Ito's calculus help us to understand those models.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 17
An Application of Ito Integrals-I

So, I am going to now speak about the use of Ito calculus, to understand interest rate models. So, do not get into this issue of why this model was built like this, and why it was built like that. So, we are going to talk about the Vasicek's model. The Vasicek's model of, so this is, we are what, we are so this is what we are going to now discuss. So, this is again the second version of application of Ito's calculus.

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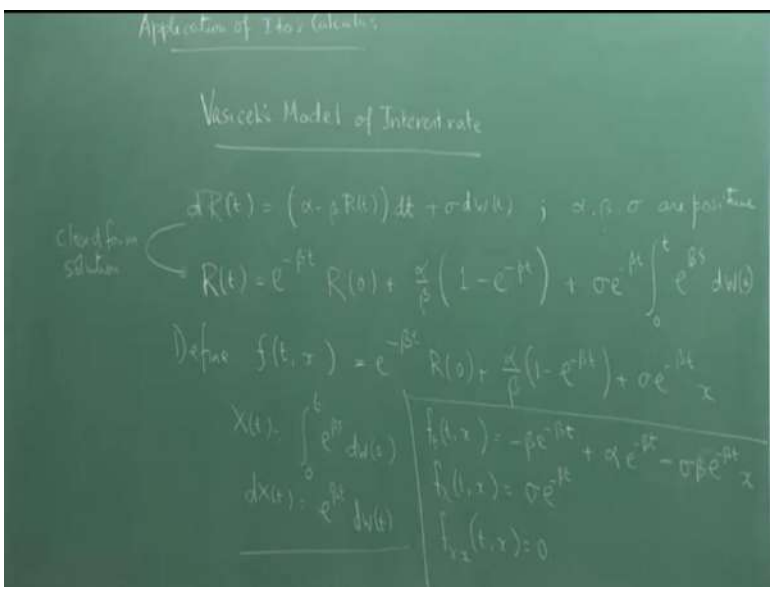

So here we will see two aspects of Ito's calculus. The first Vasicek define, the $R(t)$to be an Ito process, which is governed by the stochastic differential equation $dR(t) = (\alpha - \beta R(t))dt + \sigma dW(t)$. Of course, you know that these are only shorthand's but okay these are helpful to a certain extent. So, you observe that if I put $R(t) = \frac{\alpha}{\beta}$, there will be no drift term. It will be completely purely a random term.

Now interestingly enough, if you assume that, we will assume naturally, that α, β, γ are positive. Let us note that, every stochastic differential equation, cannot be given a closed form solution. We will not go in to show how to get a closed form solution, but it cannot be given a closed form solution every time. But luckily in the Vasicek's case, you can give it a closed form solution and the closed form solution is given in the following way $R(t) = e^{-\beta t}R(0) + \frac{\alpha}{\beta}\left(1 - e^{-\beta t}\right) + \sigma e^{-\beta t}\int_0^t e^{-\beta s}dW(s)$.

So, you will immediately see that it will tell you that this is an Ito process. So, do not think that okay this is something why $\sigma e^{-\beta t}$ is outside the integral. This was anyway outside the integral always. The integral was this was into this integral. So, this is actually a, closed form solution. Our first step of the use of Ito calculus here would be to show that this is indeed this Ito process is indeed a solution of this.

The question would be that is, Ito calculus only helpful to show that something is a solution of this, but and I do not know how to solve it, but I am telling you that, many of these equations do not have a closed form solution and when we will go to the Cox-Rand-Ingersoll model, another model of interest rate. We will see that it does not have a solution, but Ito calculus would help us, even if it does not have a solution can compute its mean and variance.

Now the interesting part or you will very soon see that we can show that $R(t)$ will satisfy a normal distribution and hence by doing so we will come to the following conclusion that with a mean 0 in fact this $R(t)$ is a random variable which can also take negative values because this is a normal distribution with mean 0. So, that something is not favorable you do not have negative interest rates. The interest rates have to be positive or 0.

So, but though it is a initially good model to understand the behavior of interest rate. Of course, I am not going to the detail of how this model was built and why it was built. That will take us too much of a field into finance. So, when we are going to talk about finance, in our next course, then we are really going to talk about these sorts of models right. That is something that we will

build up. But anybody, who would like to do this course in finance, will have an access to these lectures which I gave you in the very basics.

So, here we are trying to see how Ito calculus can be applied, what are the techniques. For example, you know when we do Ito calculus, our first idea would be to know how to choose my $f(t)$ for $f(t,x)$. So, that choosing the function f is of fundamental importance to know how to apply Ito calculus. Here of course, you want to use this as some sort of your $f(t), f(t,x)$.

So, how will you do it. So here, we define $f(t,x) = e^{-\beta t}R(0) + \frac{\alpha}{\beta}\left(1 - e^{-\beta t}\right) + \sigma e^{-\beta t}x$ as, basically here, we are bringing a process within a process. This is my, so I am getting a smaller Ito process. So, I am having $X(t) = \int_0^t e^{-\beta s}dW(s)$, so your shorthand $dX(t) = e^{-\beta t}dW(t)$, this is the shorthand right. So, shorthand is very useful. It is like game playing so you can have some fun with it always.

Now on this, I would start applying the Ito's calculus. To do this, we must first know, so you know that, I need to have the notion of three derivatives. The derivative of f with respect to x with respect to t and the second derivative with respect to x both the times. So, partial derivative second derivative with respect to x ,$\frac{\partial^2 x}{\partial x^2}$.

So, what is my f t x. So, f tx would have a larger say here, because we have terms in t, so this will become $f_t(t,x) = -\ \beta e^{-\beta t} + \alpha e^{-\beta t} - \sigma\beta e^{-\beta t}x$.

Now if you do f of x the derivative, partial derivative with respect to x all these parts would be 0 and we will simply have $f_x(t,x) = \sigma e^{-\beta t}$ and if you want to do it, $f_{xx}(t,x)$of course, it is 0 because there is no x term anymore here. So, you know one of the terms of the Ito's calculus goes away.

So, all these two classes, we are trying to see how Ito's calculus is useful. So, do not bother much about the financial implications of the model. That only would come in the next course, as why this model is important what is the issue etc., etc., etc.

So, the finance game, is a very different game, and here we are trying to learn the math associated with it, with examples coming from finance of course. So, here let me start writing the Ito's calculus. So hence by Ito's formula we have let us see what we have by Ito's formula.

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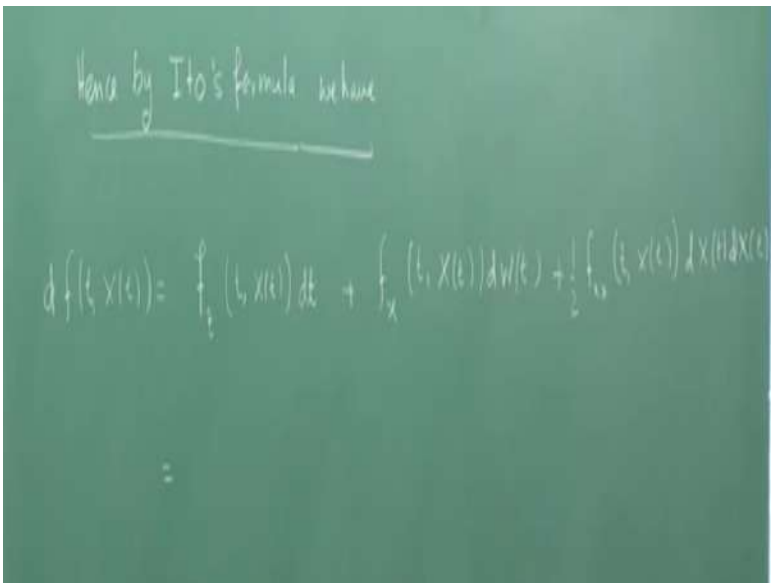

So, $df(t, X(t)) = f_t(t, X(t))dt +$ that is now I am just writing the Ito's formula. $f_x(t, X(t))dW(t)$ because you know only with this variable, this partial derivative where you have the $dW(t)$term, $\frac{+1}{2}f_{xx}(t, X(t))dX(t)dX(t)$ which is the quadratic variation term and you that this is 0. so it just goes there is nothing. So, the interesting part. which we have not told is that. all these equations that you write here. is all are in the form of almost everywhere.

These terms of Ito's inequality, what we are writing and we are not bothering to say, these are all almost everywhere type inequalities. When I am writing equality in a stochastic process setup, it is always almost everywhere type. There could be some omegas for which this does not hold

because, we are taking limits and we are taking quadratic variations and all these things are all in almost in everywhere sense. So, all these inequalities it is very important.

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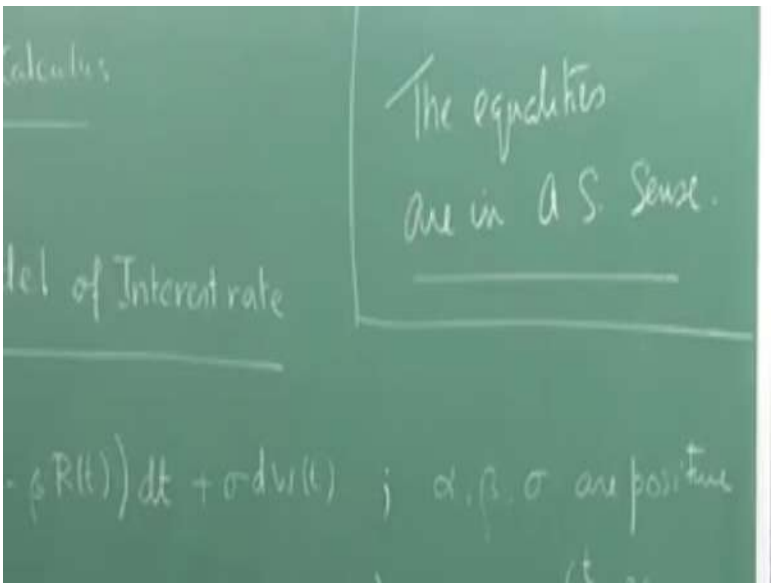

So, all inequalities are in almost everywhere sense sorry, equalities. The equalities are in almost everywhere sense. Please note this though it is very nice, so because of it only fails on a null event, we do not bother about it, we just write equality, so while doing just as a working tool, I can just write equality, but all equalities are in almost everywhere sense or it is same as they probably say almost surely sense, and this occurs with probability one and equality occurs with probability one.

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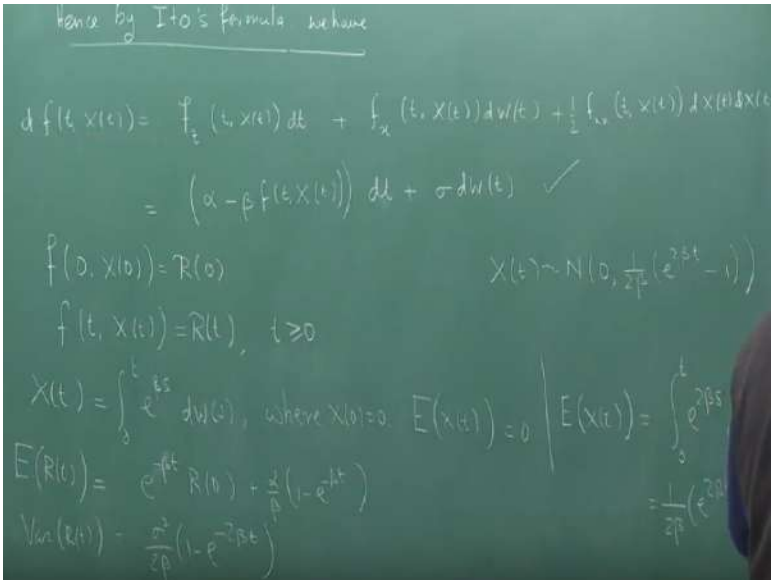

It is a random phenomenon. Do not think that it is a nonrandom phenomenon. So, then if I go and write down this and do the calculations here, what you will get is, $\alpha - \beta f(t, X(t))dt + \sigma dW(t)$ f this is 0 that part $f(t, X(t))$and you know what is this f_x. So, once you write everything the whole thing once you put everything and do the simplifications you will get this formula.

I leave it for you to do the simplifications. I am not writing the simplifications down because that will take too much of time.

So, you see here you see this is exactly the same thing. So, this we have verified by Ito's calculus rule what is the that this is actually the closed form solution of this stochastic differential equation. So, let us put $f(0, X(0))$ and $f(0, X(0)) = R(0)$. So, $f(0, X(0))$ is$R(0)$. So, the initial condition of this process and $f(t, X(t))$. $X(t)$is now this one so if you put $f(t, X(t))$ is $R(t)$.

Actually, what happens $f(t, X(t))$has the same behavior. Essentially if you look at it if I put $X(t)$ here, it is exactly your $R(t)$. So essentially, we are if you once we are trying to apply Ito's formula, we have to extract from the given Ito process, the form of the function. That is the key thing that you have to observe when we do the job.

Of course, there are certain issues. So, what about, of course you know that, what about this $X(t)$, what is the nature of this $X(t)$? $X(t)$ is also Ito process. So of course, $X(t)$ is a process where $X(0) = 0$. So, $X(t)$ is a process where $X(0)$ is 0 naturally. So, $E(X(t))$, what is $E(X(t)) = 0$. So, what is $E(R(t))$? $E(R(t))$, is nothing but $\sigma e^{-\beta t} E(X(t))$right. So, over these constants. So, expectation of this plus expectation of this. This part is it will come out.

So, let us see what is the $E(R(t))$. Now if you look at it very carefully. You can find also the variance of $E(X(t))$. So, what would be the variance of this? Again, by the Ito isometry, the variance of $E(X(t))$ would be, which is equal to $\frac{1}{2\beta}\left(e^{2\beta t} - 1\right)$

So, what we have done, what we had shown, that this $X(t)$ so W is following normal distribution right. So, $X(t)$, so scaling of a normal distribution also gives me a normal distribution. So, $X(t)$ also follows a normal distribution with mean 0 and variance this. So, $X(t)$ follows normal with mean 0 and variance. So, that shows from here this is just a constant so this shows that expectation of $R(t)$ is.

So, $R(t)$ is also normal distribution, with mean this and variance which I am not writing in detail. You can calculate out this thing. So basically, when you take the variance, the variance of this constant part is 0. You do not have variance of a constant, where the constant term does not vary.

So, your $R(t)$ also follows normal distribution with mean this and $E(R(t)) = e^{-\beta t} R(0) + \frac{\alpha}{\beta}\left(1 - e^{-\beta t}\right)$ okay.

So, what is the lesson here. The lesson is that if this is following normal distribution, and this might give me negative values, so what it cannot have negative values. So how do I modify this? The question is how do I modify this.

This modification was done by Cox-Ross-Ingersoll model. To study the Cox-Ross-Ingersoll model, we will see they have just changed the drift term slightly, sorry, not Cox-Ross-Ingersoll but Cox-Ingersoll-Rand model. I will just CIR, this is called the CIR model, or the

Cox-Ingersoll-Rand Ross model CIR. These are all big names in financial economics. Cox-Ingersoll-Ross CIR model.

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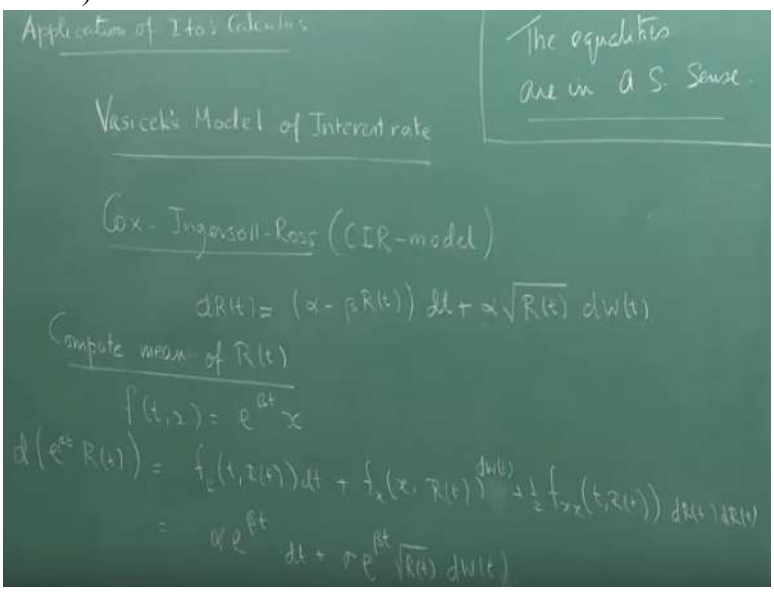

So, the CIR model, they assume that it should be of this $dR(t) = (\alpha - \beta R(t))dt + \alpha\sqrt{R(t)}dW(t)$ form. The interesting part is that this model does not have this drawback of this becoming nonnegative. $R(t)$ would never become nonnegative, but this is the example of a stochastic differential equation, which does not have a closed form solution. Even if does not have a closed form solution, you can still get some idea of its mean and variance.

If you want to compute that okay what sort of distribution $R(t)$ satisfies. Then it will be quite a difficult thing to compute the distribution, but you can at least compute its important parameters, mean and variance.

So, let us just give a hint, as how to compute mean of $R(t)$ and you will see how the Ito's calculus would be helpful here. See Ito's calculus is not just to verify some given solution our SDE. It is helpful in many-many states. So, to do this, you have to choose. Now this choice of $f(t, x)$ it must look very arbitrary to you, but it is done because you have done some work. It is

done from guess and test. So, if you always bring in exponential things because, you want to have things nonnegative. So, exponential thing always comes up usually.

So, your xt is your $R(t)$. So, your so $d\left(e^{\beta t}R(t)\right)$. So, that is again$f_t(t, R(t))dt +$. $f_x(x, R(t))dW(t) + \frac{1}{2}f_{xx}(t, R(t))dR(t)dR(t)$. So, I will ask you to calculate this thing through okay. If you calculate this thing, as you can have some fun computing $dR(t)dR(t)$ because, it will become $\alpha^2 R(t)dt$. So, if you do this calculation this will turn out to be the final form would be of this form. $\alpha e^{\beta t}dt + \sigma e^{\beta t}\sqrt{R(t)}dW(t)$ sorry here I will have d Wt.

So, when you write put in all the values, you know what is e tx ft is $\beta e^{\beta t}x$. So, you know what it is and what is x. It is $e^{\beta t}$. What is $f_{xx}f_{xx}$is, what would be f_{xx} here. Here $f_{xx} = 0$. So, once you know that fact, so you do not have to bother about the quadratic variation, and then you write down everything, $\alpha e^{\beta t}$ for example here ft would be $\beta e^{\beta t}x$ which is$R(t)$.

So, it will be $\alpha\beta e^{\beta t}R(t).dt$ So, write the full form $R(t)$ again $R(t) = \alpha e^{\beta t}dt + \sigma e^{\beta t}\sqrt{R(t)}dW(t)$ this one. So, once you write, sorry, you do not have to write the full form $R(t)$. So here you have f_x so x $e^{\beta t}$ x $e^{\beta t}$,x means 1 so $f_t R(t)e^{\beta t}dW(t)$. So, once you write this you will get back this formula. You see we are just essentially getting that.

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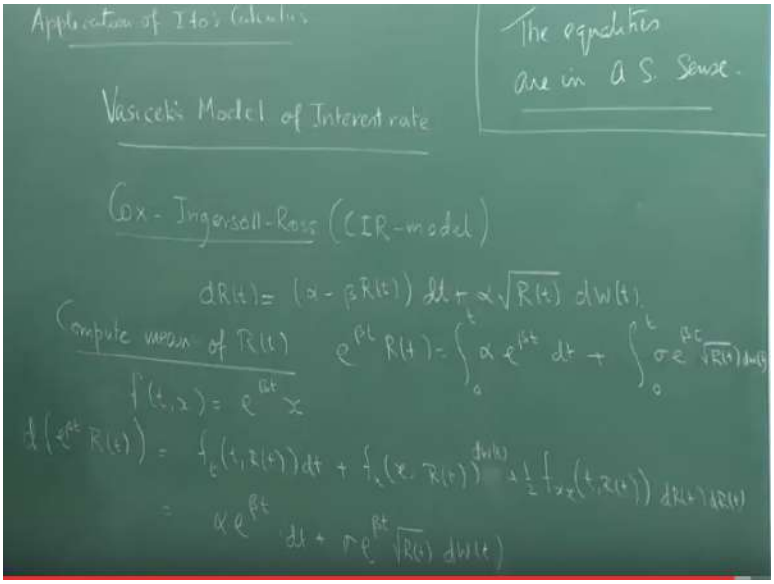

So, if you now know this formula, I would similarly ask you. You can now integrate both sides. So, you can integrate both sides take off the differential and you can write $e^{\beta t}R(t) = \int_0^t \alpha e^{\beta t} dt + \int_0^t \sigma e^{\beta t}\sqrt{R(t)}dW(t)$.

So, once you know this you can immediately use your other ideas about Ito's integral and this is a simple case. To really figure out, what is the mean, you just take the βt on the other side compute this out and then take them in. So, I will leave that to you as homework. You should try it out something yourself. If you do not try it out yourself things might not always be fine.

So, with this I end my talk to give you some idea about how things can be used. Please check up these calculations. Do not completely rely on what I have written at the end. Just checkup that you come to that conclusion.

So, thank you very much.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology Kanpur

Lecture - 18
An Application of Ito Integrals-2

Hello, so it is our third lecture in the last week, in which we are going to talk about Ito's formula in higher dimension. What we mean by that. See, we have been talking about only for example stock price of one stock, interest rate just one interest rate. Suppose, I want to talk about two stock prices. I want to monitor two stocks, at the same time and their prices and see how they are correlated, how their behavior is related to each other or not.

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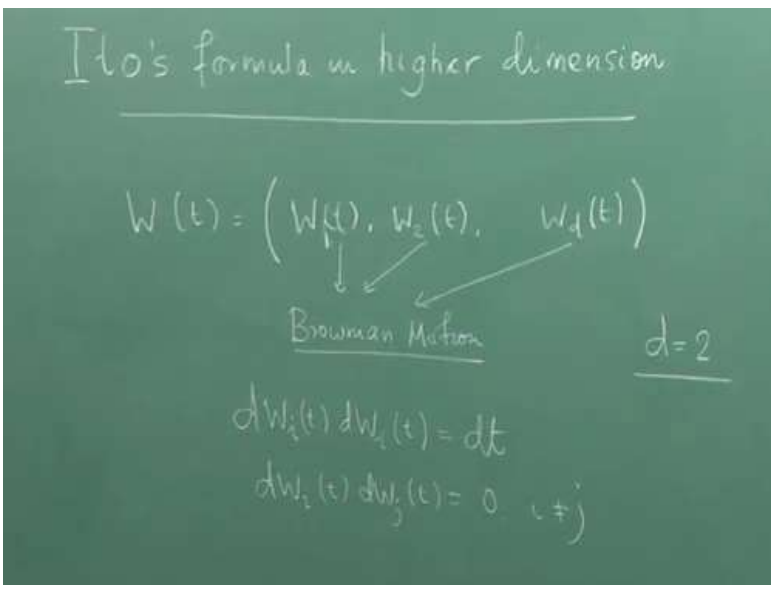

What sort of approach I should need to analyze them and then you have to talk about Ito's formula in higher dimension? In doing so, we need to talk about Brownian motion in higher dimension, that is the Brownian motion vector, that at every time t, the Brownian motion is not just a random variable, but a random vector. We had spoken about random vectors very briefly during our very first discussions of probability. So, this is what we are going to talk about $W(t) = \left(W_1(t), W_2(t),.. W_d(t)\right)$.

$W_1(t), W_2(t)$ so it is the d dimension of a vector if you assume. At every time t, so this is a higher dimensional Brownian process or Brownian motion and this consisting of d, say d components of Brownian motion, where each of them are random variables, each of them. So, now the question would be, why we need to think of such processes and what is the utility. See you can still define, a filtration adapted with $F(t)$t in the same way.

Of course, when you are talking about Brownian motion, we will write $W(t) - W(s)$ for example. This increments are independent. So, $\left(W\left(t_1\right), - W\left(t_2\right)\right), \left(W\left(t_2\right), - W\left(t_3\right)\right), \left(W\left(t_2\right), - W\left(t_1\right)\right)$ so these would be having independent increments, in the sense that, we really do not bother much about the vector. We really bother about its components.

One important thing that you have to understand about the components is the following, that if we are talking about a component i and taking its quadratic variation then, $dW_i(t)dW_i(t)$, is just dt and if you are talking about a cross variation of 1 then it is$dW_i(t)dW_j(t) = 0$ okay. So, that is what you have to keep in mind.

So, when $i \neq j$, then the cross quadratic variation is 0 and quadratic variation anyway is d t is known to you. So, what we are looking at. We are essentially looking at, not only one Ito process at the same Ito process, but we are looking at two Ito processes at the same time. For example, if you are looking at the price of two stocks they are two stocks are two different Ito processes. So, we are trying to find relations between them.

So, in general we are looking at Ito process. Suppose now you take d is equal to 2. So, we will now discuss, just look at scenario at d is equal to 2. So basically, we have two different Brownian motions right $W_1(t), W_2(t)$. So, two different scenarios.

See the Brownian motion, the W function is not a unique thing. $W(t)$is not unique. So, $W_1(t)$ and $W_2(t)$ are two different Brownian motions right. $W_1(t)$and $W_2(t)$are two different Brownian motions. Only that they have to satisfy the properties of Brownian motion. Their functional form need not be same. That is something, which you have to always keep in mind that, these are do not have unique functional forms, when it can be it depends on a what sort of a it just it is a stochastic process which would follow certain pattern.

So, if it just follows that pattern it is a Brownian motion not that it so there can be 2 different Brownian motions and has very different sample paths for a given realization omega but they are following all the properties of a Brownian motion and hence they are Brownian motions.

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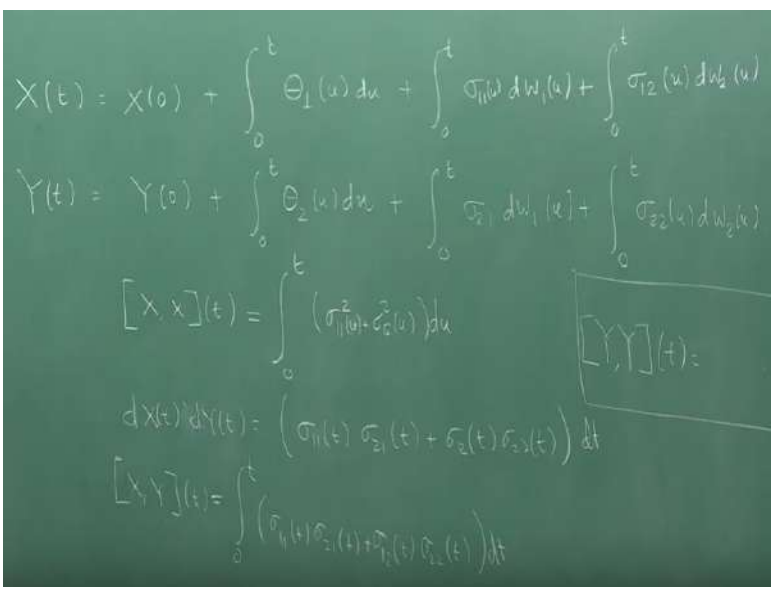

So, usually an Ito process, so for example, we are looking at two different Ito processes. The first one is given by this formula, $\sigma_{11}(u)dW_1(u) + \int_0^t \sigma_{12}(u)dW_2(u)\ \theta_1(u)du + \int_0^t X(t) = X(0) + \int_0^t$, essentially these are trying to recollect the, some sort of a core variance type thing, some sort of matrix. We will come

to it very soon. You will see soon see why I am writing like this. So, this every process xt that we are taking here, is involving both the processes. It is not just involving one process.

So, it is some sort of a core relation coefficient sort of thing and it is a function where, you are essentially describing the first Ito process, but you are talking about the second Brownian motion. Similarly, you can write $\sigma_{21}(u)dW_1(u) + \int_0^t \sigma_{22}(u)dW_2(u)\ \theta_2(u)du\ + \int_0^t\ Y(t) = Y(0) + \int_0^t$.

So, when you have two Brownian motions, so you can define this sort of processes. And then, for example, I leave it to you to figure out, what is $[X, X](t)$that is, what is the quadratic variation of this. So of course, you can write the simple formula, $\sigma_{11}(u)dW_1(t) + \int_0^t \sigma_{12}(u)dW_2(t)\ \theta_1 t(u)dudt\ + \int_0^t\ dX(t) = \int_0^t$. You can write down the shorthand's. So, using the shorthand's you can come to $\sigma_{21}^2(u) + \sigma_{22}^2(u)\ (\ \ )du \int_0^t$ conclusion, that if you take $[X, X](t)$ what will happen.

See every time, we are using, what these sorts of facts and whatever we had learnt earlier right, these shorthand notations. For example, if I want to write $dX(t)$ the cross variation, do not write the cross variation immediately as 0. Basically, the multiplying of all the things. So of course, I would not write to you, what is $dX(t)$ and what is $dY(t)$. You can write it out yourself, now because we have been doing it so much repeatedly.

Then this would turn out to be $dX(t)dY(t) = \left(\sigma_{11}(t)\sigma_{21}(t) + \sigma_{12}(t)\sigma_{22}(t)\right)dt$. Basically, if you want to write it like this. So, it is not 0 you see. See if you want to write it like this so this cross quadratic process, is a stochastic process given like $[X, Y](t) = \int_0^t \left(\sigma_{11}(t)\sigma_{21}(t) + \sigma_{12}(t)\sigma_{22}(t)\right)dt$ this.

So, I leave it to you to find, figure out what is the quadratic variation $[Y, Y](t)$. I am sure, you can just immediately write it down, without me telling you what to write. They are exactly the same thing with this, these things are replacing these things, that is all square of these things.

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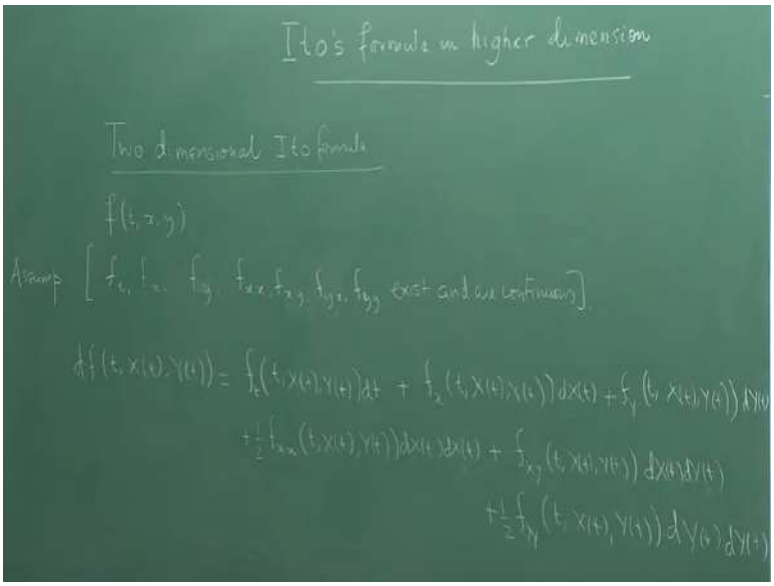

So, once I know about this, I can write a two-dimensional Ito formula. So, we are going to now talk about a two-dimensional Ito formula and then we will use it to study stock prices, two stock prices at the same time. In the two dimension Ito formula case, we do not use the function just $f(t, x)$ but $f(t, x, y)$ because, here also we have to account for the process 2nd Ito process Y.

So, we are basically now considering a function of this form and once I consider a function of this form what do I expect? I expect that $f\left[t, f_x, f_y, f_{xx}, f_{xy}, f_{yx}, f_{yy} exist \wedge arecontinuous\right]$. So, that is the assumption. This is an assumption okay and I have made this assumption, so assumption. So, once I make this assumption, you know from basic analysis that f_{xy} and f_{yx} are the same thing.

Now the formula is like this. The Ito formulas are basically stochastic Taylor's formula, Taylor's expansion.

$df(t, X(t), Y(t)) = f_t(t, X(t), Y(t))dt + f_x(t, X(t), Y(t))dX(t) + f_y(t, X(t), Y(t))dY(t)$. So, when if I had put $W(t)$ here $W(1)$ $W(2)$ say, then it will have $W(1)$ $W(2)$.

So, f_y again I should tell you all this inequality that I am writing are in the almost surely sense, but these are random variables, but do not think that, when I am writing $f(t, X(t), Y(t))$ it essentially becomes a random variable. It is no longer just an ordinary function. You are writing ordinary function but you are then looking at its random avatar, basically or looking at the random form of that.

For example, if I take the function $f(x) = \frac{1}{2}x^2$ you can write $\frac{1}{2}X^2$, where x is a random variable. The random variable at the end function, but overall behavior is slightly different. So, this is that is what I feel about random variables. This equality, I understood that okay this equality will hold with probability one, so we do not bother much about talking about this every time, so is just an equality $+ \frac{1}{2}f_{xx}(t, X(t), Y(t))dX(t)dX(t)$.

Now because f_{xy} and f_{yx} are same so we just write one of them. So, half of each will have contribute a half term, so you will have for full term so if I write $f_{xy}(t, X(t), Y(t))dX(t)dY(t)$ so the cross one it is this one, plus half f_{yy}. I again tell you that f_{xy} equal to f_{yy} so basically, I should have a term here $+ \frac{1}{2}f_{yy}(t, X(t), Y(t))dY(t)dY(t)$, but both the terms are same so that half half would give me one so, I do not write anything here.

So, this is you see, there is a pattern in the formula, with t there is d t with x there is $dX(t)$ with y there is $dY(t)$ with xx there is $dX(t)dX(t)$ with xy there is $dX(t)dY(t)$with y y there is $dY(t)dY(t)$. There is a pattern. So, once you recognize the pattern applying Ito's formula is become a child's play it becomes fun. You can just keep on applying and getting your results.

So, you have to keep in your mind this pattern that Ito's formula gives you. Once you know this pattern then everything is fine. Of course, proving this is a very different game but it is better to

first learn to use the pattern and learn to apply them. We will when we do the financial stuff we will actually go and do that we will see more of these applications.

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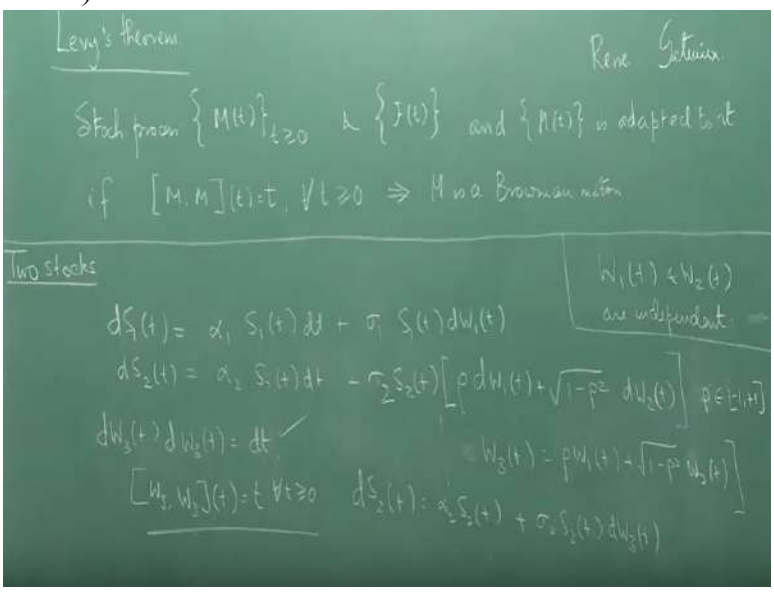

Now let us look at the stock, two stocks, price of two stocks, that will be important, to know the price of two stocks and their behavior. But it is very important to know, one of the theorems. It is called Levy's theorem. So, Levy's theorem tells us, how to recognize a Brownian motion if there is one. It tells you that if you have a stochastic process $\{M(t)\}_{t\geq 0}$. Suppose you have a stochastic process $\{M(t)\}$ it is given to you. So, this is a stochastic process$\{M(t)\}_{t\geq 0}$. That is given to you.

Now once you know that, suppose you have more additional information that, you that, there is a filtration $\{F(t)\}$, to which this stochastic process is adapted and $\{M(t)\}$ is adapted to it. And then additionally if, you have the quadratic variation to be t at time t,$[M, M](t) = t \in t \geq 0$ it accommodates the quadratic variation of the amount t, which is path independent right. If you have this then, it implies M is a Brownian motion.

So, this theorem is due to Paul Levy. It is always better to tell you some one or two stories about this guy called Paul Levy. Let us understand that Levy was really not a probabilist. He was analyst and in fact it was Rene Gateaux on whose name we have the Gateaux derivative.

Rene Gateaux, who started working on this sort of stochastic processes to look at them in a more detailed mathematical way but he was killed in the World War II, sorry World War I, and one or two, I am not, so one of the World Wars Rene Gateaux was killed. And then Paul Levy was told to see that his papers what he has written should be published. So, while reading what he has written he got interested in probability and he himself became one of the leading probabilist of the 20th century, Paul Levy.

So, the modern stuff of the Brownian motion that we see the modern theory and modern definition are essentially due to Paul Levy. So, it is very important to know the guy who had actually done stuff for us.

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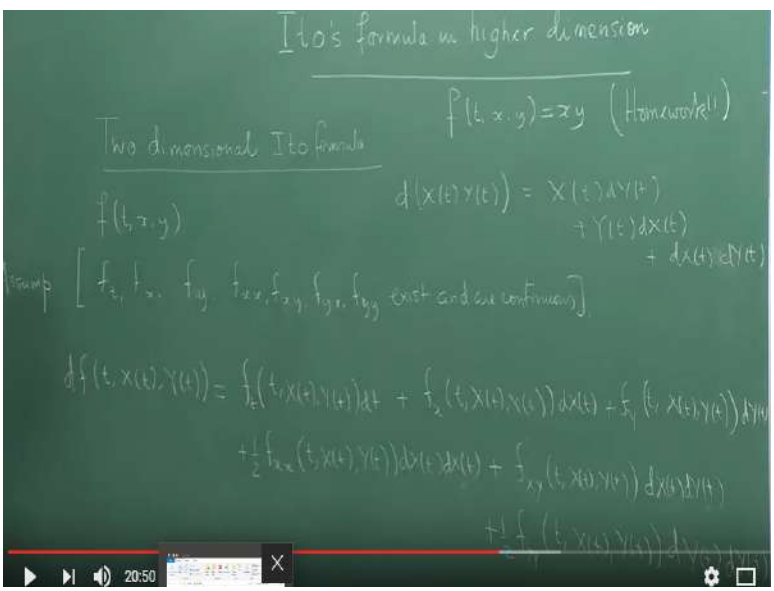

Now I give you a small homework. Thing is the following. Take a specific $f(t, x, y) = xy$ and then see what will you get. If you look at it, $f(t, x, y) = xy$ gives you the feel of a product rule, that you are talking about $X(t)$ into $Y(t)$ when you are talking about$df(t, x)$ you are talking about $d(X(t)Y(t))$. You are talking about the product. So, what would the product be, how do the product rule look like. I would like you to finish that as a homework to prove the following $d(X(t)Y(t)) = X(t)dY(t) + Y(t)dX(t) + dX(t)dY(t)$.

So, you see there is always an additional term. So, it is not the standard calculus, so it is always there is an additional term. So, you can apply this formula, to this function to prove this and this is a homework I think all of you are, so even if you do not give the exam does not matter but you should really try out this with your own, to see that, you are developing a gradual feel for applying the Ito's formula. Not that you really have to prove things etc., but just applying that.

So now I have 2 stock prices, I am analyzing two stocks. So, if I am analyzing two stocks, what should I look into. So, let us observe, let us just take, correlated stock prices. So, let us just observe or let us just say that, I have two stocks of this particular type of behavior which satisfies the following stochastic differential equation.$dS_1(t) = \alpha_1 S_1(t)dt + \sigma_1 S_1(t)dW_1(t)$ So, they are Ito processes of this type.

So, this process only depends on the first Brownian motion, but now we will define a second stock price.$dS_2(t) = \alpha_2 S_2(t)dt + \sigma_2 S_2(t)\left[\rho dW_1(t) + \sqrt{1-\rho^2}dW_2(t)\right]$, This is an example, this is just example from Shreve, it is not that it is exactly everything has to follow like that. The second price, the second one, depends on not only one Brownian motion, but of course, this $1-\rho^2$ has to be positive, so $\rho\epsilon[-1,+1]$, with rho in minus 1 plus 1.

So, this rho is some sort of a correlation coefficient. When there is an exact correlation, rho is 1 then this becomes 0. So, it just depends on $W_1(t)$ and when there is rho is 0, there is no correlation between $W_1(t)$ and $W_2(t)$ that $W_1(t)$ and $W_2(t)$ then this $S(2)$only depends on $W_2(t)$. So, that is what it is trying to capture. So, it is so of course, you can have a more feel if you have a more knowledge about statistics.

Of course, so depending on the rho, you know whether the processes are connected or not. So here it seems to be that, this process is depended on $W_1(t)$and $W_2(t)$but, this $W_1(t)$ $W_2(t)$ there is when you write this vector, W t is equal to $W_1(t)$ to $W_2(t)$ to W t this, each of this $W_1(t)$ they

are independent processes. $W_1(t)$ is independent from $W_2(t)$. $W_2(t)$is independent from $W(t)$. So, here the important thing to remember that $W_1(t)$ and $W_2(t)$ would always remain. This $W_1(t)$and $W_2(t)$are independent processes. They are not correlated here okay.

So, you are getting the $W(t)$ as $W_1(t)$ to $W(t)$, that is independent process, but it does not mean that, but this process, this whole this process could be correlated to this, but they might not be independent. So, let me figure out okay, let me, if I write this as some other Brownian motion right, then this new Brownian motion, this whole Brownian motion, may not be independent from this, but just $W_1(t)$ and $W_2(t)$are independent.

Then if that is the case, let us put $W_3(t) = \rho W_1(t) + \sqrt{1 - \rho^2} W_2(t)$ is this, so I am making a new process, a new Brownian motion and because all other properties of the Brownian motion would be satisfied, because you just have linear sums does not matter linear per combination.

So, this is my new Brownian motion, so new process, again it is the Ito process, but it is a Brownian motion fine. Basically, I want to say that okay, I want to have $dW_3(t) = \rho dW_1(t) + \sqrt{1 - \rho^2} dW_2(t)$. This is what I want. So, I had made this new process, new Brownian motion. So, this is the differential form of that okay.

So, what I will now show essentially, technically I am writing that this is a differential part, but one has to show that, see sorry sorry now, one has to show that, if I take $W_3(t)$like this the $dW_3(t)$ would actually be like this. I had just written it down, because that is what will come that, if $W_3(t)$is chosen like this then $dW_3(t)$would actually be this and there will be a correlation then between $W_1(t)$ and $W_3(t)$. You can compute the covariance and that is why, we just start looking at how to do the stuff.

So, if you look at the covariance, sorry, look at $dW_3(t).dW_3(t)$then, of course $dW_3(t)$ is this one basically. So, what I am trying to say is that, if you look at it. So, what I have done. So basically, I have now replaced this by $dW_3(t)$. So, my $S(2)$ now depend on some other Brownian motion which is a combination of $W_1(t)$ and $W_2(t)$.

So, here you have $dW_3(t)dW_3(t)$if you take, it will come out to be dt and I want you to really do this calculation. It is just basically, if you look at it, it is nothing but$\rho^2 dt + \left(1 - \rho^2\right)dt$. So, it turns out to be dt because this is the, if you do, if you do cross when a $dW_1(t)$, $W_2(t)$ is 0 so that is why the cross variation becomes 0, so you have to take only their quadratic variation.

So, their quadratic variation is $\rho^2 dt + \left(1 - \rho^2\right)dt$. So, this turns out to be so the quadratic variation is t for this and hence here $W(0) = 0$, $W(t)$is greater than equal to 0. It is adapted, it can be adapted to the same filtration because if I have full knowledge of $W_1(t)$ and $W_2(t)$, I have full knowledge of $W_3(t)$ that can be adapted to the same filtration as $W_1(t)$ $W_2(t)$ the vector W and then $W_3(t)$ is satisfying this for all t. So, the quadratic variation accumulated up to time t is t and hence by Levy's theorem $W_3(t)$ is a Brownian motion.

So, though we had been telling that okay $W_3(t)$ would be a Brownian motion it is very intuitively clear that things can just be transferred. But you see to mathematically to be rigorous we have to apply Levy's theorem here to show that it is truly a Brownian motion.

So, that is why I have written the Levy's theorem up there, to show that it is truly a Brownian motion. So basically, what you have now is that, you have written $dS_2(t) = \alpha_2 S_2(t) + \sigma_2 S_2(t) dW_3(t)$. So are these processes now independent, when are the movement of stock price 1 and stock price 2 are independent of each other, they do not depend on the price of the other.

In fact, they are independent to the extent they are not independent because we will show you can show that there is positive correlation there is correlation the covariance between $W_3(t)$ and $W_1(t)$ is not 0.

Though their means are 0, but because covariance is expectation of xy minus exy ex into ey, you can show that the covariance, which I am not going to calculate actually, you can show that expectation of $E\left[W_1(t), W_3(t)\right] = \rho t$. So, you have a nonnegative for every t unless t is greater than equal to for all t greater than equal to 0 right. You have a nonnegative correlation coefficient. You have a non-0 covariance.

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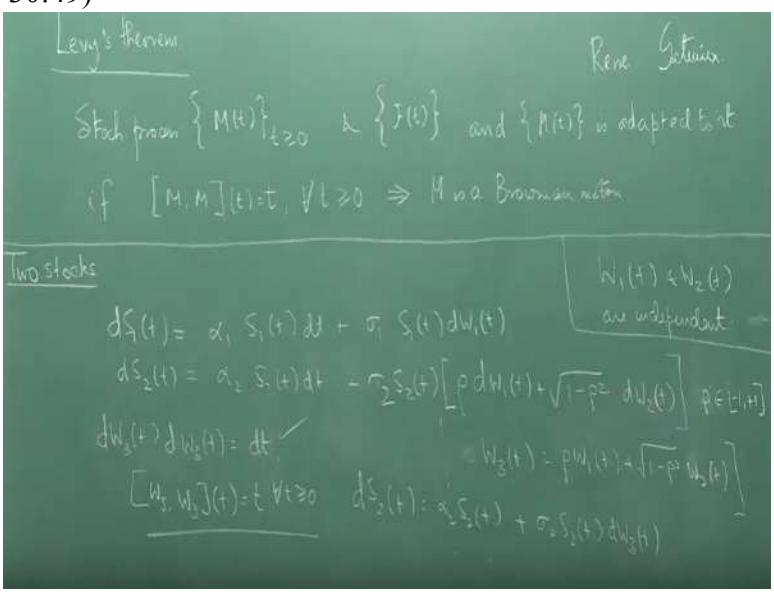

So, there is a relation between this and this. So, when you have two stocks you are essentially looking into a scenario where the stock price movements can be related to each other correlated to each other. It is not that one price is going somewhere another is going somewhere they are independent of each other.

They could be independent of each other if you possibly had just $W_2(t)$ here, then they are independent but once you involve say this sort of when I if both of them are multidimensional Ito processes like what we had done then stock price can get correlated. So, that is a much more

difficult situation and in finance, we try that is, that is not, that is something is slightly difficult to study and we will study it when we will talk about finance separately, separate course in finance so okay.

So, thank you for listening to the third lecture and then we are ending culminating in the last two lectures which is on the Black Scholes equation for which most of you must be waiting and I had myself learnt all these things, in the Black Scholes equation, and I would in this lecture I would ask my TAs to come up and join me to at least they should be able one should be able to see who are their TAs. Please come up for a minute before we start the Black Scholes lecture. Come on.

So here is Prashanth Jha, who is a PhD student in mathematics, who is working in statistics actually and she is Poonam Kesarwani, who is also a PhD student in mathematics, working in optimisation theory and both seems to have some good idea about statistics.

Maybe at the end of the course, after fifth lecture they can say, their own opinion about how they liked or enjoyed the course and we can end that. So, at least the viewer should be able to see with whom they are interacting. So, that is, that is quite important. So, it is very important to see the person with whom you are interacting, then things are things get much better.

So, thank you, go ahead. So, we will start the fourth lecture after some time. Thank you.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 19
Black Scholes formula-I

So here we are ending this course with the last 2 lectures but at the same time we are also initiating you into the world of finance. So this last 2 lectures is also an invitation to the world of amazing world of finance and when we do the course in finance we will do it very stepwise. We will do it do it in a way that a larger audience can come except the audience that you have here but we would all do these things.

These things would come and you really have to know about Brownian motion and all those things. So this course actually becomes a prerequisite or even if not a prerequisite you should be having access to this course.

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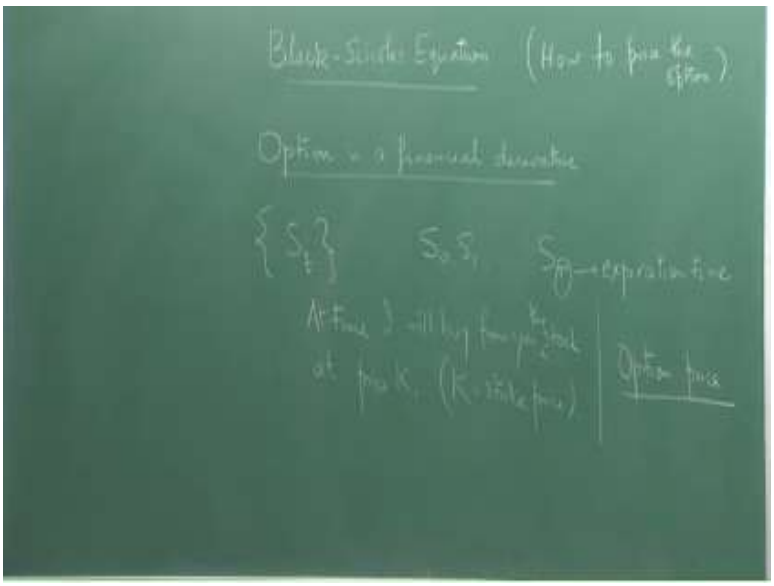

So everybody who did finance obviously wants to know about the Black Scholes formula or the Black Scholes equation. So this is Fischer Black and this is Myron Scholes. So we are going to derive a partial differential equation which has to be satisfied by the something called an option price but let us let the first lecture let me introduce to you what is the story behind all these things.

The story is the following that in a market there are lot of instruments people sell and buy right. For example I buy a stock of a company and then I say okay the stock price is increasing I sell it and make some money. This is 1 way. But I know when I buy a stock of the company I am also exposed to the risk that the stock price can fall.

There is always a time I can hold the stock too or the stock price can fall very badly and then some people want to get rid of that and get whatever little money they have. Nowadays banks almost force you to do some sort of investments where a partly or money would be kept in the fixed deposit and partly it will be traded in the market because in essentially the growth of the value of a stock if it grows is faster than the growth of a fixed deposit that you keep in a bank.

Now one of the most important derivative financial derivative is called an option. So a financial derivative is a financial instrument but its price depends on the primary instrument. For example the price of an option depends on the price of a stock. So option that is why option is called a is a secondary financial instrument it is called its price is derived from the price of a primary financial instrument like a stock.

So option and we will tell you what option is, option is a financial derivative. Remember option is such a strange financial derivative which people with not a great amount of money should ever think of actually going and using it. 95% of the option that are traded in the market are actually not realized, they are not operated and then options are used by big companies and cash rich speculators. If you are not cash rich do not bother about options.

The option is a thing like this. Suppose you have a stock and its price is $\{S_t\}$, so the stock you observe the stock price $S_0, S_1 . S_0$ is the stock price now. So you are speculating about some stock price S_T at a time capital T. So you come to somebody and get into an option contract with him, the seller of the option. You tell him that okay at time t the contract is at time t, I will buy from you a stock buy from you the stock at price K, K is called a strike price.

Means I agree in the contract that at time t, I will come to you. Suppose I have a stock so you get into a contract and you tell me at time t, I will come to you and buy a stock the share 1 share at a price K say. I said I take 1 share only do not bother about me. But there is an interesting feature of this option contract. You are not obligated to buy from me at time K at time capital T which is called expiration time this $T$is called the expiration time of the contract.

You are not obligated to come and buy from me the stock. Suppose the price of the stock so the interesting part is that you might see that okay I have got into a contract but now the stock has lesser price in the market so I will simply go the market and buy the stock at a price which is less than K, why should I come to you and pay a higher price.

See if I had been obligated then I would have to come at time T and pay you the higher price and buy the stock but since I am not obligated and you are not obligated to me and I am ready with the stock because I have to keep my stock price stock with me till time capital T to sell it to you. Suppose if the price is higher in the market than K the strike price K is lesser than the market price of the stock you definitely come to me and buy the stock.

Then what is me as a option seller who is selling this option this option is called a call option where I am selling the option and somebody is buying the option. There could be some reverse swing also but we are just bothered about call option at this moment. So suppose I am the seller of this option what is my use. So I have kept a stock in hold and now where because of the stock price is low you buy the market stock price stock from the market and you do not come to me.

So I would have no other option but to take a loss and sell the stock in the market at the market price. I was thinking I will get a money K. Now but suppose the stock price money is higher now because I am in contract with you I am obligated to give you the stock at price K at time capital T. So you come to me because you know the market price is higher you come to me and ask for the stock and I have to give it to you.

So in both cases I am at a loss. Then I could have sold the stock in the market at a higher price which I cannot because I am obligated to sell you at a lower price. So when I am so that is why

when I enter into this contract I ask you a premium that you have to pay in order to enter into that contract. That premium that you pay is called the option price. So this is not a free contract that you make and okay you are the boss and you can do whatever you want.

So but but when you make the contract you have to pay an upfront price which is called the option price. Now how do you determine that option price is the issue of Black Scholes formula. How to determine an option price in a very simplistic setting. So the whole game is to how to determine the option price.

So now at time $T = 0$ when you have come to me to buy and you are getting into a contract and your expiration day is time say capital T equal to t maybe out of 6 months then at time T equal to 0 what what price I should charge you. How to price the option, that is the question. So the question is what sort of price I should charge you. Now what should I do as a person. My goal would be that I should not be at a loss.

If I am selling you the instrument I am getting into the option contract I should not be at a loss. Then what should I do. The basic idea is the following. That I should ask you for some money. In this given period 0 to T, I will keep on trading with that money in the market buying and selling stocks, could be the same stock and then at the end of the time period T.

The worth of my portfolio that I create of buying and selling stocks the value of that portfolio would be exactly equal to the loss that I would have by selling the thing to you or the gain that you would have by taking the thing from me teek hai.

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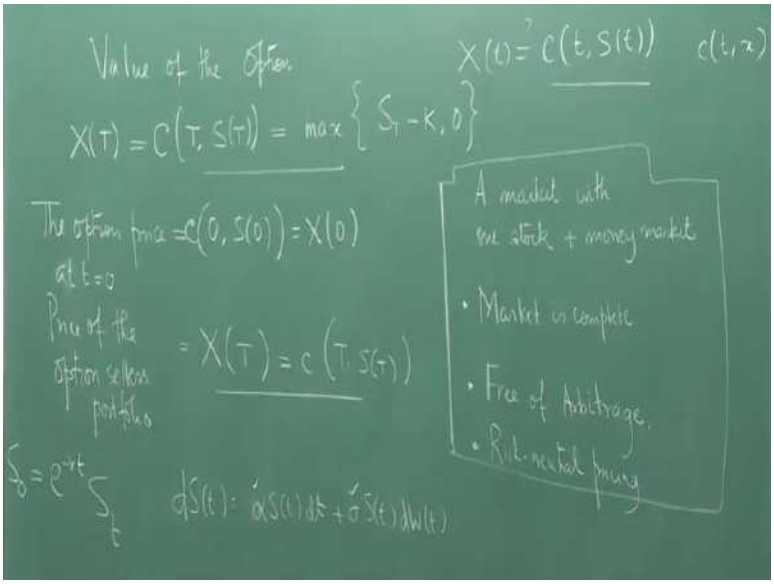

So there is something called the value of the option and that value depends on the stock price at the time T capital T. So what happens. So if at time capital T the value of the stock price is $S(T)$. I should write $S(0)$, $S(1)$because that is way I was writing the stochastic process. Please note that I should write it like this. T is the expression. So now what would be that value? So if the stock price is bigger than K then what would I do?

I will simply buy you, you will, if the stock price is K stock price is $S(T)$ and $S(T)$ is strictly bigger than K what you will do. You will simply come to me buy it at K and sell it at the market at market price $S(T)$ which is higher than K and make a gain of $S(T) - K$ and if the stock price is lesser than K you will not even come to me. So if $S(T)$ is strictly less than K or equal to K you will not come to me. So the value, the worth of the option $C(T, S(T))$ is the maximum between $\{S(T) - K, 0\}$.

The option price option price at t equal to 0 is nothing but the value of the or the worth of the option at time equal to 0 and that is $C(0, S(0))$. Our goal is to actually compute a function like this $C(T, X)$which would satisfy these sorts of properties. Under what condition can we do such computation. That is the question and it is very important to know what would I do by this $C(0, S(0))$. Suppose you pay me that $C(0, S(0))$, I would trade in the market.

Some money I will put in the risky stocks some money I will put in the bank and then I will so I have a portfolio of some stocks and bonds and stocks and fixed deposit and then I will see how my value grows I keep on trading in such a way that if $X(T)$ is my $X(T)$ is my the X determines the price of my portfolio as a seller I am doing that price of my price of the seller's portfolio, the option sellers portfolio.

So $X(T)$ price if I do not want any loss must be equal to your gain $C(T, S(T))$. In what sort of situation did of course a person can buy option on 30 stocks but here we will take a very simplistic view of 1 stock. Now in what sort of scenario you can actually find this price. You have to understand that looking at all the complex interactions of various stock prices etc., and trying to find various option values for various options.

The basket of options that you have bought is a very complicated thing. So when you model you cannot model the exact reality but you try to do some modeling try to do some sort of thinking in a way that you take the most simplistic situation and look at it as which is a as if it is a part of the more complex situation and then try to interpret that more complex situation using that simplistic idea.

So what these guys had done is they had considered a market a financial market with 1 stock plus the money market is bank or a bond 1 stock and 1 bond market. Money market means you have kept it and there will be a fixed interest rate constant and at after certain time T you will get that interest rate and in the Black Scholes world you have to assume that the market is complete. I will tell you what is the meaning of this.

And the second point is that the market is free of arbitrage and we will come to tell you what is that. In the finance course we will deal much much more details about these things, free of arbitrage. So what is arbitrage? So arbitrage is a very interesting thing. Arbitrage is making money without spending a penny, making money without spending a penny is called arbitrage.

Example, example is that suppose I go to a shop buy 10 pens of 8 Rs. each and then I come and sell among my friends each pen with 12 Rs. each. So how much money I get from my friends

120 Rs. So I have recovered my 100 Rs. I have I can take put it back in the same place so I have not made any investment actually and I have just generated 20 Rs.

But the markets immediately notice in real markets immediately notice that there is some difference of price. So that is what mainly arbitragers do in the market. They keep on for example the they keep on looking at the stock prices of some particular stock at BSE and NSE National Stock Exchange and the Bombay Stock Exchange and if 1 price is lesser they will buy it there and sell it somewhere at a higher price.

But this arbitrage condition remains for a very short time because other people start noticing it and the prices are leveled. So in levelled situation of the market the market is actually in no arbitrage situation. So the modeling has to be done in a no arbitrage condition. I and I have seen this arbitrary business even in doing in railway platforms in 1 platform I was standing and I wanted to buy cold water and someone the guy said okay this railneer bottle should cost you 12 Rs. I said but the official cost is 10.

I did not do anything I went to the other platform and I got it at 10 Rs. I came to him and asked him for a bottle. He said 12 Rs. I said I told everybody surrounding there that you see I have got his bottle at 10 Rs. I want more 1 more and he is asking for 12 Rs. You go to the other side you will get it at 10 Rs. People started leaving. So he said okay okay dus rupiah. So the price gets leveled like that.

So that is that is the sort of thing and market is complete is the means the following. See when you have taken money the option price and you are now starting to actually trade in the market so that you make up your loss and equate it to my gain. A market where this can always be done where you can always hedge, hedge means safeguarding yourself. You are exposed to the risk now, you could lose money right because you have taken the contract.

So you have to have that money to use it in the market so that you can completely hedge off your risk which is exactly this. So a market which where you can do this which allows this is called a complete market and markets are in general not complete. So that is why it is not always the fact

that whatever you compute in mathematical finance should be actually used in the market. You really have to think a bit and do it.

And when we do the option pricing very slowly step by step we will talk about these things more clearly and you will see where the arbitrage condition actually been used. Here it is not no time to really use that here. We are essentially trying to use Ito-Doeblin formula. So if this can be done always such a market is called a complete market and a Black Scholes formula is computed in this setting and his setting.

The stock has a stock price has satisfies the following differential equation $dS(t) = \alpha S(t)dt + \sigma S(t)dW(t)$, αis called the drift and σ is how much is the stock price varying captures the variation basically. So these are all constant numbers α and σ. σ is always greater than 0. Now, how will you suppose you have taken this money from me which I say the initial value of my portfolio. So I have taken this money from you right. I am the seller of the option thing and you have come in.

So you have paid me this money which I now say they are at 0. What I am supposed to do now is the following. I have supposed to start using this money $X(0)$ and start trading in the market, but my market now has 1 stock in the money market the bank. So what I do a part of the money I invest in the same stock because I have to buy that stock. I do not really have that stock. I have to buy that stock from the market.

So I am interested in buying that stock and some part I put in the bank in fixed deposit. Then I at some point I sell that stock again so I see how much money I will have the interest from the fixed deposit at that time and add it and with that I will again buy partly that money would be invested again into buying the stock and again sending back some money to the money market.

So this will go at every step so that at the end my $X(T)$ my price should be this should equal to should be exactly equal to the option value. So this is exactly what is the meaning of a complete market. So the bank will give I will assume will give me a fixed interest rate r. I will assume that during the time 0 to T it will not, 0 to capital$T$ it will not change its interest rate which is not

really true but okay. So these are the you know you are making the idealizations which do not actually happen in real world and it is not always possible to hedge it.

Hedging is this is this is a very once you are talking about hedging you are essentially this sort of pricing is called risk neutral pricing. In general when you are exposed when you have invested in a stock you are exposed to more risk than you are than you are investing in a fixed deposit. That is the fixed deposit interest rates are low because that is guaranteed because that is a sure shot income that you would have.

While this interest rates on the stock are higher they grow faster because you are exposed to the risk of having a loss. There is a downside risk which is not there in the case of fixed deposit.

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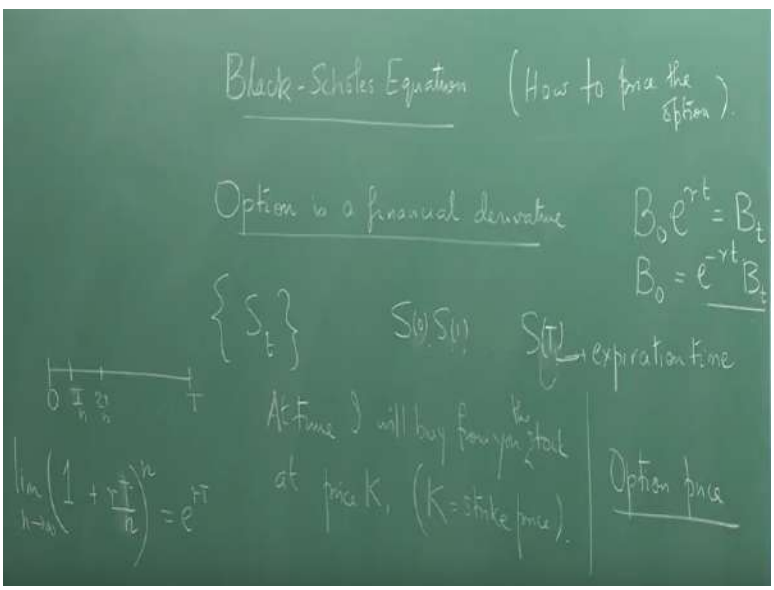

So we would start trying to learn that suppose I have this time interval 0 to T divided into small small intervals each of size $\frac{T}{n}$ okay and I have invested 1 rupee at time 0 which I will take you at that at the end at time capital T, I will take it from I will get something from you. So each I have divided into the interval $\frac{T}{n}, \frac{2T}{n}, \ldots$ and so on, the partitioning of the interval. So if I want to get some money at $\frac{T}{n}$ how much money I will get.

I have invested 1 rupee. My interest rate is r and my time duration so how much is the time duration. My time duration is. So if you repeat it for the next this $\left(1 + \frac{rT}{n}\right)$ is called the compound interest. If you repeat it for the nth interval you will have this $\left(1 + \frac{rT}{n}\right)^n$ because you had n partitions. Now because so r into 1 rupee so and $\frac{T}{n}$ so r is the interest rate over the whole interval. So in the interest in the interval T by n you will get a interest rate not the full interest rate.r is the interest rate I will give you only if take the money at time capital T.

But if you want to take off the money at time $\frac{T}{n}$ then you I will not give you the full interest rate. Of course, why should I give you the full money because I take the money from you to invest somewhere else. That is the whole game hmm. So what do you do. You give a money proportion to the interval of time. So you give $\frac{rT}{n}$. So $\left(1 + \frac{rT}{n}\right)^n$ into 1 rupee. So that if you repeat over n intervals you will have this.

So when you have a continuous system like our state scenario that we can trade at each and every movement of time. Actually we do not do that. But in reality, we can trade at every movement theoretically. So this $\left(1 + \frac{rT}{n}\right)^n$ is nothing but e^{rT}. So this is actually my instantaneous interest rate. So if I put B_0 amount of money in the stock market in the bank now then after time small t, I am supposed to get a money of this $B_t = B_0 e^{rt}$ amount.

So what is B_0. If you look at B_0 which is the price when amount of money I have kept now right. This is nothing but $B_0 = e^{-rt} B_t$. So this sort of pricing is called the discounted price of the fixed deposit at time t at time 0. So if B_t is the amount I should get at time small t what is the amount I should deposit now. Suppose I want to make this amount of money after 5 years.

What should I money I should now put in the fixed deposit given my interest rate is fixed R. This is the and I can means it is a continuous time interval. Is not that okay 0 to t fixed interval, ithna

diya gaya and $(1 + rt), r \times 1$ that is that is all. It is not that it is $(1 + r)$ just it is not like that. So then this will tell me that if this is the amount of money I require B_t is the amount of money I require this is the amount I should invest.

So this $B_0 = e^{-rt} B_t$ is called the discounted price at time t. So we will be interested in the discounted price. The interesting fact is that the discounted stock price if you talk about the S_t is the stock price at time t. So $e^{-rt} S_t$ is the discounted stock price. So what should be the stock price at time t so that I have current I have the current stock price is maybe if my current stock price is S_0 .S_0 should be $e^{-rt} S_t$ s.

Now you might ask me why are you putting r here. The stock price growth is higher. Stock grows at a faster rate. The interesting thing is that in the pricing structure of Black and Scholes we also do something called risk neutral pricing. So essentially you are not bothering whether you are investing in a stock or investing in a bond. So there is a huge amount of simplification which does not work in the market.

Actually because you want to get a understanding of how to price. So even with that the thing looks very complicated. You see so you are telling that the stock and bond you are is growing in the same rate. So you are neutral to that investment whether investing in stock or whether you are investing in a bond. Our next talk, so you have some very basic idea of what we are going to price.

Our next talk in the next and the last talk we are going to see that how we are going to get the discounted values stock prices and the discounted portfolio values and see how they evolve and we will see that we will say that every time t my value of the option $C(t, S(t))$ means if you do not buy between time 0 to capital T, if you do not come to me at time 0 but come to me at time small t which is lying between 0 and capital T then $C(T, S(T))$ should be the money you should pay me because that money I will use to and that should be at every time should be $X(t)$ essentially.

The question is in the next class we are going to figure out a differential equation, a partial differential equation which would actually solve whose solution would give me this function. Basically we have to find some function $C(t, x)$ where x is taking the place of the stock price where $C(t, x)$would solve some partial differential equation.

Of course you might ask me why do you not give me the chance to write down the exact Black Scholes formula. I would not do that now because then the fun would be gone then we would still wait for starting the actual course in finance which will give you much better which will have more siding with finance. We will do the math. We will also have a lot of pep talk on finance. So with this we end the 4th lecture of the 4th week and then in the next class we are going to actually compute and get the Black Scholes differential equations whose solution is actually the option price at any time t.

See essentially we need to know the option at any time t and knowing that this function is continuous we will get the option at time 0. Thank you. We will start quite soon.

Probability and Stochastics for finance
Prof. Joydeep Dutta
Humanities and Social Sciences
Indian Institute of Technology, Kanpur

Lecture - 20
Black Scholes formula-II

So we continue from what we were doing in the last class. We were talking about how to get an hang on the option price at any time t at a for any stock price $S(t)$. So we will start to see what should I do when I get the premium because I had I had been taking the role of the person who has sold the option and you the viewer is the person who has bought the option from me. So such an option is called a call option where you are buying the option not selling the option.

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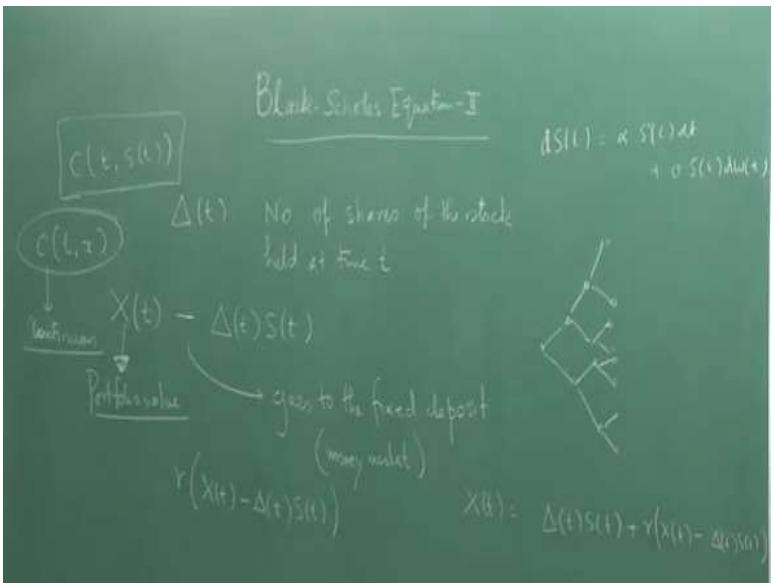

So now what I do let this is $\Delta(t)$. This $\Delta(t)$ is the number of shares I am holding at time t. So this is the number of shares I am buying of the stock that I have. There is 1 stock in the market and 1 bond 1 bank basically bank account. So number of shares of the stock held stock held at time t. So if $S(t)$ is the cost of 1 stock at time t on 1 share of the stock then total worth of my stocks is $\Delta(t)S(t)$ is the total worth of my stock.

If I sell the stock at that time this is what I will get. So my portfolio value so suppose now you have at time t I have $X(t)$ amount of money and I now buy $\Delta(t)$ amount of stocks paying $\Delta(t)S(t)$ money. So $X(t) - \Delta(t)S(t)$this is the amount of money I have left. So I came. So at

time t suppose I had $X(t)$ amount of money I bought this amount of stocks shares of $\Delta(t)$shares I bought of the stock each share costing $S(t)$so this is the amount and this is the amount left and this is the amount I am going to send in the bank.

So this $X(t) - \Delta(t)S(t)$ amount goes to the bank goes to the fixed deposit say fixed deposit or money market also this language is used in finance. Now at time t once I have invested how much would be my interest rate right. From the interest I would have the following earning. So on this amount of money this$r(X(t) - \Delta(t)S(t))$ is the amount of interest I have been given. At time t this is exactly the amount of interest I will have because that is the when he is calculating the interest from time 0 so at time t this is the exact amount of interest I will have.

This is the amount of interest I will have. Of course if I take back the money then I will have this plus this but just the interest I have is this. So I am just taking the interest. So basically, what is my $X(t)$. $X(t)$ consists of this plus this. So at any time $X(t)$is nothing but my any time my $X(t)$ should consist of how much money I have spent for the stocks plus or at basically see the idea is this. I am buying selling instantaneously.

I am selling and buying at this very moment which is very strange thing does not happen but the time interval could be very small so that you can take it almost at the same time. So basically I am holding the thing for a very short time so I have employed this money and after that very same time again taking back the interest on that money and selling the stock that I had held. So then I sell the stocks at this and get this amount of money plus I add this money which I get from the interest.

So this is very strange idea that you have instantaneous selling buying. Instantaneous selling buying does not happen in the market. You could sell now buy now and sell after 2 minutes. But we assume that the gap is so small that that is what you are doing an instantaneous selling and buying. So these things take time to seep in. So when you do the finance course gradually you will see that why we are talking why the model is so. We will do it very slowly.

I think I will plan a much longer course slowly. So that is our better idea of the financial aspect. I am not talking about the financial aspect at this moment. I am just telling that that is what can happen. So if you write the differential of this whole thing then what is this?

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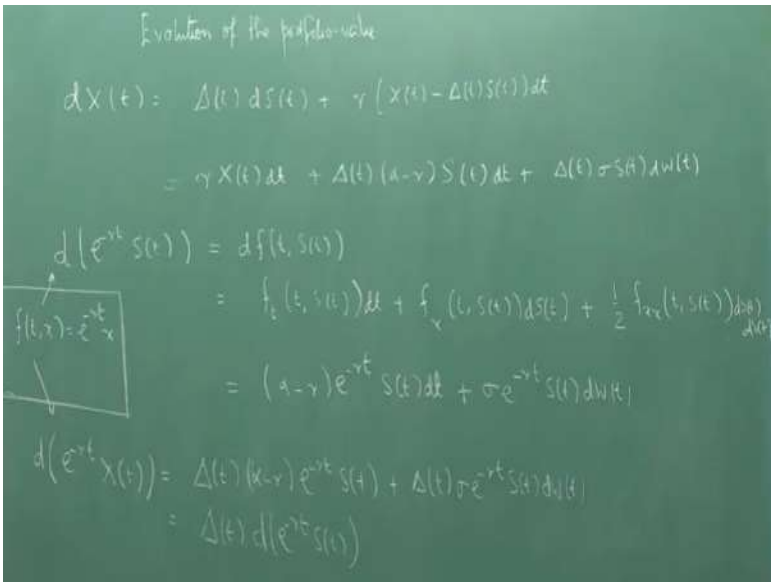

This is $dX(t) = \Delta(t)dS(t) + r(X(t) - \Delta(t)S(t))dt$. That is the differential of the stock price because $X(t)$ right$X(t) = \Delta(t)S(t) + r(X(t) - \Delta(t)S(t))$ itself is so writing the differential in that form basically. So at time d t what is the change in the interest price? At unit time the change in interest price is this. At time d t the change is this into d t. It might not be so easy to understand this unless we have seen discrete time analysis of the financial market.

That will be done in the next course. Discrete time is now you have time 0. At time 1 the price could either go up or go down. If it goes up from there it can again go up or go down like this. It is a binomial tree. So here here here and here here just like this. And actually, from there you could actually get a lot of ideas. So unless you do the binomial studies it is not really feasible to give anyone a better idea of what actually is happening.

But assume that this is the scenario basically. So then differential of this is given like this. So this $X(t) - \Delta(t)S(t)$ is the amount of interest accumulated at time dt. At every unit time the accumulated interest is r into that and at time t is and at time dt. $\Delta(t)$ is the fraction that is

accumulated. So this can be now written as opening of the $dS(t)$ actually which you know. If you have forgotten it I just make you remind you that $dS(t) = \alpha S(t)dt + \sigma S(t)dW(t)$

So this$dX(t) = rX(t)dt + \Delta(t)(\alpha - r)dS(t) + \Delta(t)\sigma S(t)dW(t)$ is the differential when I open up $S(t)$. Now you would be concerned not with just the portfolio value but by the portfolio the discounted portfolio value because I am taking decision at time 0 about what can happen at earlier times or at later times. So I want to look at everything from this point of view, current point of view. So what would be the price of my portfolio now so that after time say t the portfolio value is say $X(t)$.

So we will look at the discounted portfolio value or we will also first look at the discounted stock price and write down its differential equation or stochastic differential form and then we will also look at the discounted portfolio value. So basically, we are going to look into the differential of this $d\left(e^{-rt}S(t)\right)$ first. In order to do this we consider the function of the form $f(t,x) = e^{-rt}x$. Now we will again apply the Ito's formula. This will be nothing but $df(t,S(t))$because f(t) if you put $S(t)$ here it is this which is$f(t,S(t))$say again so again I am applying the Ito's formula $f_t(t,S(t))dt + f_x(t,S(t))dS(t) + \frac{1}{2}f_{xx}(t,S(t))dS(t)dS(t)$.

So you will see how Ito's formula is being applied where it is the quadratic variation term. So $dS(t)dS(t)$. So you see you start learning the use of of course you can find the quadratic variation term from here. So you know that will be nothing but $\sigma^2 S^2(t)dt$ and then you analyse the whole thing when you analyse and put $dS(t)$ here and write down the quadratic variation here and of course maybe here if you take the second the first derivative would give you e^{-rt} with respect to x, the second derivative is 0.

So you can basically forget this part and just have this part and when you write down what is d S t from here you will finally get the following expression which is $(\alpha - r)e^{-rt}S(t)dt + \sigma e^{-rt}S(t)dW(t)$. This is what you are going to get. Now once I know that can I now because I have $S(t)$ in this expression. Can I now know what is how my portfolio value is evolving? So, both for this and this the Ito's formula is applied using the same function.

Only instead of X in my case now when I am looking at the portfolio value I have to write $X(t)$ in the case of stock price I wrote $S(t)$. Here $X(t)$is my portfolio value. Just if you have portfolio value so I have a portfolio. My portfolio consists of some shares number of shares of a given stock and the amount I put in the fixed deposit. That is my portfolio I am holding at every time t and I am instantaneously selling buying. That really does not happen.

The selling buying happens at various periods and when the periods become very very small they tend to the towards the continuous case. So we in mathematics we always try to look at the limiting case because that is much more easier to deal with. In real life there is no such thing. It has to be only done at discrete period. You do at time after 5 minutes also it is not really instantaneous but okay but for understanding the real the whole process continuous things are much more easier to handle.

So this $X(t)$ is also called the portfolio value. So if I look at this I will go ahead and apply the same thing and then you can prove that this is nothing but $\Delta(t)(\alpha - r)e^{-rt}S(t)dt + \Delta(t)\sigma e^{-rt}S(t)dW(t)$. See if you look at it very carefully if you take the $\Delta(t)$ common outside and this is nothing but $\Delta(t)$ and if you look at this expression so there is a beautiful relation between these 2 differentials.

So the differential of the portfolio value is nothing but of the discounted portfolio value is nothing but the number of shares I am holding into the differential of the discounted stock price. So this is exactly what it is okay. Now I have this fact. Now I have to see how my option value is evolving. Option price I have already mentioned in the last lecture is a function of the form $C(t, S(t))$.

At every time t that should be my that should be the premium you should pay me to buy the when I come into the contract at time t. The expiration time capital T remains the same whether you buy at time 0 or time when I capital $\frac{T}{2}$it does not matter. So this is so essentially I am going

to look I am going to seek to find a continuous function $C(t, x)$ that should be my goal. That will and of course again I am writing continuous.

So what we now do is we now try to see how my option value evolves. So what we have here observed is the evolution of the portfolio value and then I am going to observe the evolution of the option value and what I had studied there was evolution of the portfolio value.

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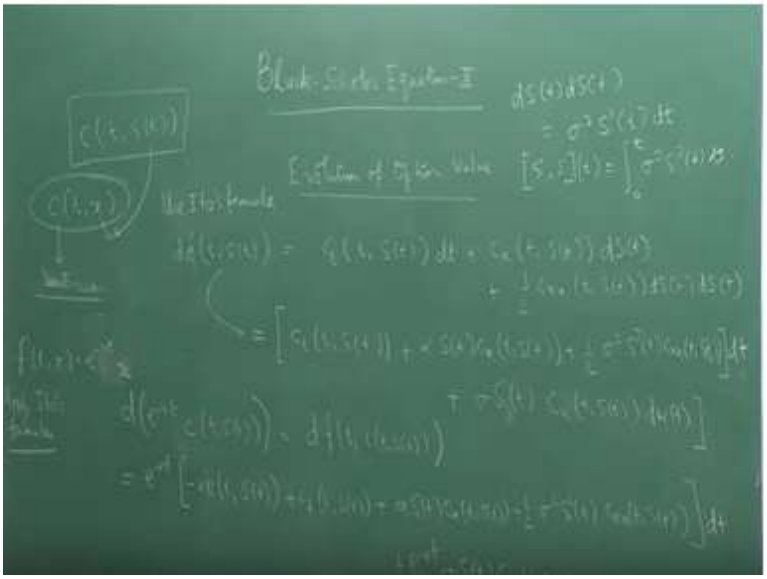

You might understand that why I am using the Shreve's book. Ya I could have run it or if I had to really do it it would have taken a long time and cannot be covered in the lecture. So I am just picking up certain parts from the book and trying to give you more explanations right. So this is to explaining more clearly which cannot be understood if you just for the first time go and read Shreve's book.

This book means to be read a lot of times to really understand what is going on. So the idea is this. If I can find the way my option value is evolving I want my portfolio value and option value to be evolved in the same way. Then only I can hedge the risk. So then I will equate the differential of these 2 and from there I will generate the equation by which c can be obtained. So actually, here I am looking at the function $C(t, x)$where x is playing the role of s t right.

So x is playing this role role of $X(t)$. So if I look at now so now again by Ito's formula so you will immediately understand so this so use Ito's formula so you see lot of finance is largely about how good or how well you are able to apply Ito's formula. So if you look at this thing this is of course $de(t, S(t)) = C_t(t, S(t))dt + C_x(t, S(t))dS(t) + \frac{1}{2}C_{xx}(t, S(t))dS(t)dS(t)$, $C_t$means of course derivative with respect to t.

Now you see here you cannot put this is equal to 0 because we do not know the form of the function. That is what we have to find out. We will be led to a some second order partial differential equation. We really have to find out what is this what is the c. So we do not know. So we cannot throw away this term. So you have to use the d the variation this quadratic variation. So quadratic variation what would be the quadratic variation of this? $dS(t)dS(t) = \sigma^2 S^2(t)dt$, $dW(t)dW(t)$which is dt.

So this $[S, S](t)$ is the quadratic variation. Essentially if you want to write it more clearly or in a more proper fashion this is $\int_0^t \sigma^2 S^2(t)dt$. So instead of so now if I put all those values all those things putting $dS(t)$ and opening up $dS(t)$then finally I get a form like this. $C\ de(t, S(t)) = \left[t(t, S(t))dt + \alpha S(t)C_x(t, S(t))dS(t) + \frac{1}{2}\sigma^2 S^2(t)C_{xx}(t, S(t))\right]dt$ this whole thing into dt maybe I should write it slightly on the side so that you get a get the writing clear. So I will write.

So this now equal to the same thing is now equal to so I have now after all the simplifications are done just plug in the $dS(t)$check at home and put this value here $\frac{+1}{2}\sigma^2 S^2(t)C_{xx}(t, S(t))dt$ This is the dt part. So this is the drift part now of this Ito process. Everything is viewed as Ito process something that has to be remembered. Some sigma S t sorry $\sigma S(t)C_x(t, S(t))dW(t)$okay. That is what you will have.

Now we have essentially spoken about the discounted stock price. See discounted the stock price itself is not a Martingale. The discounted stock price is a Martingale. So if you have a discounted

stock price you have much more stronger properties of the stock price. You can have you can have more information about it.

So we are always working with discounted stock price, discounted stock because we want to know at present what is the when a I am evaluating everything from the present scenario. So at present what is their value? So basically now I have to talk about so I am expecting the stock evaluation of the see the stock and money market evolves in the same way because the stock the option price depends on the stock price it will also evolve in the same rate.

The rate at which the stock grows the option price will vary accordingly. So we are going to look into, so without getting into, again applying, now the Ito's formula with now in this case you take $f(t,x) = e^{-rt}x$ and instead of x you put $C(t,S(t))$. This now the formula that you have to use to apply Ito's formula the f that you have to use is of this form where x you put $C(t,S(t))$..

So basically, it is$d\left(e^{-rt}, C(t,S(t))\right) = df(t, C(t,S(t)))$. and then if you actually compute the whole thing out then you will have a very rather a very complex looking stuff. Of course, because we are at the end of the course some complex looking stuff would be a good thing but no mathematics is essentially a simple thing but okay this has its own thing so this is equal to if you apply the Ito's formula. So now apply Ito's formula apply Ito's formula and then you write the following thing sorry $-\ rC(t,S(t))$.

So there will be a d t part so I am writing down the d t part. I am not doing this calculation because it will take too much then it will take I think 1 lecture would go more than 1-1/2 hours or 2 hours. Basically, when you do this you basically use again this formula here to open up the whole thing to get this final form $e^{-rt}\left[-\ rC(t,S(t)) + C_t(t,S(t))dt\ +\ \alpha S(t)C_x(t,S(t))dS(t) + \frac{1}{2}\sigma^2 S^2(t)C_{xx}(t,S(t))\right]dt\ +\ e^{-rt}\sigma S(t)C_x(t,S(t$ .Of course, you can ask me what is the real use of sigma.

These will come in the finance course where we really talk about these objects in a much more detailed way. Now what is the key idea. Let us now discuss the key idea. So we will keep this

formula at hand which is this evolution will not require we will just remove this. So we want the portfolio the evolution of the portfolio value and the evolution of the option value that or rather the evolution of the discounted portfolio value and the evolution of the discounted option value are the same.

Their nature has to be same. So we can really say that at the end what we want. We need we need a discounted thing.

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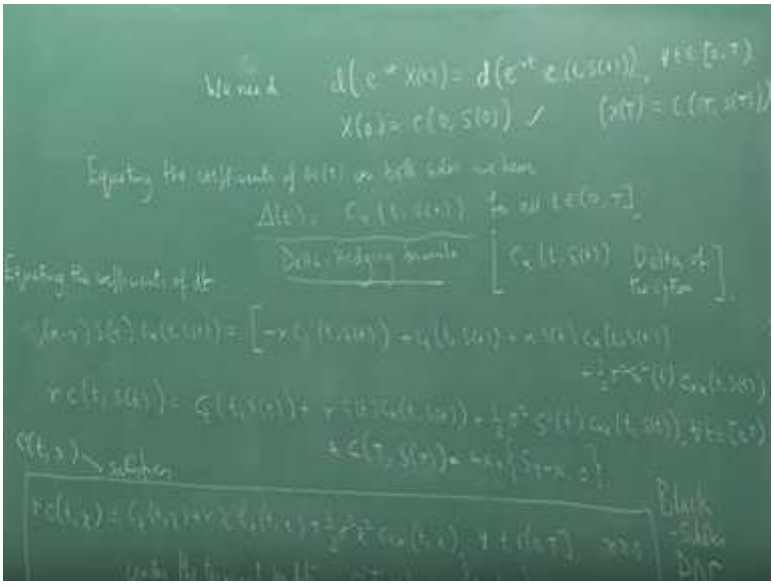

See what we are looking at is the intermediate values from time 0 to time capital T but capital T is not included there because at capital T I know what will happen because the market is complete I will be able to completely hedge. I will be able to have c capital T $S(T)$is equal to the max of that. So it will be exactly equal to $X(t)$. At that time at time t, I know exactly what is happening. So I do not have to bother about time t.

What I want that in between they should behave in a very nice fashion. They should exactly as the same behaviour. So that is some sense telling okay okay possibly if they when if it has to match at time t possibly it is a better idea to make the match at every time. So we need this $d\left(e^{-rt}, X(t)\right) = d\left(e^{-rt}, C(t, S(t))\right)$ idea. We need this idea. So basically, I want to equate this with you want to equate with this. How do you do that equation?

Of course, you know that $X(0)$is the price that you have $X(0)$ is the amount of money that you had taken in the beginning from I have taken in the beginning from you which is this$C(0, S(0))$ and $X(T)$ of course should be and $X(T)$ should be $C(T, S(T))$ and this is already guaranteed by your complete market hypothesis. This has to happen. I should be able to completely hedge. Completely offset my risk. That is the main idea.

Now let us do one thing. See if I start hedging now if I start equating so I equate this with this. So what would I get. I would like to first equate the random part with the random part and a volatility part with the volatility part and the drift part with the drift part. So if equate this part with this part then what I get is so equating the coefficients of $W(t)$ on both sides we have what do we have? We have the $\Delta(t)$.

We have $\Delta(t) = C_x(t, S(t))$. So here I want this to happen for all t element of starting time to capital T. This is already known. So what do you get? You get this. This is called the delta hedging formula. This is one of the Greeks of the option. This is called the delta hedging formula. So this formula is called the delta hedging formula. What does it say?

It tells you at every time cap T if you know the option price I can tell you how much stock shares of the stock you have to buy. So if I differentiate this function and at time 0 I will know what is the starting amount of stock I should buy. See if I am able to compute the function $C(t, x)$or $C(t, S(t))$by differentiating $C_x(t, S(t))$ and putting the appropriate time in I will know how much stock I need to buy at time t.

So that is why this is a very important formula. This is actually used in practice. This is of course the implementation is slightly different. It is called the delta hedging formula. Now we this $C_x(t, S(t))$this thing is sometimes called the delta of the option because this is the equal. Now we have to equate this part with this part and e to the power so equating the d t parts the coefficients of d t. So you know e^{-rt} will get away from both the sides and you have to equate this part with this part. I am not writing it again in detail because then you can check it out.

So if I do that after doing some manipulations this part will get cancelled you see this part will get cancelled with some part you have $S(t)$ here you have $S(t)$ here e^{-rt}will be taken off. So you will write $(\alpha - r)S(t)$ and $(\alpha - r)S(t)$. So I am just equating this part the dt part sorry there should be a dt here. Please note there should be a dt here.

If I missed it in the earlier thing please make it sure. So I am equating this part. So $(\alpha - r)S(t)\Delta(t)$ should be equal to this whole thing. Now if I do that so what I would have but $\Delta(t) = C_x(t, S(t))$. So I can write $(\alpha - r)S(t)C_x(t, S(t)) =- rC(t, S(t)) + C_t(t, S(t)) + \alpha S(t)C_x(t, S(t))dS(t) + \frac{1}{2}\sigma^2 S^2(t)C_{xx}(t, S(t))$.

So we have equated the dt part by knowing that $\Delta(t)$ here is this. So you have found out $\Delta(t)$ using the other one and then you have done this. Now if you look at it very carefully $\alpha S(t)C_x(t, S(t))$ here you also have $\alpha S(t)C_x(t, S(t))$here same thing. So when cancel that out and you can take$- rS(t)C_x(t, S(t))$ to that side and keep this the other minus to this side so you will have an equation of this form $- rC(t, S(t)) = C_t(t, S(t)) + rS(t)C_x(t, S(t)) + \frac{1}{2}\sigma^2 S^2(t)C_{xx}(t, S(t))$.

And this should be true for all t between 0 to capital T and you know that with the fact and c T S capital T. $C(T, S(T))$ is the max of $\{S(T) - K, 0\}$ because that is exactly the hedging that I have to offset my risk. Now which means that the options price at every time t is can be found by trying to find a continuous function $C(t, x)$ where $C(t, x)$satisfies the following partial differential equation$- rC(t, x) = C_t(t, x) + rxC_x(t, x) + \frac{1}{2}\sigma^2 x^2 C_{xx}(t, x)$. $S(t)$ is playing the role of x.

So this is for all t element of 0 to T this equation must be satisfied and x must be greater than equal to 0 because the stock price must be nonnegative and there must be a terminal condition which is this and the and under the terminal condition under the terminal condition and under the

terminal condition the following this which is $C(T, x)$. T is there terminal time which must be equal to max of$\{x - K, 0\}$, where x is actually S of capital T right.

So this thing the last this one is called the Black Scholes PDE or the Black Scholes equation. Of course what is this price. So we will show in our finance class finance lectures that the price at time 0 can be evaluated not just by solving this equation because solving this equation numerically is a huge piece of research.

A lot of work has been done how to solve the Black Scholes equation. It can be always written and converted into a type of heat equation and then try to solve it but lot of work has been done on the numeric of it. But there are other ways to forget the partial differential equation and using the risk neutral idea and doing something called the Girsanov's using something called change of measure going from one type of Brownian motion to the other using Girsanov's theorem.

And using basic probability theory one can actually compute what should be my price at any given time t, $C(t, x)$can be computed. So that can only come in a finance class where lot of lot more information about finance has been given to you. So it is called the Black Scholes partial differential equation PDE and with the as promised we have ended our discussion by coming to the Black Scholes equation and this ends the course and hoping that you have enjoyed it even if you have not enjoyed it you have got some information.

In a 10-hour course it is not really just it is not one cannot do too much justice to this huge fascinating subject called stochastic calculus. You think that you are sitting down there and you are really understanding what is going on no because possibly I also do not understand what is really going on. So you might ask me that this equation is in almost everywhere sense, yes but this equation when you do it is not really almost everywhere sense. It is the standard PDE.

So hope that you have enjoyed here and hope that many of you would join the finance class but which will be done in a much simpler straightforward way and which will be done in a with not coming to these equations, not too much bombarding you with Ito's formula.

The Ito's formula would require in some cases but essentially giving you the basics of finance. There finance would play a major role, the financial intuition will play a major role and also you will see how the mathematics can be naturally evolved out. That would be the course. But many tools from this course would be required in that course and that would not be repeated here.

We will come with the assumption that you have certain understanding. So with this I thank all of you for being a part of this course and I hope you had enjoyed as I had enjoyed in giving these lectures. Thank you very much.

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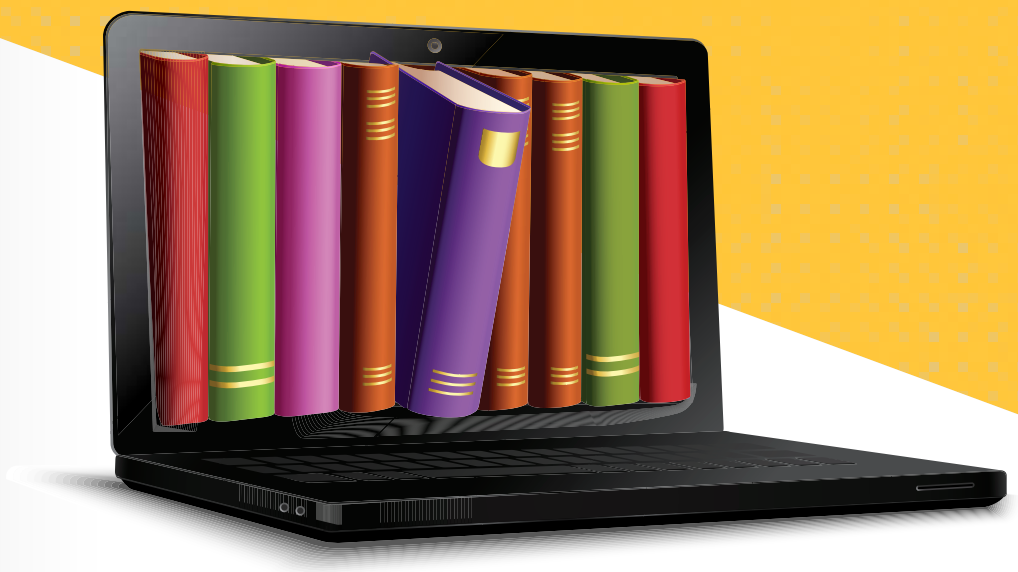

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